Bilderwand – layout candidates

9 search paradigms, 144 layouts

25 October 2026

What this document is

The 28 picture frames listed on the right (measured outer dimensions, orientation as listed) are to be hung as one collage of about 180 × 120 cm (± 10 cm) with gaps as uniform as possible and, ideally, a sharp rectangular outer border. Not all frames have to be used.

The same search was run with 9 different settings (“paradigms”: free-gap, per-run-gap, proportional, gap-3, linear-cost, free-gap-s1, large-apart, entropy, large-entropy). Section 1 explains what distinguishes them and compares their best layouts, section 2 ranks all 144 distinct layouts of this document together, section 3 covers the sub-analysis with large-frame constraints and the entropy reward, and the remaining sections show every paradigm’s 16 best layouts, one per page at scale 1:10, with frame labels, red annotations for gaps that deviate from the nominal gap, and a coordinate table for hanging.

Reading the numbers

Nominal gap: the spacing the layout is built on. *Max. / mean deviation*: largest and average difference between an actual gap and the nominal gap, in cm. *Notches*: total length of the outer border that is not covered by a frame or a regular gap. *Corners*: how many of the four corners of the bounding rectangle are filled by a frame. *Score* = max. deviation $+0.5\cdot$ mean deviation $+0.15\cdot$ notch length $+2$ per missing corner (lower is better). Layouts are ranked by score rounded to whole cm, then by number of frames (more is better), then by exact score, then by distance from the target size. *Entropy*: irregularity of the constellation in $[0, 1]$ (mix of formats, different neighbours, staggered edges, few through-lines; see the sub-analysis section). *Large frames*: the 52.5 × 42.5 and 42.5 × 52.5 frames; a layout page says how many are used, whether two of them touch and how far apart the closest two are.

Inventory

	outer w × h (cm)		have
	20.5×15.5	L	9
	15.5×20.5	P	2
	32.5×42.5	P	2
	42.5×32.5	L	1
	32.5×23.5	L	5
	23.5×32.5	P	1
	25.5×25.5		5
	42.5×52.5	P	1
	52.5×42.5	L	2

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1 Search paradigms

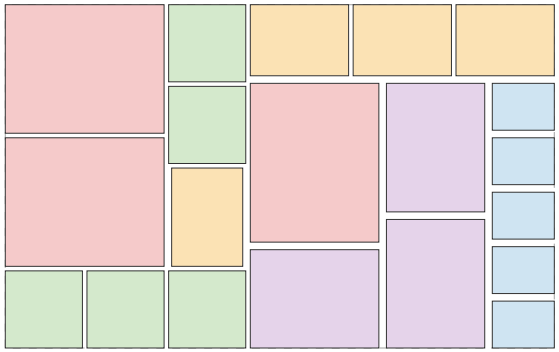
All paradigms share the same model and search. A layout is a *slicing tree* over a subset of the frames: frames are recursively placed side by side or on top of each other with a nominal gap. Where two siblings differ in height (or width) the smaller one has slack, which is absorbed by widening the gaps inside it; if it has no inner gaps the slack becomes dead space, which counts as a notch when it touches the outer border. Simulated annealing over the tree, the subset of frames used and (unless fixed) the nominal gap minimises a cost made of gap deviations, border notches, the distance from the target size and a small bonus per frame used. Every paradigm ran 256 independent annealing runs, 20 min wall-clock budget on all cores; its candidate states were then placed, measured exactly, de-duplicated (mirror images count as equal) and ranked by the score described on the title page.

paradigm	what differs	parameters that differ from the baseline
free-gap	Baseline. The nominal gap is an annealing variable between 1.5 and 6 cm (steps of 0.5 cm). Where a strip of frames is shorter than its neighbour, the slack is spread evenly over the gaps inside that strip. Gap deviations and border notches are penalised with a Huber shape: quadratic up to 2 cm, linear beyond, so a few large deviations are not punished out of proportion.	baseline
per-run-gap	Every annealing run keeps one fixed nominal gap from the grid 1.5, 2, …, 6 cm (the runs are spread over the grid) instead of changing the gap during the search. Each run explores its gap value more thoroughly, which gives a more diverse set of results across gap values.	gap_mode = per-run
proportional	Unavoidable slack is distributed in proportion to the size of the neighbouring frames: the weight of a gap is the mean extent of the two frames (or groups) it separates, so gaps next to big frames widen more than gaps between small ones. The deviations are therefore systematic (bigger frame, bigger gap) rather than equal.	slack_rule = proportional
gap-3	The nominal gap is fixed at 3 cm; the search only changes the arrangement and the subset of frames. Actual gaps still deviate from 3 cm wherever slack has to be absorbed.	gap = 3
linear-cost	Gap deviations and notches are penalised linearly instead of with the Huber shape. The cost landscape is smoother and small deviations count relatively more, which favours layouts whose deviations are spread thinly instead of concentrated in one gap.	cost_shape = linear
free-gap-s1	Identical to the baseline but with another random seed. Comparing it with the baseline shows how reproducible the results of a 20 minute search are.	seed = 1
large-apart	Sub-analysis. Like the baseline, but the layout must contain 2 to 3 large frames (outer area of at least 2000 cm^2, i.e. the 52.5×42.5 and 42.5×52.5 frames) and they are kept apart: two large frames that are direct neighbours are penalised in proportion to the edge they share (barely touching costs little), and the closest pair of large frames is pulled towards a centre distance of at least 70 cm.	large_area = 2000, large_count = 2-3, large_apart = True, large_min_dist = 70, entropy = 0
entropy	Sub-analysis. Like the baseline, but the cost also rewards the irregularity (“entropy”) of the constellation: a mix of many formats, different formats next to each other, frame edges that do not line up, and few straight gap lines running across the whole collage.	large_apart = False, large_min_dist = 70, entropy = 40
large-entropy	Sub-analysis. Both of the above: 2 to 3 large frames kept apart from each other, and the irregularity reward.	large_area = 2000, large_count = 2-3, large_apart = True, large_min_dist = 70, entropy = 40

Best-ranked layout of each paradigm (all measurements in cm)

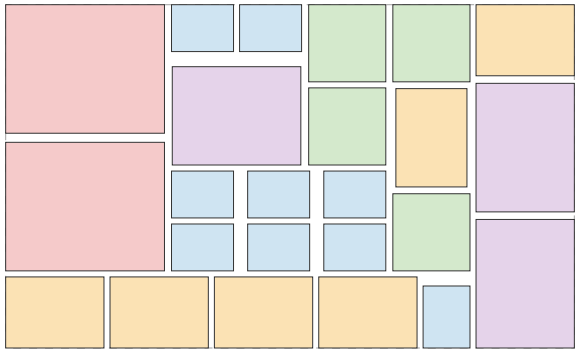
paradigm	layout	size	frames	nominal gap	mean dev	max dev	notches	corners	entropy	score
free-gap	1	181.5 × 113.5	20	1.5	0.59	1.0	none	4/4	0.71	1.29
per-run-gap	1	188.0 × 113.5	24	2	0.62	3.0	none	4/4	0.76	3.31
proportional	1	185.5 × 116.5	23	2.5	0.51	2.0	none	4/4	0.77	2.25
gap-3	1	179.0 × 126.5	24	3	0.44	2.2	none	4/4	0.64	2.47
linear-cost	1	172.5 × 115.5	24	2.5	0.30	2.0	none	4/4	0.71	2.15
free-gap-s1	1	170.0 × 114.5	24	2	0.33	1.8	4 cm	4/4	0.66	2.44
large-apart	1	182.0 × 129.0	25	4	0.60	3.0	none	4/4	0.65	3.30
entropy	1	172.5 × 116.5	24	2.5	0.43	2.0	none	4/4	0.72	2.21
large-entropy	1	182.0 × 129.0	24	4	0.56	1.8	none	4/4	0.65	2.03

Best-ranked layout of each paradigm at a glance (scale 1:25; details on the linked pages)



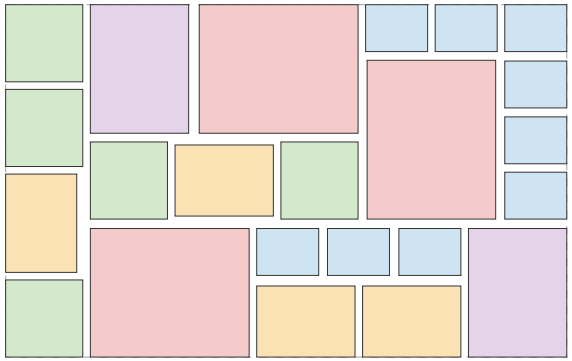
free-gap – layout 1

181.5 × 113.5 cm, 20 frames, gap 1.5 cm, max. dev. 1.0 cm, notches none, entropy 0.71



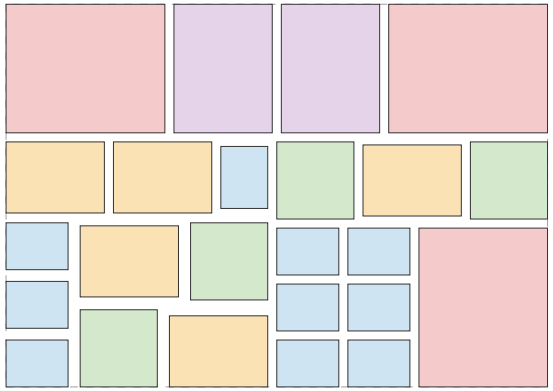
per-run-gap – layout 1

188.0 × 113.5 cm, 24 frames, gap 2 cm, max. dev. 3.0 cm, notches none, entropy 0.76



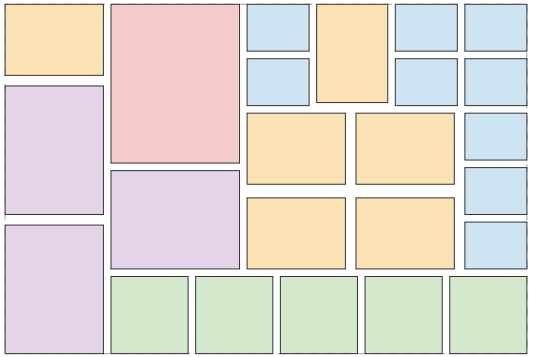
proportional – layout 1

185.5 × 116.5 cm, 23 frames, gap 2.5 cm, max. dev. 2.0 cm, notches none, entropy 0.77



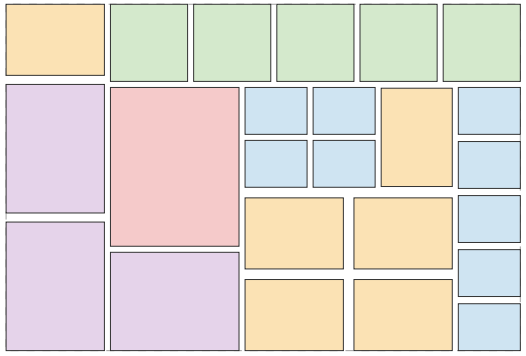
gap-3 – layout 1

179.0 × 126.5 cm, 24 frames, gap 3 cm, max. dev. 2.2 cm, notches none, entropy 0.64



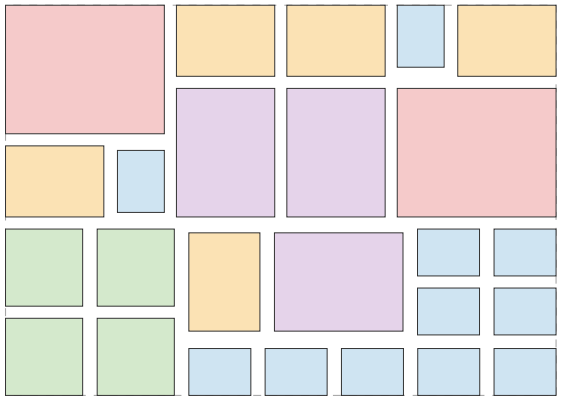
linear-cost – layout 1

172.5 × 115.5 cm, 24 frames, gap 2.5 cm, max. dev. 2.0 cm, notches none, entropy 0.71



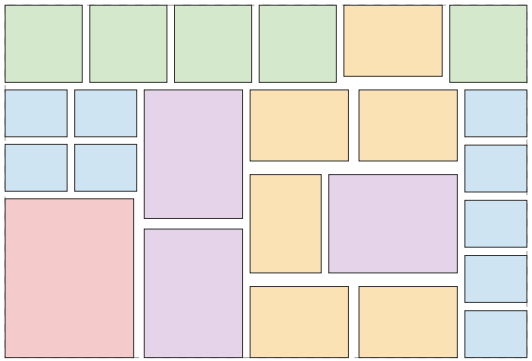
free-gap-s1 – layout 1

170.0 × 114.5 cm, 24 frames, gap 2 cm, max. dev. 1.8 cm, notches 4 cm, entropy 0.66



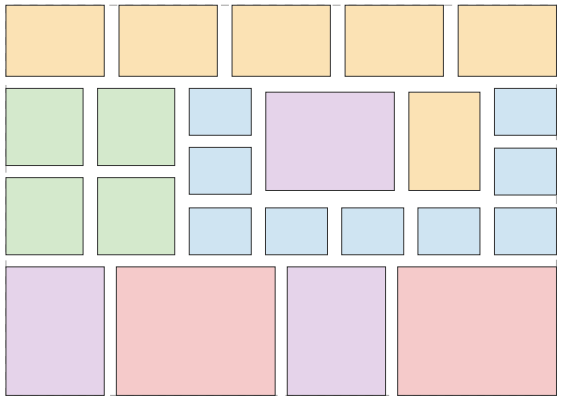
large-apart – layout 1

182.0 × 129.0 cm, 25 frames, gap 4 cm, max. dev. 3.0 cm, notches none, entropy 0.65



entropy – layout 1

172.5 × 116.5 cm, 24 frames, gap 2.5 cm, max. dev. 2.0 cm, notches none, entropy 0.72



large-entropy – layout 1

182.0 × 129.0 cm, 24 frames, gap 4 cm, max. dev. 1.8 cm, notches none, entropy 0.65

2 Combined ranking

The 144 layouts shown in this document (144 distinct ones; a layout found by several paradigms is listed once, with the other paradigms in the third column) ranked together by the same score (gap uniformity and border only; the sub-analysis constraints are not part of this score). Click a layout to jump to its page.

#	paradigm / layout	also found by	size (cm)	gap	frames	max dev	mean dev	notches	corners	entropy	score
1	free-gap / 1	–	181.5 × 113.5	1.5	20	1.0	0.59	none	4/4	0.71	1.29
2	free-gap / 2	–	182.0 × 129.0	4	24	1.8	0.53	none	4/4	0.64	2.01
3	large-entropy / 1	–	182.0 × 129.0	4	24	1.8	0.56	none	4/4	0.65	2.03
4	linear-cost / 1	–	172.5 × 115.5	2.5	24	2.0	0.30	none	4/4	0.71	2.15
5	entropy / 1	–	172.5 × 116.5	2.5	24	2.0	0.43	none	4/4	0.72	2.21
6	free-gap-s1 / 1	–	170.0 × 114.5	2	24	1.8	0.33	4 cm	4/4	0.66	2.44
7	free-gap-s1 / 2	–	170.0 × 114.5	2	24	1.8	0.35	4 cm	4/4	0.71	2.45
8	gap-3 / 1	–	179.0 × 126.5	3	24	2.2	0.44	none	4/4	0.64	2.47
9	proportional / 1	–	185.5 × 116.5	2.5	23	2.0	0.51	none	4/4	0.77	2.25
10	linear-cost / 2	–	181.5 × 113.5	1.5	23	2.0	0.52	none	4/4	0.57	2.26
11	linear-cost / 3	–	181.5 × 113.5	1.5	23	2.0	0.53	none	4/4	0.72	2.27
12	free-gap / 3	–	184.0 × 115.0	2	23	2.0	0.56	none	4/4	0.76	2.28
13	free-gap-s1 / 3	–	184.0 × 115.0	2	23	2.0	0.57	none	4/4	0.74	2.29
14	proportional / 2	–	184.0 × 115.0	2	23	2.0	0.57	none	4/4	0.69	2.29
15	free-gap-s1 / 4	–	183.5 × 115.0	2	23	2.2	0.48	none	4/4	0.76	2.49
16	free-gap-s1 / 5	–	190.5 × 115.5	2.5	22	2.0	0.41	none	4/4	0.68	2.20
17	linear-cost / 4	–	189.5 × 109.5	3.5	22	2.0	0.40	none	4/4	0.76	2.20
18	free-gap-s1 / 6	–	171.5 × 116.5	2.5	22	2.0	0.61	none	4/4	0.74	2.30
19	free-gap / 4	–	177.5 × 113.5	1.5	21	2.0	0.51	none	4/4	0.66	2.26
20	linear-cost / 5	–	181.5 × 113.5	1.5	20	2.0	0.55	none	4/4	0.78	2.28

3 Sub-analysis: large frames apart, entropy

The following paradigms add constraints on top of the baseline search: large-apart, entropy, large-entropy. The gap and border measures stay the same; the extra measures are shown on every layout page and in the tables below.

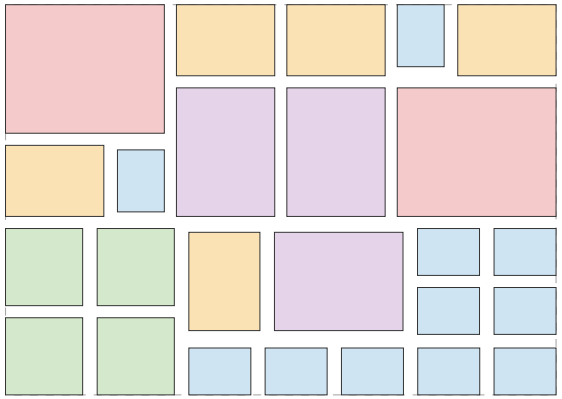
Large frames apart. Frames with an outer area of at least 2000 cm^2 count as large (here the 52.5 $\times$ 42.5 and 42.5 $\times$ 52.5 frames). The layout must contain between 2 and 3 of them (each one missing or too many costs as much as a 1.5 cm notch across the whole width). Two large frames that face each other across a regular gap are “touching”; the penalty grows with the edge they share, so barely touching is almost free. In addition the closest pair of large frames is pulled towards a centre distance of at least 70 cm. The tables report the number of large frames, how many pairs touch (with the total shared edge) and the smallest centre distance.

Entropy. A number in $[0, 1]$ for how irregular (“random”) a constellation looks, the mean of four shares: *mix* – the Shannon entropy of the formats used, relative to using all formats equally; *variety* – the share of neighbouring pairs (across a regular gap) that have different formats; *stagger* – the share of distinct frame edge positions (in a grid most edges line up, so it scores low); *lines* – 1 for a single straight gap line running across the whole collage, 0 for five or more. The search adds $w \cdot (1 - \mathrm{entropy})$ to the cost, so it trades a little gap uniformity for irregularity; layouts with equal rounded score are ranked by entropy.

Best-ranked layout of each sub-analysis paradigm (cm; large = count, touching pairs, min. centre distance; entropy components = mix / variety / stagger / lines)

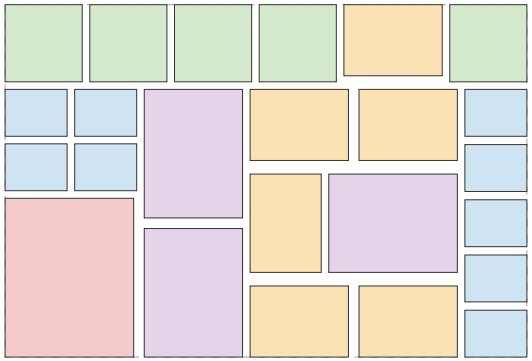
paradigm	layout	size	frames	gap	max dev	notches	large frames	entropy	components	score
large-apart	1	182.0 $\times$ 129.0	25	4	3.0	none	2, apart, 132 cm	0.65	0.83 / 0.54 / 0.48 / 0.75	3.30
entropy	1	172.5 $\times$ 116.5	24	2.5	2.0	none	1	0.72	0.74 / 0.65 / 0.49 / 1.00	2.21
large-entropy	1	182.0 $\times$ 129.0	24	4	1.8	none	2, apart, 93 cm	0.65	0.76 / 0.60 / 0.48 / 0.75	2.03

Best-ranked layout of each sub-analysis paradigm at a glance (scale 1:25)



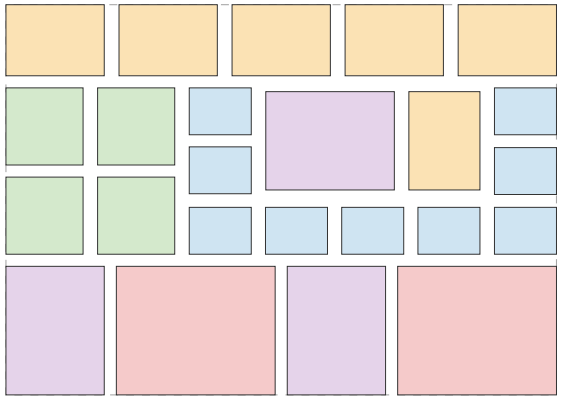
large-apart – layout 1

182.0 × 129.0 cm, 25 frames, gap 4 cm, max. dev. 3.0 cm, notches none, entropy 0.65



entropy – layout 1

172.5 × 116.5 cm, 24 frames, gap 2.5 cm, max. dev. 2.0 cm, notches none, entropy 0.72



large-entropy – layout 1

182.0 × 129.0 cm, 24 frames, gap 4 cm, max. dev. 1.8 cm, notches none, entropy 0.65

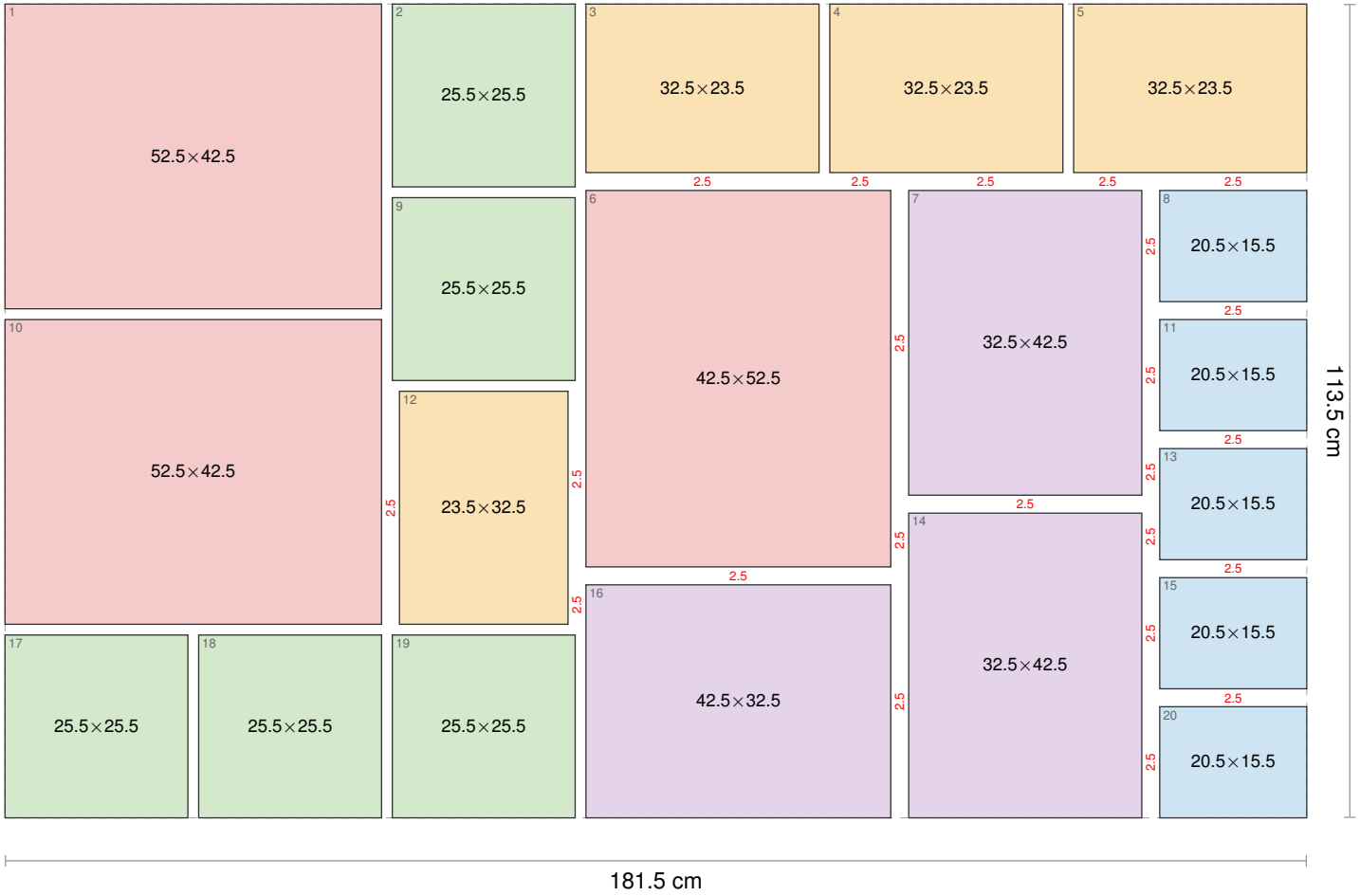
4 free-gap

Baseline. The nominal gap is an annealing variable between 1.5 and 6 cm (steps of 0.5 cm). Where a strip of frames is shorter than its neighbour, the slack is spread evenly over the gaps inside that strip. Gap deviations and border notches are penalised with a Huber shape: quadratic up to 2 cm, linear beyond, so a few large deviations are not punished out of proportion.

This is the baseline the other paradigms are compared with.
256 runs, 20 min wall-clock budget, 542 candidate states, the 16 best distinct layouts are shown.

#	size (cm)	gap	frames	max dev	mean dev	notches	corners	entropy
1	181.5 × 113.5	1.5	20	1.0	0.59	none	4/4	0.71
2	182.0 × 129.0	4	24	1.8	0.53	none	4/4	0.64
3	184.0 × 115.0	2	23	2.0	0.56	none	4/4	0.76
4	177.5 × 113.5	1.5	21	2.0	0.51	none	4/4	0.66
5	181.5 × 126.5	2.5	25	3.0	0.76	none	4/4	0.79
6	186.5 × 115.0	2	24	2.2	0.46	4 cm	4/4	0.76
7	180.5 × 128.5	3.5	24	3.0	0.61	none	4/4	0.66
8	169.5 × 113.5	1.5	23	2.5	0.68	none	4/4	0.76
9	184.0 × 115.0	2	23	2.2	0.53	4 cm	4/4	0.68
10	180.0 × 117.0	4	23	3.0	0.29	none	4/4	0.58
11	182.5 × 112.5	4.5	22	3.0	0.64	none	4/4	0.67
12	170.5 × 114.5	2	22	3.0	0.81	none	4/4	0.72
13	177.5 × 113.5	1.5	21	2.0	0.54	6 cm	4/4	0.68
14	171.0 × 120.5	6	14	3.0	0.29	none	4/4	0.81
15	187.5 × 130.0	2	26	3.5	0.78	4 cm	4/4	0.83
16	180.0 × 114.5	2	25	3.0	0.64	4 cm	4/4	0.66

Layout 1 181.5 × 113.5 cm gap 1.5 cm 20 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	181.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 2.5 cm
Max. deviation	1.0 cm
Mean deviation	0.59 cm
Border notches	none
Corners filled	4 / 4
Frames used	20 of 28
Large frames	3, touching 1× (52 cm), min. dist. 44 cm
Entropy	0.71
mix / variety	0.86 / 0.72
stagger / lines	0.51 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	5	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	25.5 × 25.5	54.0	0.0
3	32.5 × 23.5	81.0	0.0
4	32.5 × 23.5	115.0	0.0
5	32.5 × 23.5	149.0	0.0
6	42.5 × 52.5	81.0	26.0
7	32.5 × 42.5	126.0	26.0
8	20.5 × 15.5	161.0	26.0
9	25.5 × 25.5	54.0	27.0
10	52.5 × 42.5	0.0	44.0
11	20.5 × 15.5	161.0	44.0
12	23.5 × 32.5	55.0	54.0
13	20.5 × 15.5	161.0	62.0
14	32.5 × 42.5	126.0	71.0
15	20.5 × 15.5	161.0	80.0
16	42.5 × 32.5	81.0	81.0
17	25.5 × 25.5	0.0	88.0
18	25.5 × 25.5	27.0	88.0
19	25.5 × 25.5	54.0	88.0
20	20.5 × 15.5	161.0	98.0

Layout 2 182.0 × 129.0 cm gap 4 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

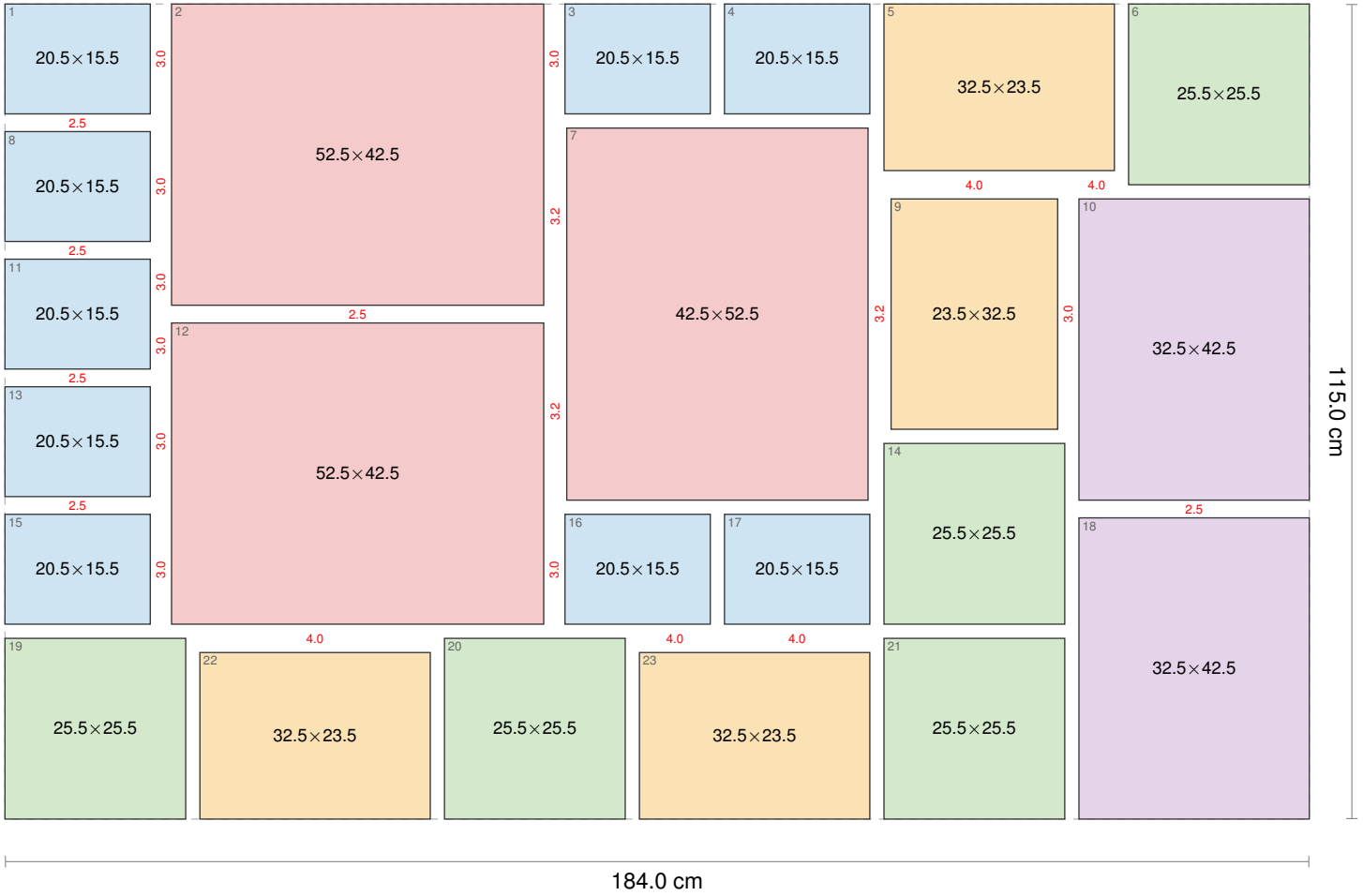
Outer size	182.0 × 129.0 cm
Nominal gap	4 cm
Actual gaps	4.0 – 5.8 cm
Max. deviation	1.8 cm
Mean deviation	0.53 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, touching 1× (42 cm), min. dist. 56 cm
Entropy	0.64
mix / variety	0.76 / 0.56
stagger / lines	0.48 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	0	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	0	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	36.5	0.0
3	52.5 × 42.5	93.0	0.0
4	32.5 × 42.5	149.5	0.0
5	25.5 × 25.5	0.0	46.5
6	20.5 × 15.5	30.3	46.5
7	20.5 × 15.5	55.5	46.5
8	20.5 × 15.5	80.8	46.5
9	20.5 × 15.5	106.0	46.5
10	20.5 × 15.5	131.2	46.5
11	25.5 × 25.5	156.5	46.5
12	20.5 × 15.5	77.6	66.5
13	20.5 × 15.5	102.9	66.5
14	42.5 × 32.5	30.3	67.8
15	23.5 × 32.5	128.2	67.8
16	25.5 × 25.5	0.0	76.0
17	25.5 × 25.5	156.5	76.0
18	20.5 × 15.5	77.6	86.0
19	20.5 × 15.5	102.9	86.0
20	32.5 × 23.5	0.0	105.5
21	32.5 × 23.5	37.4	105.5
22	32.5 × 23.5	74.8	105.5
23	32.5 × 23.5	112.1	105.5
24	32.5 × 23.5	149.5	105.5

Layout 3 184.0 × 115.0 cm gap 2 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

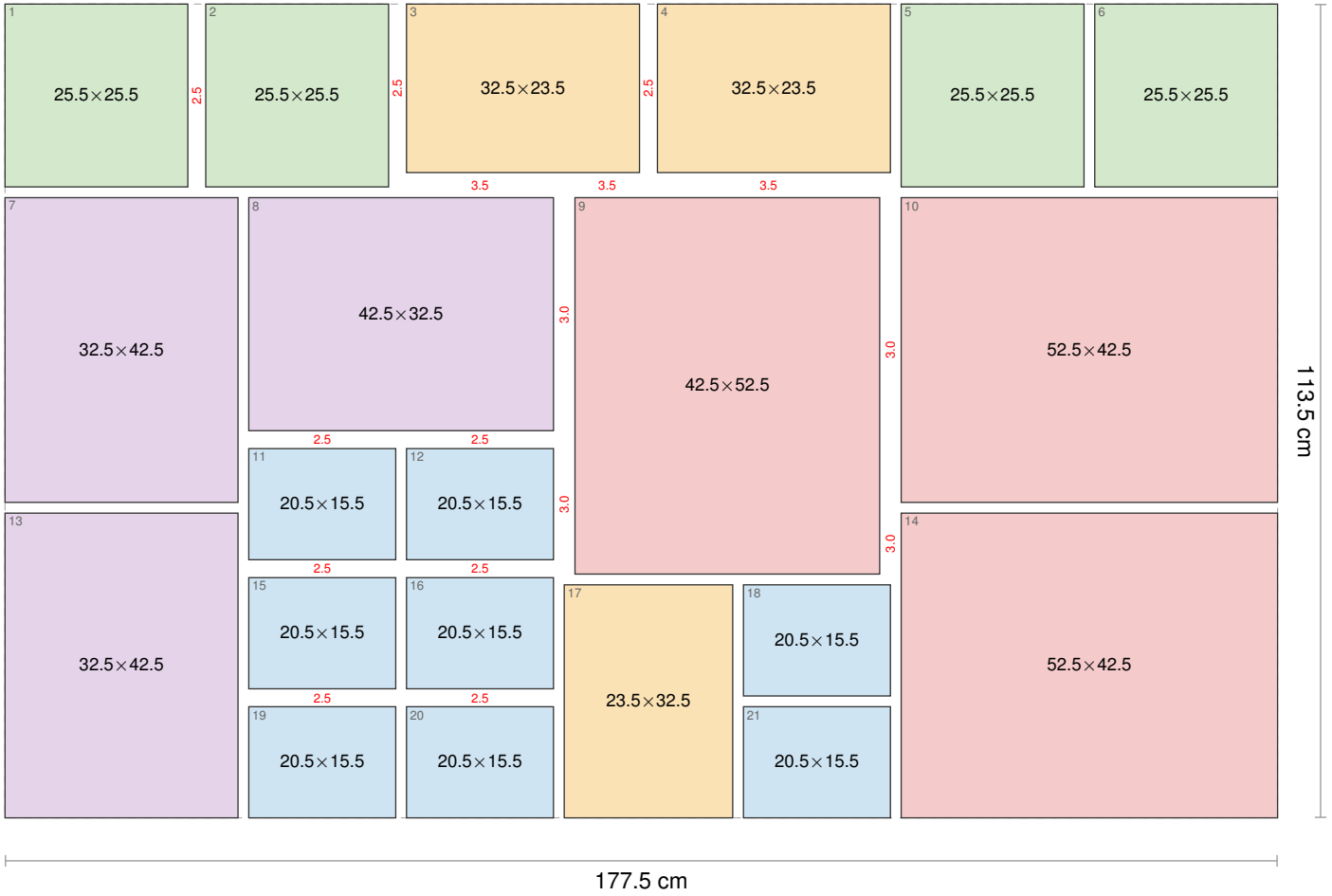
Outer size	184.0 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.0 cm
Max. deviation	2.0 cm
Mean deviation	0.56 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 1× (52 cm), min. dist. 45 cm
Entropy	0.76
mix / variety	0.76 / 0.77
stagger / lines	0.51 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	52.5 × 42.5	23.5	0.0
3	20.5 × 15.5	79.0	0.0
4	20.5 × 15.5	101.5	0.0
5	32.5 × 23.5	124.0	0.0
6	25.5 × 25.5	158.5	0.0
7	42.5 × 52.5	79.2	17.5
8	20.5 × 15.5	0.0	18.0
9	23.5 × 32.5	125.0	27.5
10	32.5 × 42.5	151.5	27.5
11	20.5 × 15.5	0.0	36.0
12	52.5 × 42.5	23.5	45.0
13	20.5 × 15.5	0.0	54.0
14	25.5 × 25.5	124.0	62.0
15	20.5 × 15.5	0.0	72.0
16	20.5 × 15.5	79.0	72.0
17	20.5 × 15.5	101.5	72.0
18	32.5 × 42.5	151.5	72.5
19	25.5 × 25.5	0.0	89.5
20	25.5 × 25.5	62.0	89.5
21	25.5 × 25.5	124.0	89.5
22	32.5 × 23.5	27.5	91.5
23	32.5 × 23.5	89.5	91.5

Layout 4 177.5 × 113.5 cm gap 1.5 cm 21 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

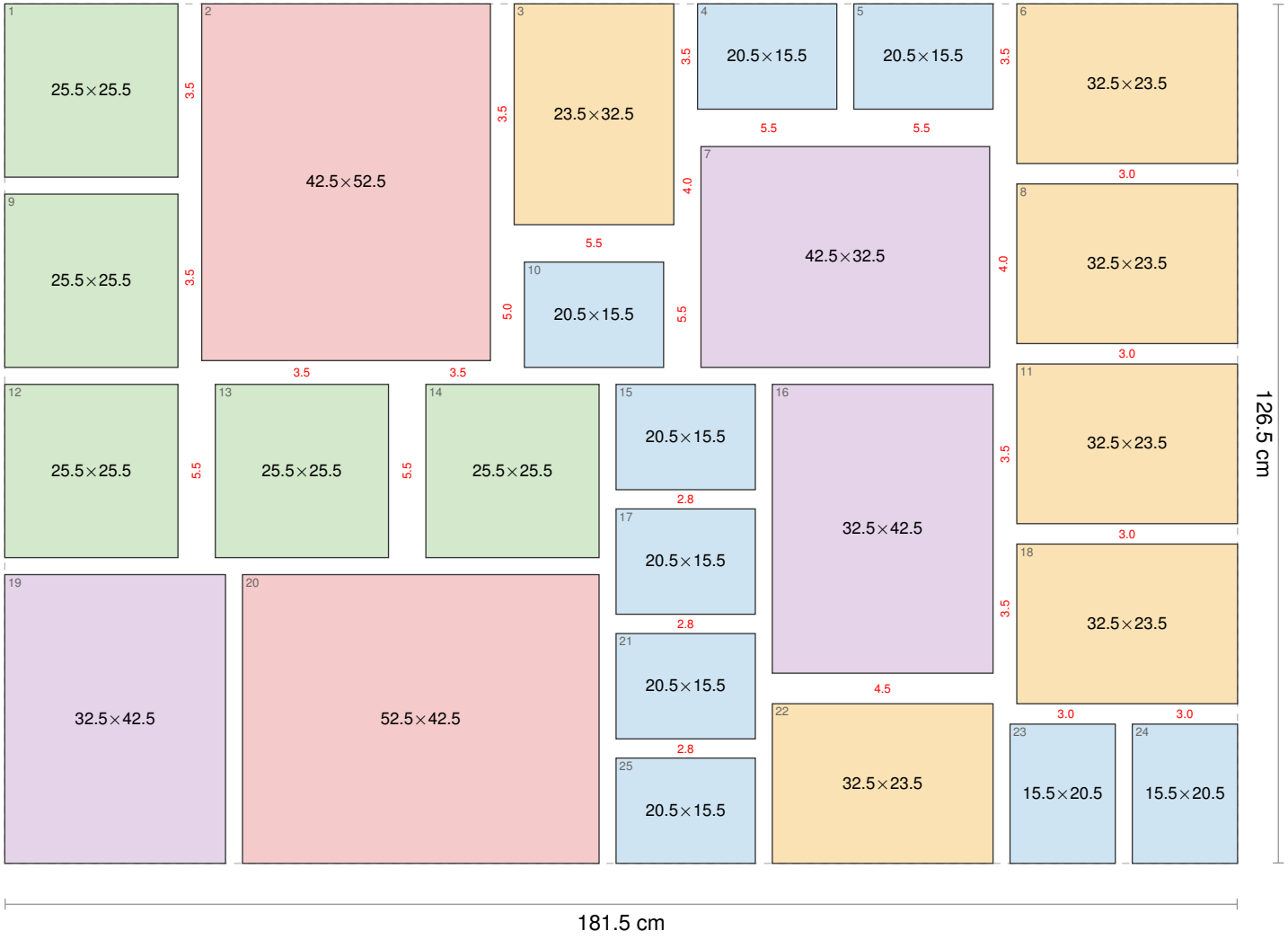
Outer size	177.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.51 cm
Border notches	none
Corners filled	4 / 4
Frames used	21 of 28
Large frames	3, touching 1× (52 cm), min. dist. 44 cm
Entropy	0.66
mix / variety	0.81 / 0.62
stagger / lines	0.44 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	2	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	28.0	0.0
3	32.5 × 23.5	56.0	0.0
4	32.5 × 23.5	91.0	0.0
5	25.5 × 25.5	125.0	0.0
6	25.5 × 25.5	152.0	0.0
7	32.5 × 42.5	0.0	27.0
8	42.5 × 32.5	34.0	27.0
9	42.5 × 52.5	79.5	27.0
10	52.5 × 42.5	125.0	27.0
11	20.5 × 15.5	34.0	62.0
12	20.5 × 15.5	56.0	62.0
13	32.5 × 42.5	0.0	71.0
14	52.5 × 42.5	125.0	71.0
15	20.5 × 15.5	34.0	80.0
16	20.5 × 15.5	56.0	80.0
17	23.5 × 32.5	78.0	81.0
18	20.5 × 15.5	103.0	81.0
19	20.5 × 15.5	34.0	98.0
20	20.5 × 15.5	56.0	98.0
21	20.5 × 15.5	103.0	98.0

Layout 5 181.5 × 126.5 cm gap 2.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

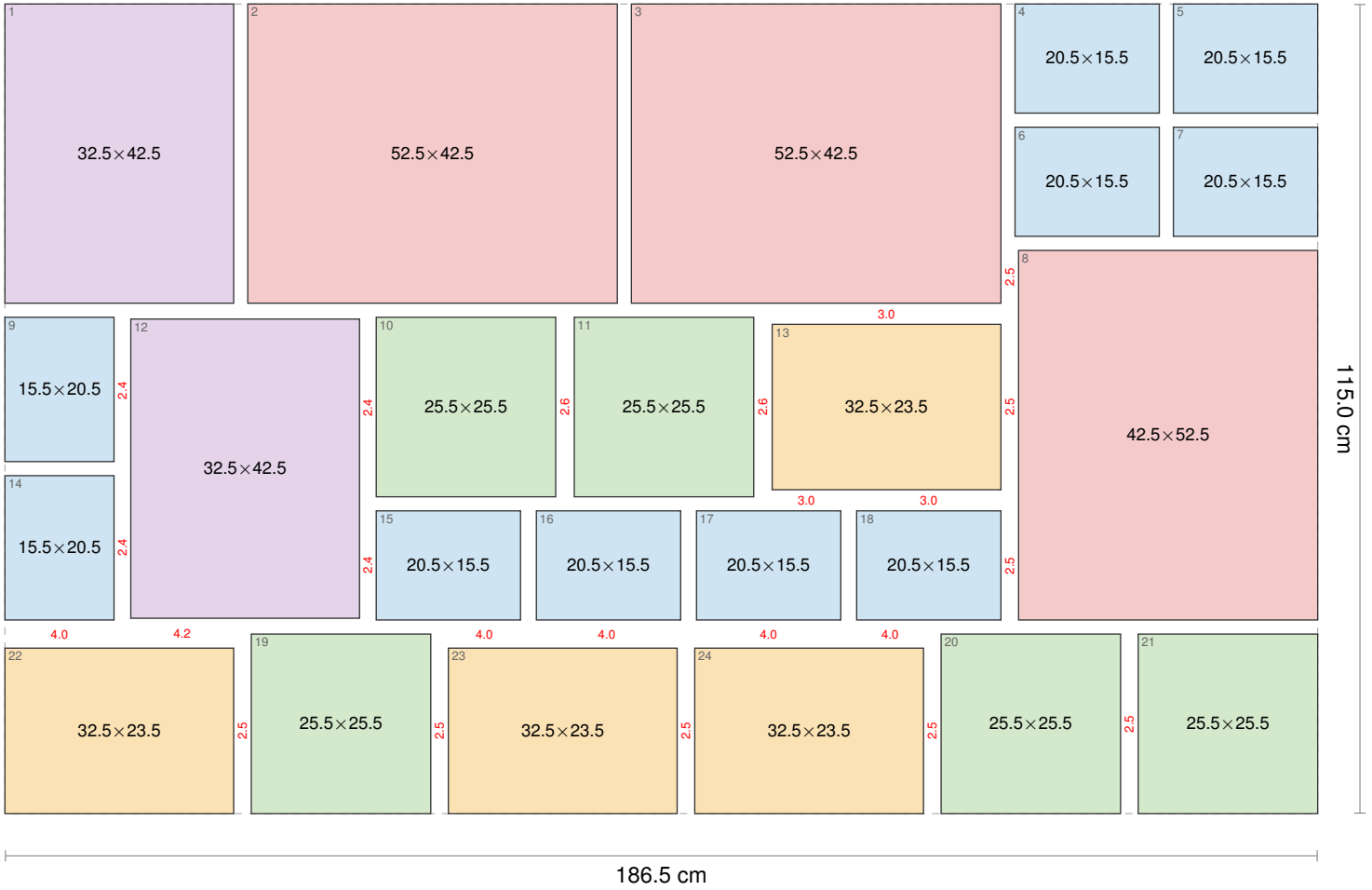
Outer size	181.5 × 126.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.5 cm
Max. deviation	3.0 cm
Mean deviation	0.76 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 80 cm
Entropy	0.79
mix / variety	0.87 / 0.72
stagger / lines	0.56 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	42.5 × 52.5	29.0	0.0
3	23.5 × 32.5	75.0	0.0
4	20.5 × 15.5	102.0	0.0
5	20.5 × 15.5	125.0	0.0
6	32.5 × 23.5	149.0	0.0
7	42.5 × 32.5	102.5	21.0
8	32.5 × 23.5	149.0	26.5
9	25.5 × 25.5	0.0	28.0
10	20.5 × 15.5	76.5	38.0
11	32.5 × 23.5	149.0	53.0
12	25.5 × 25.5	0.0	56.0
13	25.5 × 25.5	31.0	56.0
14	25.5 × 25.5	62.0	56.0
15	20.5 × 15.5	90.0	56.0
16	32.5 × 42.5	113.0	56.0
17	20.5 × 15.5	90.0	74.3
18	32.5 × 23.5	149.0	79.5
19	32.5 × 42.5	0.0	84.0
20	52.5 × 42.5	35.0	84.0
21	20.5 × 15.5	90.0	92.7
22	32.5 × 23.5	113.0	103.0
23	15.5 × 20.5	148.0	106.0
24	15.5 × 20.5	166.0	106.0
25	20.5 × 15.5	90.0	111.0

Layout 6 186.5 × 115.0 cm gap 2 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	186.5 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.2 cm
Max. deviation	2.2 cm
Mean deviation	0.46 cm
Border notches	left 4 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	3, touching 2× (50 cm), min. dist. 54 cm
Entropy	0.76
mix / variety	0.79 / 0.71
stagger / lines	0.55 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	4	5
	23.5×32.5	P	0	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	34.5	0.0
3	52.5 × 42.5	89.0	0.0
4	20.5 × 15.5	143.5	0.0
5	20.5 × 15.5	166.0	0.0
6	20.5 × 15.5	143.5	17.5
7	20.5 × 15.5	166.0	17.5
8	42.5 × 52.5	144.0	35.0
9	15.5 × 20.5	0.0	44.5
10	25.5 × 25.5	52.8	44.5
11	25.5 × 25.5	80.9	44.5
12	32.5 × 42.5	17.9	44.8
13	32.5 × 23.5	109.0	45.5
14	15.5 × 20.5	0.0	67.0
15	20.5 × 15.5	52.8	72.0
16	20.5 × 15.5	75.5	72.0
17	20.5 × 15.5	98.2	72.0
18	20.5 × 15.5	121.0	72.0
19	25.5 × 25.5	35.0	89.5
20	25.5 × 25.5	133.0	89.5
21	25.5 × 25.5	161.0	89.5
22	32.5 × 23.5	0.0	91.5
23	32.5 × 23.5	63.0	91.5
24	32.5 × 23.5	98.0	91.5

Layout 7 180.5 × 128.5 cm gap 3.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

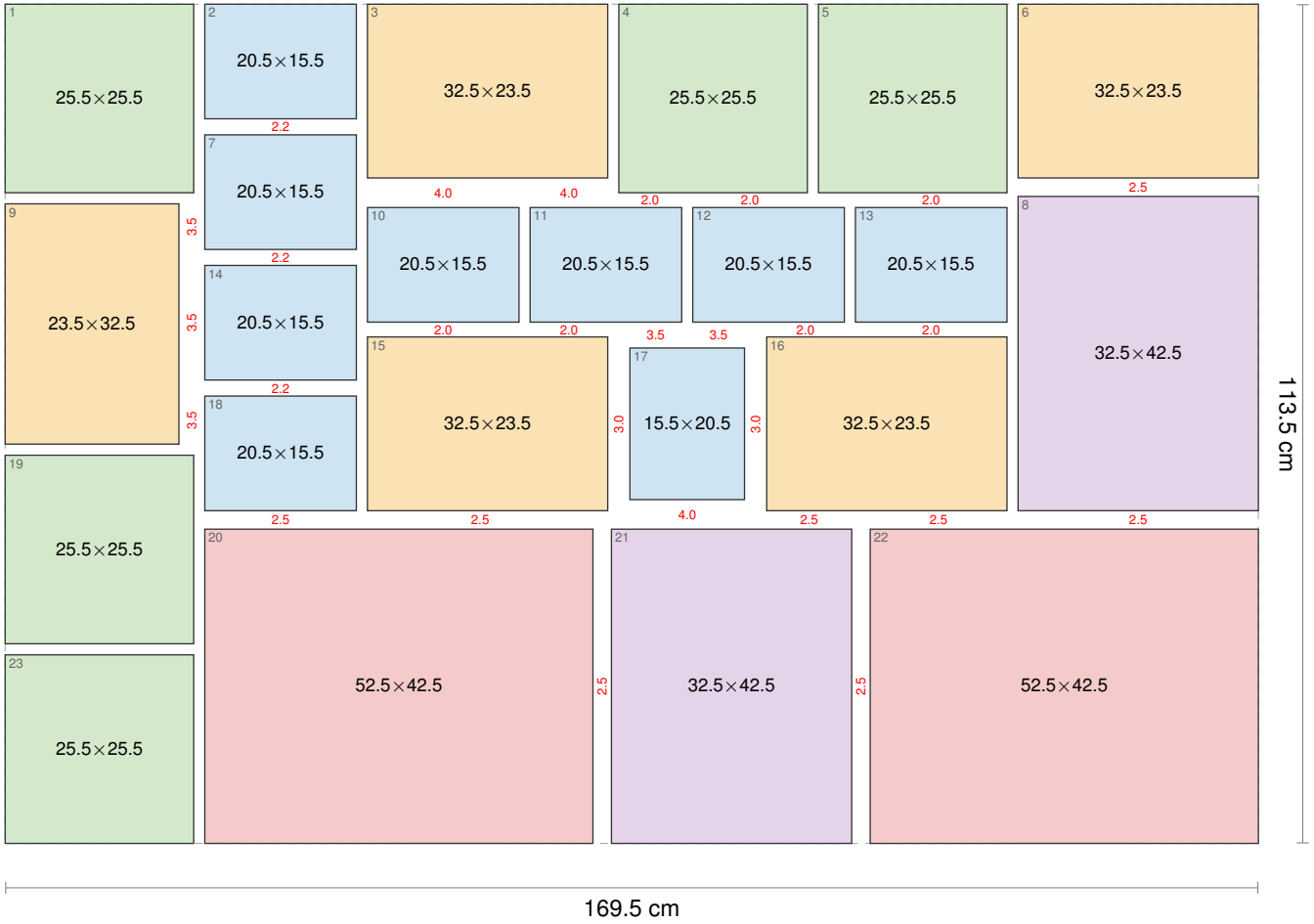
Outer size	180.5 × 128.5 cm
Nominal gap	3.5 cm
Actual gaps	3.5 – 6.5 cm
Max. deviation	3.0 cm
Mean deviation	0.61 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	3, touching 2× (85 cm), min. dist. 52 cm
Entropy	0.66
mix / variety	0.74 / 0.63
stagger / lines	0.50 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	0	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	52.5 × 42.5	56.0	0.0
3	32.5 × 23.5	112.0	0.0
4	32.5 × 23.5	148.0	0.0
5	20.5 × 15.5	112.0	27.0
6	20.5 × 15.5	136.0	27.0
7	20.5 × 15.5	160.0	27.0
8	20.5 × 15.5	112.0	46.0
9	20.5 × 15.5	136.0	46.0
10	20.5 × 15.5	160.0	46.0
11	42.5 × 52.5	0.0	47.0
12	23.5 × 32.5	47.5	47.0
13	32.5 × 23.5	76.0	47.0
14	20.5 × 15.5	160.0	65.0
15	42.5 × 32.5	112.0	66.0
16	32.5 × 23.5	76.0	76.0
17	20.5 × 15.5	49.0	84.0
18	20.5 × 15.5	160.0	84.0
19	25.5 × 25.5	0.0	103.0
20	25.5 × 25.5	29.6	103.0
21	25.5 × 25.5	59.2	103.0
22	25.5 × 25.5	88.8	103.0
23	25.5 × 25.5	155.0	103.0
24	32.5 × 23.5	118.4	105.0

Layout 8 169.5 × 113.5 cm gap 1.5 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

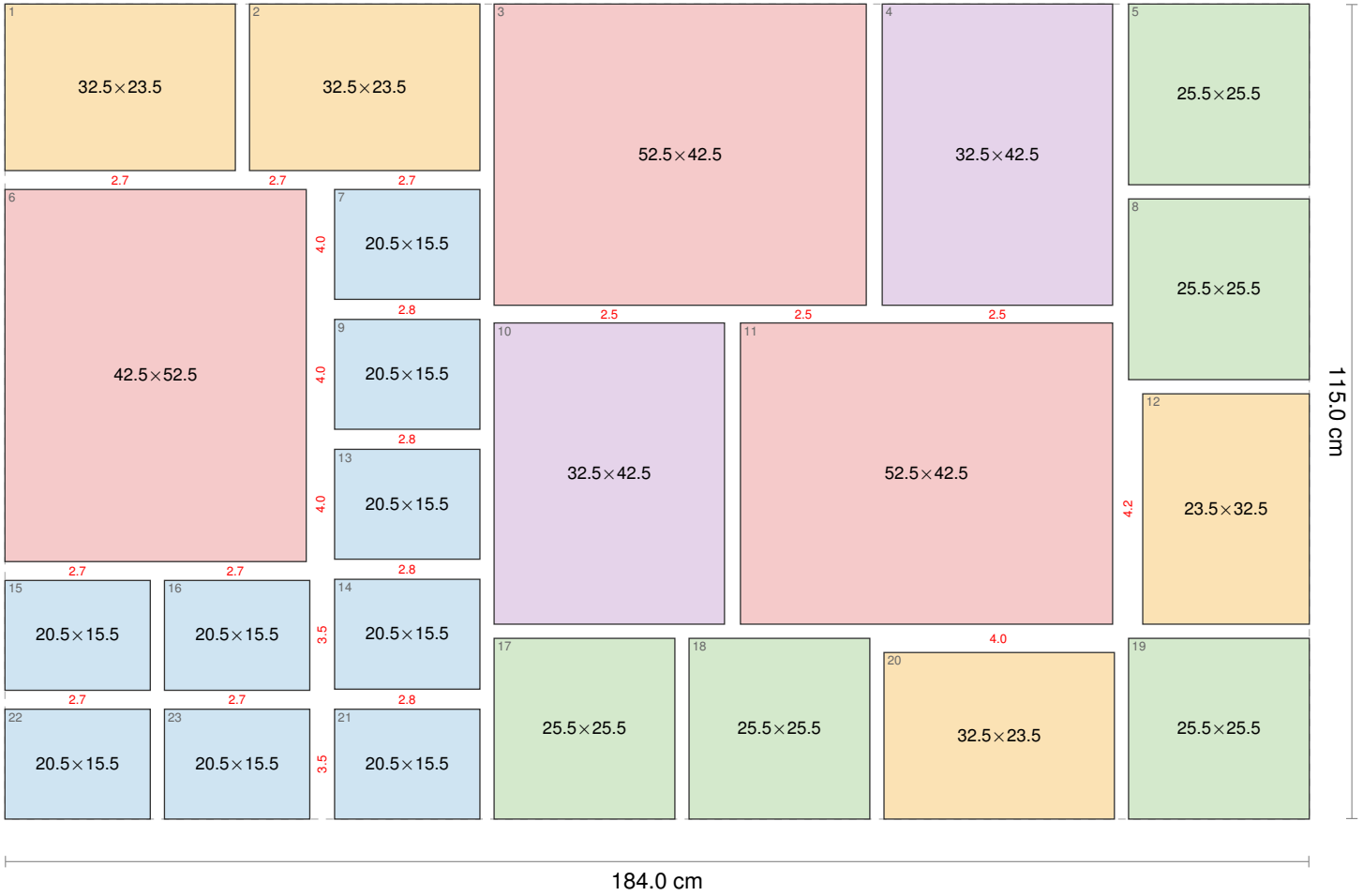
Outer size	169.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.68 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	2, not touching, min. dist. 90 cm
Entropy	0.76
mix / variety	0.77 / 0.75
stagger / lines	0.53 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	8	9
15.5×20.5	P	1	2
32.5×42.5	P	2	2
42.5×32.5	L	0	1
32.5×23.5	L	4	5
23.5×32.5	P	1	1
25.5×25.5		5	5
42.5×52.5	P	0	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	20.5 × 15.5	27.0	0.0
3	32.5 × 23.5	49.0	0.0
4	25.5 × 25.5	83.0	0.0
5	25.5 × 25.5	110.0	0.0
6	32.5 × 23.5	137.0	0.0
7	20.5 × 15.5	27.0	17.7
8	32.5 × 42.5	137.0	26.0
9	23.5 × 32.5	0.0	27.0
10	20.5 × 15.5	49.0	27.5
11	20.5 × 15.5	71.0	27.5
12	20.5 × 15.5	93.0	27.5
13	20.5 × 15.5	115.0	27.5
14	20.5 × 15.5	27.0	35.3
15	32.5 × 23.5	49.0	45.0
16	32.5 × 23.5	103.0	45.0
17	15.5 × 20.5	84.5	46.5
18	20.5 × 15.5	27.0	53.0
19	25.5 × 25.5	0.0	61.0
20	52.5 × 42.5	27.0	71.0
21	32.5 × 42.5	82.0	71.0
22	52.5 × 42.5	117.0	71.0
23	25.5 × 25.5	0.0	88.0

Layout 9 184.0 × 115.0 cm gap 2 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

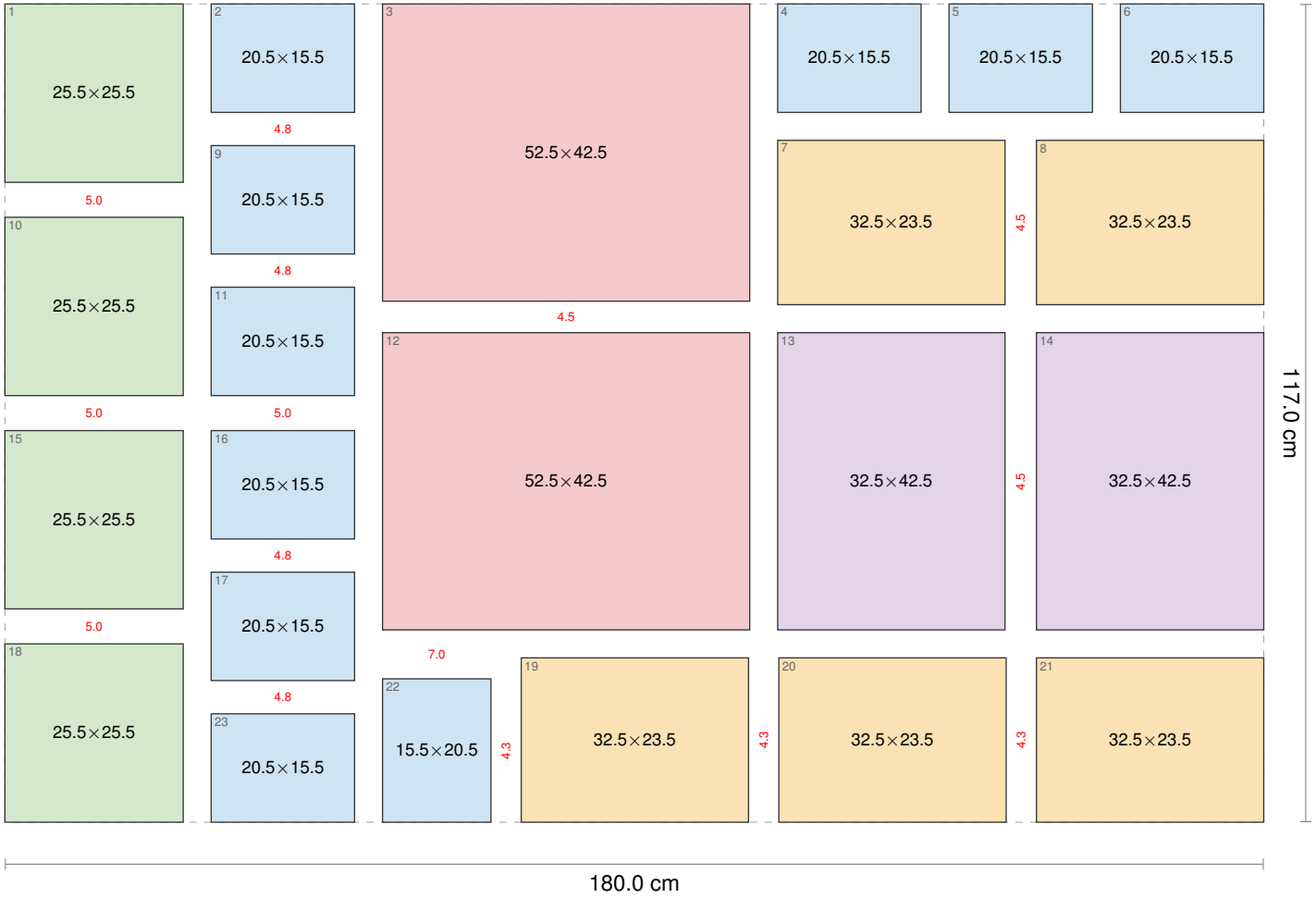
Outer size	184.0 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.2 cm
Max. deviation	2.2 cm
Mean deviation	0.53 cm
Border notches	bottom 4 cm
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 1× (18 cm), min. dist. 57 cm
Entropy	0.68
mix / variety	0.76 / 0.68
stagger / lines	0.51 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	0	2
32.5×42.5	P	2	2
42.5×32.5	L	0	1
32.5×23.5	L	3	5
23.5×32.5	P	1	1
25.5×25.5		5	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	32.5 × 23.5	34.5	0.0
3	52.5 × 42.5	69.0	0.0
4	32.5 × 42.5	123.8	0.0
5	25.5 × 25.5	158.5	0.0
6	42.5 × 52.5	0.0	26.2
7	20.5 × 15.5	46.5	26.2
8	25.5 × 25.5	158.5	27.5
9	20.5 × 15.5	46.5	44.5
10	32.5 × 42.5	69.0	45.0
11	52.5 × 42.5	103.8	45.0
12	23.5 × 32.5	160.5	55.0
13	20.5 × 15.5	46.5	62.8
14	20.5 × 15.5	46.5	81.2
15	20.5 × 15.5	0.0	81.3
16	20.5 × 15.5	22.5	81.3
17	25.5 × 25.5	69.0	89.5
18	25.5 × 25.5	96.5	89.5
19	25.5 × 25.5	158.5	89.5
20	32.5 × 23.5	124.0	91.5
21	20.5 × 15.5	46.5	99.5
22	20.5 × 15.5	0.0	99.5
23	20.5 × 15.5	22.5	99.5

Layout 10 180.0 × 117.0 cm gap 4 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

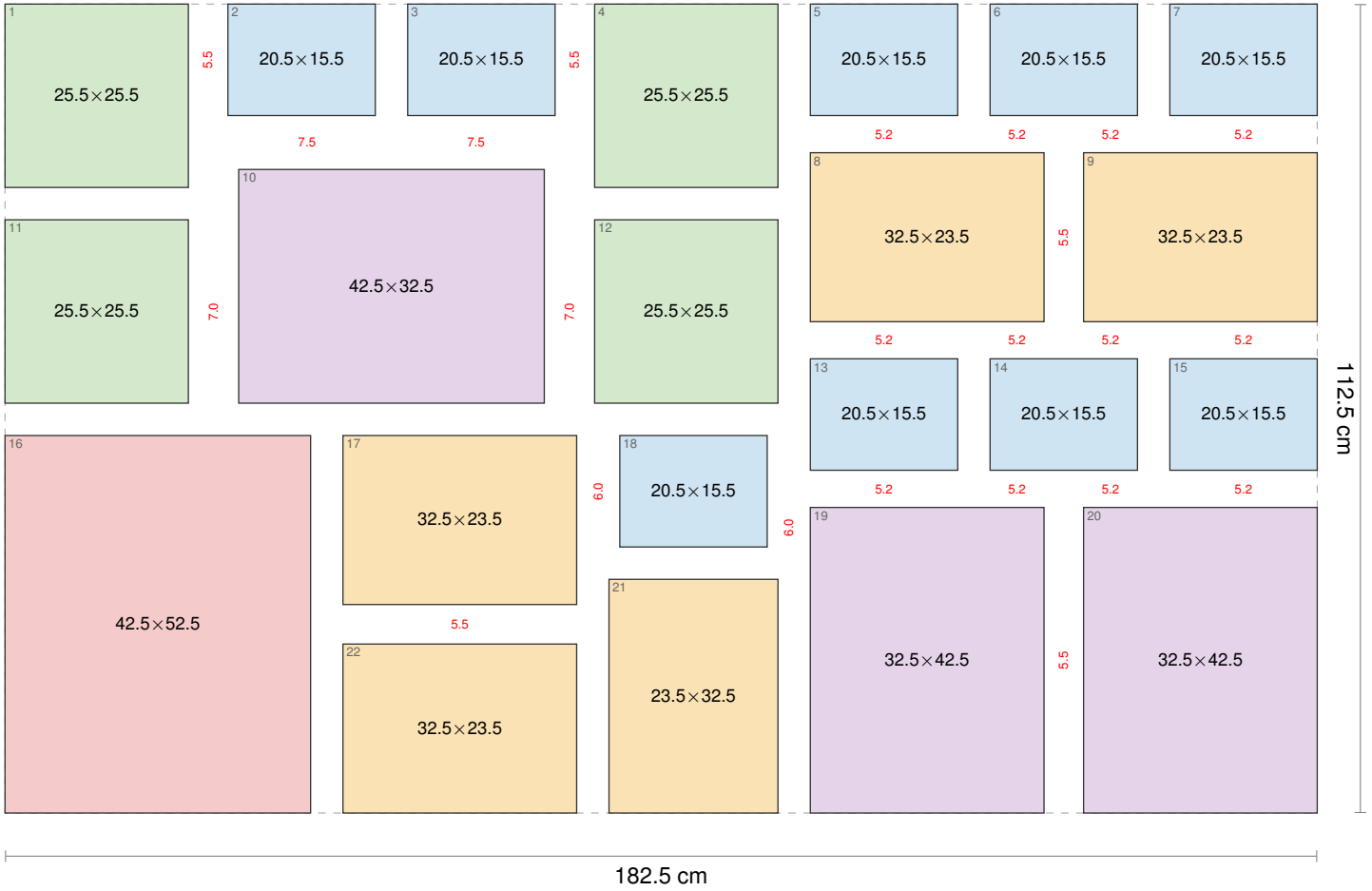
Outer size	180.0 × 117.0 cm
Nominal gap	4 cm
Actual gaps	4.0 – 7.0 cm
Max. deviation	3.0 cm
Mean deviation	0.29 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	2, touching 1× (52 cm), min. dist. 47 cm
Entropy	0.58
mix / variety	0.71 / 0.64
stagger / lines	0.46 / 0.50 (3 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	5	5
	23.5×32.5	P	0	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	20.5 × 15.5	29.5	0.0
3	52.5 × 42.5	54.0	0.0
4	20.5 × 15.5	110.5	0.0
5	20.5 × 15.5	135.0	0.0
6	20.5 × 15.5	159.5	0.0
7	32.5 × 23.5	110.5	19.5
8	32.5 × 23.5	147.5	19.5
9	20.5 × 15.5	29.5	20.2
10	25.5 × 25.5	0.0	30.5
11	20.5 × 15.5	29.5	40.5
12	52.5 × 42.5	54.0	47.0
13	32.5 × 42.5	110.5	47.0
14	32.5 × 42.5	147.5	47.0
15	25.5 × 25.5	0.0	61.0
16	20.5 × 15.5	29.5	61.0
17	20.5 × 15.5	29.5	81.2
18	25.5 × 25.5	0.0	91.5
19	32.5 × 23.5	73.8	93.5
20	32.5 × 23.5	110.7	93.5
21	32.5 × 23.5	147.5	93.5
22	15.5 × 20.5	54.0	96.5
23	20.5 × 15.5	29.5	101.5

Layout 11 182.5 × 112.5 cm gap 4.5 cm 22 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

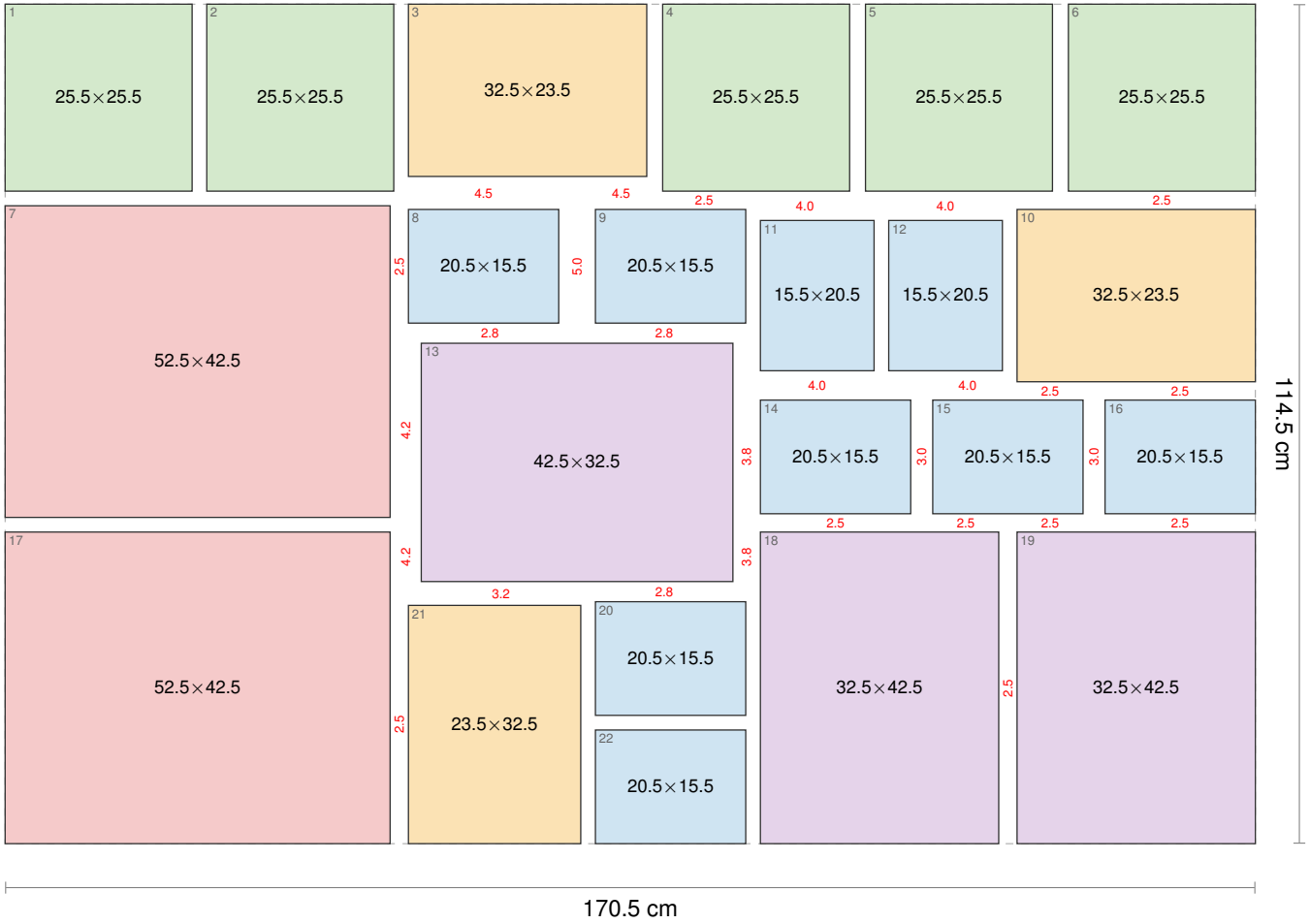
Outer size	182.5 × 112.5 cm
Nominal gap	4.5 cm
Actual gaps	4.5 – 7.5 cm
Max. deviation	3.0 cm
Mean deviation	0.64 cm
Border notches	none
Corners filled	4 / 4
Frames used	22 of 28
Large frames	1
Entropy	0.67
mix / variety	0.74 / 0.72
stagger / lines	0.47 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	20.5 × 15.5	31.0	0.0
3	20.5 × 15.5	56.0	0.0
4	25.5 × 25.5	82.0	0.0
5	20.5 × 15.5	112.0	0.0
6	20.5 × 15.5	137.0	0.0
7	20.5 × 15.5	162.0	0.0
8	32.5 × 23.5	112.0	20.7
9	32.5 × 23.5	150.0	20.7
10	42.5 × 32.5	32.5	23.0
11	25.5 × 25.5	0.0	30.0
12	25.5 × 25.5	82.0	30.0
13	20.5 × 15.5	112.0	49.3
14	20.5 × 15.5	137.0	49.3
15	20.5 × 15.5	162.0	49.3
16	42.5 × 52.5	0.0	60.0
17	32.5 × 23.5	47.0	60.0
18	20.5 × 15.5	85.5	60.0
19	32.5 × 42.5	112.0	70.0
20	32.5 × 42.5	150.0	70.0
21	23.5 × 32.5	84.0	80.0
22	32.5 × 23.5	47.0	89.0

Layout 12 170.5 × 114.5 cm gap 2 cm 22 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

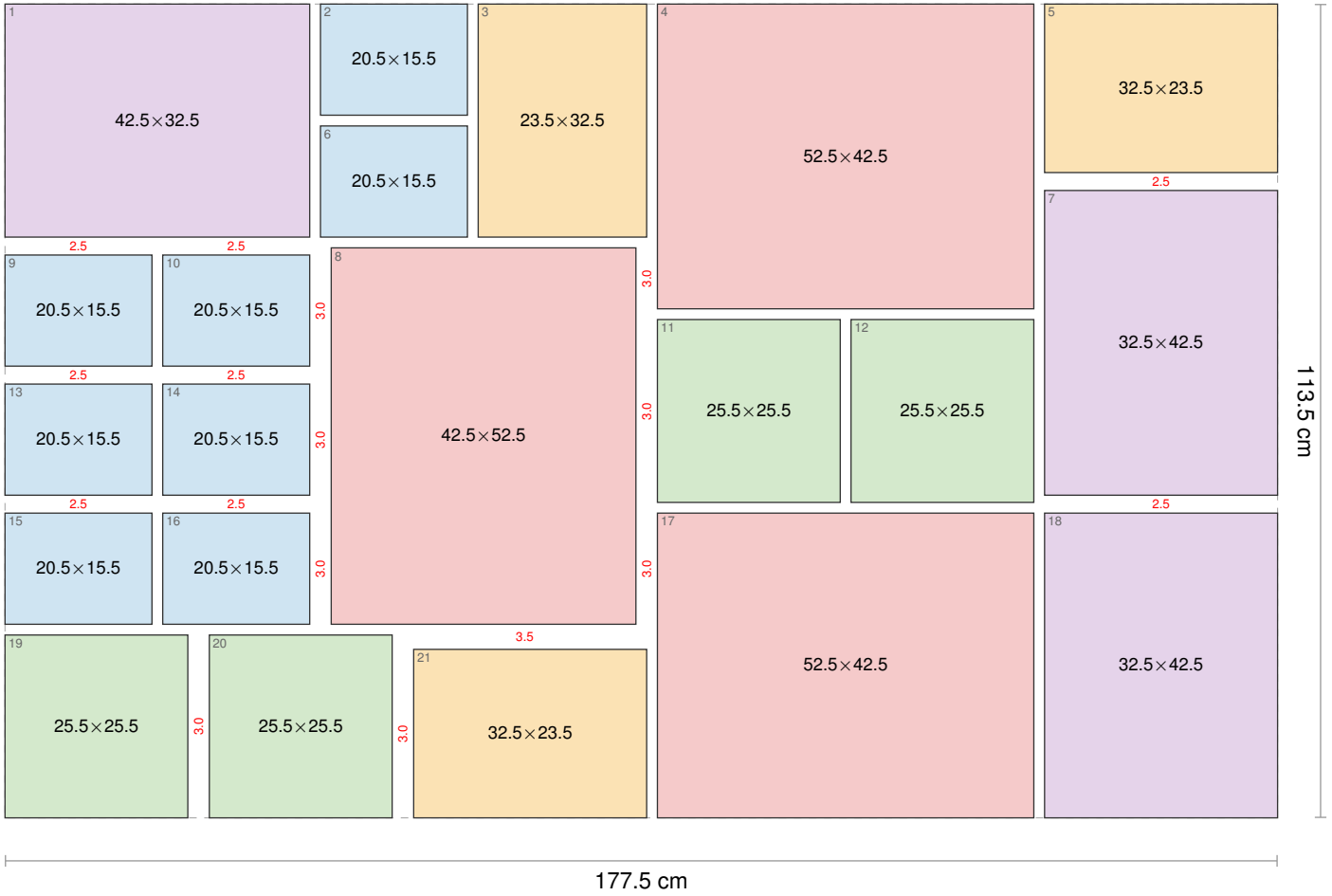
Outer size	170.5 × 114.5 cm
Nominal gap	2 cm
Actual gaps	2.0 – 5.0 cm
Max. deviation	3.0 cm
Mean deviation	0.81 cm
Border notches	none
Corners filled	4 / 4
Frames used	22 of 28
Large frames	2, touching 1× (52 cm), min. dist. 44 cm
Entropy	0.72
mix / variety	0.84 / 0.72
stagger / lines	0.56 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	2	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	27.5	0.0
3	32.5 × 23.5	55.0	0.0
4	25.5 × 25.5	89.7	0.0
5	25.5 × 25.5	117.3	0.0
6	25.5 × 25.5	145.0	0.0
7	52.5 × 42.5	0.0	27.5
8	20.5 × 15.5	55.0	28.0
9	20.5 × 15.5	80.5	28.0
10	32.5 × 23.5	138.0	28.0
11	15.5 × 20.5	103.0	29.5
12	15.5 × 20.5	120.5	29.5
13	42.5 × 32.5	56.8	46.2
14	20.5 × 15.5	103.0	54.0
15	20.5 × 15.5	126.5	54.0
16	20.5 × 15.5	150.0	54.0
17	52.5 × 42.5	0.0	72.0
18	32.5 × 42.5	103.0	72.0
19	32.5 × 42.5	138.0	72.0
20	20.5 × 15.5	80.5	81.5
21	23.5 × 32.5	55.0	82.0
22	20.5 × 15.5	80.5	99.0

Layout 13 177.5 × 113.5 cm gap 1.5 cm 21 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

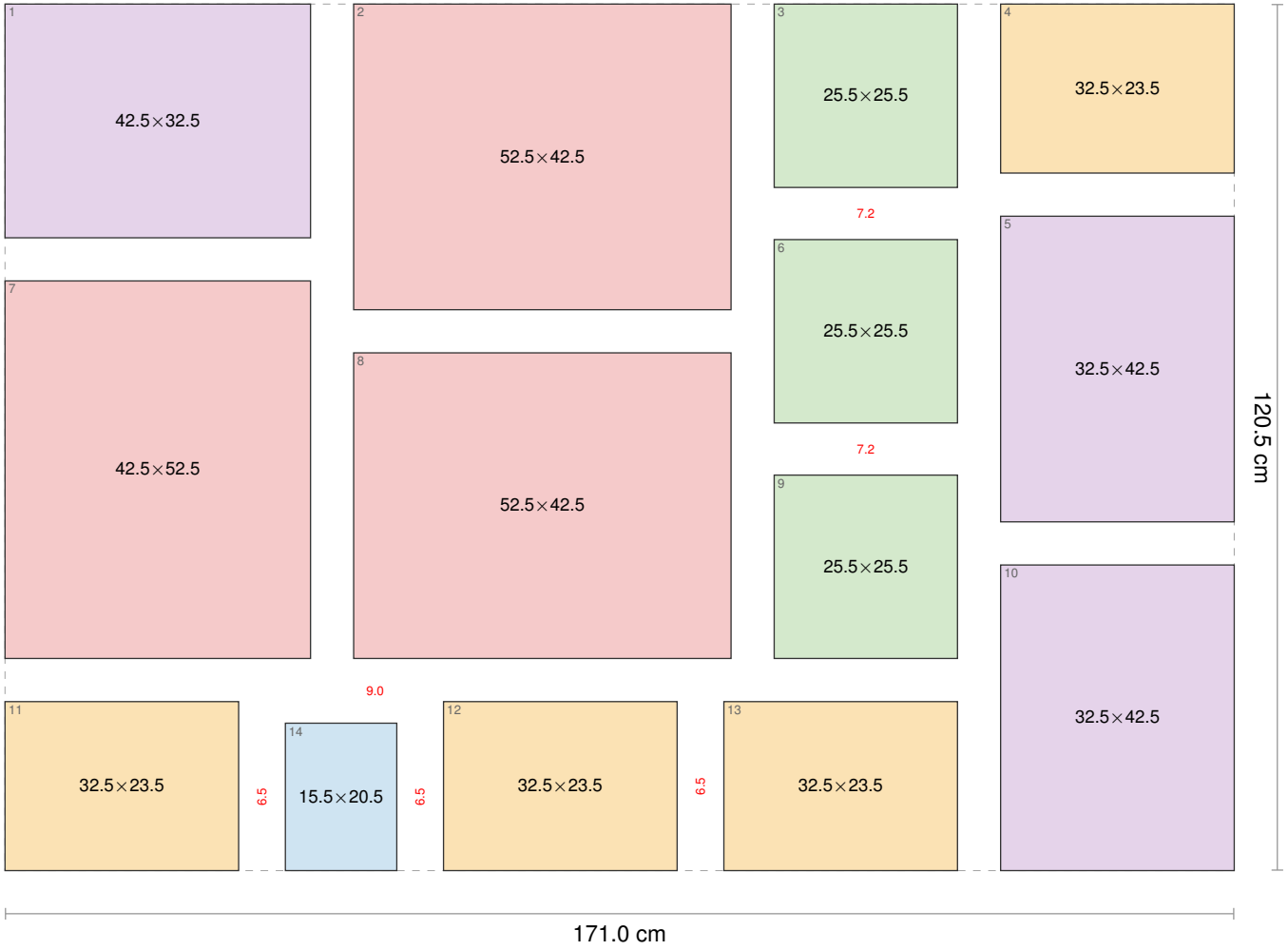
Outer size	177.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.54 cm
Border notches	bottom 6 cm
Corners filled	4 / 4
Frames used	21 of 28
Large frames	3, not touching, min. dist. 60 cm
Entropy	0.68
mix / variety	0.81 / 0.69
stagger / lines	0.46 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	2	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	42.5 × 32.5	0.0	0.0
2	20.5 × 15.5	44.0	0.0
3	23.5 × 32.5	66.0	0.0
4	52.5 × 42.5	91.0	0.0
5	32.5 × 23.5	145.0	0.0
6	20.5 × 15.5	44.0	17.0
7	32.5 × 42.5	145.0	26.0
8	42.5 × 52.5	45.5	34.0
9	20.5 × 15.5	0.0	35.0
10	20.5 × 15.5	22.0	35.0
11	25.5 × 25.5	91.0	44.0
12	25.5 × 25.5	118.0	44.0
13	20.5 × 15.5	0.0	53.0
14	20.5 × 15.5	22.0	53.0
15	20.5 × 15.5	0.0	71.0
16	20.5 × 15.5	22.0	71.0
17	52.5 × 42.5	91.0	71.0
18	32.5 × 42.5	145.0	71.0
19	25.5 × 25.5	0.0	88.0
20	25.5 × 25.5	28.5	88.0
21	32.5 × 23.5	57.0	90.0

Layout 14 171.0 × 120.5 cm gap 6 cm 14 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	171.0 × 120.5 cm
Nominal gap	6 cm
Actual gaps	6.0 – 9.0 cm
Max. deviation	3.0 cm
Mean deviation	0.29 cm
Border notches	none
Corners filled	4 / 4
Frames used	14 of 28
Large frames	3, touching 2× (95 cm), min. dist. 48 cm
Entropy	0.81
mix / variety	0.82 / 0.86
stagger / lines	0.55 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	0	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	0	1
	25.5×25.5		3	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	42.5 × 32.5	0.0	0.0
2	52.5 × 42.5	48.5	0.0
3	25.5 × 25.5	107.0	0.0
4	32.5 × 23.5	138.5	0.0
5	32.5 × 42.5	138.5	29.5
6	25.5 × 25.5	107.0	32.8
7	42.5 × 52.5	0.0	38.5
8	52.5 × 42.5	48.5	48.5
9	25.5 × 25.5	107.0	65.5
10	32.5 × 42.5	138.5	78.0
11	32.5 × 23.5	0.0	97.0
12	32.5 × 23.5	61.0	97.0
13	32.5 × 23.5	100.0	97.0
14	15.5 × 20.5	39.0	100.0

Layout 15 187.5 × 130.0 cm gap 2 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

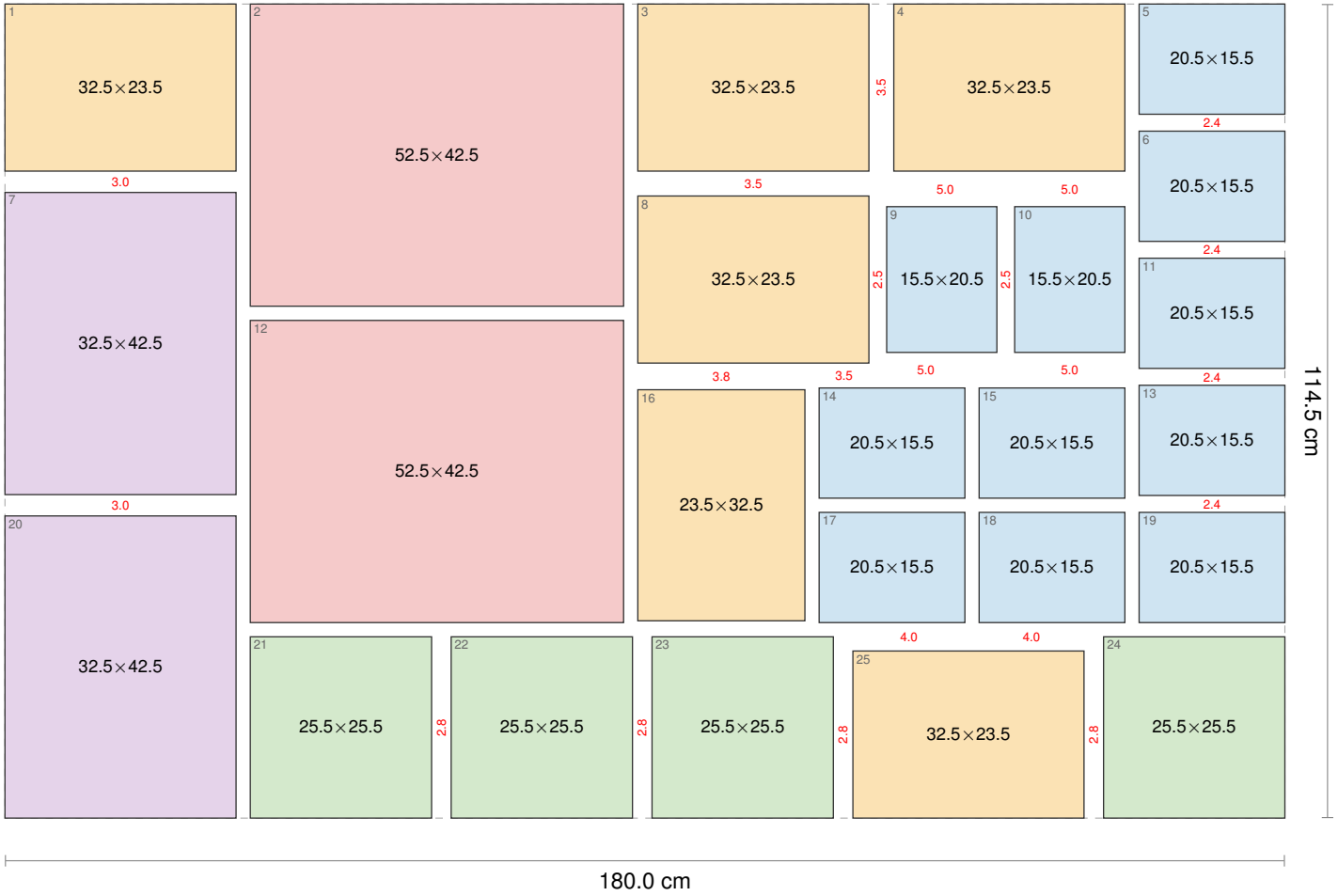
Outer size	187.5 × 130.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 5.5 cm
Max. deviation	3.5 cm
Mean deviation	0.78 cm
Border notches	bottom 4 cm
Corners filled	4 / 4
Frames used	26 of 28
Large frames	3, not touching, min. dist. 94 cm
Entropy	0.83
mix / variety	0.88 / 0.89
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	35.5	0.0
3	25.5 × 25.5	90.0	0.0
4	25.5 × 25.5	117.5	0.0
5	42.5 × 52.5	145.0	0.0
6	32.5 × 23.5	90.0	29.0
7	15.5 × 20.5	127.5	30.5
8	20.5 × 15.5	0.0	44.5
9	20.5 × 15.5	22.5	44.5
10	20.5 × 15.5	45.0	44.5
11	20.5 × 15.5	67.5	44.5
12	20.5 × 15.5	92.5	54.5
13	23.5 × 32.5	118.5	54.5
14	42.5 × 32.5	145.0	54.5
15	32.5 × 23.5	0.0	62.0
16	32.5 × 23.5	55.5	62.0
17	15.5 × 20.5	36.2	63.5
18	25.5 × 25.5	90.0	74.5
19	52.5 × 42.5	0.0	87.5
20	32.5 × 42.5	55.5	87.5
21	20.5 × 15.5	118.5	89.0
22	20.5 × 15.5	142.8	89.0
23	20.5 × 15.5	167.0	89.0
24	25.5 × 25.5	90.0	104.5
25	32.5 × 23.5	118.5	106.5
26	32.5 × 23.5	155.0	106.5

Layout 16 180.0 × 114.5 cm gap 2 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	180.0 × 114.5 cm
Nominal gap	2 cm
Actual gaps	2.0 – 5.0 cm
Max. deviation	3.0 cm
Mean deviation	0.64 cm
Border notches	top 4 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, touching 1× (52 cm), min. dist. 44 cm
Entropy	0.66
mix / variety	0.78 / 0.59
stagger / lines	0.52 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	2	2
32.5×42.5	P	2	2
42.5×32.5	L	0	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	0	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	52.5 × 42.5	34.5	0.0
3	32.5 × 23.5	89.0	0.0
4	32.5 × 23.5	125.0	0.0
5	20.5 × 15.5	159.5	0.0
6	20.5 × 15.5	159.5	17.9
7	32.5 × 42.5	0.0	26.5
8	32.5 × 23.5	89.0	27.0
9	15.5 × 20.5	124.0	28.5
10	15.5 × 20.5	142.0	28.5
11	20.5 × 15.5	159.5	35.8
12	52.5 × 42.5	34.5	44.5
13	20.5 × 15.5	159.5	53.6
14	20.5 × 15.5	114.5	54.0
15	20.5 × 15.5	137.0	54.0
16	23.5 × 32.5	89.0	54.2
17	20.5 × 15.5	114.5	71.5
18	20.5 × 15.5	137.0	71.5
19	20.5 × 15.5	159.5	71.5
20	32.5 × 42.5	0.0	72.0
21	25.5 × 25.5	34.5	89.0
22	25.5 × 25.5	62.8	89.0
23	25.5 × 25.5	91.0	89.0
24	25.5 × 25.5	154.5	89.0
25	32.5 × 23.5	119.2	91.0

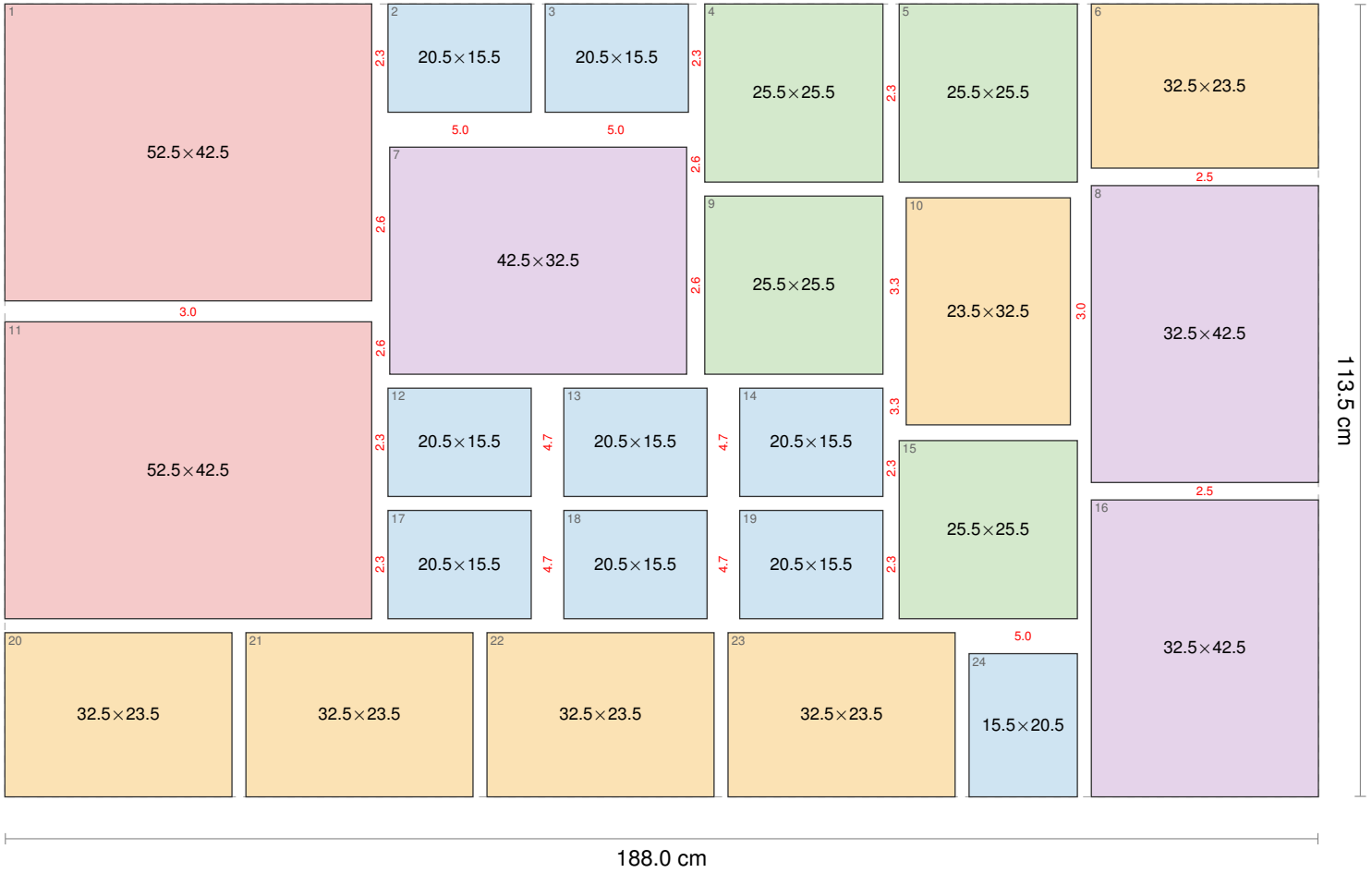
5 per-run-gap

Every annealing run keeps one fixed nominal gap from the grid 1.5, 2, . . . , 6 cm (the runs are spread over the grid) instead of changing the gap during the search. Each run explores its gap value more thoroughly, which gives a more diverse set of results across gap values.

Parameters that differ from the baseline: gap_mode = per-run.
256 runs, 20 min wall-clock budget, 542 candidate states, the 16 best distinct layouts are shown.

#	size (cm)	gap	frames	max dev	mean dev	notches	corners	entropy
1	188.0 × 113.5	2	24	3.0	0.62	none	4/4	0.76
2	167.5 × 130.5	3.5	24	3.0	0.72	none	4/4	0.75
3	183.5 × 115.0	2	23	2.2	0.53	4 cm	4/4	0.62
4	169.5 × 117.5	4.5	21	3.0	0.57	none	4/4	0.72
5	179.5 × 123.5	2	25	2.5	0.87	9 cm	4/4	0.73
6	177.5 × 126.5	1.5	24	3.0	0.63	4 cm	4/4	0.75
7	175.5 × 123.5	2.5	24	4.0	0.74	none	4/4	0.77
8	189.0 × 118.0	3	23	2.0	0.62	9 cm	4/4	0.75
9	183.5 × 130.5	4.5	23	3.6	0.60	none	4/4	0.76
10	189.0 × 119.0	5	22	3.5	0.54	none	4/4	0.75
11	189.5 × 116.5	2.5	21	2.0	0.59	9 cm	4/4	0.78
12	177.5 × 113.5	1.5	21	2.0	0.61	10 cm	4/4	0.66
13	179.5 × 126.5	5.5	21	2.7	0.92	8 cm	4/4	0.76
14	181.5 × 113.5	1.5	20	2.0	0.67	10 cm	4/4	0.64
15	191.0 × 103.5	6	16	3.5	0.50	none	4/4	0.67
16	182.5 × 123.5	1.5	26	5.0	0.67	none	4/4	0.78

Layout 1 188.0 × 113.5 cm gap 2 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

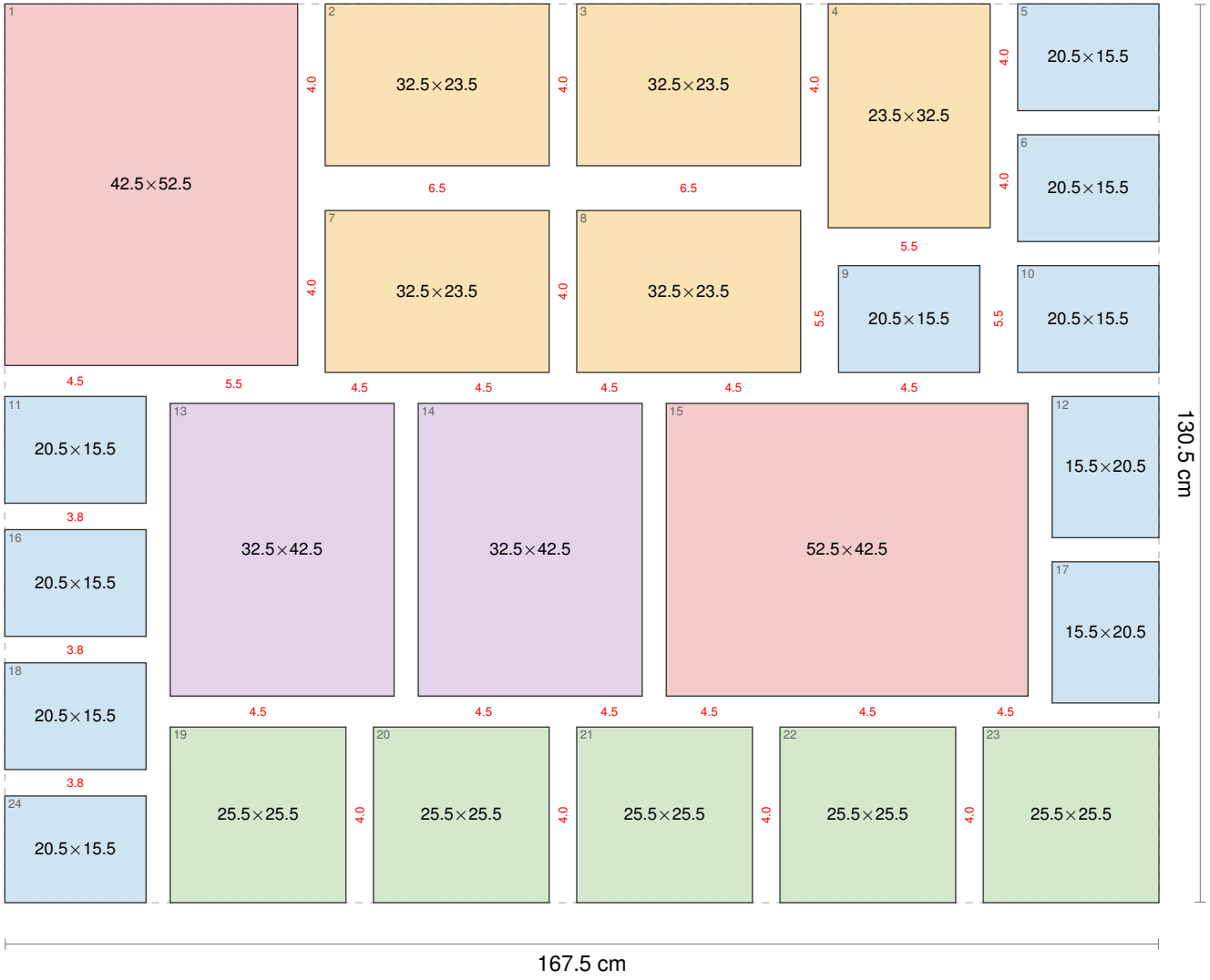
Outer size	188.0 × 113.5 cm
Nominal gap	2 cm
Actual gaps	2.0 – 5.0 cm
Max. deviation	3.0 cm
Mean deviation	0.62 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, touching 1× (52 cm), min. dist. 46 cm
Entropy	0.76
mix / variety	0.82 / 0.72
stagger / lines	0.51 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	8	9
15.5×20.5	P	1	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	0	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	20.5 × 15.5	54.8	0.0
3	20.5 × 15.5	77.3	0.0
4	25.5 × 25.5	100.2	0.0
5	25.5 × 25.5	128.0	0.0
6	32.5 × 23.5	155.5	0.0
7	42.5 × 32.5	55.1	20.5
8	32.5 × 42.5	155.5	26.0
9	25.5 × 25.5	100.2	27.5
10	23.5 × 32.5	129.0	27.8
11	52.5 × 42.5	0.0	45.5
12	20.5 × 15.5	54.8	55.0
13	20.5 × 15.5	80.0	55.0
14	20.5 × 15.5	105.2	55.0
15	25.5 × 25.5	128.0	62.5
16	32.5 × 42.5	155.5	71.0
17	20.5 × 15.5	54.8	72.5
18	20.5 × 15.5	80.0	72.5
19	20.5 × 15.5	105.2	72.5
20	32.5 × 23.5	0.0	90.0
21	32.5 × 23.5	34.5	90.0
22	32.5 × 23.5	69.0	90.0
23	32.5 × 23.5	103.5	90.0
24	15.5 × 20.5	138.0	93.0

Layout 2 167.5 × 130.5 cm gap 3.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

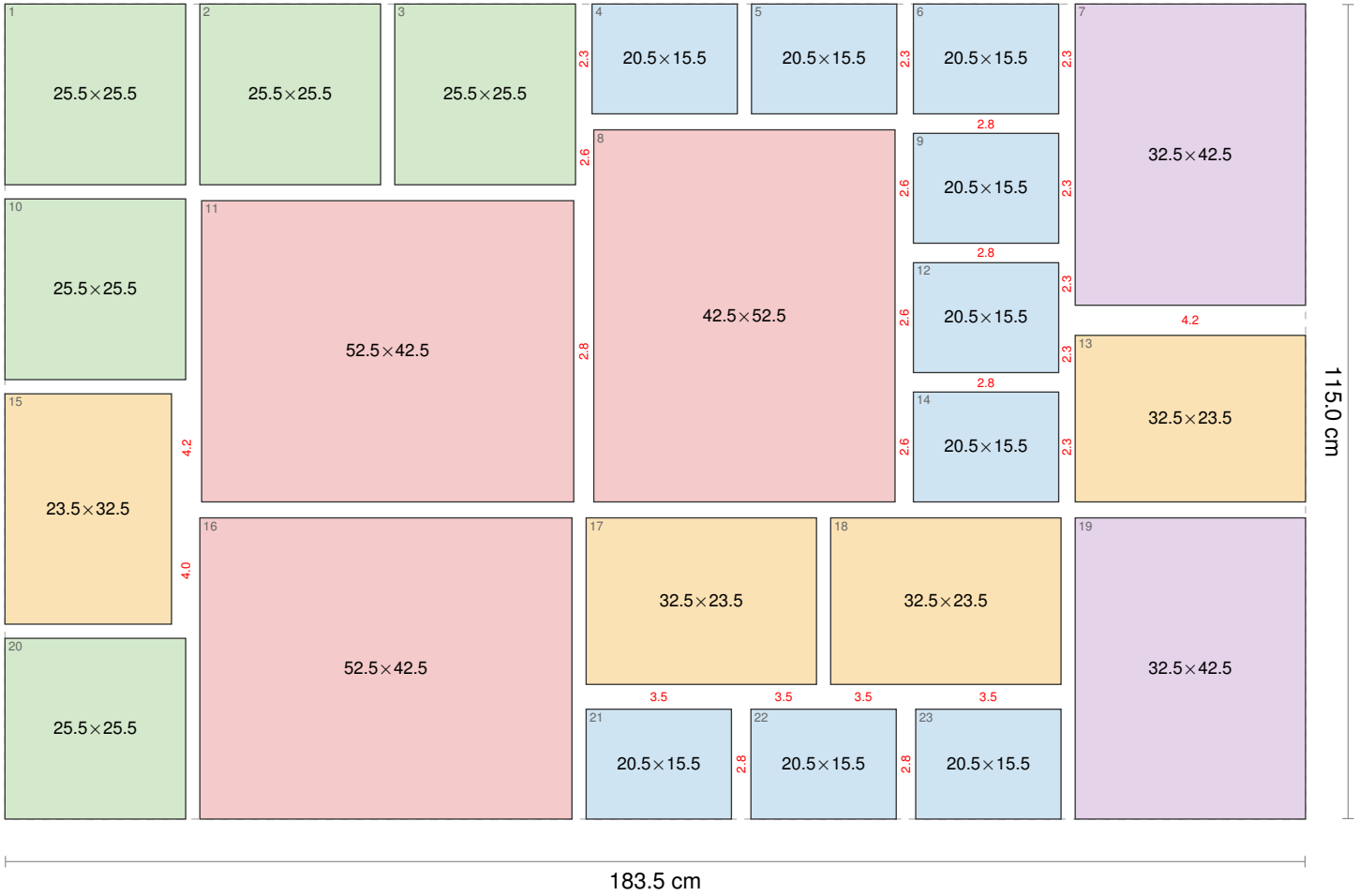
Outer size	167.5 × 130.5 cm
Nominal gap	3.5 cm
Actual gaps	3.5 – 6.5 cm
Max. deviation	3.0 cm
Mean deviation	0.72 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 114 cm
Entropy	0.75
mix / variety	0.82 / 0.67
stagger / lines	0.52 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	8	9
15.5×20.5	P	2	2
32.5×42.5	P	2	2
42.5×32.5	L	0	1
32.5×23.5	L	4	5
23.5×32.5	P	1	1
25.5×25.5		5	5
42.5×52.5	P	1	1
52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	42.5 × 52.5	0.0	0.0
2	32.5 × 23.5	46.5	0.0
3	32.5 × 23.5	83.0	0.0
4	23.5 × 32.5	119.5	0.0
5	20.5 × 15.5	147.0	0.0
6	20.5 × 15.5	147.0	19.0
7	32.5 × 23.5	46.5	30.0
8	32.5 × 23.5	83.0	30.0
9	20.5 × 15.5	121.0	38.0
10	20.5 × 15.5	147.0	38.0
11	20.5 × 15.5	0.0	57.0
12	15.5 × 20.5	152.0	57.0
13	32.5 × 42.5	24.0	58.0
14	32.5 × 42.5	60.0	58.0
15	52.5 × 42.5	96.0	58.0
16	20.5 × 15.5	0.0	76.3
17	15.5 × 20.5	152.0	81.0
18	20.5 × 15.5	0.0	95.7
19	25.5 × 25.5	24.0	105.0
20	25.5 × 25.5	53.5	105.0
21	25.5 × 25.5	83.0	105.0
22	25.5 × 25.5	112.5	105.0
23	25.5 × 25.5	142.0	105.0
24	20.5 × 15.5	0.0	115.0

Layout 3 183.5 × 115.0 cm gap 2 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

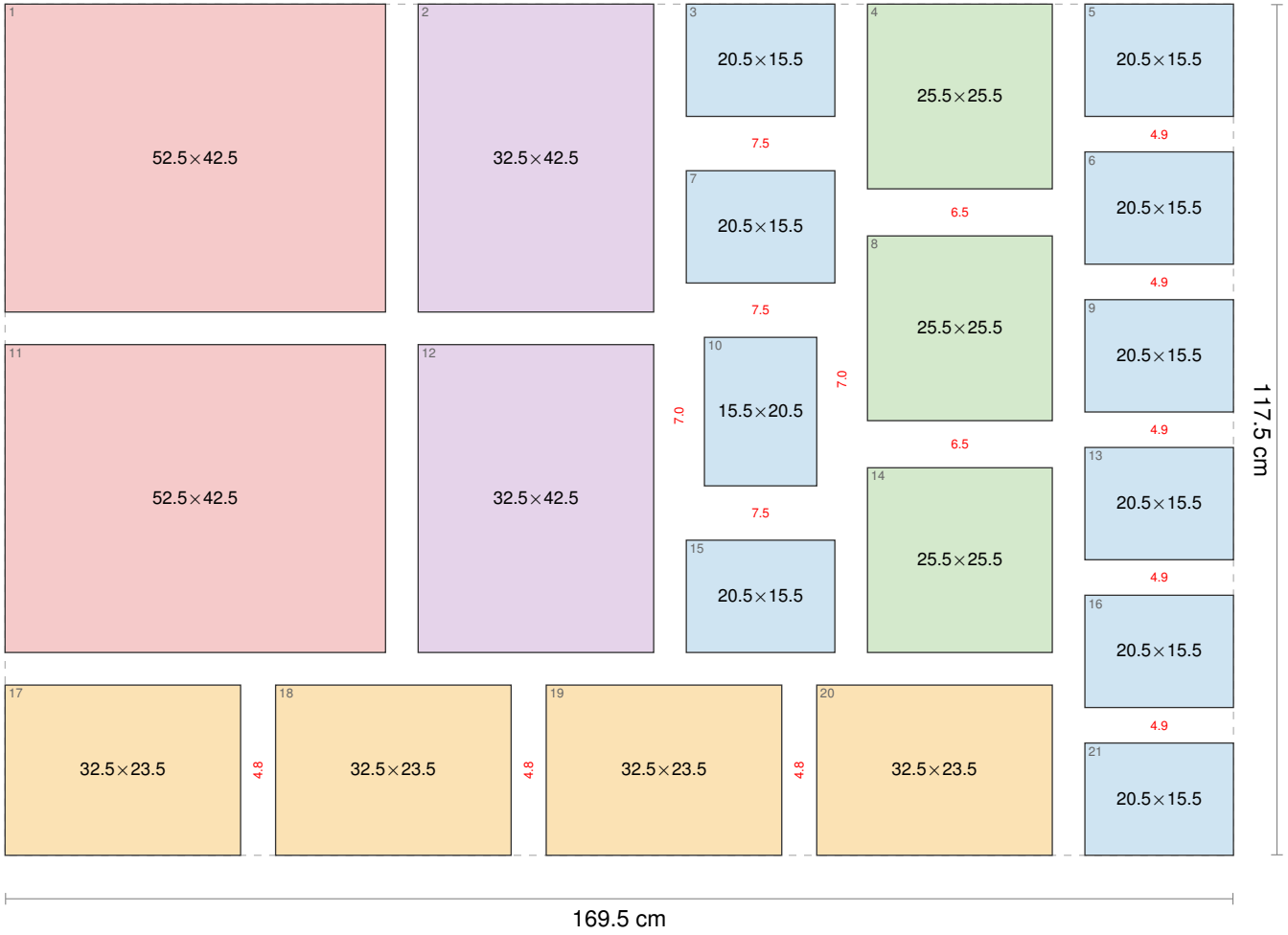
Outer size	183.5 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.2 cm
Max. deviation	2.2 cm
Mean deviation	0.53 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 2× (95 cm), min. dist. 45 cm
Entropy	0.62
mix / variety	0.76 / 0.69
stagger / lines	0.53 / 0.50 (3 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	27.5	0.0
3	25.5 × 25.5	55.0	0.0
4	20.5 × 15.5	82.8	0.0
5	20.5 × 15.5	105.3	0.0
6	20.5 × 15.5	128.2	0.0
7	32.5 × 42.5	151.0	0.0
8	42.5 × 52.5	83.1	17.8
9	20.5 × 15.5	128.2	18.2
10	25.5 × 25.5	0.0	27.5
11	52.5 × 42.5	27.8	27.8
12	20.5 × 15.5	128.2	36.5
13	32.5 × 23.5	151.0	46.8
14	20.5 × 15.5	128.2	54.8
15	23.5 × 32.5	0.0	55.0
16	52.5 × 42.5	27.5	72.5
17	32.5 × 23.5	82.0	72.5
18	32.5 × 23.5	116.5	72.5
19	32.5 × 42.5	151.0	72.5
20	25.5 × 25.5	0.0	89.5
21	20.5 × 15.5	82.0	99.5
22	20.5 × 15.5	105.2	99.5
23	20.5 × 15.5	128.5	99.5

Layout 4 169.5 × 117.5 cm gap 4.5 cm 21 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

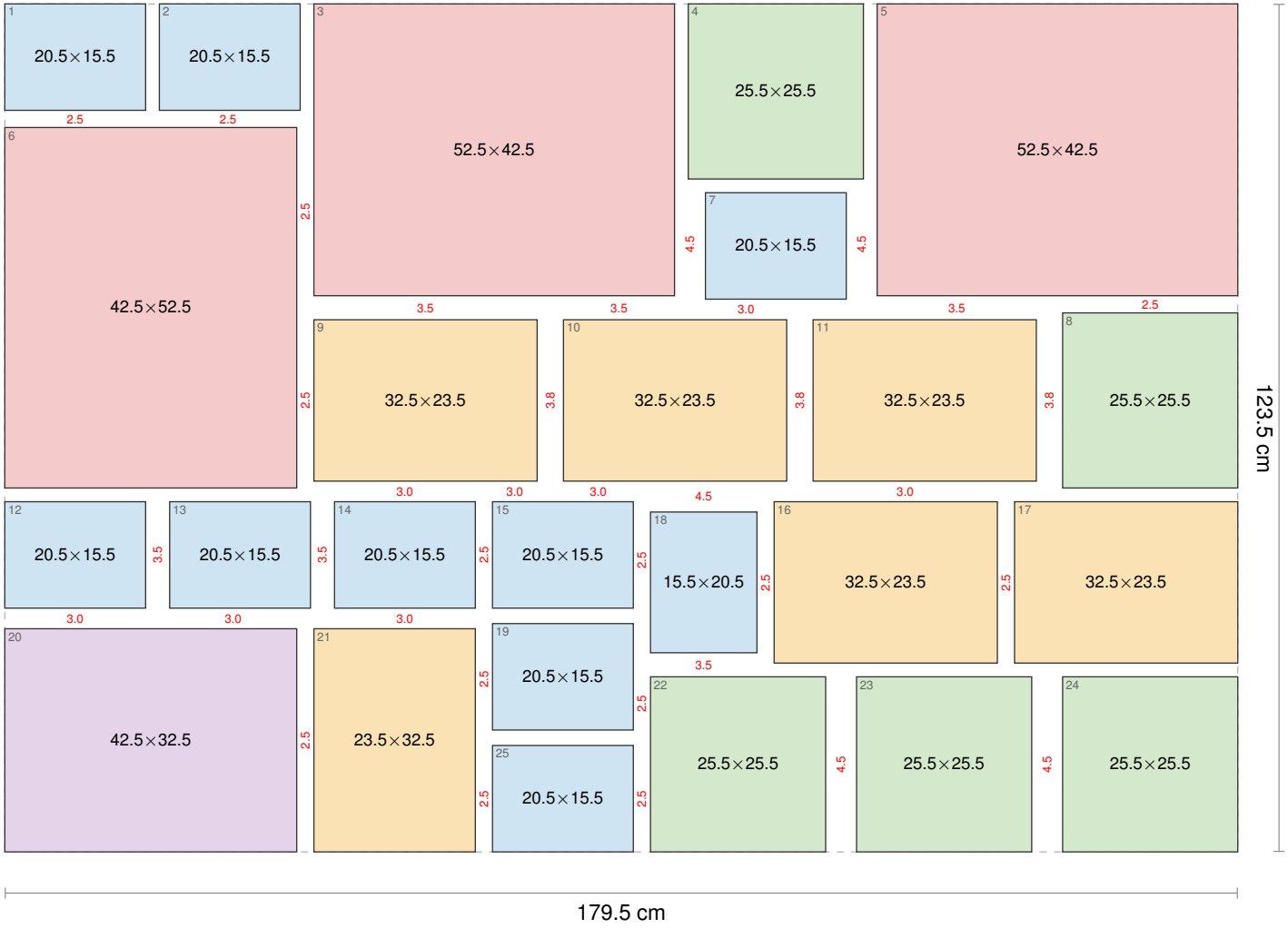
Outer size	169.5 × 117.5 cm
Nominal gap	4.5 cm
Actual gaps	4.5 – 7.5 cm
Max. deviation	3.0 cm
Mean deviation	0.57 cm
Border notches	none
Corners filled	4 / 4
Frames used	21 of 28
Large frames	2, touching 1× (52 cm), min. dist. 47 cm
Entropy	0.72
mix / variety	0.71 / 0.67
stagger / lines	0.50 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	4	5
	23.5×32.5	P	0	1
	25.5×25.5		3	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 42.5	57.0	0.0
3	20.5 × 15.5	94.0	0.0
4	25.5 × 25.5	119.0	0.0
5	20.5 × 15.5	149.0	0.0
6	20.5 × 15.5	149.0	20.4
7	20.5 × 15.5	94.0	23.0
8	25.5 × 25.5	119.0	32.0
9	20.5 × 15.5	149.0	40.8
10	15.5 × 20.5	96.5	46.0
11	52.5 × 42.5	0.0	47.0
12	32.5 × 42.5	57.0	47.0
13	20.5 × 15.5	149.0	61.2
14	25.5 × 25.5	119.0	64.0
15	20.5 × 15.5	94.0	74.0
16	20.5 × 15.5	149.0	81.6
17	32.5 × 23.5	0.0	94.0
18	32.5 × 23.5	37.3	94.0
19	32.5 × 23.5	74.7	94.0
20	32.5 × 23.5	112.0	94.0
21	20.5 × 15.5	149.0	102.0

Layout 5 179.5 × 123.5 cm gap 2 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

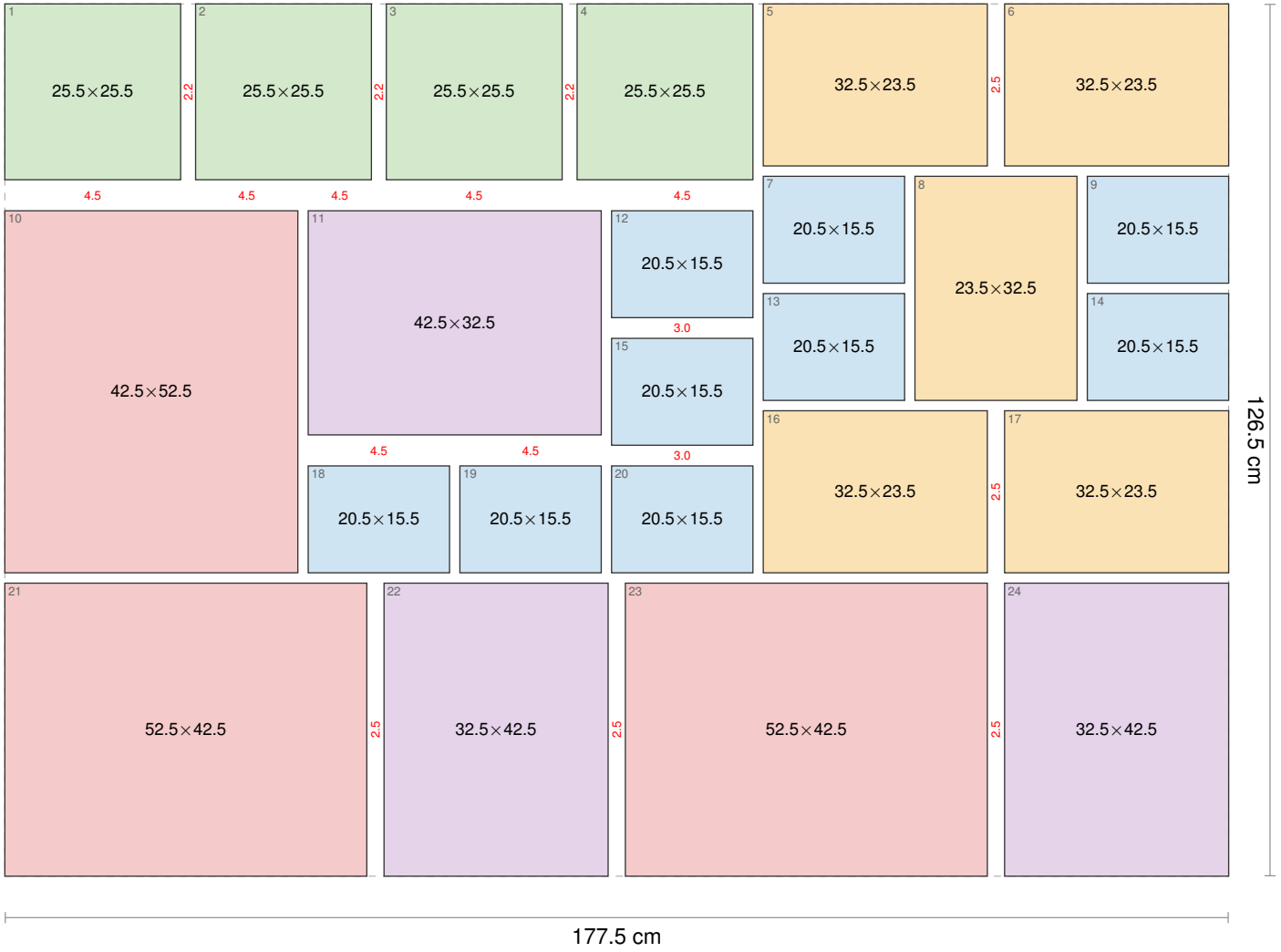
Outer size	179.5 × 123.5 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.5 cm
Max. deviation	2.5 cm
Mean deviation	0.87 cm
Border notches	bottom 9 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, touching 1× (24 cm), min. dist. 55 cm
Entropy	0.73
mix / variety	0.79 / 0.83
stagger / lines	0.55 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	0	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	22.5	0.0
3	52.5 × 42.5	45.0	0.0
4	25.5 × 25.5	99.5	0.0
5	52.5 × 42.5	127.0	0.0
6	42.5 × 52.5	0.0	18.0
7	20.5 × 15.5	102.0	27.5
8	25.5 × 25.5	154.0	45.0
9	32.5 × 23.5	45.0	46.0
10	32.5 × 23.5	81.3	46.0
11	32.5 × 23.5	117.7	46.0
12	20.5 × 15.5	0.0	72.5
13	20.5 × 15.5	24.0	72.5
14	20.5 × 15.5	48.0	72.5
15	20.5 × 15.5	71.0	72.5
16	32.5 × 23.5	112.0	72.5
17	32.5 × 23.5	147.0	72.5
18	15.5 × 20.5	94.0	74.0
19	20.5 × 15.5	71.0	90.2
20	42.5 × 32.5	0.0	91.0
21	23.5 × 32.5	45.0	91.0
22	25.5 × 25.5	94.0	98.0
23	25.5 × 25.5	124.0	98.0
24	25.5 × 25.5	154.0	98.0
25	20.5 × 15.5	71.0	108.0

Layout 6 177.5 × 126.5 cm gap 1.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

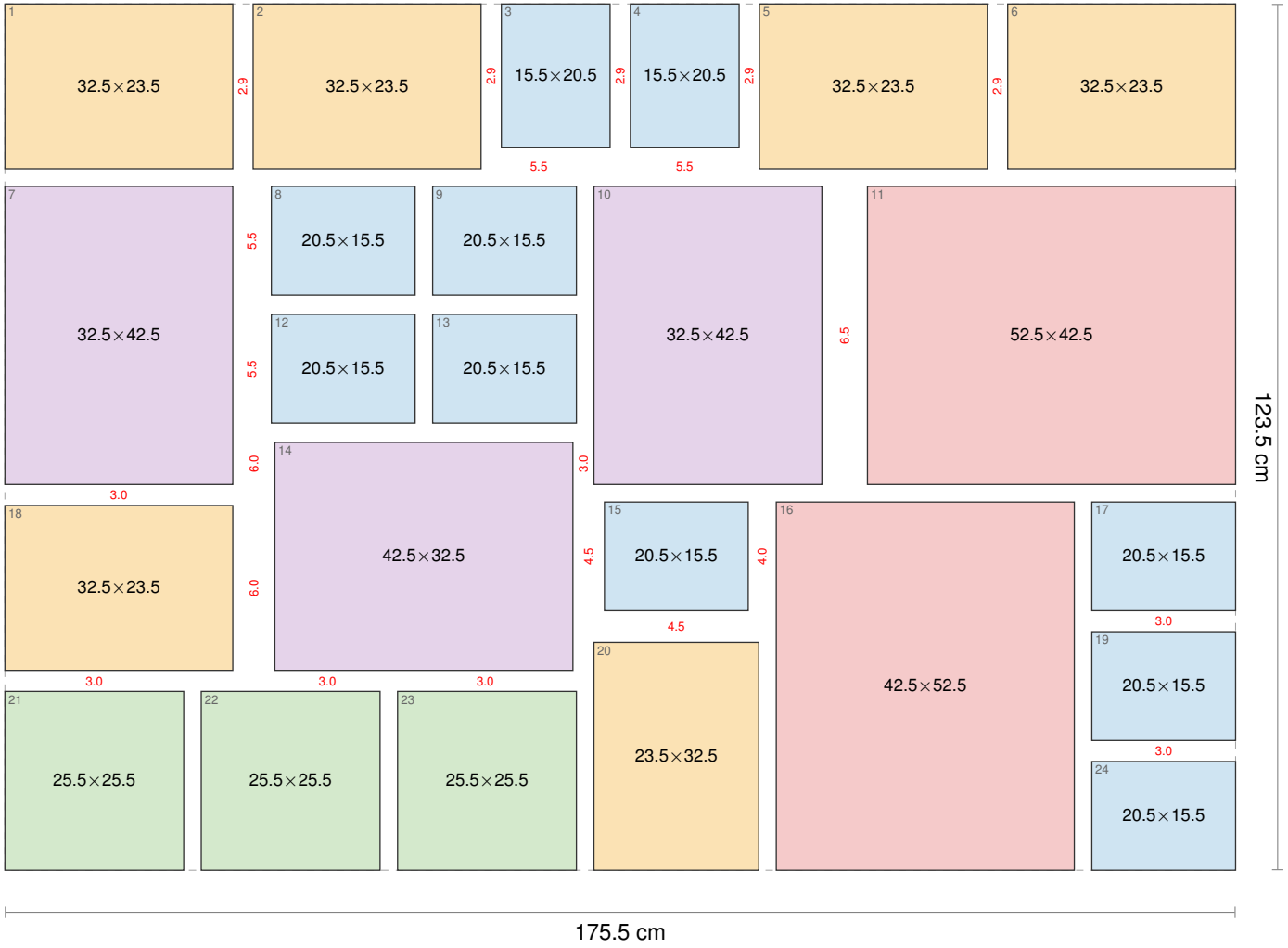
Outer size	177.5 × 126.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.5 cm
Max. deviation	3.0 cm
Mean deviation	0.63 cm
Border notches	left 4 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	3, touching 1× (42 cm), min. dist. 49 cm
Entropy	0.75
mix / variety	0.81 / 0.72
stagger / lines	0.45 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	27.7	0.0
3	25.5 × 25.5	55.3	0.0
4	25.5 × 25.5	83.0	0.0
5	32.5 × 23.5	110.0	0.0
6	32.5 × 23.5	145.0	0.0
7	20.5 × 15.5	110.0	25.0
8	23.5 × 32.5	132.0	25.0
9	20.5 × 15.5	157.0	25.0
10	42.5 × 52.5	0.0	30.0
11	42.5 × 32.5	44.0	30.0
12	20.5 × 15.5	88.0	30.0
13	20.5 × 15.5	110.0	42.0
14	20.5 × 15.5	157.0	42.0
15	20.5 × 15.5	88.0	48.5
16	32.5 × 23.5	110.0	59.0
17	32.5 × 23.5	145.0	59.0
18	20.5 × 15.5	44.0	67.0
19	20.5 × 15.5	66.0	67.0
20	20.5 × 15.5	88.0	67.0
21	52.5 × 42.5	0.0	84.0
22	32.5 × 42.5	55.0	84.0
23	52.5 × 42.5	90.0	84.0
24	32.5 × 42.5	145.0	84.0

Layout 7 175.5 × 123.5 cm gap 2.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

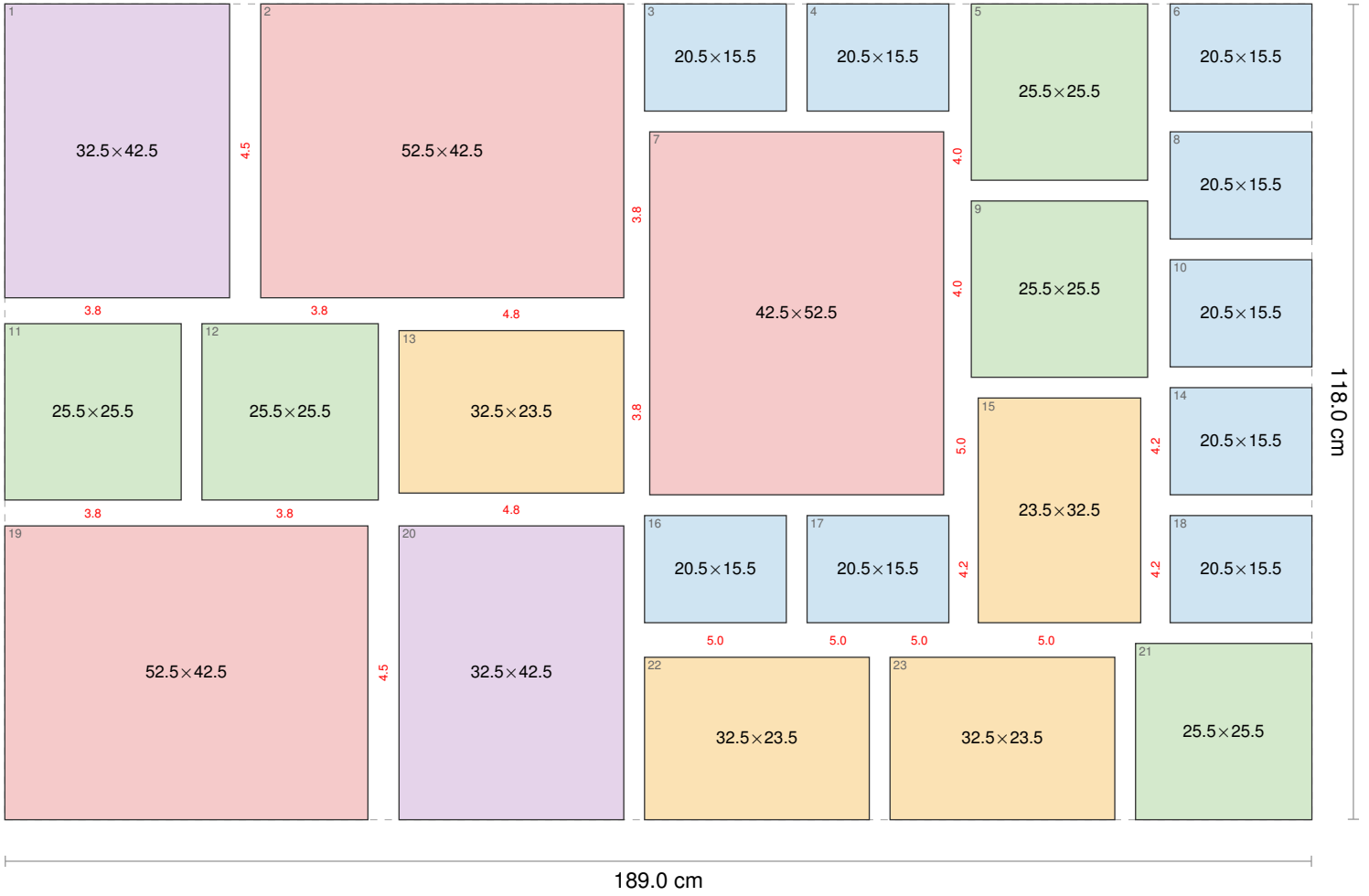
Outer size	175.5 × 123.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 6.5 cm
Max. deviation	4.0 cm
Mean deviation	0.74 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, touching 1× (30 cm), min. dist. 53 cm
Entropy	0.77
mix / variety	0.86 / 0.70
stagger / lines	0.52 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		3	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	32.5 × 23.5	35.4	0.0
3	15.5 × 20.5	70.8	0.0
4	15.5 × 20.5	89.2	0.0
5	32.5 × 23.5	107.6	0.0
6	32.5 × 23.5	143.0	0.0
7	32.5 × 42.5	0.0	26.0
8	20.5 × 15.5	38.0	26.0
9	20.5 × 15.5	61.0	26.0
10	32.5 × 42.5	84.0	26.0
11	52.5 × 42.5	123.0	26.0
12	20.5 × 15.5	38.0	44.2
13	20.5 × 15.5	61.0	44.2
14	42.5 × 32.5	38.5	62.5
15	20.5 × 15.5	85.5	71.0
16	42.5 × 52.5	110.0	71.0
17	20.5 × 15.5	155.0	71.0
18	32.5 × 23.5	0.0	71.5
19	20.5 × 15.5	155.0	89.5
20	23.5 × 32.5	84.0	91.0
21	25.5 × 25.5	0.0	98.0
22	25.5 × 25.5	28.0	98.0
23	25.5 × 25.5	56.0	98.0
24	20.5 × 15.5	155.0	108.0

Layout 8 189.0 × 118.0 cm gap 3 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	189.0 × 118.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 5.0 cm
Max. deviation	2.0 cm
Mean deviation	0.62 cm
Border notches	top 4 cm, bottom 4 cm
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 1× (24 cm), min. dist. 56 cm
Entropy	0.75
mix / variety	0.76 / 0.73
stagger / lines	0.52 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	37.0	0.0
3	20.5 × 15.5	92.5	0.0
4	20.5 × 15.5	116.0	0.0
5	25.5 × 25.5	139.8	0.0
6	20.5 × 15.5	168.5	0.0
7	42.5 × 52.5	93.2	18.5
8	20.5 × 15.5	168.5	18.5
9	25.5 × 25.5	139.8	28.5
10	20.5 × 15.5	168.5	37.0
11	25.5 × 25.5	0.0	46.2
12	25.5 × 25.5	28.5	46.2
13	32.5 × 23.5	57.0	47.2
14	20.5 × 15.5	168.5	55.5
15	23.5 × 32.5	140.8	57.0
16	20.5 × 15.5	92.5	74.0
17	20.5 × 15.5	116.0	74.0
18	20.5 × 15.5	168.5	74.0
19	52.5 × 42.5	0.0	75.5
20	32.5 × 42.5	57.0	75.5
21	25.5 × 25.5	163.5	92.5
22	32.5 × 23.5	92.5	94.5
23	32.5 × 23.5	128.0	94.5

Layout 9 183.5 × 130.5 cm gap 4.5 cm 23 frames



Outer size	183.5 × 130.5 cm
Nominal gap	4.5 cm
Actual gaps	4.5 – 8.1 cm
Max. deviation	3.6 cm
Mean deviation	0.60 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 1× (42 cm), min. dist. 57 cm
Entropy	0.76
mix / variety	0.86 / 0.75
stagger / lines	0.45 / 1.00 (1 through)

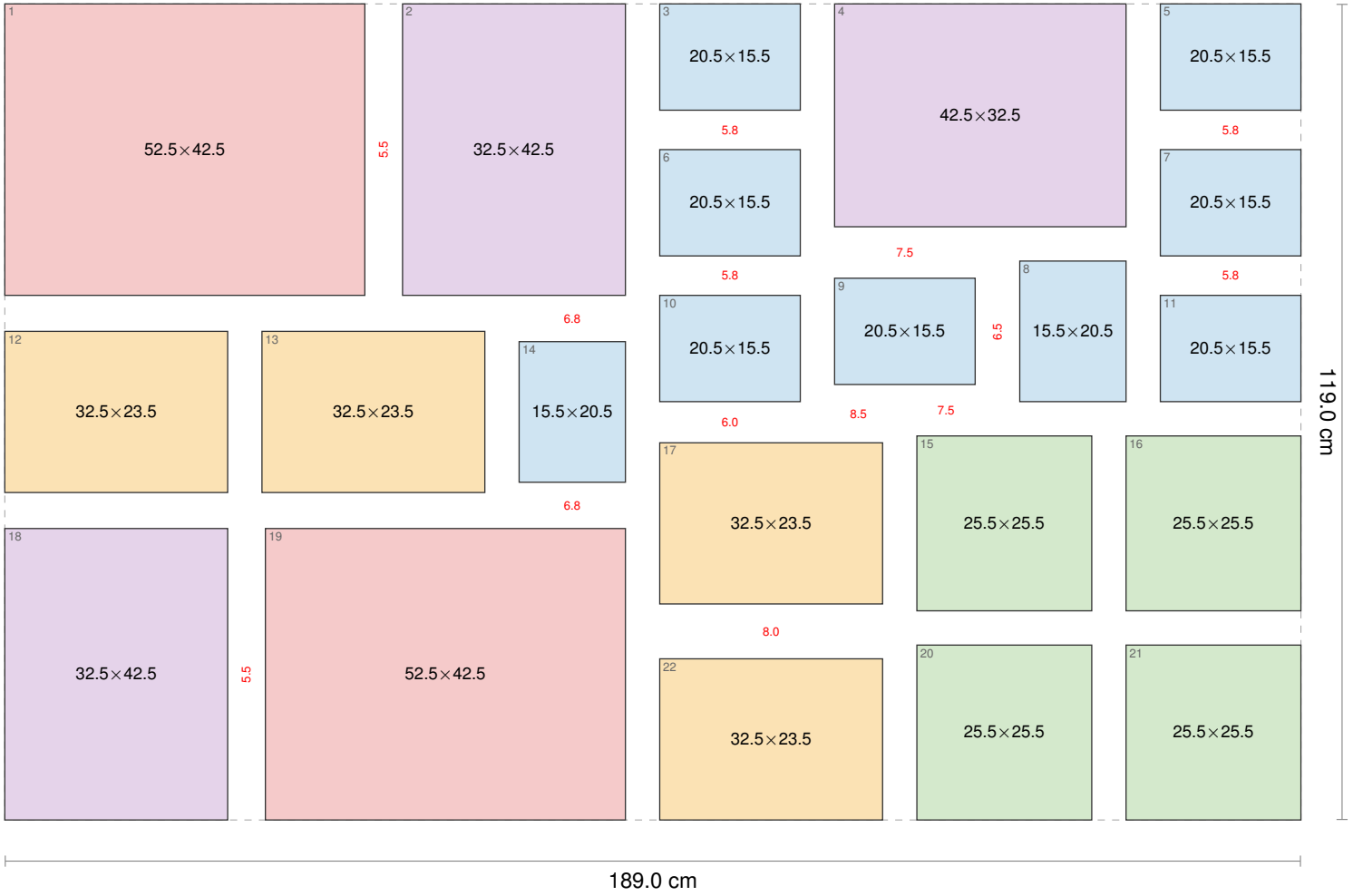
	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	52.5 × 42.5	37.0	0.0
3	52.5 × 42.5	94.0	0.0
4	32.5 × 23.5	151.0	0.0
5	32.5 × 42.5	0.0	28.0
6	32.5 × 42.5	151.0	28.0
7	32.5 × 23.5	37.0	47.0
8	32.5 × 23.5	94.0	47.0
9	15.5 × 20.5	74.0	48.5
10	15.5 × 20.5	131.0	48.5
11	20.5 × 15.5	0.0	75.0
12	20.5 × 15.5	73.2	75.0
13	20.5 × 15.5	101.8	75.0
14	25.5 × 25.5	127.4	75.0
15	25.5 × 25.5	158.0	75.0
16	42.5 × 52.5	25.6	78.0
17	20.5 × 15.5	0.0	95.0
18	20.5 × 15.5	101.8	95.0
19	23.5 × 32.5	73.2	98.0
20	25.5 × 25.5	127.4	105.0
21	25.5 × 25.5	158.0	105.0
22	20.5 × 15.5	0.0	115.0
23	20.5 × 15.5	101.8	115.0

Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Layout 10 189.0 × 119.0 cm gap 5 cm 22 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	189.0 × 119.0 cm
Nominal gap	5 cm
Actual gaps	5.0 – 8.5 cm
Max. deviation	3.5 cm
Mean deviation	0.54 cm
Border notches	none
Corners filled	4 / 4
Frames used	22 of 28
Large frames	2, not touching, min. dist. 85 cm
Entropy	0.75
mix / variety	0.81 / 0.71
stagger / lines	0.48 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	7	9
15.5×20.5	P	2	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	4	5
23.5×32.5	P	0	1
25.5×25.5		4	5
42.5×52.5	P	0	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 42.5	58.0	0.0
3	20.5 × 15.5	95.5	0.0
4	42.5 × 32.5	121.0	0.0
5	20.5 × 15.5	168.5	0.0
6	20.5 × 15.5	95.5	21.2
7	20.5 × 15.5	168.5	21.2
8	15.5 × 20.5	148.0	37.5
9	20.5 × 15.5	121.0	40.0
10	20.5 × 15.5	95.5	42.5
11	20.5 × 15.5	168.5	42.5
12	32.5 × 23.5	0.0	47.8
13	32.5 × 23.5	37.5	47.8
14	15.5 × 20.5	75.0	49.2
15	25.5 × 25.5	133.0	63.0
16	25.5 × 25.5	163.5	63.0
17	32.5 × 23.5	95.5	64.0
18	32.5 × 42.5	0.0	76.5
19	52.5 × 42.5	38.0	76.5
20	25.5 × 25.5	133.0	93.5
21	25.5 × 25.5	163.5	93.5
22	32.5 × 23.5	95.5	95.5

Layout 11 189.5 × 116.5 cm gap 2.5 cm 21 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

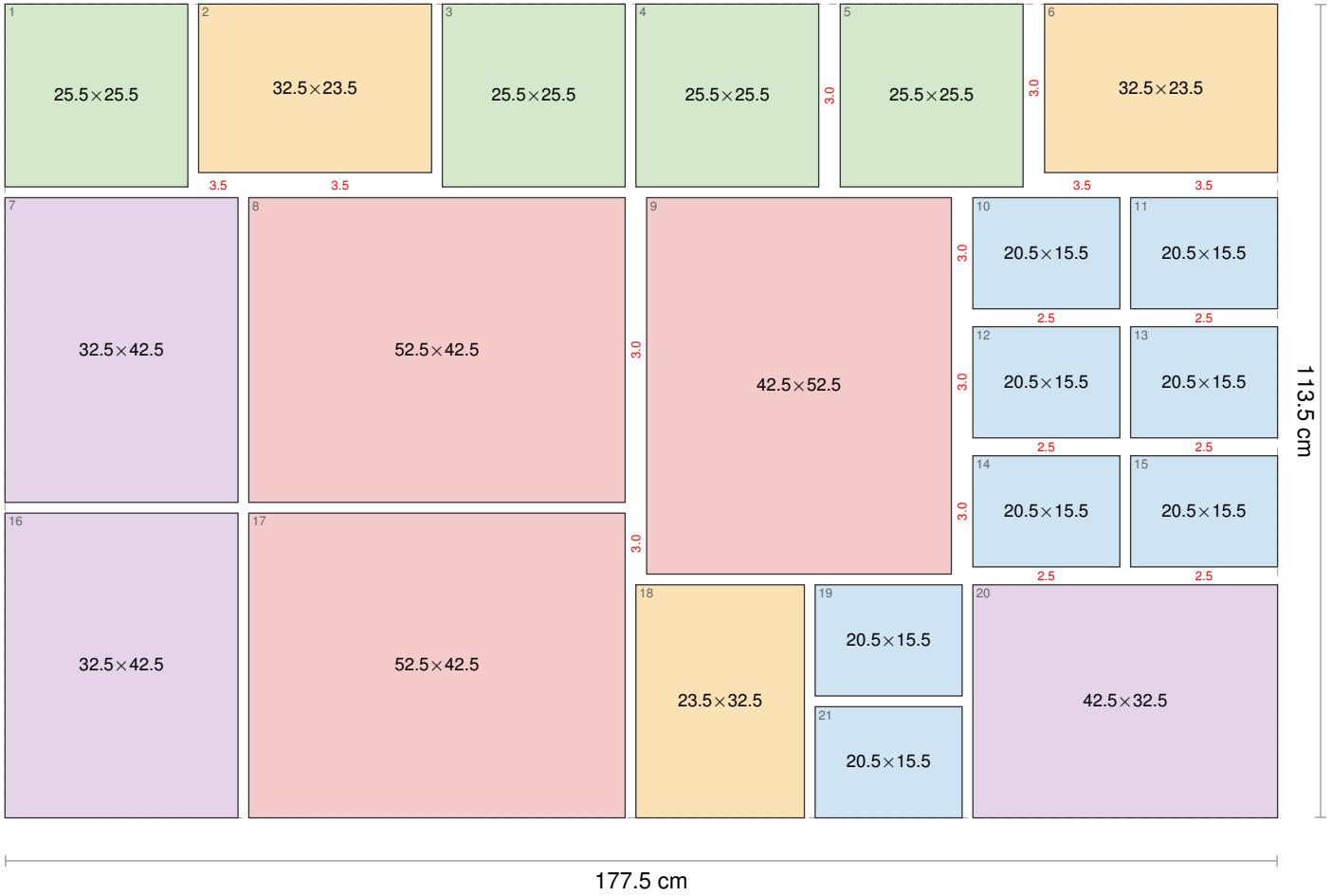
Outer size	189.5 × 116.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 4.5 cm
Max. deviation	2.0 cm
Mean deviation	0.59 cm
Border notches	top 4 cm, bottom 4 cm
Corners filled	4 / 4
Frames used	21 of 28
Large frames	3, touching 2× (21 cm), min. dist. 62 cm
Entropy	0.78
mix / variety	0.88 / 0.75
stagger / lines	0.50 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	4	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	0	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	29.0	0.0
3	32.5 × 23.5	65.0	0.0
4	52.5 × 42.5	100.0	0.0
5	32.5 × 42.5	157.0	0.0
6	42.5 × 32.5	0.0	28.0
7	52.5 × 42.5	45.0	28.0
8	25.5 × 25.5	100.0	45.0
9	25.5 × 25.5	164.0	45.0
10	32.5 × 23.5	128.5	46.0
11	42.5 × 52.5	0.0	64.0
12	15.5 × 20.5	45.0	73.0
13	25.5 × 25.5	100.0	73.0
14	25.5 × 25.5	164.0	73.0
15	32.5 × 42.5	65.0	74.0
16	32.5 × 23.5	128.5	74.0
17	15.5 × 20.5	45.0	96.0
18	20.5 × 15.5	100.0	101.0
19	20.5 × 15.5	123.0	101.0
20	20.5 × 15.5	146.0	101.0
21	20.5 × 15.5	169.0	101.0

Layout 12 177.5 × 113.5 cm gap 1.5 cm 21 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	177.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.61 cm
Border notches	top 6 cm, right 4 cm
Corners filled	4 / 4
Frames used	21 of 28
Large frames	3, touching 1× (52 cm), min. dist. 44 cm
Entropy	0.66
mix / variety	0.81 / 0.62
stagger / lines	0.45 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	2	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	27.0	0.0
3	25.5 × 25.5	61.0	0.0
4	25.5 × 25.5	88.0	0.0
5	25.5 × 25.5	116.5	0.0
6	32.5 × 23.5	145.0	0.0
7	32.5 × 42.5	0.0	27.0
8	52.5 × 42.5	34.0	27.0
9	42.5 × 52.5	89.5	27.0
10	20.5 × 15.5	135.0	27.0
11	20.5 × 15.5	157.0	27.0
12	20.5 × 15.5	135.0	45.0
13	20.5 × 15.5	157.0	45.0
14	20.5 × 15.5	135.0	63.0
15	20.5 × 15.5	157.0	63.0
16	32.5 × 42.5	0.0	71.0
17	52.5 × 42.5	34.0	71.0
18	23.5 × 32.5	88.0	81.0
19	20.5 × 15.5	113.0	81.0
20	42.5 × 32.5	135.0	81.0
21	20.5 × 15.5	113.0	98.0

Layout 13 179.5 × 126.5 cm gap 5.5 cm 21 frames



Outer size	179.5 × 126.5 cm
Nominal gap	5.5 cm
Actual gaps	5.5 – 8.2 cm
Max. deviation	2.7 cm
Mean deviation	0.92 cm
Border notches	bottom 8 cm
Corners filled	4 / 4
Frames used	21 of 28
Large frames	3, not touching, min. dist. 79 cm
Entropy	0.76
mix / variety	0.80 / 0.71
stagger / lines	0.52 / 1.00 (1 through)

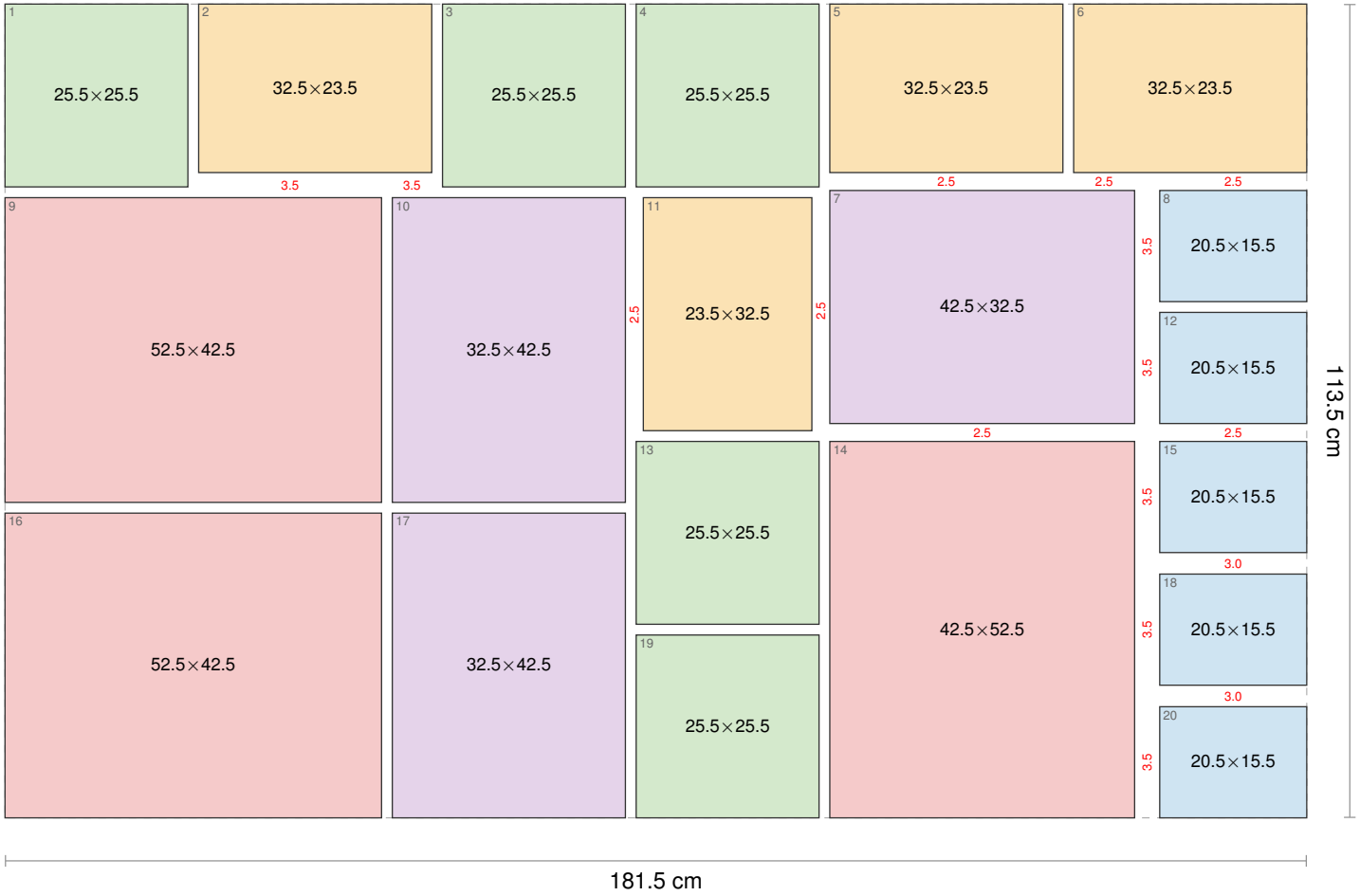
	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	1	2
	32.5×42.5	P	1	2
	42.5×32.5	L	0	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		3	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	52.5 × 42.5	31.0	0.0
3	32.5 × 42.5	89.0	0.0
4	52.5 × 42.5	127.0	0.0
5	25.5 × 25.5	0.0	31.3
6	20.5 × 15.5	31.0	48.0
7	20.5 × 15.5	57.7	48.0
8	20.5 × 15.5	84.0	48.0
9	20.5 × 15.5	110.3	48.0
10	15.5 × 20.5	164.0	48.0
11	20.5 × 15.5	137.0	50.5
12	23.5 × 32.5	0.0	62.7
13	20.5 × 15.5	31.0	69.0
14	32.5 × 23.5	57.7	71.5
15	32.5 × 23.5	98.3	71.5
16	42.5 × 52.5	137.0	74.0
17	20.5 × 15.5	31.0	90.0
18	25.5 × 25.5	0.0	101.0
19	32.5 × 23.5	57.7	103.0
20	32.5 × 23.5	98.3	103.0
21	20.5 × 15.5	31.0	111.0

Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Layout 14 181.5 × 113.5 cm gap 1.5 cm 20 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

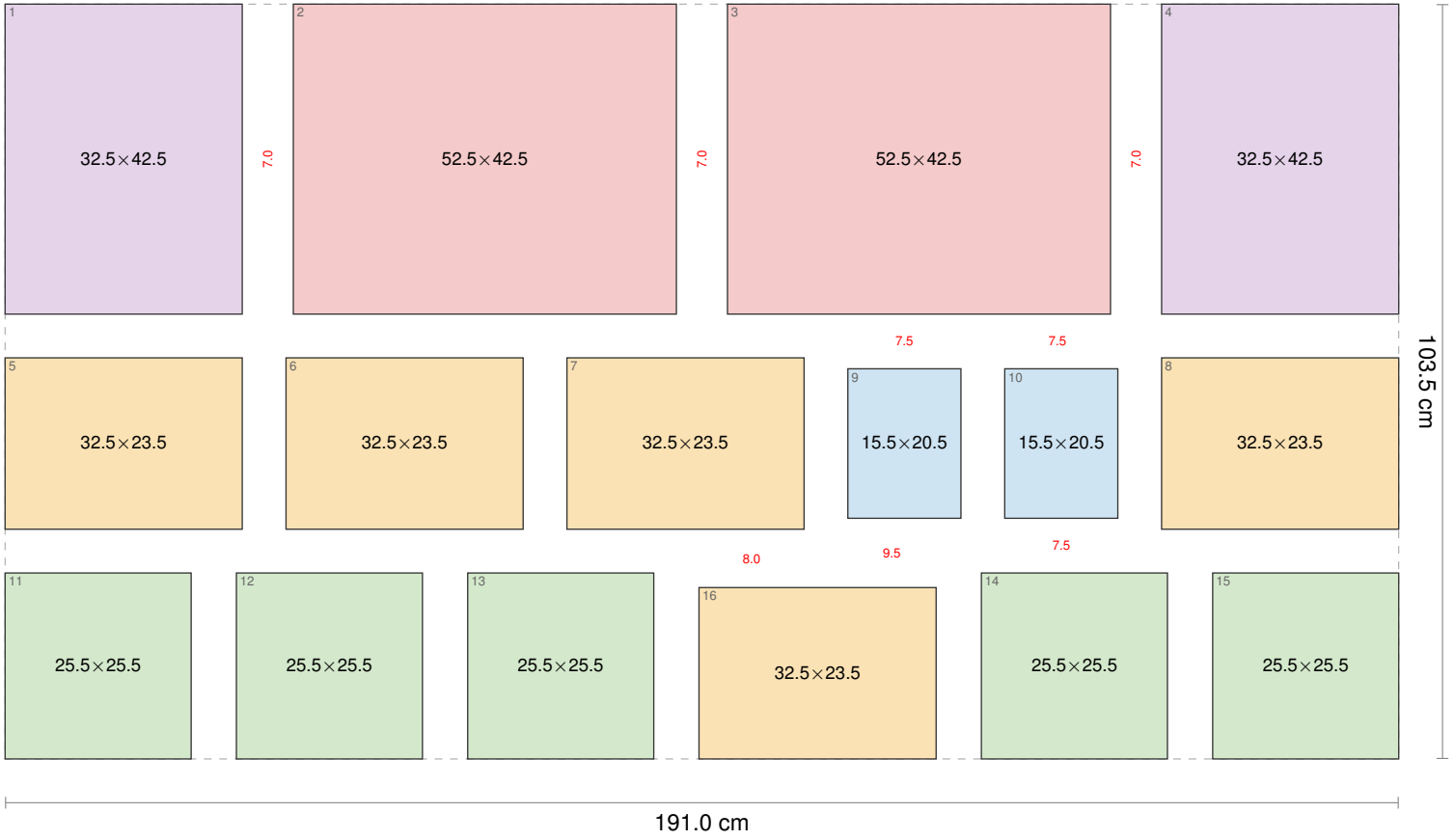
Outer size	181.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.67 cm
Border notches	bottom 4 cm, right 6 cm
Corners filled	4 / 4
Frames used	20 of 28
Large frames	3, touching 1× (52 cm), min. dist. 44 cm
Entropy	0.64
mix / variety	0.86 / 0.74
stagger / lines	0.46 / 0.50 (3 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	5	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	27.0	0.0
3	25.5 × 25.5	61.0	0.0
4	25.5 × 25.5	88.0	0.0
5	32.5 × 23.5	115.0	0.0
6	32.5 × 23.5	149.0	0.0
7	42.5 × 32.5	115.0	26.0
8	20.5 × 15.5	161.0	26.0
9	52.5 × 42.5	0.0	27.0
10	32.5 × 42.5	54.0	27.0
11	23.5 × 32.5	89.0	27.0
12	20.5 × 15.5	161.0	43.0
13	25.5 × 25.5	88.0	61.0
14	42.5 × 52.5	115.0	61.0
15	20.5 × 15.5	161.0	61.0
16	52.5 × 42.5	0.0	71.0
17	32.5 × 42.5	54.0	71.0
18	20.5 × 15.5	161.0	79.5
19	25.5 × 25.5	88.0	88.0
20	20.5 × 15.5	161.0	98.0

Layout 15 191.0 × 103.5 cm gap 6 cm 16 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

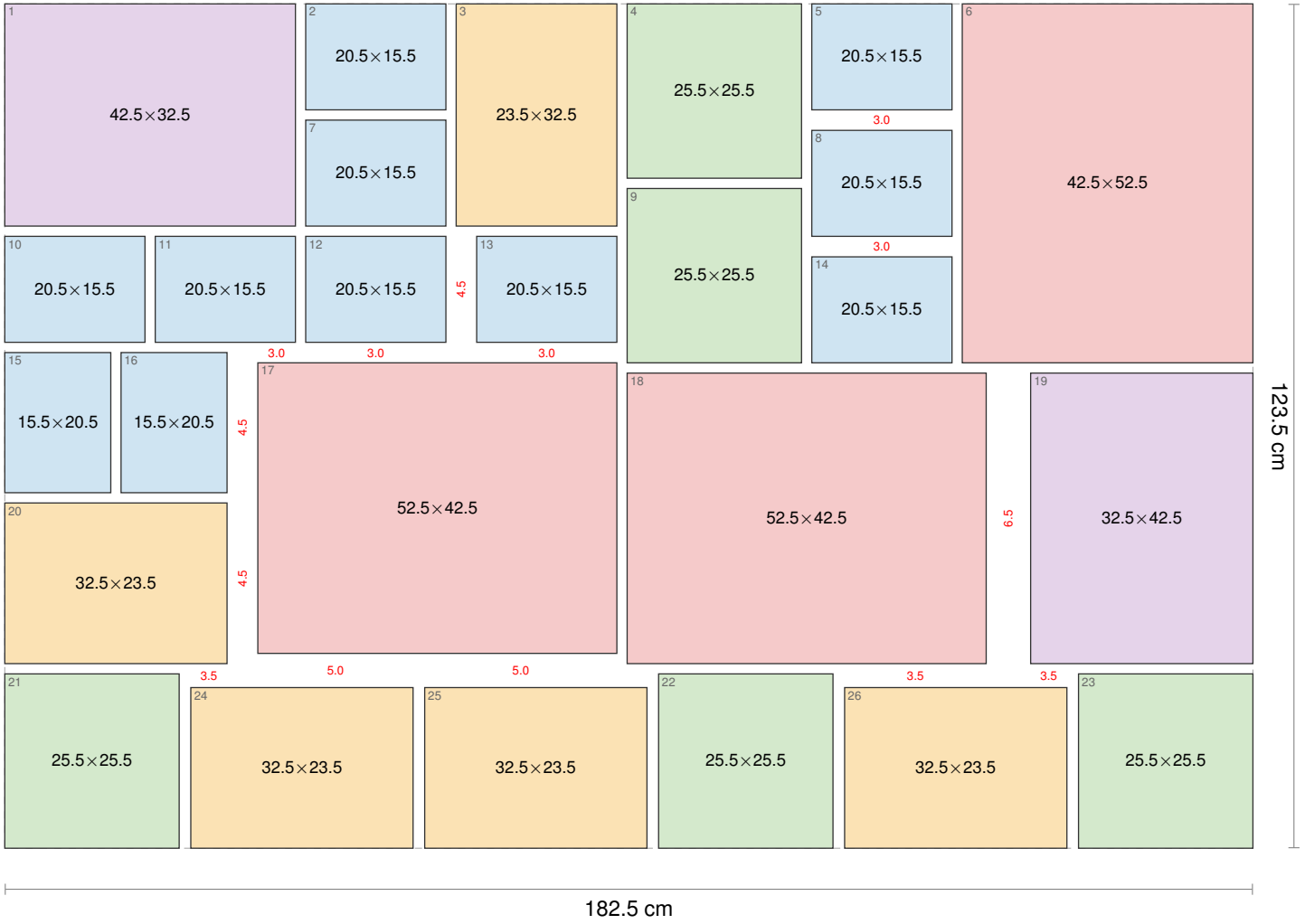
Outer size	191.0 × 103.5 cm
Nominal gap	6 cm
Actual gaps	6.0 – 9.5 cm
Max. deviation	3.5 cm
Mean deviation	0.50 cm
Border notches	none
Corners filled	4 / 4
Frames used	16 of 28
Large frames	2, touching 1× (42 cm), min. dist. 60 cm
Entropy	0.67
mix / variety	0.69 / 0.70
stagger / lines	0.55 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	0	9
15.5×20.5	P	2	2
32.5×42.5	P	2	2
42.5×32.5	L	0	1
32.5×23.5	L	5	5
23.5×32.5	P	0	1
25.5×25.5		5	5
42.5×52.5	P	0	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	39.5	0.0
3	52.5 × 42.5	99.0	0.0
4	32.5 × 42.5	158.5	0.0
5	32.5 × 23.5	0.0	48.5
6	32.5 × 23.5	38.5	48.5
7	32.5 × 23.5	77.0	48.5
8	32.5 × 23.5	158.5	48.5
9	15.5 × 20.5	115.5	50.0
10	15.5 × 20.5	137.0	50.0
11	25.5 × 25.5	0.0	78.0
12	25.5 × 25.5	31.7	78.0
13	25.5 × 25.5	63.4	78.0
14	25.5 × 25.5	133.8	78.0
15	25.5 × 25.5	165.5	78.0
16	32.5 × 23.5	95.1	80.0

Layout 16 182.5 × 123.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	182.5 × 123.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 6.5 cm
Max. deviation	5.0 cm
Mean deviation	0.67 cm
Border notches	none
Corners filled	4 / 4
Frames used	26 of 28
Large frames	3, touching 1× (41 cm), min. dist. 54 cm
Entropy	0.78
mix / variety	0.85 / 0.79
stagger / lines	0.49 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	1	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	42.5 × 32.5	0.0	0.0
2	20.5 × 15.5	44.0	0.0
3	23.5 × 32.5	66.0	0.0
4	25.5 × 25.5	91.0	0.0
5	20.5 × 15.5	118.0	0.0
6	42.5 × 52.5	140.0	0.0
7	20.5 × 15.5	44.0	17.0
8	20.5 × 15.5	118.0	18.5
9	25.5 × 25.5	91.0	27.0
10	20.5 × 15.5	0.0	34.0
11	20.5 × 15.5	22.0	34.0
12	20.5 × 15.5	44.0	34.0
13	20.5 × 15.5	69.0	34.0
14	20.5 × 15.5	118.0	37.0
15	15.5 × 20.5	0.0	51.0
16	15.5 × 20.5	17.0	51.0
17	52.5 × 42.5	37.0	52.5
18	52.5 × 42.5	91.0	54.0
19	32.5 × 42.5	150.0	54.0
20	32.5 × 23.5	0.0	73.0
21	25.5 × 25.5	0.0	98.0
22	25.5 × 25.5	95.6	98.0
23	25.5 × 25.5	157.0	98.0
24	32.5 × 23.5	27.2	100.0
25	32.5 × 23.5	61.4	100.0
26	32.5 × 23.5	122.8	100.0

6 proportional

Unavoidable slack is distributed in proportion to the size of the neighbouring frames: the weight of a gap is the mean extent of the two frames (or groups) it separates, so gaps next to big frames widen more than gaps between small ones. The deviations are therefore systematic (bigger frame, bigger gap) rather than equal.

Parameters that differ from the baseline: slack_rule = proportional.
256 runs, 20 min wall-clock budget, 538 candidate states, the 16 best distinct layouts are shown.

#	size (cm)	gap	frames	max dev	mean dev	notches	corners	entropy
1	185.5 × 116.5	2.5	23	2.0	0.51	none	4/4	0.77
2	184.0 × 115.0	2	23	2.0	0.57	none	4/4	0.69
3	191.5 × 113.5	1.5	26	2.6	0.78	3 cm	4/4	0.68
4	172.0 × 113.5	2	25	2.8	0.48	none	4/4	0.77
5	177.0 × 123.0	2	25	3.0	0.60	none	4/4	0.78
6	184.0 × 115.0	2	23	2.2	0.57	4 cm	4/4	0.59
7	170.5 × 123.5	2	23	3.0	0.63	none	4/4	0.65
8	184.0 × 115.0	2	20	2.0	0.68	3 cm	4/4	0.78
9	181.5 × 113.5	1.5	20	2.0	0.51	5 cm	4/4	0.78
10	186.5 × 120.5	6	20	3.1	0.38	none	4/4	0.64
11	175.5 × 124.5	2.5	26	3.0	0.50	4 cm	4/4	0.78
12	175.5 × 126.0	3	25	2.5	0.72	5 cm	4/4	0.63
13	191.5 × 113.5	1.5	25	2.0	0.72	9 cm	4/4	0.75
14	190.5 × 126.0	3	25	3.0	0.90	4 cm	4/4	0.64
15	182.5 × 113.5	1.5	25	3.0	0.95	7 cm	4/4	0.79
16	170.0 × 115.0	2	24	2.2	0.44	8 cm	4/4	0.72

Layout 1 185.5 × 116.5 cm gap 2.5 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

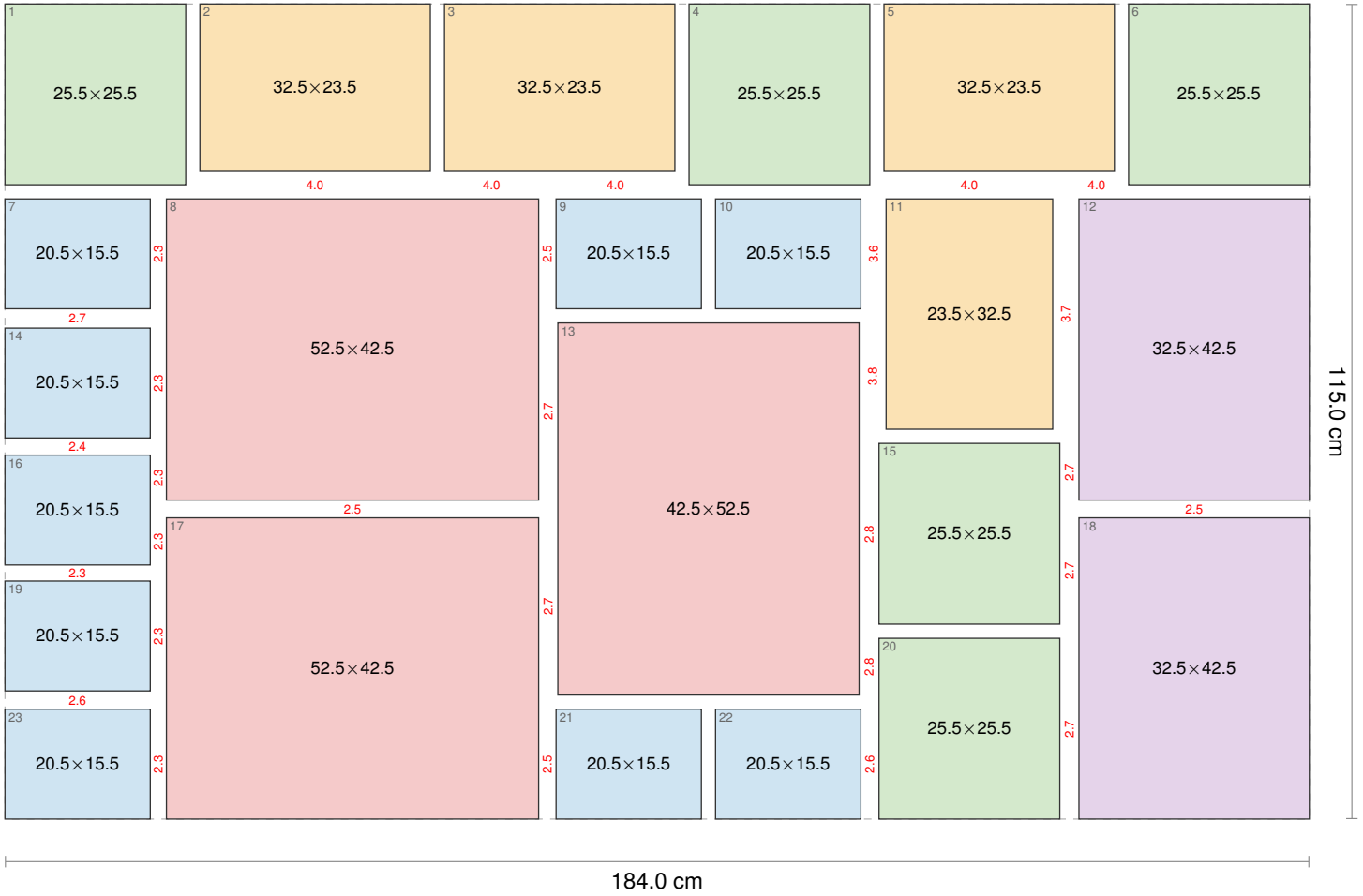
Outer size	185.5 × 116.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 4.5 cm
Max. deviation	2.0 cm
Mean deviation	0.51 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 1× (24 cm), min. dist. 56 cm
Entropy	0.77
mix / variety	0.76 / 0.77
stagger / lines	0.57 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 42.5	28.0	0.0
3	52.5 × 42.5	64.0	0.0
4	20.5 × 15.5	119.0	0.0
5	20.5 × 15.5	142.0	0.0
6	20.5 × 15.5	165.0	0.0
7	42.5 × 52.5	119.5	18.4
8	20.5 × 15.5	165.0	18.6
9	25.5 × 25.5	0.0	28.0
10	20.5 × 15.5	165.0	37.1
11	25.5 × 25.5	28.0	45.4
12	25.5 × 25.5	91.0	45.4
13	32.5 × 23.5	56.0	46.4
14	20.5 × 15.5	165.0	55.4
15	23.5 × 32.5	0.0	56.0
16	52.5 × 42.5	28.0	74.0
17	20.5 × 15.5	83.0	74.0
18	20.5 × 15.5	106.4	74.0
19	20.5 × 15.5	130.0	74.0
20	32.5 × 42.5	153.0	74.0
21	25.5 × 25.5	0.0	91.0
22	32.5 × 23.5	83.0	93.0
23	32.5 × 23.5	118.0	93.0

Layout 2 184.0 × 115.0 cm gap 2 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

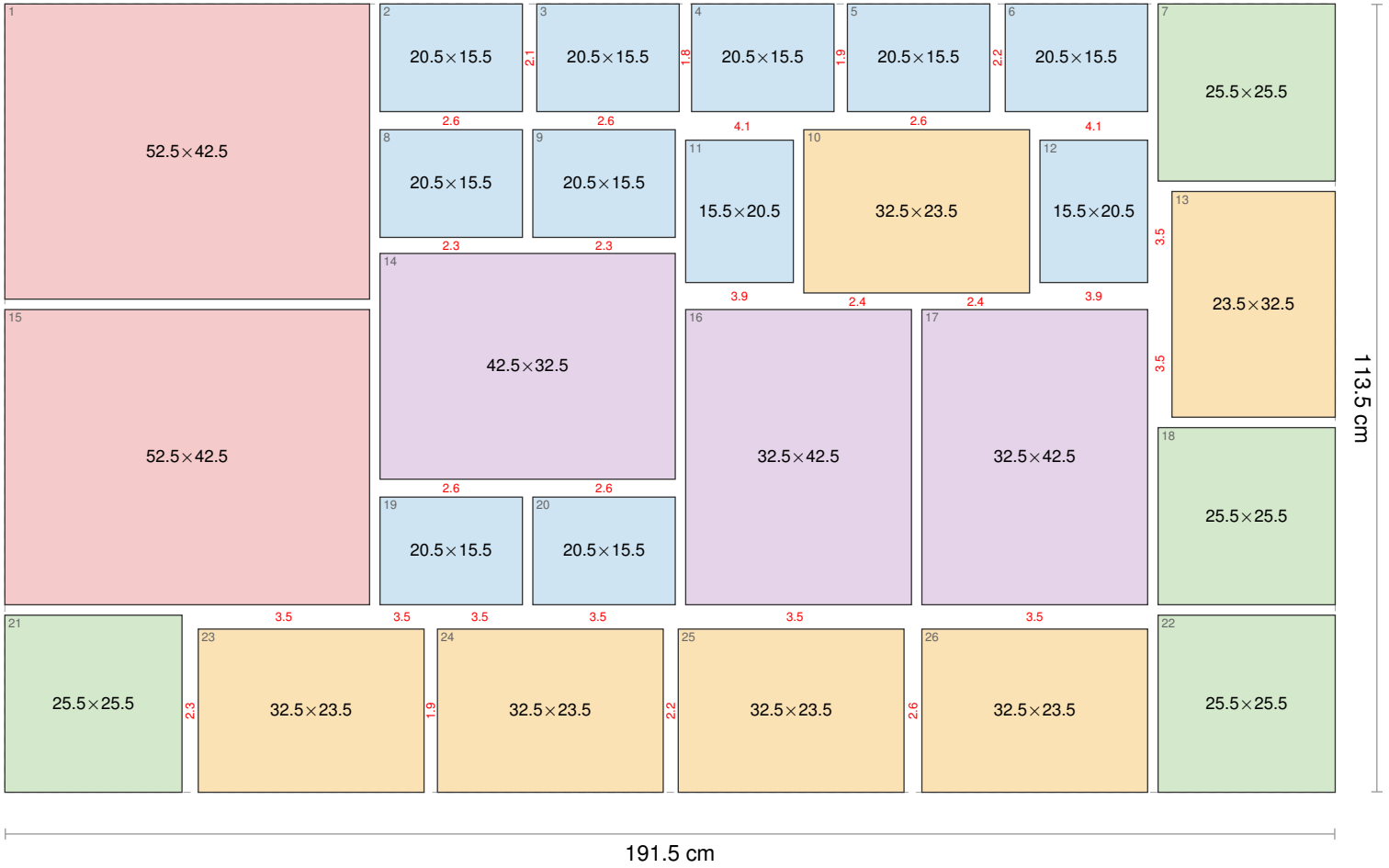
Outer size	184.0 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.0 cm
Max. deviation	2.0 cm
Mean deviation	0.57 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 3× (102 cm), min. dist. 45 cm
Entropy	0.69
mix / variety	0.76 / 0.74
stagger / lines	0.51 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	0	2
32.5×42.5	P	2	2
42.5×32.5	L	0	1
32.5×23.5	L	3	5
23.5×32.5	P	1	1
25.5×25.5		5	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	27.5	0.0
3	32.5 × 23.5	62.0	0.0
4	25.5 × 25.5	96.5	0.0
5	32.5 × 23.5	124.0	0.0
6	25.5 × 25.5	158.5	0.0
7	20.5 × 15.5	0.0	27.5
8	52.5 × 42.5	22.8	27.5
9	20.5 × 15.5	77.7	27.5
10	20.5 × 15.5	100.2	27.5
11	23.5 × 32.5	124.3	27.5
12	32.5 × 42.5	151.5	27.5
13	42.5 × 52.5	78.0	45.0
14	20.5 × 15.5	0.0	45.7
15	25.5 × 25.5	123.3	62.0
16	20.5 × 15.5	0.0	63.7
17	52.5 × 42.5	22.8	72.5
18	32.5 × 42.5	151.5	72.5
19	20.5 × 15.5	0.0	81.4
20	25.5 × 25.5	123.3	89.5
21	20.5 × 15.5	77.7	99.5
22	20.5 × 15.5	100.2	99.5
23	20.5 × 15.5	0.0	99.5

Layout 3 191.5 × 113.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

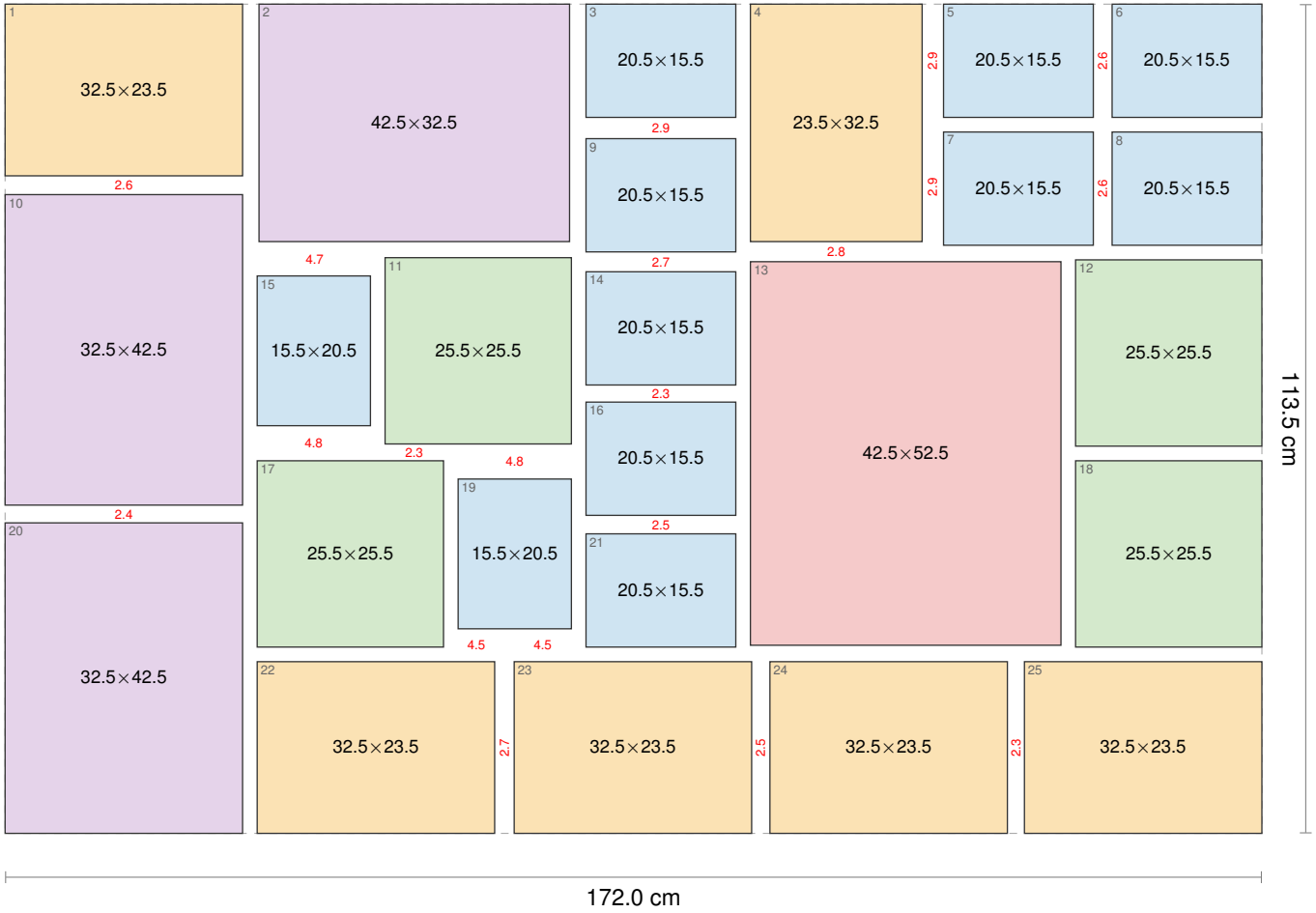
Outer size	191.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.1 cm
Max. deviation	2.6 cm
Mean deviation	0.78 cm
Border notches	bottom 3 cm
Corners filled	4 / 4
Frames used	26 of 28
Large frames	2, touching 1× (52 cm), min. dist. 44 cm
Entropy	0.68
mix / variety	0.83 / 0.67
stagger / lines	0.49 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	20.5 × 15.5	54.0	0.0
3	20.5 × 15.5	76.6	0.0
4	20.5 × 15.5	98.9	0.0
5	20.5 × 15.5	121.3	0.0
6	20.5 × 15.5	144.0	0.0
7	25.5 × 25.5	166.0	0.0
8	20.5 × 15.5	54.0	18.1
9	20.5 × 15.5	76.0	18.1
10	32.5 × 23.5	115.0	18.1
11	15.5 × 20.5	98.0	19.6
12	15.5 × 20.5	149.0	19.6
13	23.5 × 32.5	168.0	27.0
14	42.5 × 32.5	54.0	35.9
15	52.5 × 42.5	0.0	44.0
16	32.5 × 42.5	98.0	44.0
17	32.5 × 42.5	132.0	44.0
18	25.5 × 25.5	166.0	61.0
19	20.5 × 15.5	54.0	71.0
20	20.5 × 15.5	76.0	71.0
21	25.5 × 25.5	0.0	88.0
22	25.5 × 25.5	166.0	88.0
23	32.5 × 23.5	27.8	90.0
24	32.5 × 23.5	62.3	90.0
25	32.5 × 23.5	96.9	90.0
26	32.5 × 23.5	132.0	90.0

Layout 4 172.0 × 113.5 cm gap 2 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

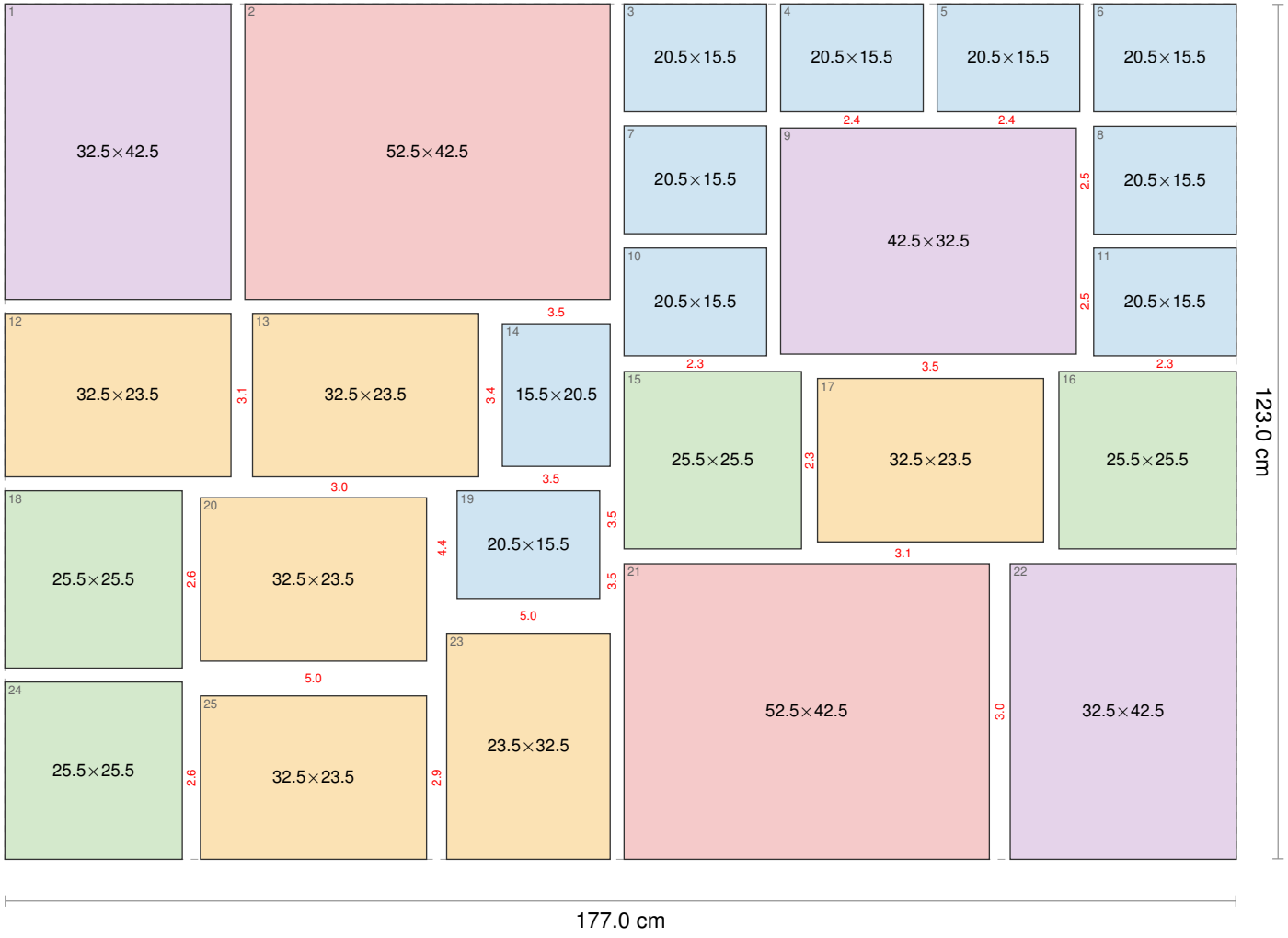
Outer size	172.0 × 113.5 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.8 cm
Max. deviation	2.8 cm
Mean deviation	0.48 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	1
Entropy	0.77
mix / variety	0.81 / 0.70
stagger / lines	0.56 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	42.5 × 32.5	34.8	0.0
3	20.5 × 15.5	79.5	0.0
4	23.5 × 32.5	102.0	0.0
5	20.5 × 15.5	128.4	0.0
6	20.5 × 15.5	151.5	0.0
7	20.5 × 15.5	128.4	17.5
8	20.5 × 15.5	151.5	17.5
9	20.5 × 15.5	79.5	18.4
10	32.5 × 42.5	0.0	26.1
11	25.5 × 25.5	52.0	34.7
12	25.5 × 25.5	146.5	35.0
13	42.5 × 52.5	102.0	35.2
14	20.5 × 15.5	79.5	36.6
15	15.5 × 20.5	34.5	37.2
16	20.5 × 15.5	79.5	54.5
17	25.5 × 25.5	34.5	62.5
18	25.5 × 25.5	146.5	62.5
19	15.5 × 20.5	62.0	65.0
20	32.5 × 42.5	0.0	71.0
21	20.5 × 15.5	79.5	72.5
22	32.5 × 23.5	34.5	90.0
23	32.5 × 23.5	69.7	90.0
24	32.5 × 23.5	104.7	90.0
25	32.5 × 23.5	139.5	90.0

Layout 5 177.0 × 123.0 cm gap 2 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

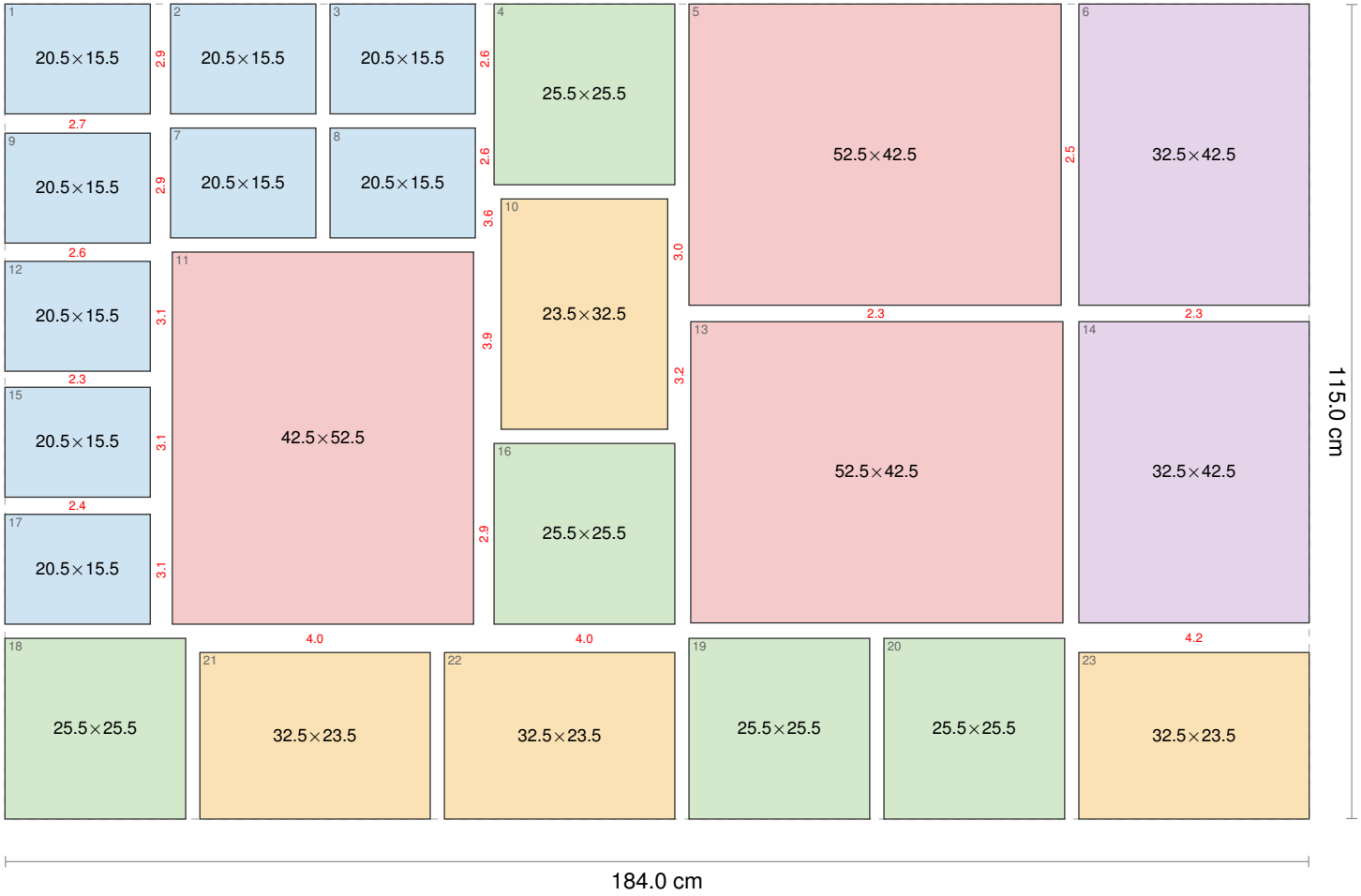
Outer size	177.0 × 123.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 5.0 cm
Max. deviation	3.0 cm
Mean deviation	0.60 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 97 cm
Entropy	0.78
mix / variety	0.81 / 0.74
stagger / lines	0.57 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	34.5	0.0
3	20.5 × 15.5	89.0	0.0
4	20.5 × 15.5	111.5	0.0
5	20.5 × 15.5	134.0	0.0
6	20.5 × 15.5	156.5	0.0
7	20.5 × 15.5	89.0	17.5
8	20.5 × 15.5	156.5	17.6
9	42.5 × 32.5	111.5	17.9
10	20.5 × 15.5	89.0	35.1
11	20.5 × 15.5	156.5	35.1
12	32.5 × 23.5	0.0	44.5
13	32.5 × 23.5	35.6	44.5
14	15.5 × 20.5	71.5	46.0
15	25.5 × 25.5	89.0	52.9
16	25.5 × 25.5	151.5	52.9
17	32.5 × 23.5	116.8	53.9
18	25.5 × 25.5	0.0	70.0
19	20.5 × 15.5	65.0	70.0
20	32.5 × 23.5	28.1	71.0
21	52.5 × 42.5	89.0	80.5
22	32.5 × 42.5	144.5	80.5
23	23.5 × 32.5	63.5	90.5
24	25.5 × 25.5	0.0	97.5
25	32.5 × 23.5	28.1	99.5

Layout 6 184.0 × 115.0 cm gap 2 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

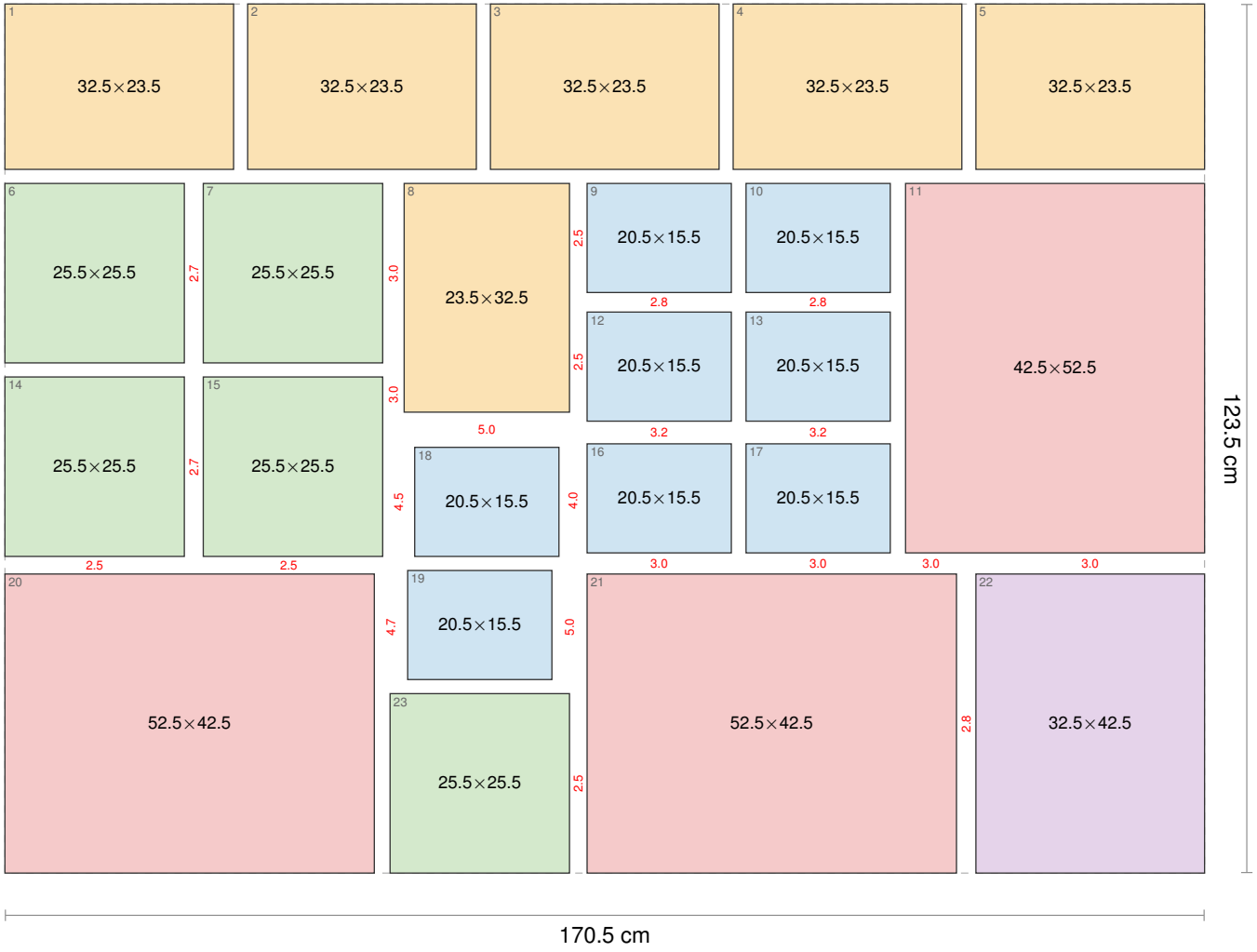
Outer size	184.0 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.2 cm
Max. deviation	2.2 cm
Mean deviation	0.57 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 1× (52 cm), min. dist. 45 cm
Entropy	0.59
mix / variety	0.76 / 0.56
stagger / lines	0.52 / 0.50 (3 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	23.4	0.0
3	20.5 × 15.5	45.9	0.0
4	25.5 × 25.5	69.0	0.0
5	52.5 × 42.5	96.5	0.0
6	32.5 × 42.5	151.5	0.0
7	20.5 × 15.5	23.4	17.5
8	20.5 × 15.5	45.9	17.5
9	20.5 × 15.5	0.0	18.2
10	23.5 × 32.5	70.0	27.5
11	42.5 × 52.5	23.6	35.0
12	20.5 × 15.5	0.0	36.3
13	52.5 × 42.5	96.8	44.8
14	32.5 × 42.5	151.5	44.8
15	20.5 × 15.5	0.0	54.1
16	25.5 × 25.5	69.0	62.0
17	20.5 × 15.5	0.0	72.0
18	25.5 × 25.5	0.0	89.5
19	25.5 × 25.5	96.5	89.5
20	25.5 × 25.5	124.0	89.5
21	32.5 × 23.5	27.5	91.5
22	32.5 × 23.5	62.0	91.5
23	32.5 × 23.5	151.5	91.5

Layout 7 170.5 × 123.5 cm gap 2 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

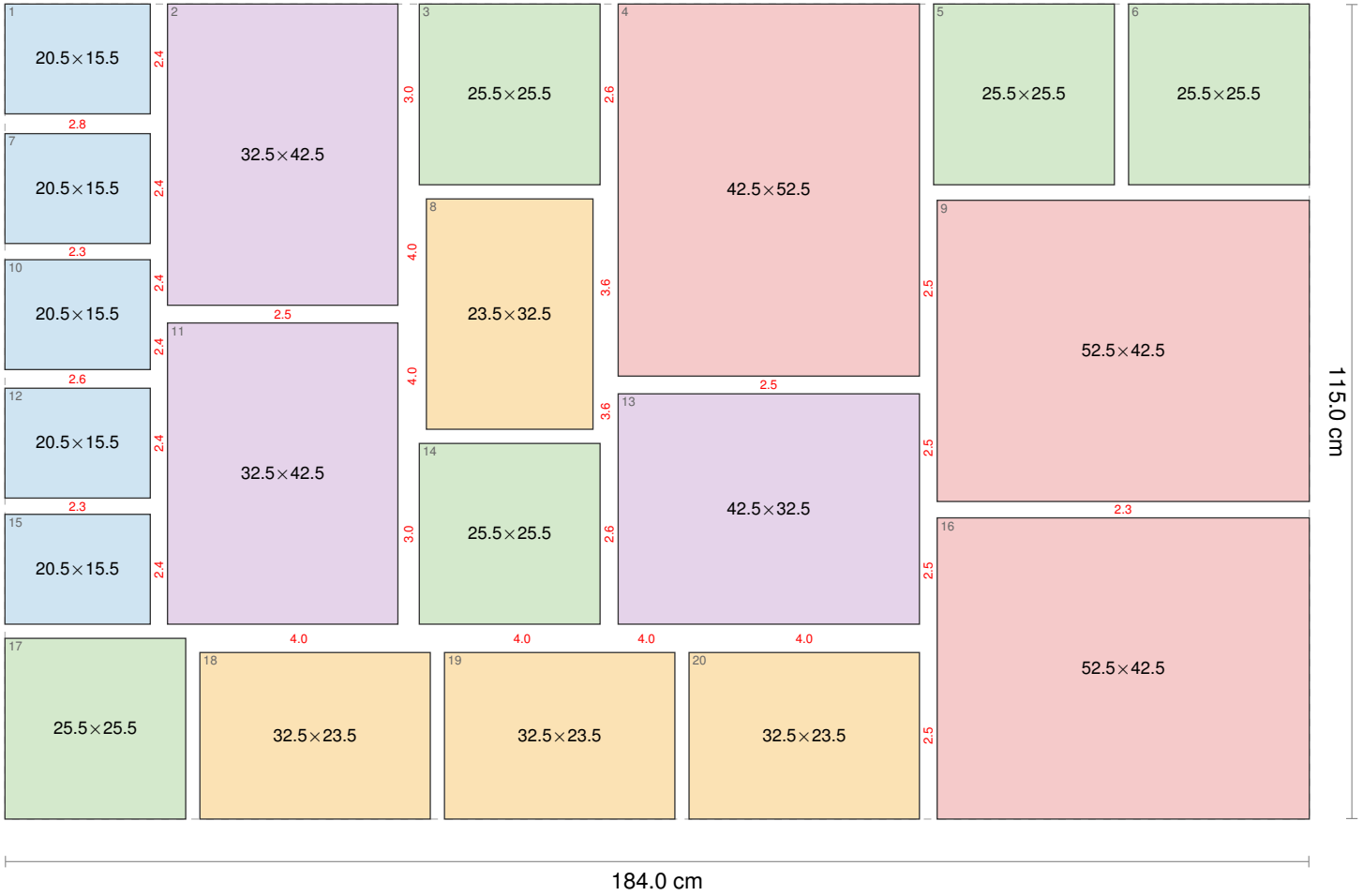
Outer size	170.5 × 123.5 cm
Nominal gap	2 cm
Actual gaps	2.0 – 5.0 cm
Max. deviation	3.0 cm
Mean deviation	0.63 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 1× (7 cm), min. dist. 65 cm
Entropy	0.65
mix / variety	0.75 / 0.62
stagger / lines	0.49 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	0	2
	32.5×42.5	P	1	2
	42.5×32.5	L	0	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	32.5 × 23.5	34.5	0.0
3	32.5 × 23.5	69.0	0.0
4	32.5 × 23.5	103.5	0.0
5	32.5 × 23.5	138.0	0.0
6	25.5 × 25.5	0.0	25.5
7	25.5 × 25.5	28.2	25.5
8	23.5 × 32.5	56.7	25.5
9	20.5 × 15.5	82.7	25.5
10	20.5 × 15.5	105.3	25.5
11	42.5 × 52.5	128.0	25.5
12	20.5 × 15.5	82.7	43.8
13	20.5 × 15.5	105.3	43.8
14	25.5 × 25.5	0.0	53.0
15	25.5 × 25.5	28.2	53.0
16	20.5 × 15.5	82.7	62.5
17	20.5 × 15.5	105.3	62.5
18	20.5 × 15.5	58.2	63.0
19	20.5 × 15.5	57.2	80.5
20	52.5 × 42.5	0.0	81.0
21	52.5 × 42.5	82.7	81.0
22	32.5 × 42.5	138.0	81.0
23	25.5 × 25.5	54.7	98.0

Layout 8 184.0 × 115.0 cm gap 2 cm 20 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

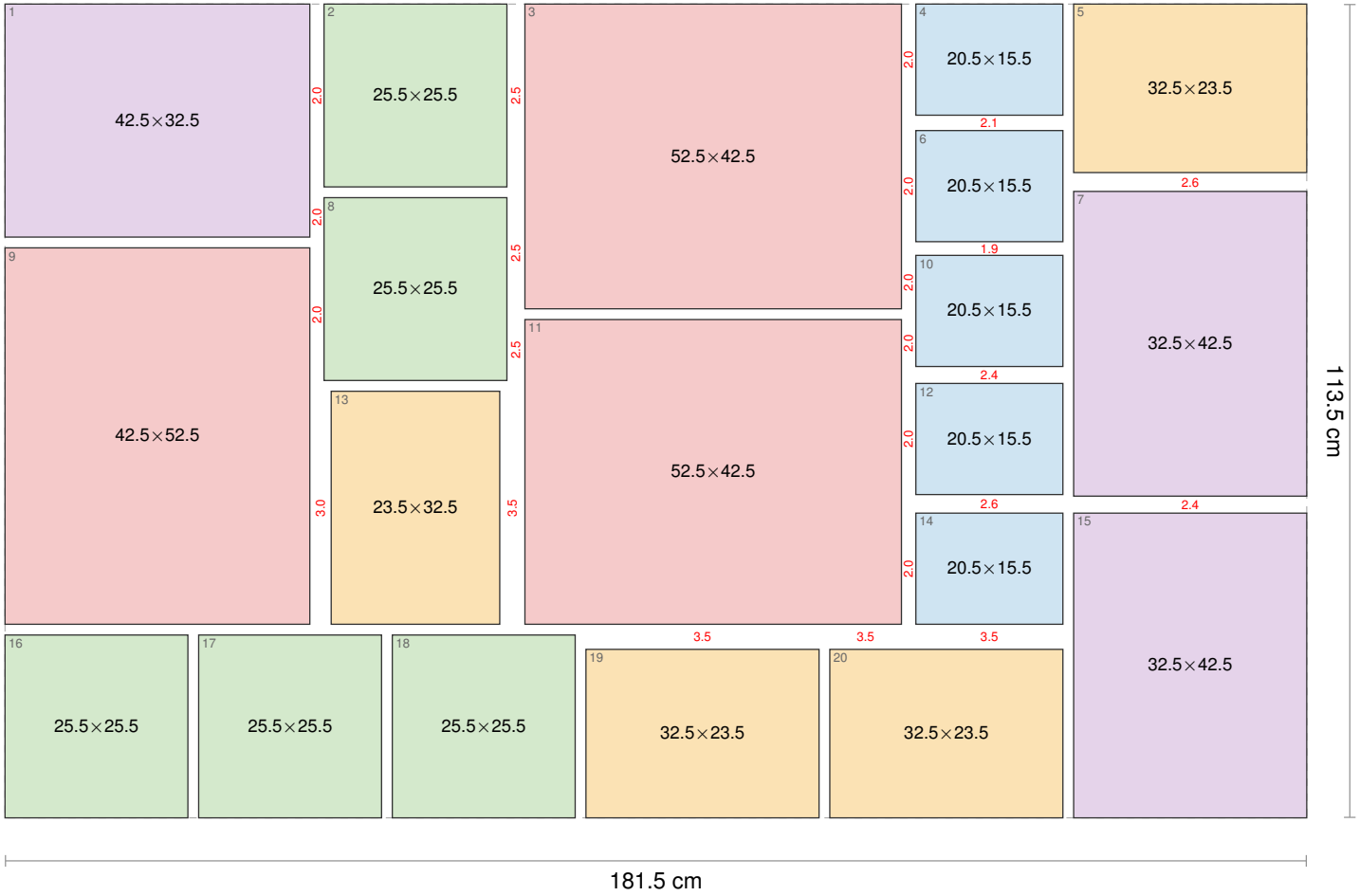
Outer size	184.0 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.0 cm
Max. deviation	2.0 cm
Mean deviation	0.68 cm
Border notches	top 3 cm
Corners filled	4 / 4
Frames used	20 of 28
Large frames	3, touching 2× (77 cm), min. dist. 45 cm
Entropy	0.78
mix / variety	0.86 / 0.69
stagger / lines	0.56 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	5	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	32.5 × 42.5	22.9	0.0
3	25.5 × 25.5	58.4	0.0
4	42.5 × 52.5	86.5	0.0
5	25.5 × 25.5	131.0	0.0
6	25.5 × 25.5	158.5	0.0
7	20.5 × 15.5	0.0	18.3
8	23.5 × 32.5	59.4	27.5
9	52.5 × 42.5	131.5	27.7
10	20.5 × 15.5	0.0	36.1
11	32.5 × 42.5	22.9	45.0
12	20.5 × 15.5	0.0	54.2
13	42.5 × 32.5	86.5	55.0
14	25.5 × 25.5	58.4	62.0
15	20.5 × 15.5	0.0	72.0
16	52.5 × 42.5	131.5	72.5
17	25.5 × 25.5	0.0	89.5
18	32.5 × 23.5	27.5	91.5
19	32.5 × 23.5	62.0	91.5
20	32.5 × 23.5	96.5	91.5

Layout 9 181.5 × 113.5 cm gap 1.5 cm 20 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

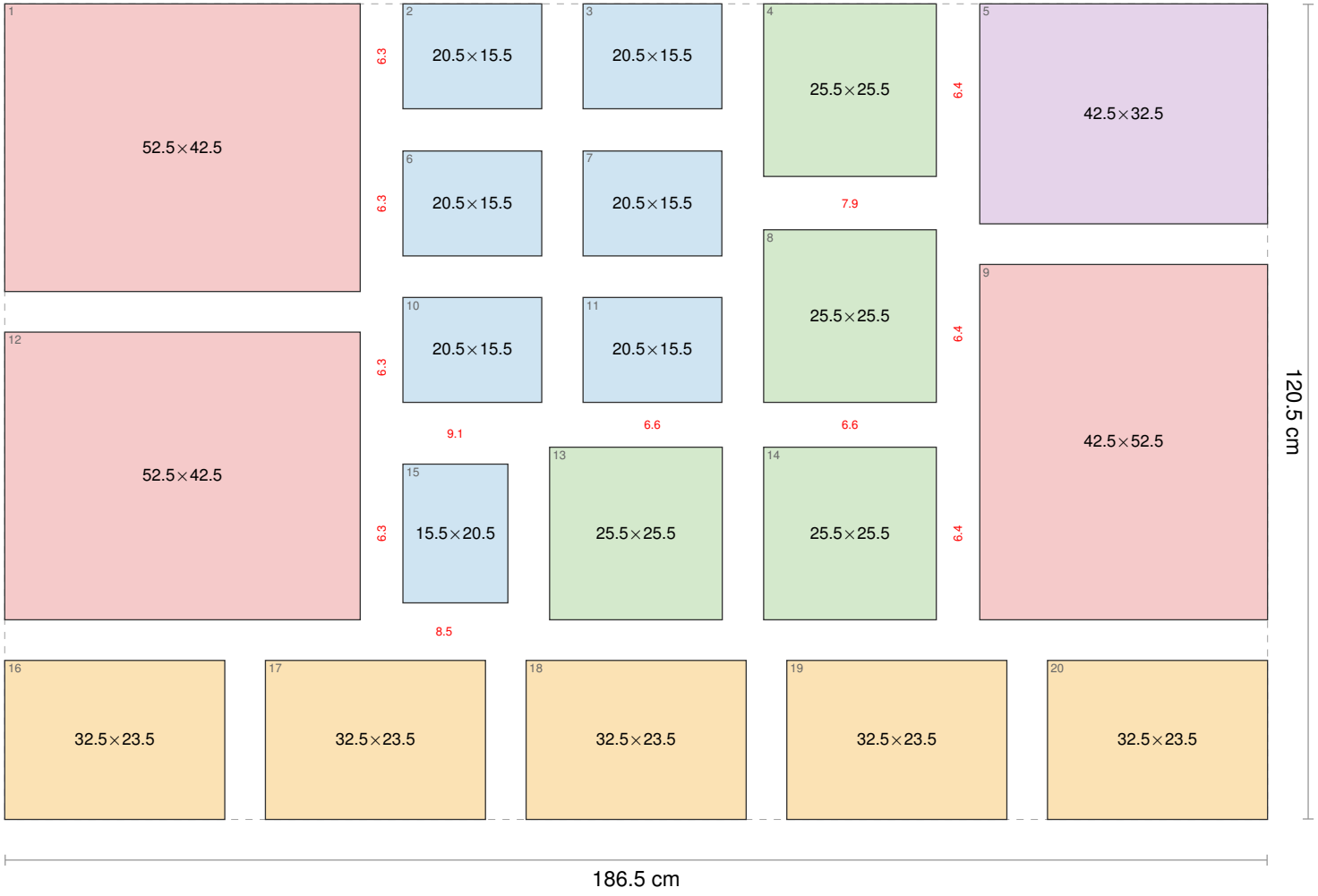
Outer size	181.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.51 cm
Border notches	top 3 cm, right 3 cm
Corners filled	4 / 4
Frames used	20 of 28
Large frames	3, touching 1× (52 cm), min. dist. 44 cm
Entropy	0.78
mix / variety	0.86 / 0.73
stagger / lines	0.55 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	5	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	42.5 × 32.5	0.0	0.0
2	25.5 × 25.5	44.5	0.0
3	52.5 × 42.5	72.5	0.0
4	20.5 × 15.5	127.0	0.0
5	32.5 × 23.5	149.0	0.0
6	20.5 × 15.5	127.0	17.6
7	32.5 × 42.5	149.0	26.1
8	25.5 × 25.5	44.5	27.0
9	42.5 × 52.5	0.0	34.0
10	20.5 × 15.5	127.0	35.0
11	52.5 × 42.5	72.5	44.0
12	20.5 × 15.5	127.0	52.9
13	23.5 × 32.5	45.5	54.0
14	20.5 × 15.5	127.0	71.0
15	32.5 × 42.5	149.0	71.0
16	25.5 × 25.5	0.0	88.0
17	25.5 × 25.5	27.0	88.0
18	25.5 × 25.5	54.0	88.0
19	32.5 × 23.5	81.0	90.0
20	32.5 × 23.5	115.0	90.0

Layout 10 186.5 × 120.5 cm gap 6 cm 20 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	186.5 × 120.5 cm
Nominal gap	6 cm
Actual gaps	6.0 – 9.1 cm
Max. deviation	3.1 cm
Mean deviation	0.38 cm
Border notches	none
Corners filled	4 / 4
Frames used	20 of 28
Large frames	3, touching 1× (52 cm), min. dist. 48 cm
Entropy	0.64
mix / variety	0.78 / 0.55
stagger / lines	0.49 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	6	9
15.5×20.5	P	1	2
32.5×42.5	P	0	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	0	1
25.5×25.5		4	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	20.5 × 15.5	58.8	0.0
3	20.5 × 15.5	85.4	0.0
4	25.5 × 25.5	112.1	0.0
5	42.5 × 32.5	144.0	0.0
6	20.5 × 15.5	58.8	21.7
7	20.5 × 15.5	85.4	21.7
8	25.5 × 25.5	112.1	33.4
9	42.5 × 52.5	144.0	38.5
10	20.5 × 15.5	58.8	43.4
11	20.5 × 15.5	85.4	43.4
12	52.5 × 42.5	0.0	48.5
13	25.5 × 25.5	80.5	65.5
14	25.5 × 25.5	112.1	65.5
15	15.5 × 20.5	58.8	68.0
16	32.5 × 23.5	0.0	97.0
17	32.5 × 23.5	38.5	97.0
18	32.5 × 23.5	77.0	97.0
19	32.5 × 23.5	115.5	97.0
20	32.5 × 23.5	154.0	97.0

Layout 11 175.5 × 124.5 cm gap 2.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

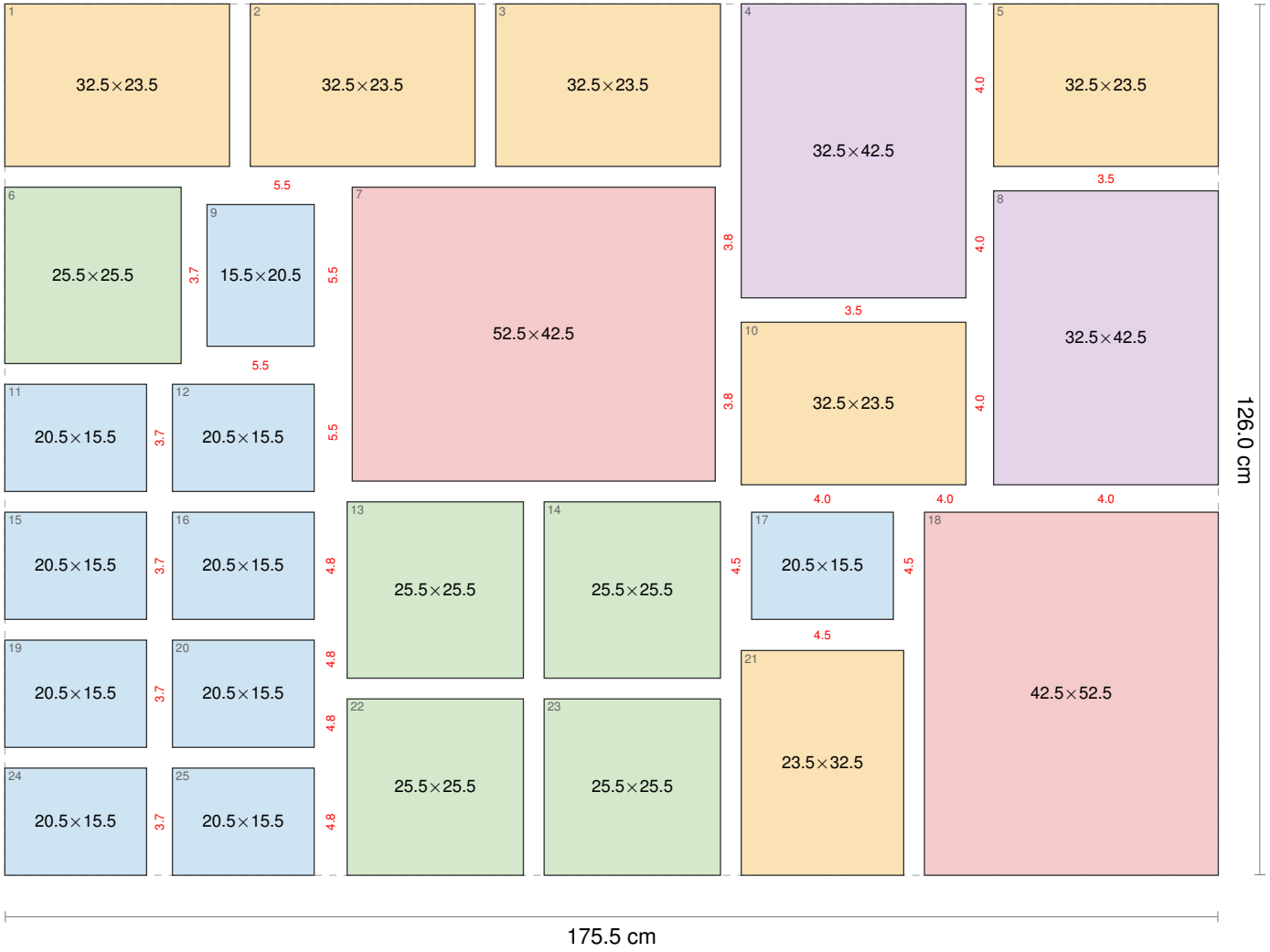
Outer size	175.5 × 124.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.5 cm
Max. deviation	3.0 cm
Mean deviation	0.50 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	26 of 28
Large frames	2, touching 1× (17 cm), min. dist. 59 cm
Entropy	0.78
mix / variety	0.83 / 0.74
stagger / lines	0.57 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	1	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	28.3	0.0
3	25.5 × 25.5	63.8	0.0
4	25.5 × 25.5	92.4	0.0
5	25.5 × 25.5	121.1	0.0
6	25.5 × 25.5	150.0	0.0
7	32.5 × 23.5	0.0	28.0
8	32.5 × 23.5	35.0	28.0
9	15.5 × 20.5	125.0	28.0
10	32.5 × 23.5	143.0	28.0
11	52.5 × 42.5	70.0	28.5
12	15.5 × 20.5	125.0	51.0
13	20.5 × 15.5	0.0	54.0
14	42.5 × 52.5	24.5	54.0
15	32.5 × 42.5	143.0	54.8
16	20.5 × 15.5	0.0	72.3
17	20.5 × 15.5	71.5	74.0
18	42.5 × 32.5	97.5	74.0
19	20.5 × 15.5	0.0	90.5
20	23.5 × 32.5	70.0	92.0
21	32.5 × 23.5	143.0	101.0
22	20.5 × 15.5	0.0	109.0
23	20.5 × 15.5	24.0	109.0
24	20.5 × 15.5	47.0	109.0
25	20.5 × 15.5	97.0	109.0
26	20.5 × 15.5	120.0	109.0

Layout 12 175.5 × 126.0 cm gap 3 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

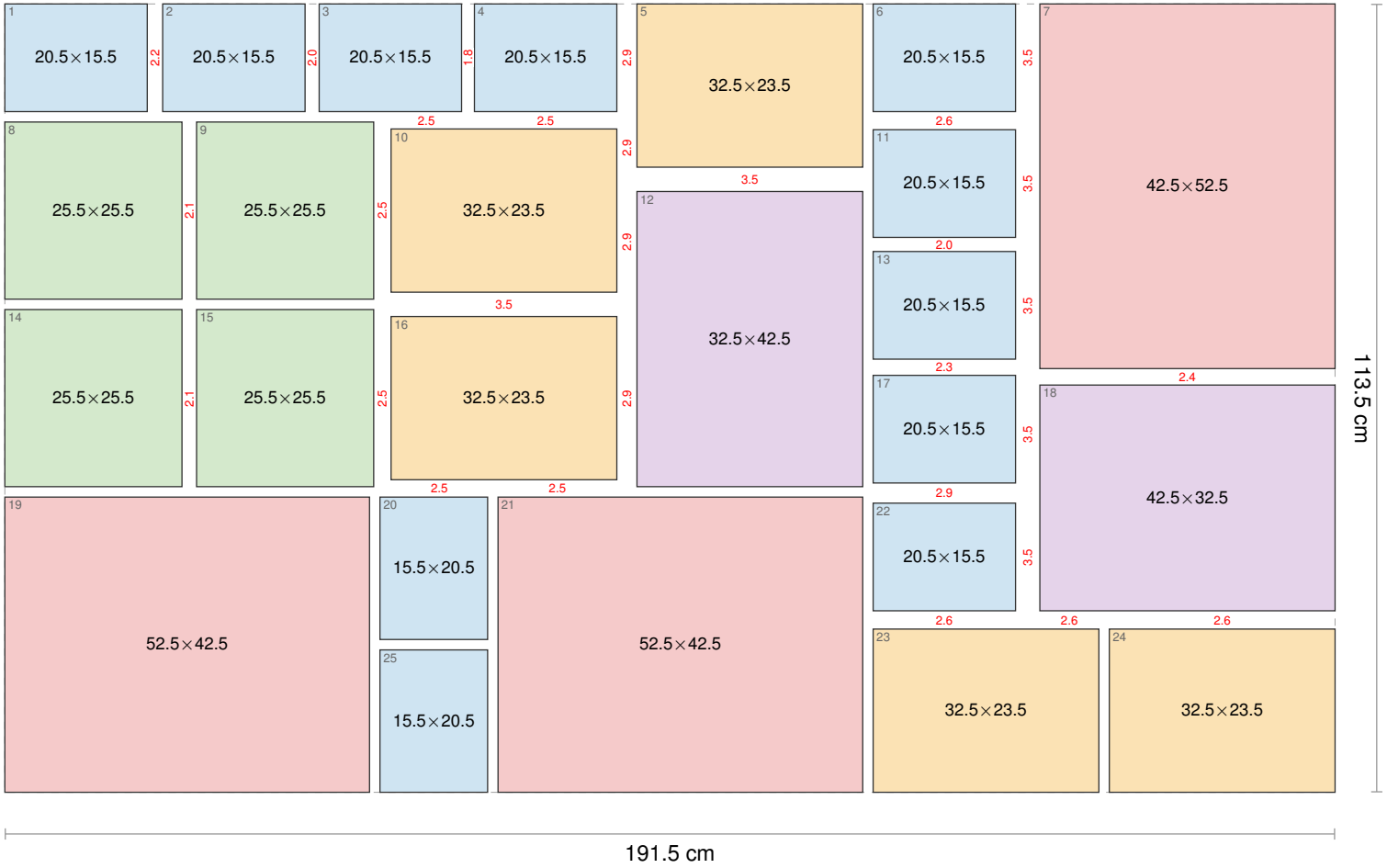
Outer size	175.5 × 126.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 5.5 cm
Max. deviation	2.5 cm
Mean deviation	0.72 cm
Border notches	bottom 5 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 94 cm
Entropy	0.63
mix / variety	0.79 / 0.53
stagger / lines	0.47 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	1	2
32.5×42.5	P	2	2
42.5×32.5	L	0	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		5	5
42.5×52.5	P	1	1
52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	32.5 × 23.5	35.5	0.0
3	32.5 × 23.5	71.0	0.0
4	32.5 × 42.5	106.5	0.0
5	32.5 × 23.5	143.0	0.0
6	25.5 × 25.5	0.0	26.5
7	52.5 × 42.5	50.2	26.5
8	32.5 × 42.5	143.0	27.0
9	15.5 × 20.5	29.2	29.0
10	32.5 × 23.5	106.5	46.0
11	20.5 × 15.5	0.0	55.0
12	20.5 × 15.5	24.2	55.0
13	25.5 × 25.5	49.5	72.0
14	25.5 × 25.5	78.0	72.0
15	20.5 × 15.5	0.0	73.5
16	20.5 × 15.5	24.2	73.5
17	20.5 × 15.5	108.0	73.5
18	42.5 × 52.5	133.0	73.5
19	20.5 × 15.5	0.0	92.0
20	20.5 × 15.5	24.2	92.0
21	23.5 × 32.5	106.5	93.5
22	25.5 × 25.5	49.5	100.5
23	25.5 × 25.5	78.0	100.5
24	20.5 × 15.5	0.0	110.5
25	20.5 × 15.5	24.2	110.5

Layout 13 191.5 × 113.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

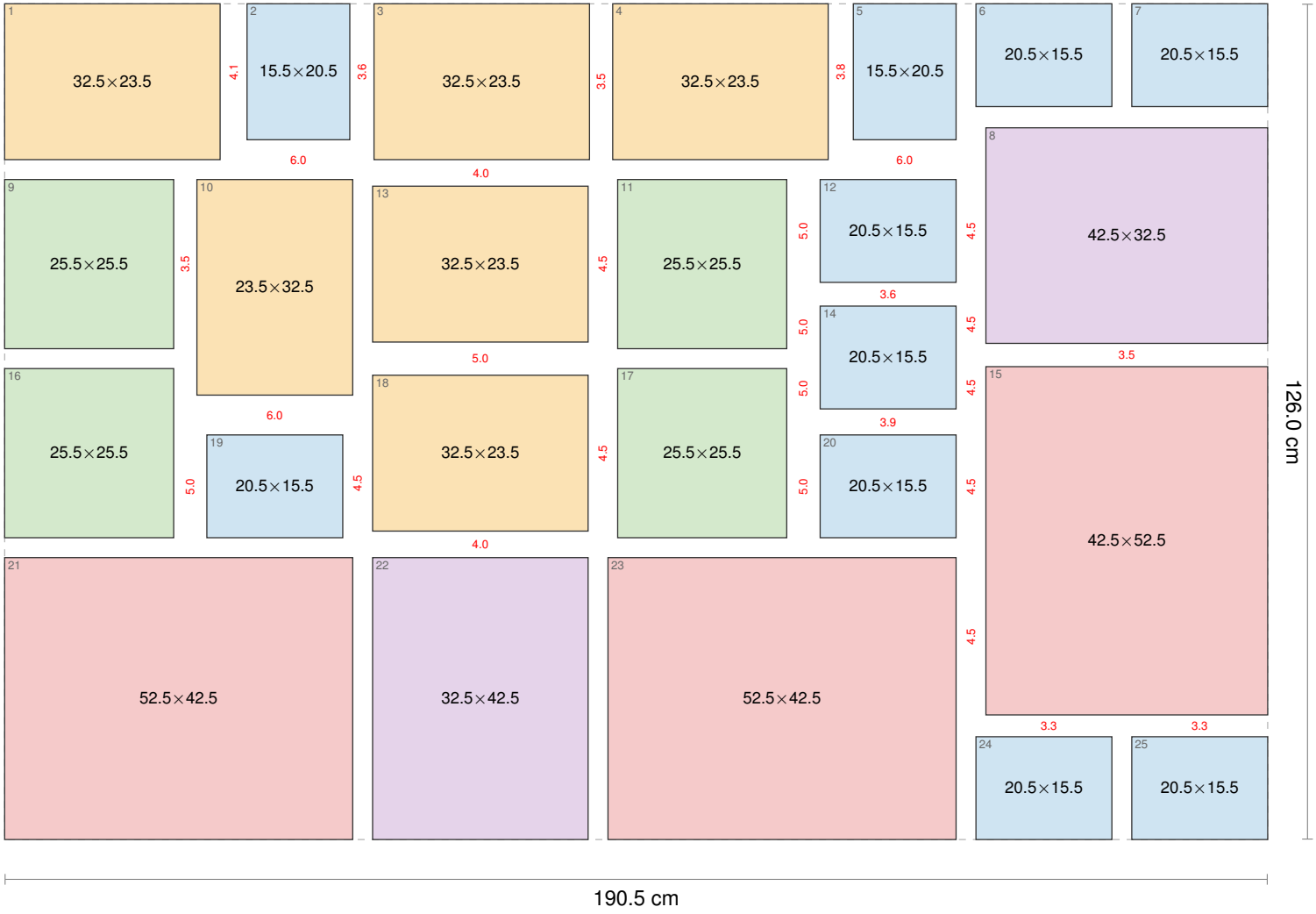
Outer size	191.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.72 cm
Border notches	top 6 cm, right 3 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, not touching, min. dist. 71 cm
Entropy	0.75
mix / variety	0.81 / 0.69
stagger / lines	0.51 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	2	2
32.5×42.5	P	1	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	0	1
25.5×25.5		4	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	22.7	0.0
3	20.5 × 15.5	45.3	0.0
4	20.5 × 15.5	67.6	0.0
5	32.5 × 23.5	91.0	0.0
6	20.5 × 15.5	125.0	0.0
7	42.5 × 52.5	149.0	0.0
8	25.5 × 25.5	0.0	17.0
9	25.5 × 25.5	27.6	17.0
10	32.5 × 23.5	55.6	18.0
11	20.5 × 15.5	125.0	18.1
12	32.5 × 42.5	91.0	27.0
13	20.5 × 15.5	125.0	35.6
14	25.5 × 25.5	0.0	44.0
15	25.5 × 25.5	27.6	44.0
16	32.5 × 23.5	55.6	45.0
17	20.5 × 15.5	125.0	53.5
18	42.5 × 32.5	149.0	54.9
19	52.5 × 42.5	0.0	71.0
20	15.5 × 20.5	54.0	71.0
21	52.5 × 42.5	71.0	71.0
22	20.5 × 15.5	125.0	71.9
23	32.5 × 23.5	125.0	90.0
24	32.5 × 23.5	159.0	90.0
25	15.5 × 20.5	54.0	93.0

Layout 14 190.5 × 126.0 cm gap 3 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

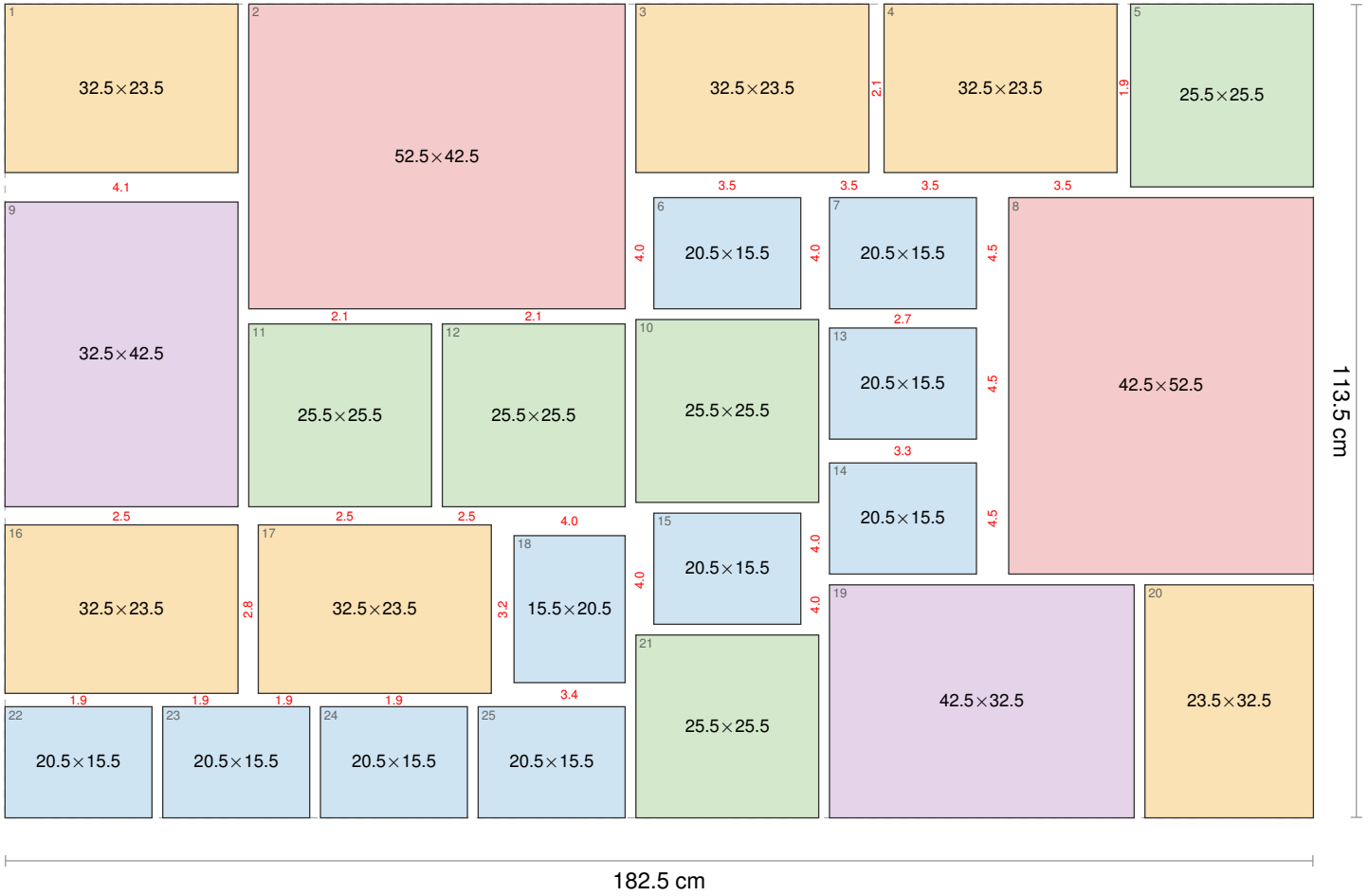
Outer size	190.5 × 126.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 6.0 cm
Max. deviation	3.0 cm
Mean deviation	0.90 cm
Border notches	top 4 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, not touching, min. dist. 57 cm
Entropy	0.64
mix / variety	0.86 / 0.71
stagger / lines	0.50 / 0.50 (3 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	1	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	15.5 × 20.5	36.6	0.0
3	32.5 × 23.5	55.7	0.0
4	32.5 × 23.5	91.7	0.0
5	15.5 × 20.5	128.0	0.0
6	20.5 × 15.5	146.5	0.0
7	20.5 × 15.5	170.0	0.0
8	42.5 × 32.5	148.0	18.7
9	25.5 × 25.5	0.0	26.5
10	23.5 × 32.5	29.0	26.5
11	25.5 × 25.5	92.5	26.5
12	20.5 × 15.5	123.0	26.5
13	32.5 × 23.5	55.5	27.5
14	20.5 × 15.5	123.0	45.6
15	42.5 × 52.5	148.0	54.7
16	25.5 × 25.5	0.0	55.0
17	25.5 × 25.5	92.5	55.0
18	32.5 × 23.5	55.5	56.0
19	20.5 × 15.5	30.5	65.0
20	20.5 × 15.5	123.0	65.0
21	52.5 × 42.5	0.0	83.5
22	32.5 × 42.5	55.5	83.5
23	52.5 × 42.5	91.0	83.5
24	20.5 × 15.5	146.5	110.5
25	20.5 × 15.5	170.0	110.5

Layout 15 182.5 × 113.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

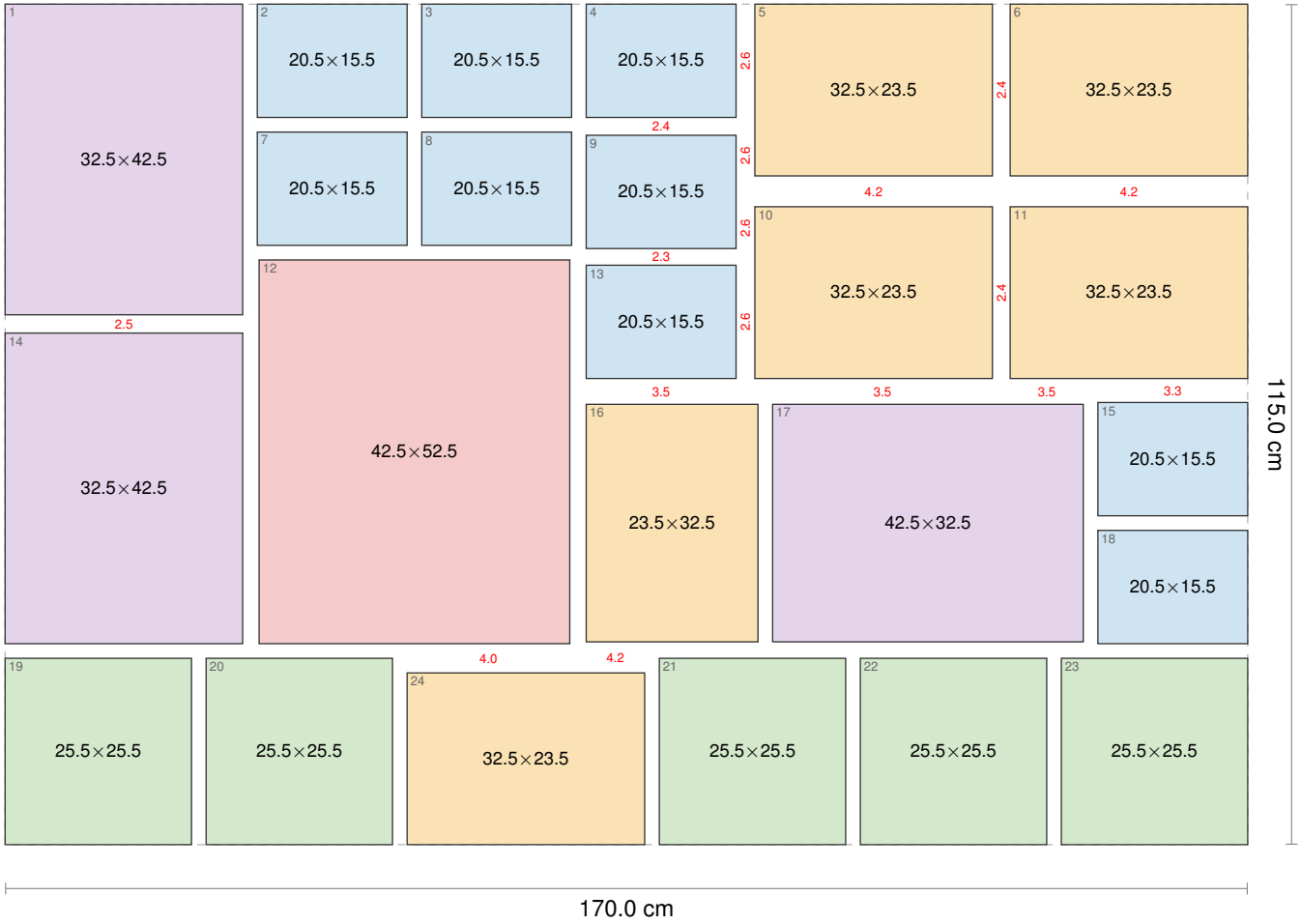
Outer size	182.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.5 cm
Max. deviation	3.0 cm
Mean deviation	0.95 cm
Border notches	left 7 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 106 cm
Entropy	0.79
mix / variety	0.81 / 0.80
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	1	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	52.5 × 42.5	34.0	0.0
3	32.5 × 23.5	88.0	0.0
4	32.5 × 23.5	122.6	0.0
5	25.5 × 25.5	157.0	0.0
6	20.5 × 15.5	90.5	27.0
7	20.5 × 15.5	115.0	27.0
8	42.5 × 52.5	140.0	27.0
9	32.5 × 42.5	0.0	27.6
10	25.5 × 25.5	88.0	44.0
11	25.5 × 25.5	34.0	44.6
12	25.5 × 25.5	61.0	44.6
13	20.5 × 15.5	115.0	45.2
14	20.5 × 15.5	115.0	64.0
15	20.5 × 15.5	90.5	71.0
16	32.5 × 23.5	0.0	72.6
17	32.5 × 23.5	35.3	72.6
18	15.5 × 20.5	71.0	74.1
19	42.5 × 32.5	115.0	81.0
20	23.5 × 32.5	159.0	81.0
21	25.5 × 25.5	88.0	88.0
22	20.5 × 15.5	0.0	98.0
23	20.5 × 15.5	22.0	98.0
24	20.5 × 15.5	44.0	98.0
25	20.5 × 15.5	66.0	98.0

Layout 16 170.0 × 115.0 cm gap 2 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	170.0 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.2 cm
Max. deviation	2.2 cm
Mean deviation	0.44 cm
Border notches	right 8 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	1
Entropy	0.72
mix / variety	0.74 / 0.62
stagger / lines	0.51 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	20.5 × 15.5	34.5	0.0
3	20.5 × 15.5	57.0	0.0
4	20.5 × 15.5	79.5	0.0
5	32.5 × 23.5	102.6	0.0
6	32.5 × 23.5	137.5	0.0
7	20.5 × 15.5	34.5	17.5
8	20.5 × 15.5	57.0	17.5
9	20.5 × 15.5	79.5	17.9
10	32.5 × 23.5	102.6	27.7
11	32.5 × 23.5	137.5	27.7
12	42.5 × 52.5	34.8	35.0
13	20.5 × 15.5	79.5	35.7
14	32.5 × 42.5	0.0	45.0
15	20.5 × 15.5	149.5	54.5
16	23.5 × 32.5	79.5	54.8
17	42.5 × 32.5	105.0	54.8
18	20.5 × 15.5	149.5	72.0
19	25.5 × 25.5	0.0	89.5
20	25.5 × 25.5	27.5	89.5
21	25.5 × 25.5	89.5	89.5
22	25.5 × 25.5	117.0	89.5
23	25.5 × 25.5	144.5	89.5
24	32.5 × 23.5	55.0	91.5

7 gap-3

The nominal gap is fixed at 3 cm; the search only changes the arrangement and the subset of frames. Actual gaps still deviate from 3 cm wherever slack has to be absorbed.

Parameters that differ from the baseline: gap = 3.
256 runs, 20 min wall-clock budget, 533 candidate states, the 16 best distinct layouts are shown.

#	size (cm)	gap	frames	max dev	mean dev	notches	corners	entropy
1	179.0 × 126.5	3	24	2.2	0.44	none	4/4	0.64
2	180.0 × 128.0	3	25	3.0	0.78	none	4/4	0.66
3	175.0 × 118.0	3	24	2.0	0.47	4 cm	4/4	0.70
4	188.5 × 116.5	3	24	2.8	0.61	none	4/4	0.68
5	179.0 × 128.0	3	24	3.0	0.64	none	4/4	0.76
6	189.0 × 109.0	3	24	3.0	0.85	none	4/4	0.74
7	173.5 × 118.0	3	23	2.5	0.42	4 cm	4/4	0.65
8	187.0 × 108.0	3	22	2.5	0.39	4 cm	4/4	0.73
9	183.5 × 118.0	3	21	2.0	0.66	4 cm	4/4	0.72
10	179.0 × 126.0	3	25	3.0	0.85	6 cm	4/4	0.69
11	181.5 × 125.0	3	25	2.5	0.66	10 cm	4/4	0.69
12	181.5 × 128.0	3	25	4.0	0.98	none	4/4	0.77
13	175.0 × 118.0	3	24	2.8	0.45	4 cm	4/4	0.72
14	178.5 × 126.5	3	24	3.2	0.86	5 cm	4/4	0.77
15	189.0 × 118.0	3	23	2.8	0.58	4 cm	4/4	0.57
16	181.0 × 126.5	3	23	3.0	0.52	5 cm	4/4	0.78

Layout 1 179.0 × 126.5 cm gap 3 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	179.0 × 126.5 cm
Nominal gap	3 cm
Actual gaps	3.0 – 5.2 cm
Max. deviation	2.2 cm
Mean deviation	0.44 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	3, not touching, min. dist. 79 cm
Entropy	0.64
mix / variety	0.76 / 0.74
stagger / lines	0.54 / 0.50 (3 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	1	2
32.5×42.5	P	2	2
42.5×32.5	L	0	1
32.5×23.5	L	5	5
23.5×32.5	P	0	1
25.5×25.5		4	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 42.5	55.5	0.0
3	32.5 × 42.5	91.0	0.0
4	52.5 × 42.5	126.5	0.0
5	32.5 × 23.5	0.0	45.5
6	32.5 × 23.5	35.5	45.5
7	25.5 × 25.5	89.5	45.5
8	25.5 × 25.5	153.5	45.5
9	32.5 × 23.5	118.0	46.5
10	15.5 × 20.5	71.0	47.0
11	20.5 × 15.5	0.0	72.2
12	25.5 × 25.5	61.0	72.2
13	32.5 × 23.5	24.5	73.2
14	20.5 × 15.5	89.5	74.0
15	20.5 × 15.5	113.0	74.0
16	42.5 × 52.5	136.5	74.0
17	20.5 × 15.5	0.0	91.6
18	20.5 × 15.5	89.5	92.5
19	20.5 × 15.5	113.0	92.5
20	25.5 × 25.5	24.5	101.0
21	32.5 × 23.5	54.0	103.0
22	20.5 × 15.5	0.0	111.0
23	20.5 × 15.5	89.5	111.0
24	20.5 × 15.5	113.0	111.0

Layout 2 180.0 × 128.0 cm gap 3 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

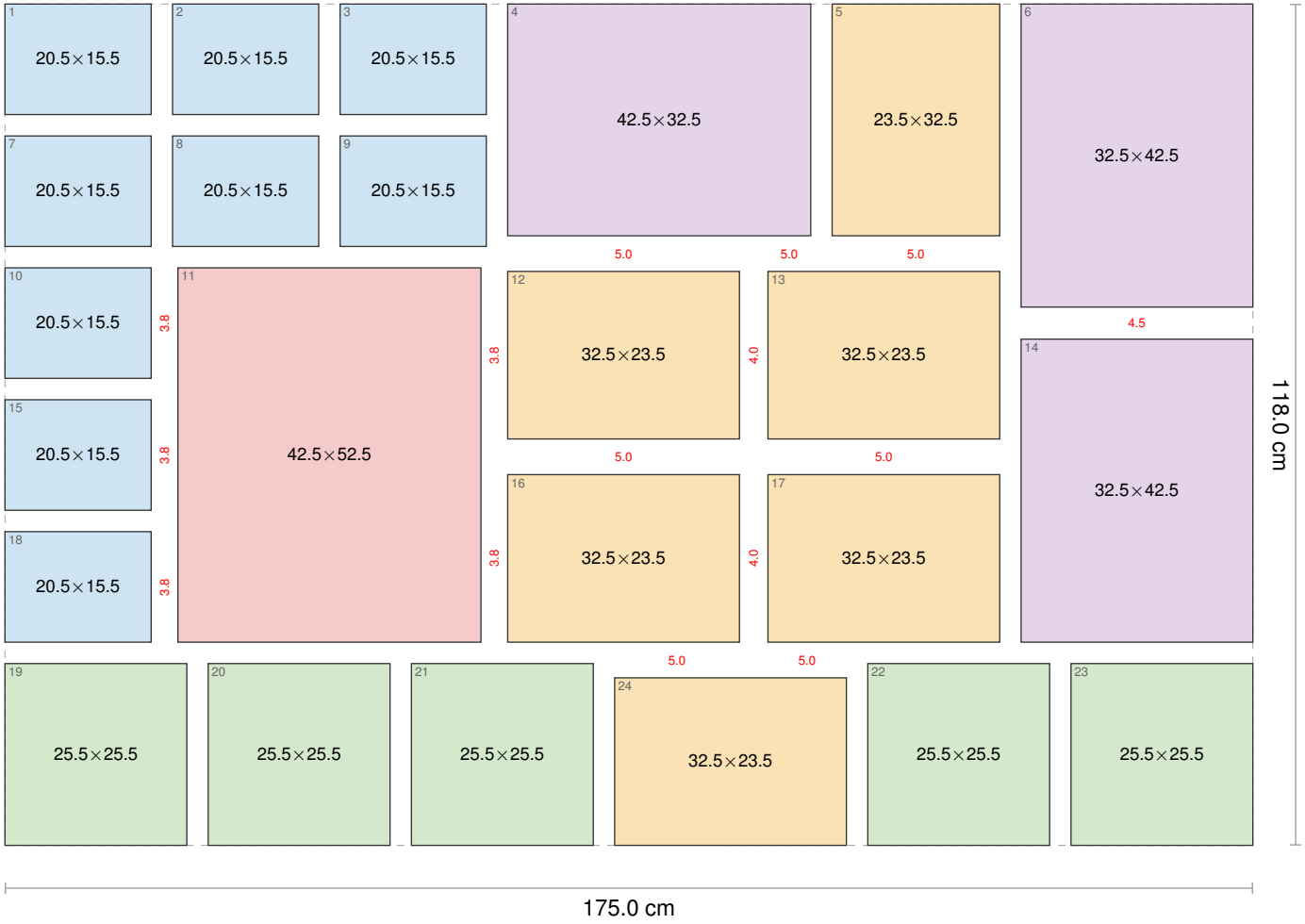
Outer size	180.0 × 128.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 6.0 cm
Max. deviation	3.0 cm
Mean deviation	0.78 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 92 cm
Entropy	0.66
mix / variety	0.81 / 0.59
stagger / lines	0.48 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	1	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	0	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	35.8	0.0
3	32.5 × 42.5	91.7	0.0
4	52.5 × 42.5	127.5	0.0
5	25.5 × 25.5	0.0	45.5
6	32.5 × 23.5	64.0	45.5
7	32.5 × 23.5	100.5	45.5
8	25.5 × 25.5	154.5	45.5
9	32.5 × 23.5	28.5	46.5
10	15.5 × 20.5	136.0	48.0
11	25.5 × 25.5	0.0	74.0
12	20.5 × 15.5	64.0	74.5
13	20.5 × 15.5	88.2	74.5
14	20.5 × 15.5	112.5	74.5
15	20.5 × 15.5	136.0	74.5
16	20.5 × 15.5	159.5	74.5
17	32.5 × 23.5	28.5	75.0
18	20.5 × 15.5	136.0	93.5
19	20.5 × 15.5	159.5	93.5
20	23.5 × 32.5	64.0	95.5
21	42.5 × 32.5	90.5	95.5
22	25.5 × 25.5	0.0	102.5
23	32.5 × 23.5	28.5	104.5
24	20.5 × 15.5	136.0	112.5
25	20.5 × 15.5	159.5	112.5

Layout 3 175.0 × 118.0 cm gap 3 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

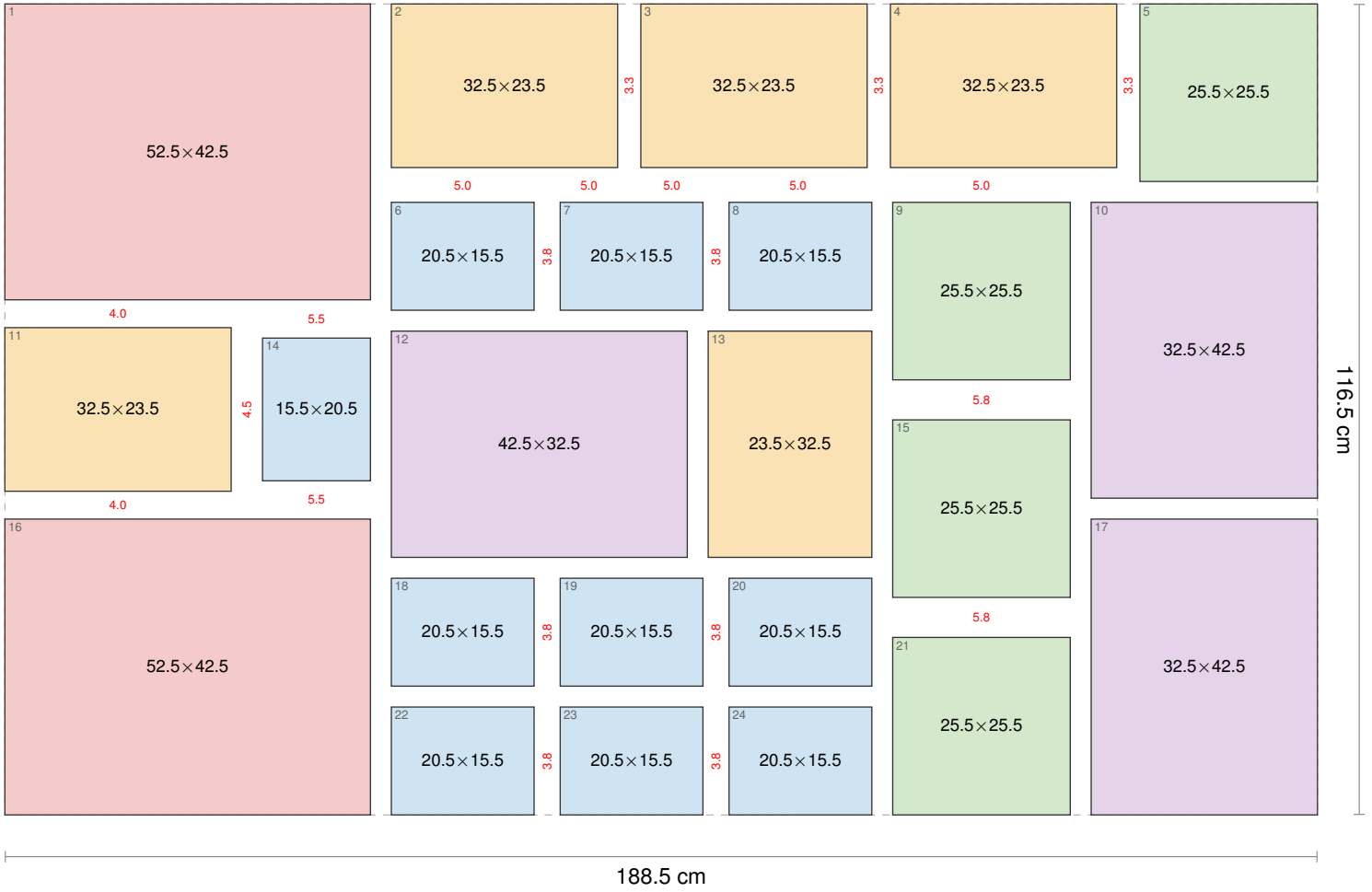
Outer size	175.0 × 118.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 5.0 cm
Max. deviation	2.0 cm
Mean deviation	0.47 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	1
Entropy	0.70
mix / variety	0.74 / 0.59
stagger / lines	0.47 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	23.5	0.0
3	20.5 × 15.5	47.0	0.0
4	42.5 × 32.5	70.5	0.0
5	23.5 × 32.5	116.0	0.0
6	32.5 × 42.5	142.5	0.0
7	20.5 × 15.5	0.0	18.5
8	20.5 × 15.5	23.5	18.5
9	20.5 × 15.5	47.0	18.5
10	20.5 × 15.5	0.0	37.0
11	42.5 × 52.5	24.2	37.0
12	32.5 × 23.5	70.5	37.5
13	32.5 × 23.5	107.0	37.5
14	32.5 × 42.5	142.5	47.0
15	20.5 × 15.5	0.0	55.5
16	32.5 × 23.5	70.5	66.0
17	32.5 × 23.5	107.0	66.0
18	20.5 × 15.5	0.0	74.0
19	25.5 × 25.5	0.0	92.5
20	25.5 × 25.5	28.5	92.5
21	25.5 × 25.5	57.0	92.5
22	25.5 × 25.5	121.0	92.5
23	25.5 × 25.5	149.5	92.5
24	32.5 × 23.5	85.5	94.5

Layout 4 188.5 × 116.5 cm gap 3 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

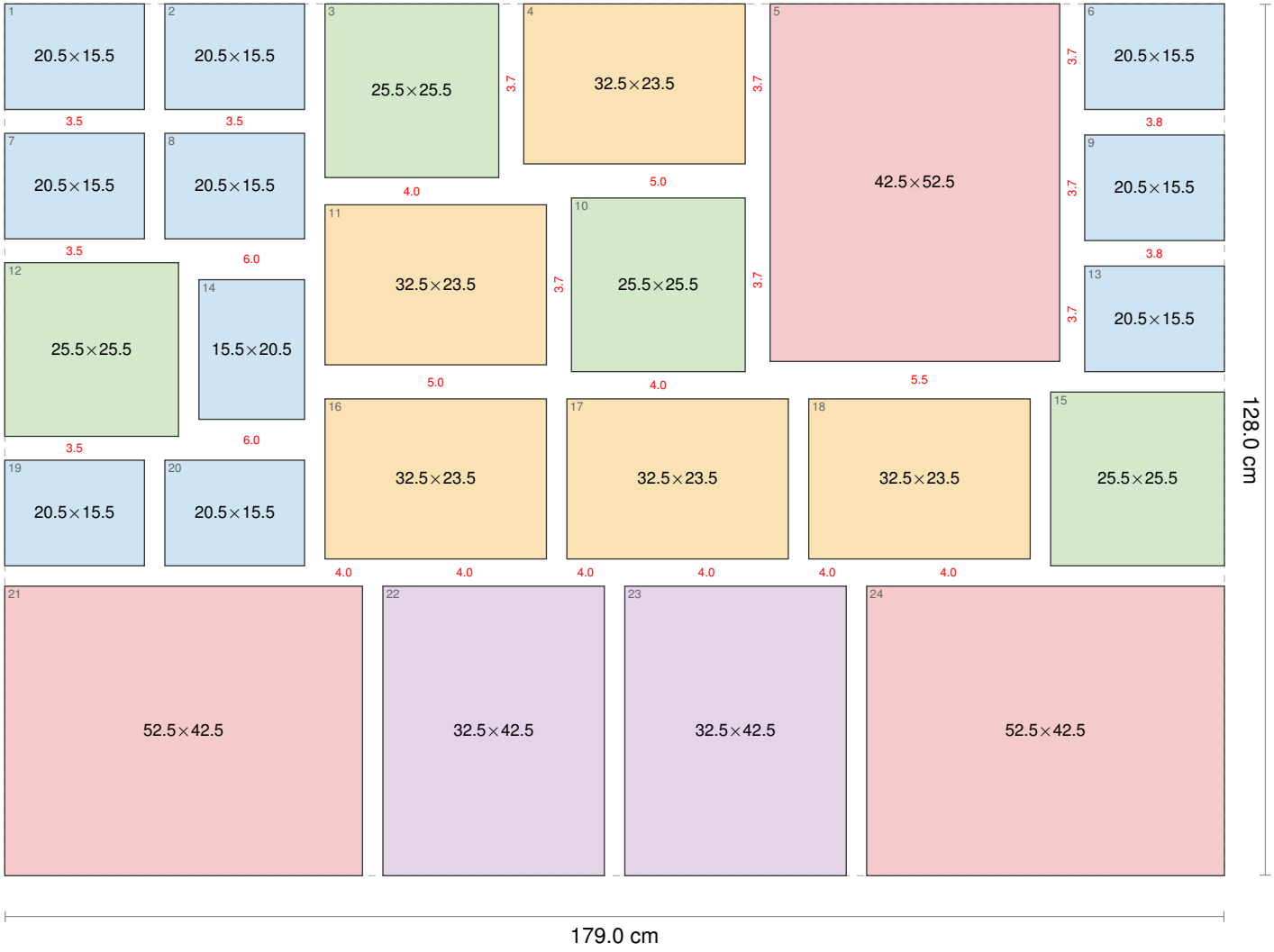
Outer size	188.5 × 116.5 cm
Nominal gap	3 cm
Actual gaps	3.0 – 5.8 cm
Max. deviation	2.8 cm
Mean deviation	0.61 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 74 cm
Entropy	0.68
mix / variety	0.81 / 0.68
stagger / lines	0.46 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 23.5	55.5	0.0
3	32.5 × 23.5	91.3	0.0
4	32.5 × 23.5	127.2	0.0
5	25.5 × 25.5	163.0	0.0
6	20.5 × 15.5	55.5	28.5
7	20.5 × 15.5	79.8	28.5
8	20.5 × 15.5	104.0	28.5
9	25.5 × 25.5	127.5	28.5
10	32.5 × 42.5	156.0	28.5
11	32.5 × 23.5	0.0	46.5
12	42.5 × 32.5	55.5	47.0
13	23.5 × 32.5	101.0	47.0
14	15.5 × 20.5	37.0	48.0
15	25.5 × 25.5	127.5	59.8
16	52.5 × 42.5	0.0	74.0
17	32.5 × 42.5	156.0	74.0
18	20.5 × 15.5	55.5	82.5
19	20.5 × 15.5	79.8	82.5
20	20.5 × 15.5	104.0	82.5
21	25.5 × 25.5	127.5	91.0
22	20.5 × 15.5	55.5	101.0
23	20.5 × 15.5	79.8	101.0
24	20.5 × 15.5	104.0	101.0

Layout 5 179.0 × 128.0 cm gap 3 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

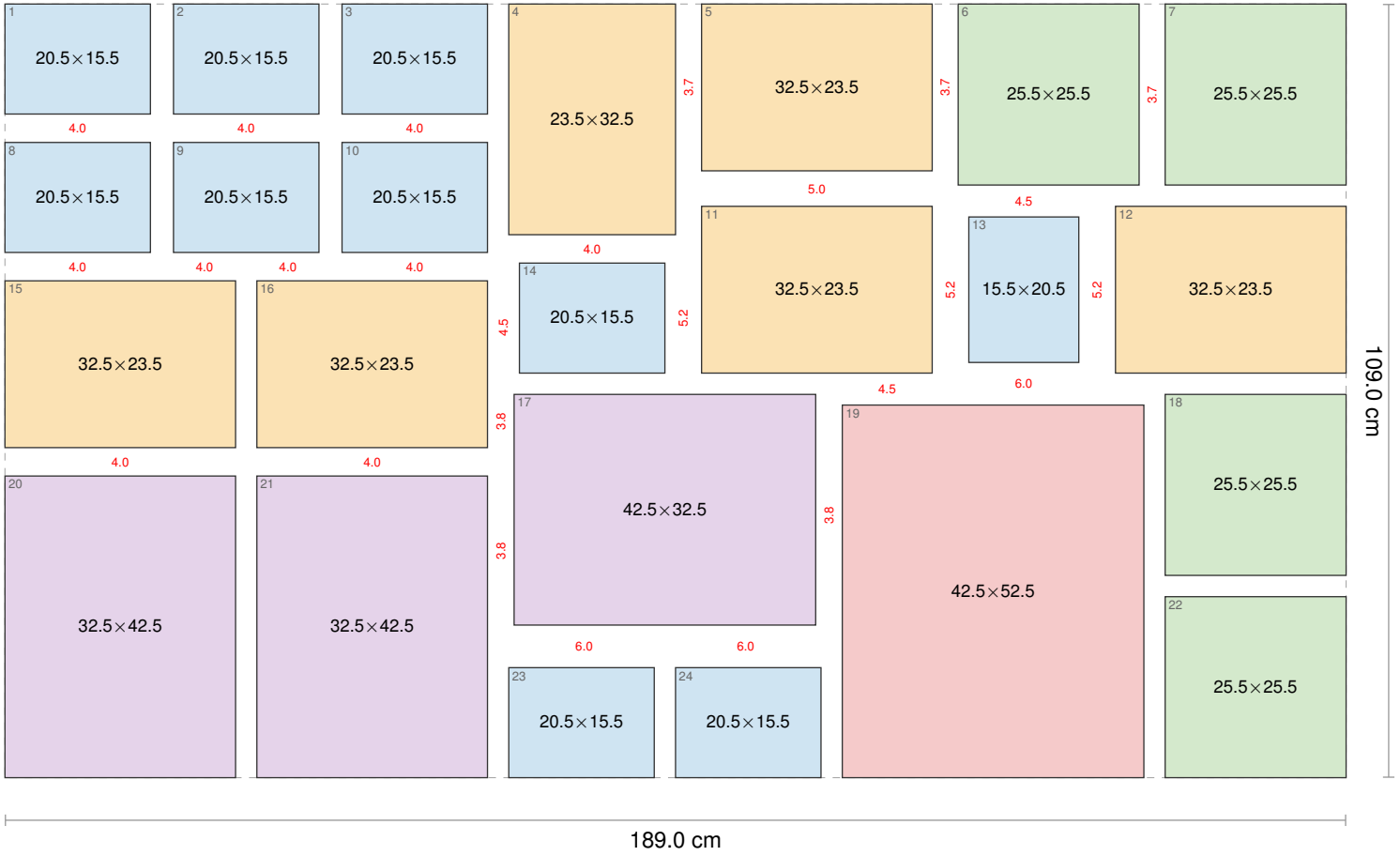
Outer size	179.0 × 128.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 6.0 cm
Max. deviation	3.0 cm
Mean deviation	0.64 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	3, not touching, min. dist. 83 cm
Entropy	0.76
mix / variety	0.76 / 0.75
stagger / lines	0.54 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	5	5
	23.5×32.5	P	0	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	23.5	0.0
3	25.5 × 25.5	47.0	0.0
4	32.5 × 23.5	76.2	0.0
5	42.5 × 52.5	112.3	0.0
6	20.5 × 15.5	158.5	0.0
7	20.5 × 15.5	0.0	19.0
8	20.5 × 15.5	23.5	19.0
9	20.5 × 15.5	158.5	19.2
10	25.5 × 25.5	83.2	28.5
11	32.5 × 23.5	47.0	29.5
12	25.5 × 25.5	0.0	38.0
13	20.5 × 15.5	158.5	38.5
14	15.5 × 20.5	28.5	40.5
15	25.5 × 25.5	153.5	57.0
16	32.5 × 23.5	47.0	58.0
17	32.5 × 23.5	82.5	58.0
18	32.5 × 23.5	118.0	58.0
19	20.5 × 15.5	0.0	67.0
20	20.5 × 15.5	23.5	67.0
21	52.5 × 42.5	0.0	85.5
22	32.5 × 42.5	55.5	85.5
23	32.5 × 42.5	91.0	85.5
24	52.5 × 42.5	126.5	85.5

Layout 6 189.0 × 109.0 cm gap 3 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

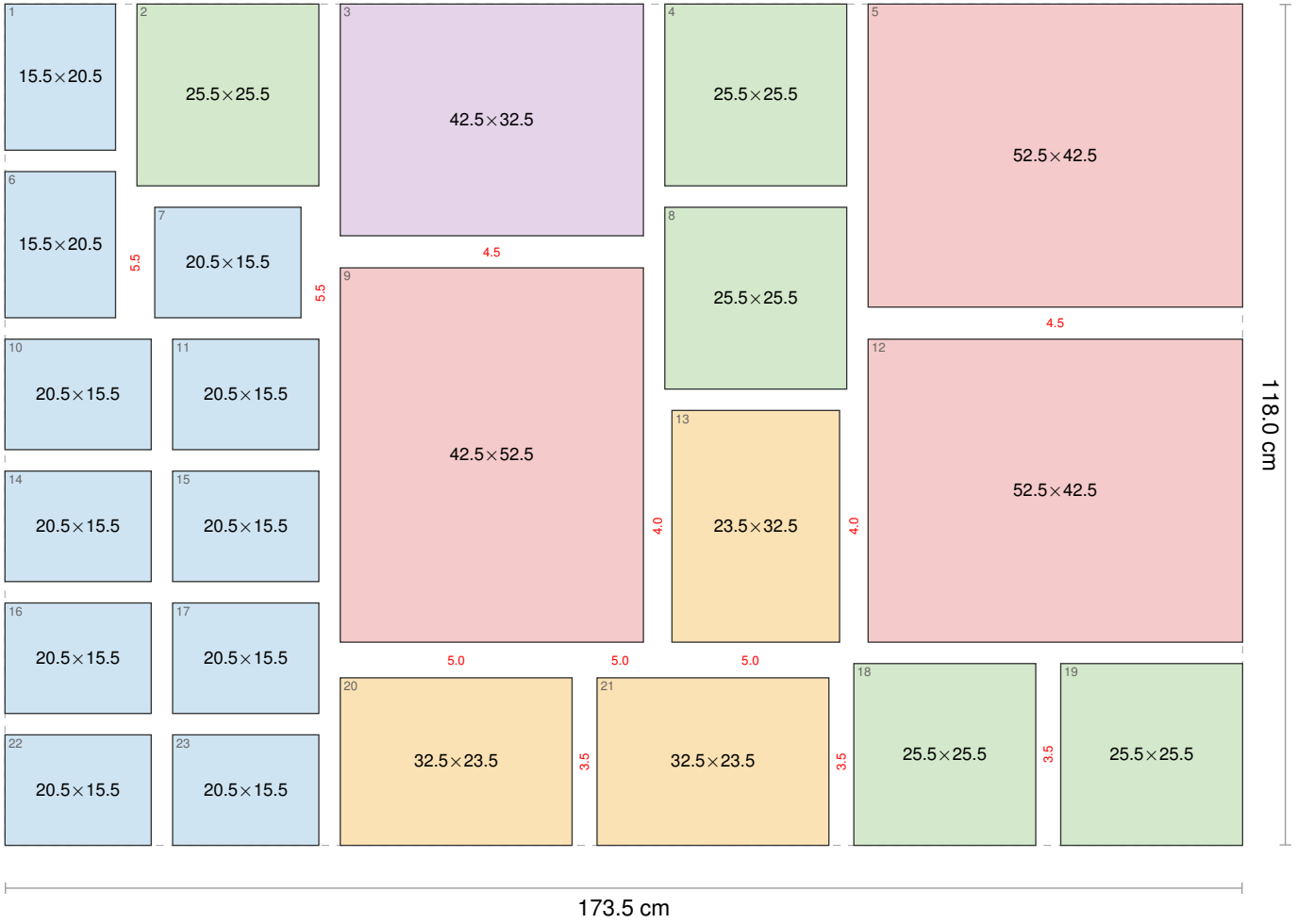
Outer size	189.0 × 109.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 6.0 cm
Max. deviation	3.0 cm
Mean deviation	0.85 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	1
Entropy	0.74
mix / variety	0.79 / 0.65
stagger / lines	0.51 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	23.8	0.0
3	20.5 × 15.5	47.5	0.0
4	23.5 × 32.5	71.0	0.0
5	32.5 × 23.5	98.2	0.0
6	25.5 × 25.5	134.3	0.0
7	25.5 × 25.5	163.5	0.0
8	20.5 × 15.5	0.0	19.5
9	20.5 × 15.5	23.8	19.5
10	20.5 × 15.5	47.5	19.5
11	32.5 × 23.5	98.2	28.5
12	32.5 × 23.5	156.5	28.5
13	15.5 × 20.5	135.8	30.0
14	20.5 × 15.5	72.5	36.5
15	32.5 × 23.5	0.0	39.0
16	32.5 × 23.5	35.5	39.0
17	42.5 × 32.5	71.8	55.0
18	25.5 × 25.5	163.5	55.0
19	42.5 × 52.5	118.0	56.5
20	32.5 × 42.5	0.0	66.5
21	32.5 × 42.5	35.5	66.5
22	25.5 × 25.5	163.5	83.5
23	20.5 × 15.5	71.0	93.5
24	20.5 × 15.5	94.5	93.5

Layout 7 173.5 × 118.0 cm gap 3 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

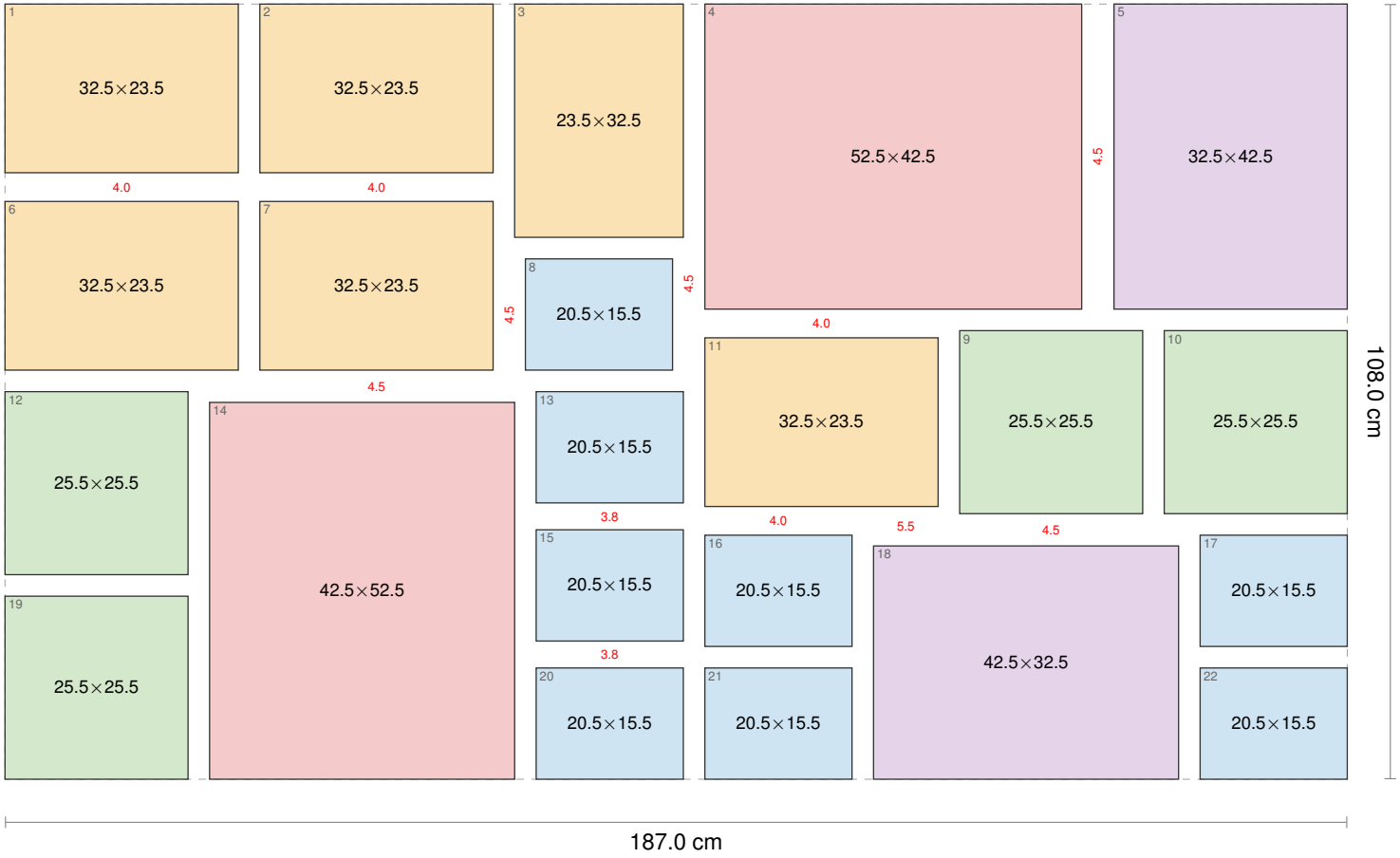
Outer size	173.5 × 118.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 5.5 cm
Max. deviation	2.5 cm
Mean deviation	0.42 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, not touching, min. dist. 47 cm
Entropy	0.65
mix / variety	0.79 / 0.57
stagger / lines	0.48 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	0	2
	42.5×32.5	L	1	1
	32.5×23.5	L	2	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	15.5 × 20.5	0.0	0.0
2	25.5 × 25.5	18.5	0.0
3	42.5 × 32.5	47.0	0.0
4	25.5 × 25.5	92.5	0.0
5	52.5 × 42.5	121.0	0.0
6	15.5 × 20.5	0.0	23.5
7	20.5 × 15.5	21.0	28.5
8	25.5 × 25.5	92.5	28.5
9	42.5 × 52.5	47.0	37.0
10	20.5 × 15.5	0.0	47.0
11	20.5 × 15.5	23.5	47.0
12	52.5 × 42.5	121.0	47.0
13	23.5 × 32.5	93.5	57.0
14	20.5 × 15.5	0.0	65.5
15	20.5 × 15.5	23.5	65.5
16	20.5 × 15.5	0.0	84.0
17	20.5 × 15.5	23.5	84.0
18	25.5 × 25.5	119.0	92.5
19	25.5 × 25.5	148.0	92.5
20	32.5 × 23.5	47.0	94.5
21	32.5 × 23.5	83.0	94.5
22	20.5 × 15.5	0.0	102.5
23	20.5 × 15.5	23.5	102.5

Layout 8 187.0 × 108.0 cm gap 3 cm 22 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	187.0 × 108.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 5.5 cm
Max. deviation	2.5 cm
Mean deviation	0.39 cm
Border notches	top 4 cm
Corners filled	4 / 4
Frames used	22 of 28
Large frames	2, not touching, min. dist. 96 cm
Entropy	0.73
mix / variety	0.78 / 0.62
stagger / lines	0.52 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	8	9
15.5×20.5	P	0	2
32.5×42.5	P	1	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	1	1
52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	32.5 × 23.5	35.5	0.0
3	23.5 × 32.5	71.0	0.0
4	52.5 × 42.5	97.5	0.0
5	32.5 × 42.5	154.5	0.0
6	32.5 × 23.5	0.0	27.5
7	32.5 × 23.5	35.5	27.5
8	20.5 × 15.5	72.5	35.5
9	25.5 × 25.5	133.0	45.5
10	25.5 × 25.5	161.5	45.5
11	32.5 × 23.5	97.5	46.5
12	25.5 × 25.5	0.0	54.0
13	20.5 × 15.5	74.0	54.0
14	42.5 × 52.5	28.5	55.5
15	20.5 × 15.5	74.0	73.2
16	20.5 × 15.5	97.5	74.0
17	20.5 × 15.5	166.5	74.0
18	42.5 × 32.5	121.0	75.5
19	25.5 × 25.5	0.0	82.5
20	20.5 × 15.5	74.0	92.5
21	20.5 × 15.5	97.5	92.5
22	20.5 × 15.5	166.5	92.5

Layout 9 183.5 × 118.0 cm gap 3 cm 21 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	183.5 × 118.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 5.0 cm
Max. deviation	2.0 cm
Mean deviation	0.66 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	21 of 28
Large frames	3, not touching, min. dist. 47 cm
Entropy	0.72
mix / variety	0.81 / 0.61
stagger / lines	0.45 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	2	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	28.8	0.0
3	25.5 × 25.5	57.6	0.0
4	32.5 × 23.5	86.4	0.0
5	32.5 × 23.5	122.2	0.0
6	25.5 × 25.5	158.0	0.0
7	20.5 × 15.5	0.0	28.5
8	20.5 × 15.5	25.0	28.5
9	42.5 × 52.5	50.0	28.5
10	32.5 × 42.5	95.5	28.5
11	52.5 × 42.5	131.0	28.5
12	20.5 × 15.5	0.0	47.0
13	20.5 × 15.5	25.0	47.0
14	20.5 × 15.5	0.0	65.5
15	20.5 × 15.5	25.0	65.5
16	32.5 × 42.5	95.5	75.5
17	52.5 × 42.5	131.0	75.5
18	20.5 × 15.5	0.0	84.0
19	23.5 × 32.5	23.5	85.5
20	42.5 × 32.5	50.0	85.5
21	20.5 × 15.5	0.0	102.5

Layout 10 179.0 × 126.0 cm gap 3 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

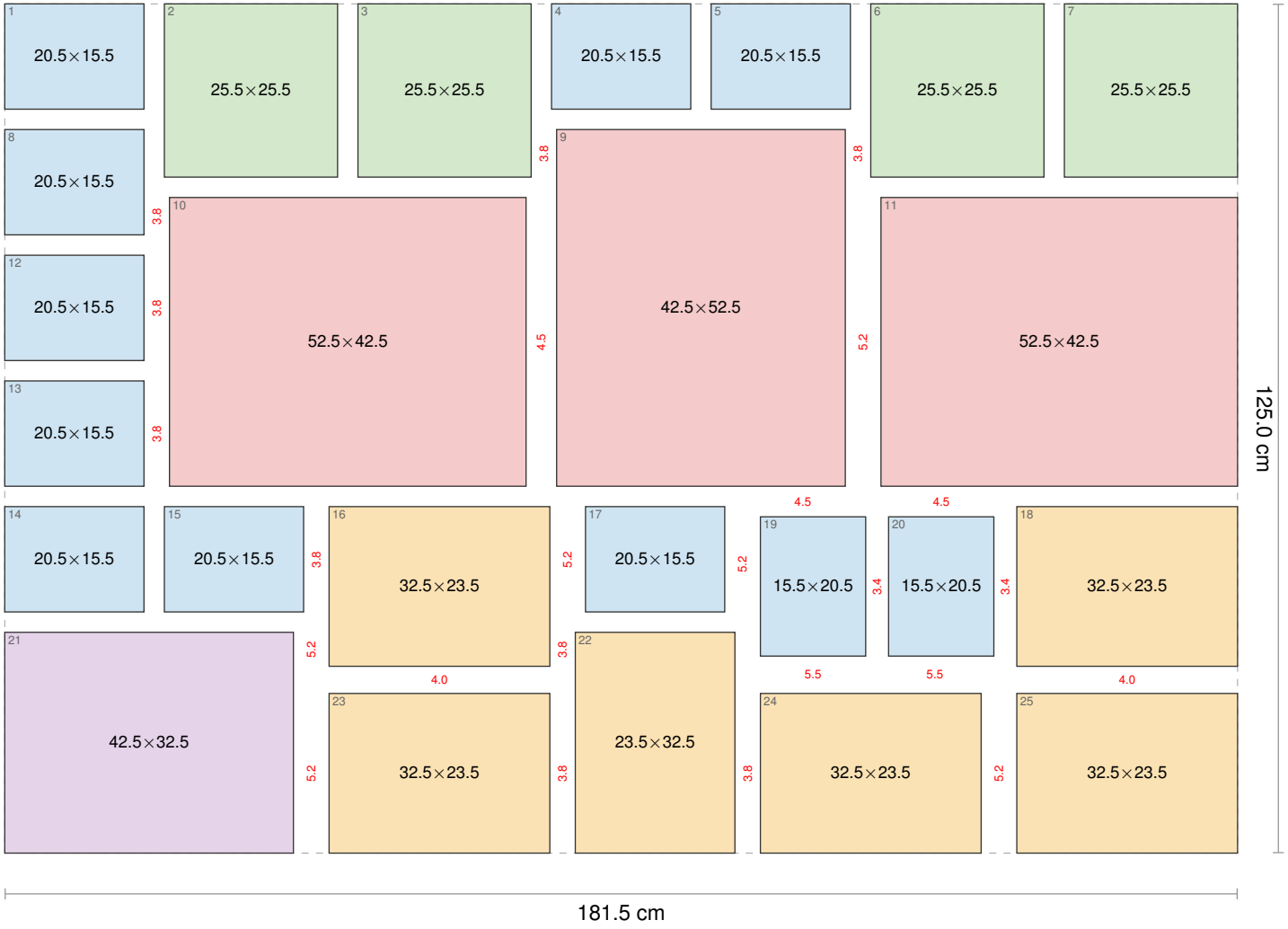
Outer size	179.0 × 126.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 6.0 cm
Max. deviation	3.0 cm
Mean deviation	0.85 cm
Border notches	left 6 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 126 cm
Entropy	0.69
mix / variety	0.83 / 0.70
stagger / lines	0.50 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 42.5	55.5	0.0
3	32.5 × 42.5	91.0	0.0
4	52.5 × 42.5	126.5	0.0
5	20.5 × 15.5	0.0	45.5
6	20.5 × 15.5	23.5	45.5
7	20.5 × 15.5	48.3	45.5
8	25.5 × 25.5	73.2	45.5
9	20.5 × 15.5	103.0	45.5
10	32.5 × 23.5	146.5	45.5
11	15.5 × 20.5	126.5	47.0
12	20.5 × 15.5	48.3	64.8
13	20.5 × 15.5	103.0	64.8
14	42.5 × 32.5	0.0	67.0
15	20.5 × 15.5	128.0	72.0
16	25.5 × 25.5	153.5	72.0
17	25.5 × 25.5	73.2	74.0
18	20.5 × 15.5	48.3	84.0
19	20.5 × 15.5	103.0	84.0
20	23.5 × 32.5	126.5	93.5
21	25.5 × 25.5	153.5	100.5
22	32.5 × 23.5	0.0	102.5
23	32.5 × 23.5	36.0	102.5
24	32.5 × 23.5	91.0	102.5
25	15.5 × 20.5	72.0	105.5

Layout 11 181.5 × 125.0 cm gap 3 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	181.5 × 125.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 5.5 cm
Max. deviation	2.5 cm
Mean deviation	0.66 cm
Border notches	bottom 10 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, not touching, min. dist. 52 cm
Entropy	0.69
mix / variety	0.81 / 0.71
stagger / lines	0.51 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	0	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	25.5 × 25.5	23.5	0.0
3	25.5 × 25.5	52.0	0.0
4	20.5 × 15.5	80.5	0.0
5	20.5 × 15.5	104.0	0.0
6	25.5 × 25.5	127.5	0.0
7	25.5 × 25.5	156.0	0.0
8	20.5 × 15.5	0.0	18.5
9	42.5 × 52.5	81.2	18.5
10	52.5 × 42.5	24.2	28.5
11	52.5 × 42.5	129.0	28.5
12	20.5 × 15.5	0.0	37.0
13	20.5 × 15.5	0.0	55.5
14	20.5 × 15.5	0.0	74.0
15	20.5 × 15.5	23.5	74.0
16	32.5 × 23.5	47.8	74.0
17	20.5 × 15.5	85.5	74.0
18	32.5 × 23.5	149.0	74.0
19	15.5 × 20.5	111.2	75.5
20	15.5 × 20.5	130.1	75.5
21	42.5 × 32.5	0.0	92.5
22	23.5 × 32.5	84.0	92.5
23	32.5 × 23.5	47.8	101.5
24	32.5 × 23.5	111.2	101.5
25	32.5 × 23.5	149.0	101.5

Layout 12 181.5 × 128.0 cm gap 3 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

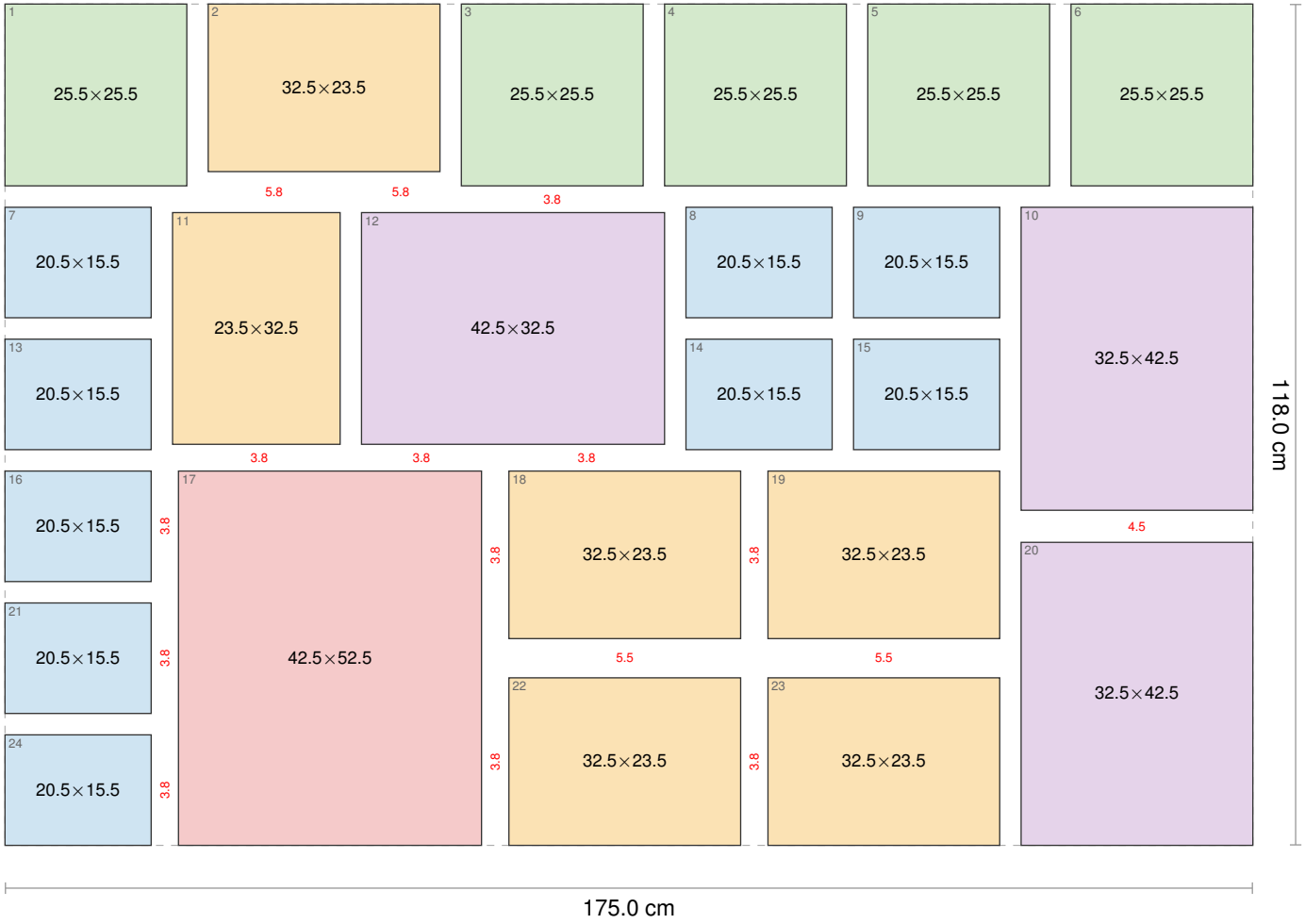
Outer size	181.5 × 128.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 7.0 cm
Max. deviation	4.0 cm
Mean deviation	0.98 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, touching 1× (42 cm), min. dist. 56 cm
Entropy	0.77
mix / variety	0.84 / 0.68
stagger / lines	0.56 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	8	9
15.5×20.5	P	2	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	4	5
23.5×32.5	P	1	1
25.5×25.5		5	5
42.5×52.5	P	0	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	32.5 × 42.5	36.3	0.0
3	52.5 × 42.5	72.7	0.0
4	52.5 × 42.5	129.0	0.0
5	25.5 × 25.5	0.0	45.5
6	25.5 × 25.5	30.0	45.5
7	15.5 × 20.5	135.0	45.5
8	25.5 × 25.5	156.0	45.5
9	32.5 × 23.5	60.0	46.5
10	32.5 × 23.5	97.0	46.5
11	15.5 × 20.5	135.0	69.5
12	25.5 × 25.5	156.0	74.0
13	20.5 × 15.5	0.0	74.8
14	42.5 × 32.5	24.2	74.8
15	32.5 × 23.5	70.5	74.8
16	20.5 × 15.5	107.5	74.8
17	20.5 × 15.5	132.5	93.5
18	20.5 × 15.5	0.0	93.6
19	23.5 × 32.5	106.0	95.5
20	25.5 × 25.5	156.0	102.5
21	32.5 × 23.5	70.5	104.5
22	20.5 × 15.5	0.0	112.5
23	20.5 × 15.5	23.5	112.5
24	20.5 × 15.5	47.0	112.5
25	20.5 × 15.5	132.5	112.5

Layout 13 175.0 × 118.0 cm gap 3 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

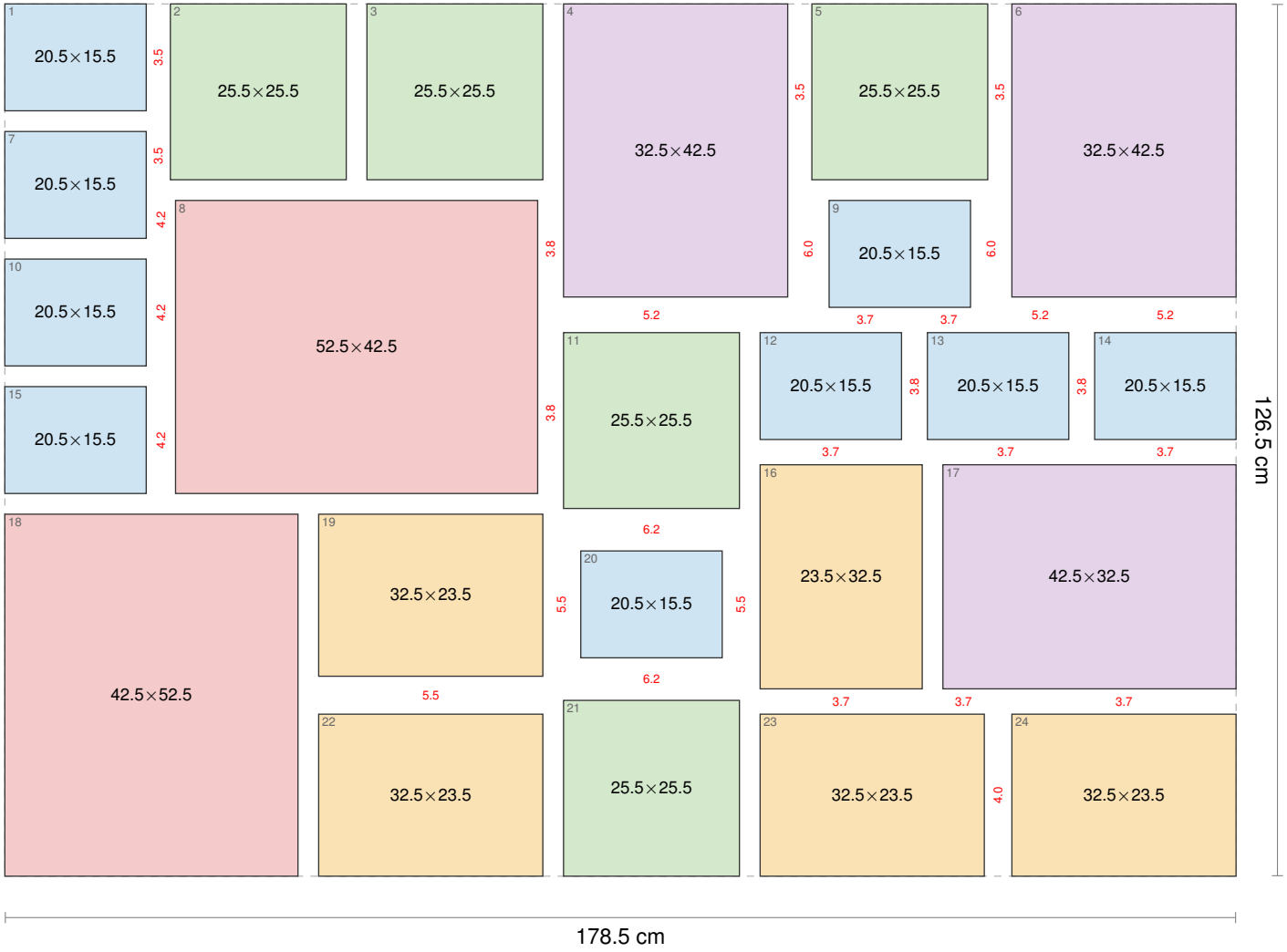
Outer size	175.0 × 118.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 5.8 cm
Max. deviation	2.8 cm
Mean deviation	0.45 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	1
Entropy	0.72
mix / variety	0.74 / 0.68
stagger / lines	0.47 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	28.5	0.0
3	25.5 × 25.5	64.0	0.0
4	25.5 × 25.5	92.5	0.0
5	25.5 × 25.5	121.0	0.0
6	25.5 × 25.5	149.5	0.0
7	20.5 × 15.5	0.0	28.5
8	20.5 × 15.5	95.5	28.5
9	20.5 × 15.5	119.0	28.5
10	32.5 × 42.5	142.5	28.5
11	23.5 × 32.5	23.5	29.2
12	42.5 × 32.5	50.0	29.2
13	20.5 × 15.5	0.0	47.0
14	20.5 × 15.5	95.5	47.0
15	20.5 × 15.5	119.0	47.0
16	20.5 × 15.5	0.0	65.5
17	42.5 × 52.5	24.3	65.5
18	32.5 × 23.5	70.7	65.5
19	32.5 × 23.5	107.0	65.5
20	32.5 × 42.5	142.5	75.5
21	20.5 × 15.5	0.0	84.0
22	32.5 × 23.5	70.7	94.5
23	32.5 × 23.5	107.0	94.5
24	20.5 × 15.5	0.0	102.5

Layout 14 178.5 × 126.5 cm gap 3 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

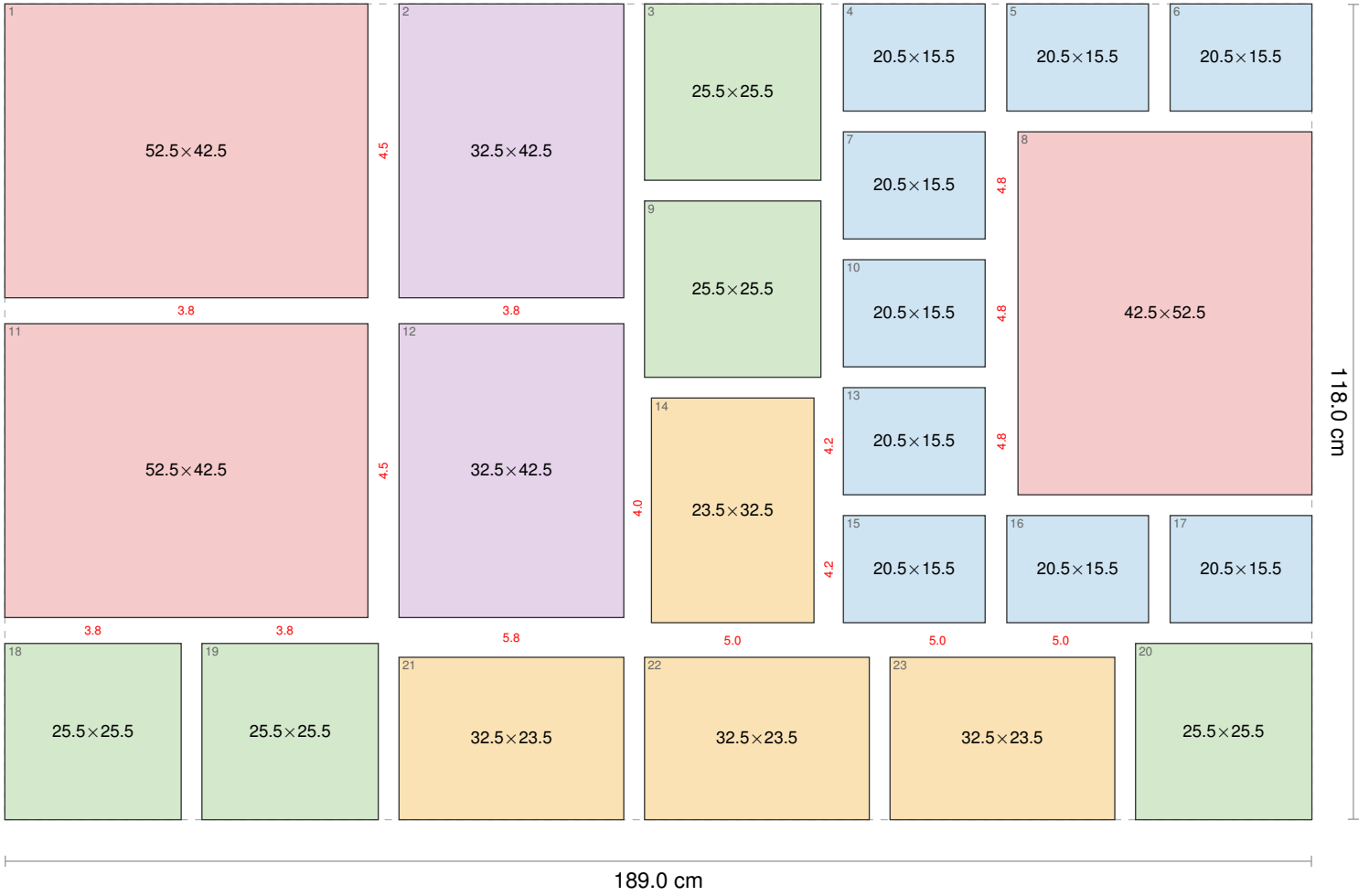
Outer size	178.5 × 126.5 cm
Nominal gap	3 cm
Actual gaps	3.0 – 6.2 cm
Max. deviation	3.2 cm
Mean deviation	0.86 cm
Border notches	right 5 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, touching 1× (18 cm), min. dist. 59 cm
Entropy	0.77
mix / variety	0.79 / 0.74
stagger / lines	0.55 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	25.5 × 25.5	24.0	0.0
3	25.5 × 25.5	52.5	0.0
4	32.5 × 42.5	81.0	0.0
5	25.5 × 25.5	117.0	0.0
6	32.5 × 42.5	146.0	0.0
7	20.5 × 15.5	0.0	18.5
8	52.5 × 42.5	24.8	28.5
9	20.5 × 15.5	119.5	28.5
10	20.5 × 15.5	0.0	37.0
11	25.5 × 25.5	81.0	47.7
12	20.5 × 15.5	109.5	47.7
13	20.5 × 15.5	133.8	47.7
14	20.5 × 15.5	158.0	47.7
15	20.5 × 15.5	0.0	55.5
16	23.5 × 32.5	109.5	66.8
17	42.5 × 32.5	136.0	66.8
18	42.5 × 52.5	0.0	74.0
19	32.5 × 23.5	45.5	74.0
20	20.5 × 15.5	83.5	79.3
21	25.5 × 25.5	81.0	101.0
22	32.5 × 23.5	45.5	103.0
23	32.5 × 23.5	109.5	103.0
24	32.5 × 23.5	146.0	103.0

Layout 15 189.0 × 118.0 cm gap 3 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

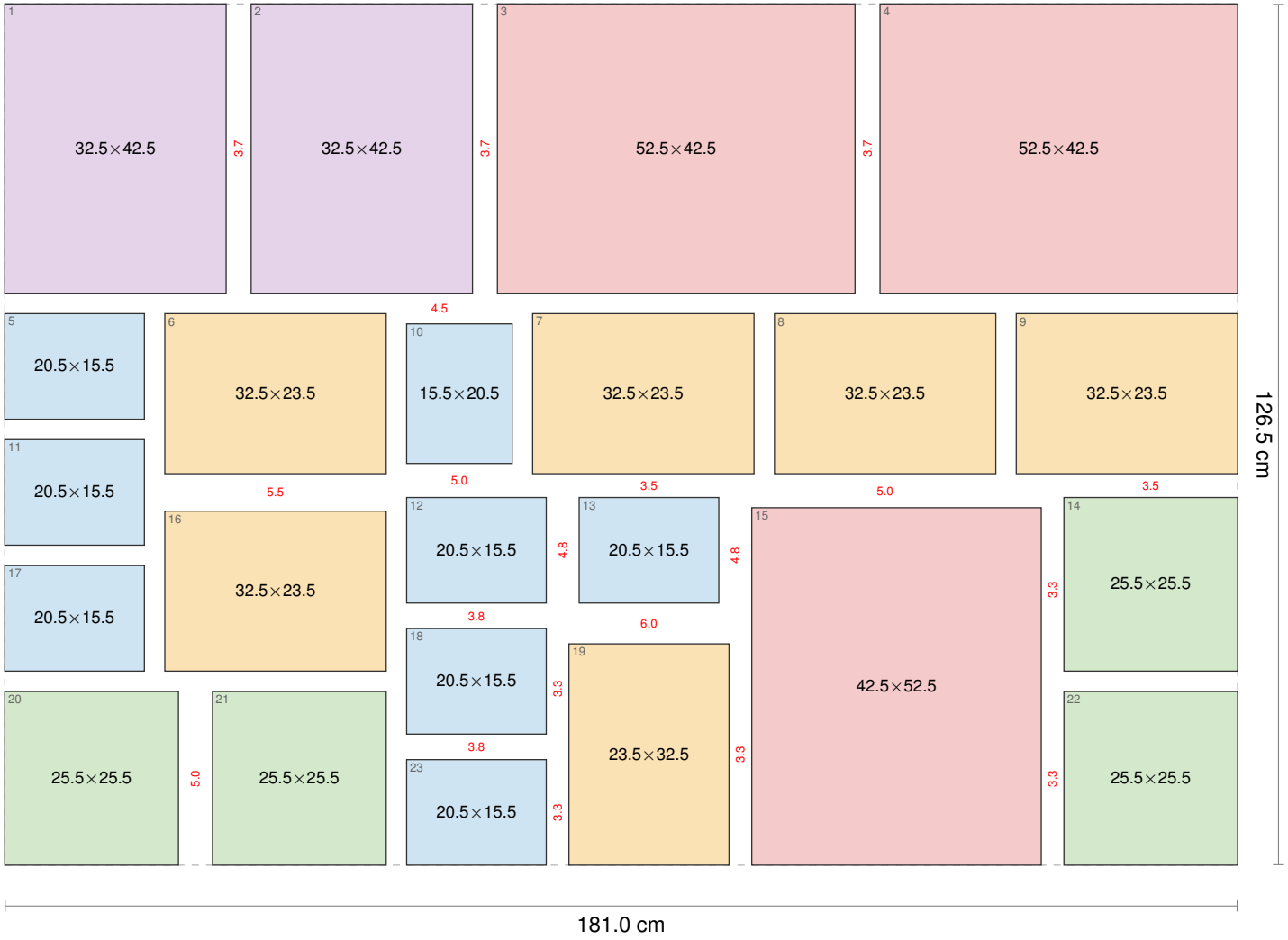
Outer size	189.0 × 118.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 5.8 cm
Max. deviation	2.8 cm
Mean deviation	0.58 cm
Border notches	top 4 cm
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 1× (52 cm), min. dist. 46 cm
Entropy	0.57
mix / variety	0.76 / 0.56
stagger / lines	0.46 / 0.50 (3 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 42.5	57.0	0.0
3	25.5 × 25.5	92.5	0.0
4	20.5 × 15.5	121.2	0.0
5	20.5 × 15.5	144.9	0.0
6	20.5 × 15.5	168.5	0.0
7	20.5 × 15.5	121.2	18.5
8	42.5 × 52.5	146.5	18.5
9	25.5 × 25.5	92.5	28.5
10	20.5 × 15.5	121.2	37.0
11	52.5 × 42.5	0.0	46.2
12	32.5 × 42.5	57.0	46.2
13	20.5 × 15.5	121.2	55.5
14	23.5 × 32.5	93.5	57.0
15	20.5 × 15.5	121.2	74.0
16	20.5 × 15.5	144.9	74.0
17	20.5 × 15.5	168.5	74.0
18	25.5 × 25.5	0.0	92.5
19	25.5 × 25.5	28.5	92.5
20	25.5 × 25.5	163.5	92.5
21	32.5 × 23.5	57.0	94.5
22	32.5 × 23.5	92.5	94.5
23	32.5 × 23.5	128.0	94.5

Layout 16 181.0 × 126.5 cm gap 3 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	181.0 × 126.5 cm
Nominal gap	3 cm
Actual gaps	3.0 – 6.0 cm
Max. deviation	3.0 cm
Mean deviation	0.52 cm
Border notches	bottom 5 cm
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 1× (42 cm), min. dist. 56 cm
Entropy	0.78
mix / variety	0.83 / 0.75
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	32.5 × 42.5	36.2	0.0
3	52.5 × 42.5	72.3	0.0
4	52.5 × 42.5	128.5	0.0
5	20.5 × 15.5	0.0	45.5
6	32.5 × 23.5	23.5	45.5
7	32.5 × 23.5	77.5	45.5
8	32.5 × 23.5	113.0	45.5
9	32.5 × 23.5	148.5	45.5
10	15.5 × 20.5	59.0	47.0
11	20.5 × 15.5	0.0	64.0
12	20.5 × 15.5	59.0	72.5
13	20.5 × 15.5	84.3	72.5
14	25.5 × 25.5	155.5	72.5
15	42.5 × 52.5	109.7	74.0
16	32.5 × 23.5	23.5	74.5
17	20.5 × 15.5	0.0	82.5
18	20.5 × 15.5	59.0	91.8
19	23.5 × 32.5	82.8	94.0
20	25.5 × 25.5	0.0	101.0
21	25.5 × 25.5	30.5	101.0
22	25.5 × 25.5	155.5	101.0
23	20.5 × 15.5	59.0	111.0

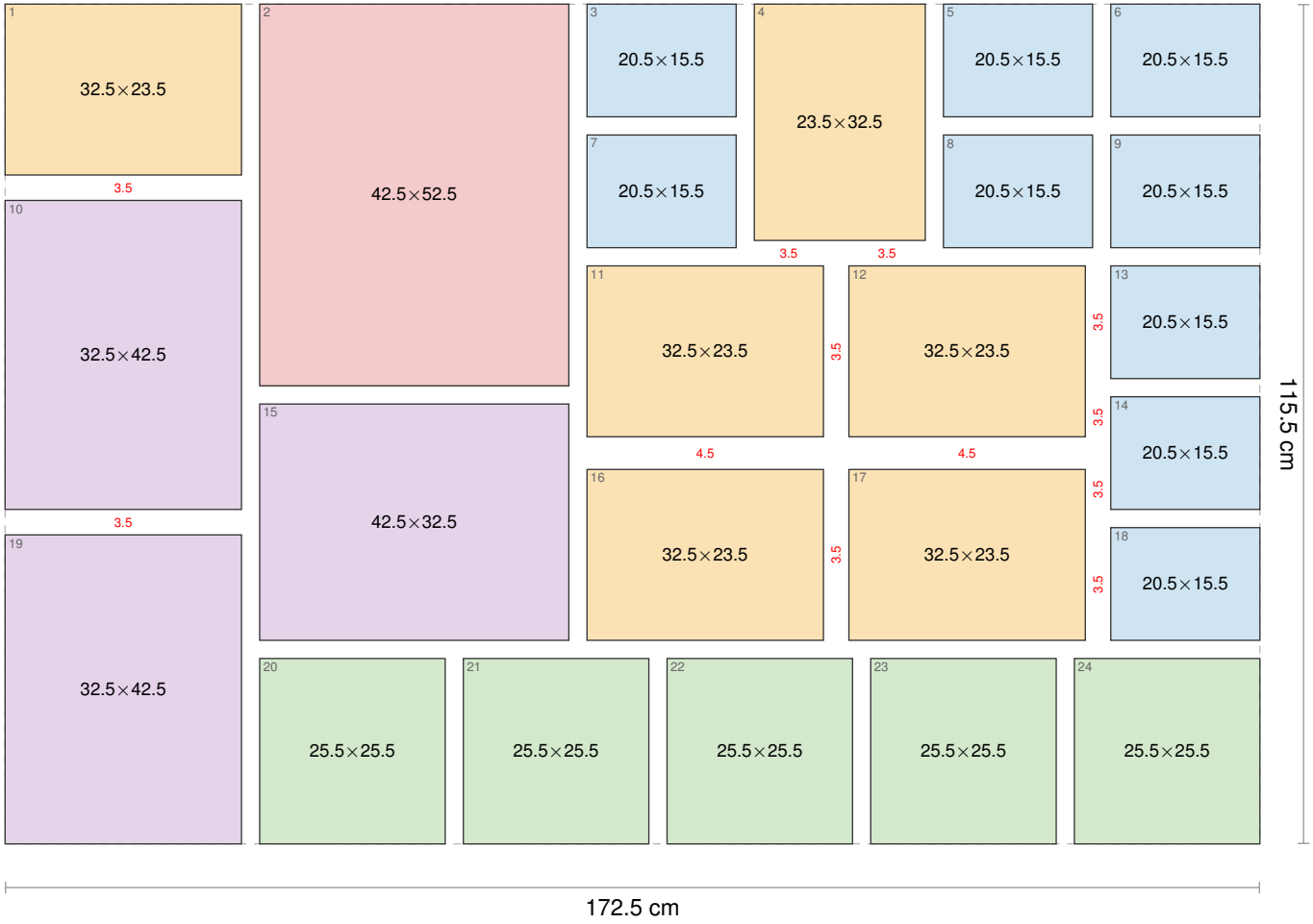
8 linear-cost

Gap deviations and notches are penalised linearly instead of with the Huber shape. The cost landscape is smoother and small deviations count relatively more, which favours layouts whose deviations are spread thinly instead of concentrated in one gap.

Parameters that differ from the baseline: cost_shape = linear.
256 runs, 20 min wall-clock budget, 729 candidate states, the 16 best distinct layouts are shown.

#	size (cm)	gap	frames	max dev	mean dev	notches	corners	entropy
1	172.5 × 115.5	2.5	24	2.0	0.30	none	4/4	0.71
2	181.5 × 113.5	1.5	23	2.0	0.52	none	4/4	0.57
3	181.5 × 113.5	1.5	23	2.0	0.53	none	4/4	0.72
4	189.5 × 109.5	3.5	22	2.0	0.40	none	4/4	0.76
5	181.5 × 113.5	1.5	20	2.0	0.55	none	4/4	0.78
6	181.5 × 113.5	1.5	20	2.0	0.57	none	4/4	0.72
7	181.5 × 113.5	1.5	20	2.0	0.59	none	4/4	0.73
8	181.5 × 113.5	1.5	20	2.0	0.68	none	4/4	0.64
9	179.5 × 120.5	1.5	26	3.0	0.55	none	4/4	0.78
10	181.5 × 114.5	1.5	25	2.5	0.49	none	4/4	0.73
11	181.5 × 114.5	1.5	25	2.5	0.60	none	4/4	0.76
12	181.5 × 114.5	1.5	25	2.5	0.62	none	4/4	0.67
13	185.5 × 113.5	1.5	24	2.0	0.51	4 cm	4/4	0.74
14	172.5 × 115.5	2.5	24	2.0	0.31	4 cm	4/4	0.65
15	177.5 × 118.5	3.5	24	1.5	0.35	10 cm	4/4	0.64
16	172.5 × 116.5	2.5	24	2.0	0.43	8 cm	4/4	0.64

Layout 1 172.5 × 115.5 cm gap 2.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

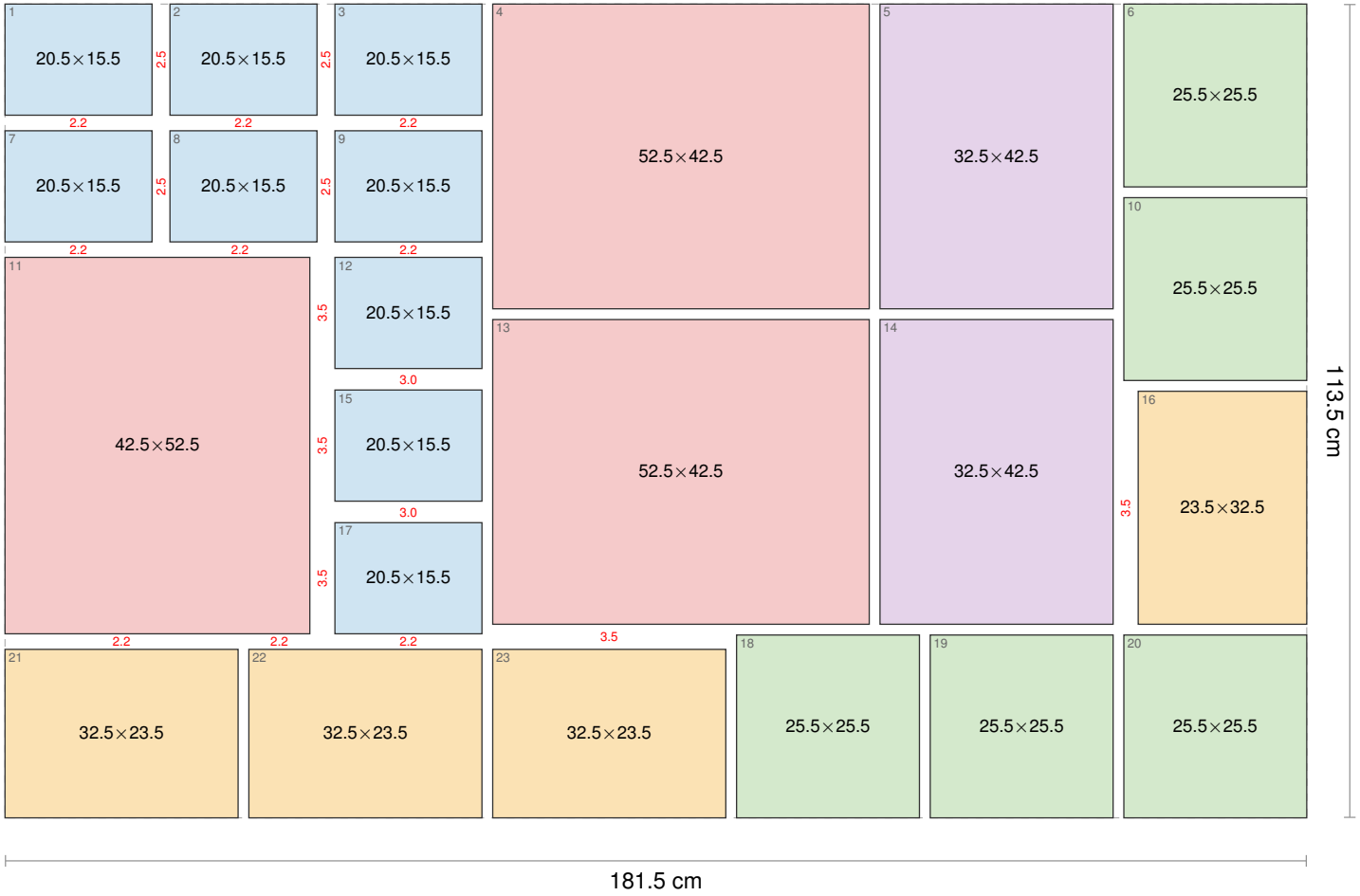
Outer size	172.5 × 115.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 4.5 cm
Max. deviation	2.0 cm
Mean deviation	0.30 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	1
Entropy	0.71
mix / variety	0.74 / 0.66
stagger / lines	0.45 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	42.5 × 52.5	35.0	0.0
3	20.5 × 15.5	80.0	0.0
4	23.5 × 32.5	103.0	0.0
5	20.5 × 15.5	129.0	0.0
6	20.5 × 15.5	152.0	0.0
7	20.5 × 15.5	80.0	18.0
8	20.5 × 15.5	129.0	18.0
9	20.5 × 15.5	152.0	18.0
10	32.5 × 42.5	0.0	27.0
11	32.5 × 23.5	80.0	36.0
12	32.5 × 23.5	116.0	36.0
13	20.5 × 15.5	152.0	36.0
14	20.5 × 15.5	152.0	54.0
15	42.5 × 32.5	35.0	55.0
16	32.5 × 23.5	80.0	64.0
17	32.5 × 23.5	116.0	64.0
18	20.5 × 15.5	152.0	72.0
19	32.5 × 42.5	0.0	73.0
20	25.5 × 25.5	35.0	90.0
21	25.5 × 25.5	63.0	90.0
22	25.5 × 25.5	91.0	90.0
23	25.5 × 25.5	119.0	90.0
24	25.5 × 25.5	147.0	90.0

Layout 2 181.5 × 113.5 cm gap 1.5 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	181.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.52 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 1× (52 cm), min. dist. 44 cm
Entropy	0.57
mix / variety	0.76 / 0.59
stagger / lines	0.43 / 0.50 (3 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	23.0	0.0
3	20.5 × 15.5	46.0	0.0
4	52.5 × 42.5	68.0	0.0
5	32.5 × 42.5	122.0	0.0
6	25.5 × 25.5	156.0	0.0
7	20.5 × 15.5	0.0	17.7
8	20.5 × 15.5	23.0	17.7
9	20.5 × 15.5	46.0	17.7
10	25.5 × 25.5	156.0	27.0
11	42.5 × 52.5	0.0	35.3
12	20.5 × 15.5	46.0	35.3
13	52.5 × 42.5	68.0	44.0
14	32.5 × 42.5	122.0	44.0
15	20.5 × 15.5	46.0	53.8
16	23.5 × 32.5	158.0	54.0
17	20.5 × 15.5	46.0	72.3
18	25.5 × 25.5	102.0	88.0
19	25.5 × 25.5	129.0	88.0
20	25.5 × 25.5	156.0	88.0
21	32.5 × 23.5	0.0	90.0
22	32.5 × 23.5	34.0	90.0
23	32.5 × 23.5	68.0	90.0

Layout 3 181.5 × 113.5 cm gap 1.5 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

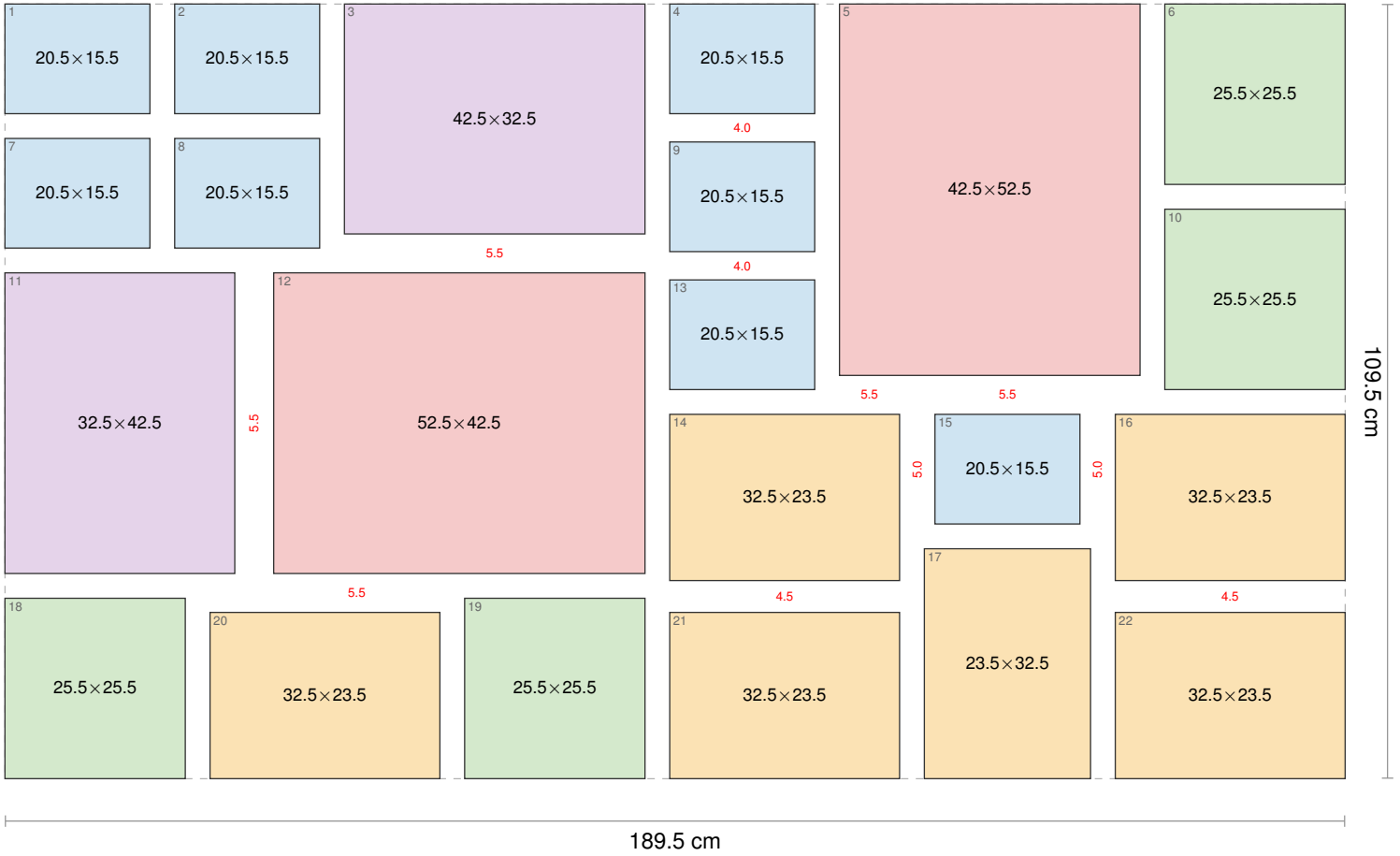
Outer size	181.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.53 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 3× (104 cm), min. dist. 44 cm
Entropy	0.72
mix / variety	0.76 / 0.68
stagger / lines	0.47 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	27.0	0.0
3	32.5 × 23.5	61.0	0.0
4	32.5 × 23.5	95.0	0.0
5	25.5 × 25.5	129.0	0.0
6	25.5 × 25.5	156.0	0.0
7	52.5 × 42.5	0.0	27.0
8	42.5 × 52.5	54.5	27.0
9	25.5 × 25.5	99.0	27.0
10	20.5 × 15.5	126.5	27.0
11	32.5 × 42.5	149.0	27.0
12	20.5 × 15.5	126.5	45.5
13	25.5 × 25.5	99.0	54.0
14	20.5 × 15.5	126.5	64.0
15	52.5 × 42.5	0.0	71.0
16	32.5 × 42.5	149.0	71.0
17	20.5 × 15.5	54.5	81.0
18	20.5 × 15.5	77.5	81.0
19	20.5 × 15.5	100.5	81.0
20	23.5 × 32.5	123.5	81.0
21	20.5 × 15.5	54.5	98.0
22	20.5 × 15.5	77.5	98.0
23	20.5 × 15.5	100.5	98.0

Layout 4 189.5 × 109.5 cm gap 3.5 cm 22 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	189.5 × 109.5 cm
Nominal gap	3.5 cm
Actual gaps	3.5 – 5.5 cm
Max. deviation	2.0 cm
Mean deviation	0.40 cm
Border notches	none
Corners filled	4 / 4
Frames used	22 of 28
Large frames	2, not touching, min. dist. 82 cm
Entropy	0.76
mix / variety	0.78 / 0.73
stagger / lines	0.51 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	8	9
15.5×20.5	P	0	2
32.5×42.5	P	1	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	1	1
52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	24.0	0.0
3	42.5 × 32.5	48.0	0.0
4	20.5 × 15.5	94.0	0.0
5	42.5 × 52.5	118.0	0.0
6	25.5 × 25.5	164.0	0.0
7	20.5 × 15.5	0.0	19.0
8	20.5 × 15.5	24.0	19.0
9	20.5 × 15.5	94.0	19.5
10	25.5 × 25.5	164.0	29.0
11	32.5 × 42.5	0.0	38.0
12	52.5 × 42.5	38.0	38.0
13	20.5 × 15.5	94.0	39.0
14	32.5 × 23.5	94.0	58.0
15	20.5 × 15.5	131.5	58.0
16	32.5 × 23.5	157.0	58.0
17	23.5 × 32.5	130.0	77.0
18	25.5 × 25.5	0.0	84.0
19	25.5 × 25.5	65.0	84.0
20	32.5 × 23.5	29.0	86.0
21	32.5 × 23.5	94.0	86.0
22	32.5 × 23.5	157.0	86.0

Layout 5 181.5 × 113.5 cm gap 1.5 cm 20 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

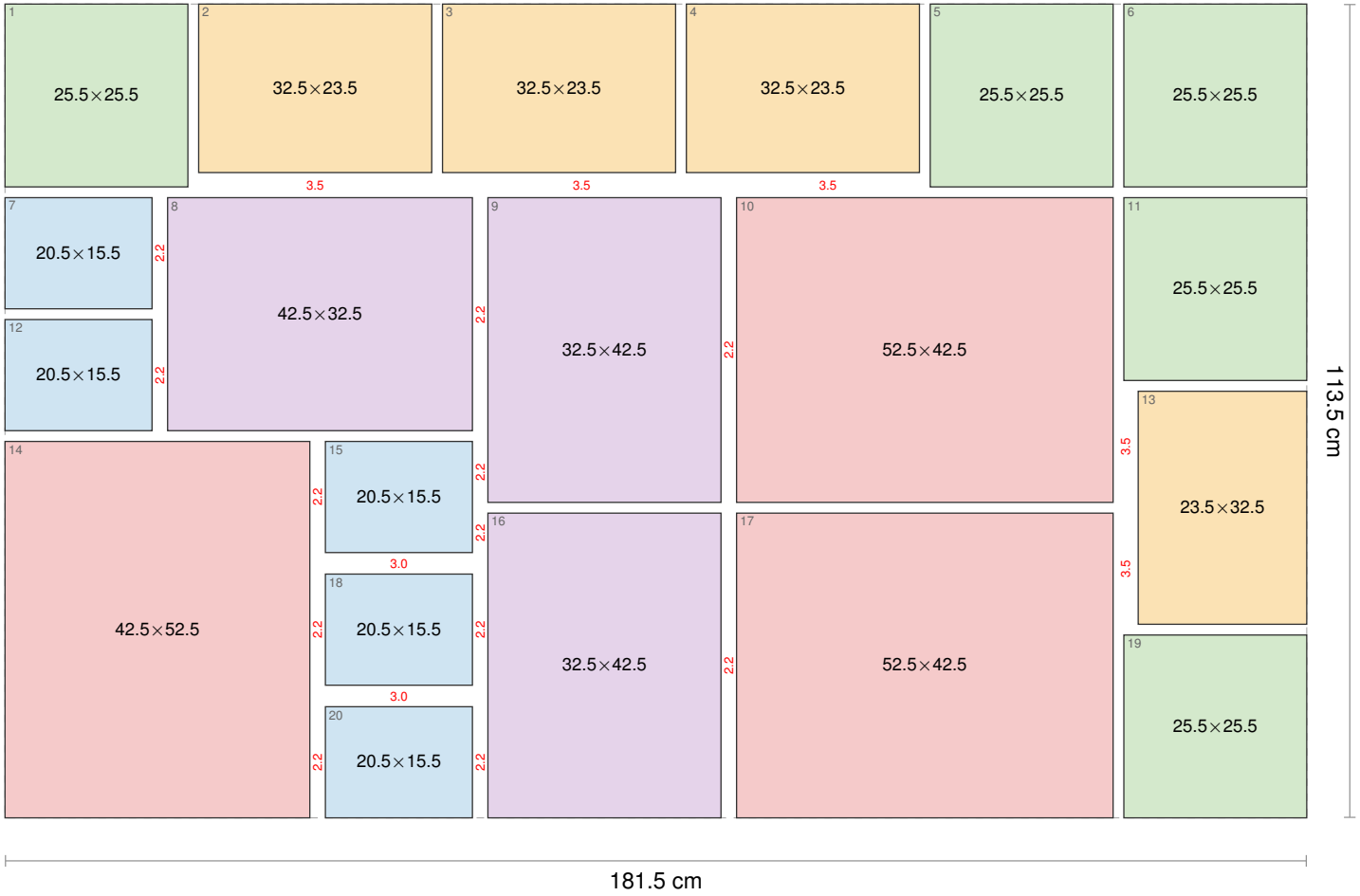
Outer size	181.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.55 cm
Border notches	none
Corners filled	4 / 4
Frames used	20 of 28
Large frames	3, not touching, min. dist. 71 cm
Entropy	0.78
mix / variety	0.86 / 0.73
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	5	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	27.0	0.0
3	32.5 × 23.5	61.0	0.0
4	32.5 × 23.5	95.0	0.0
5	52.5 × 42.5	129.0	0.0
6	20.5 × 15.5	0.0	27.0
7	42.5 × 32.5	22.7	27.0
8	32.5 × 42.5	67.3	27.0
9	25.5 × 25.5	102.0	27.0
10	25.5 × 25.5	129.0	44.0
11	25.5 × 25.5	156.0	44.0
12	20.5 × 15.5	0.0	44.8
13	23.5 × 32.5	103.0	54.0
14	42.5 × 52.5	22.7	61.0
15	20.5 × 15.5	0.0	62.5
16	32.5 × 42.5	67.3	71.0
17	52.5 × 42.5	129.0	71.0
18	20.5 × 15.5	0.0	80.2
19	25.5 × 25.5	102.0	88.0
20	20.5 × 15.5	0.0	98.0

Layout 6 181.5 × 113.5 cm gap 1.5 cm 20 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

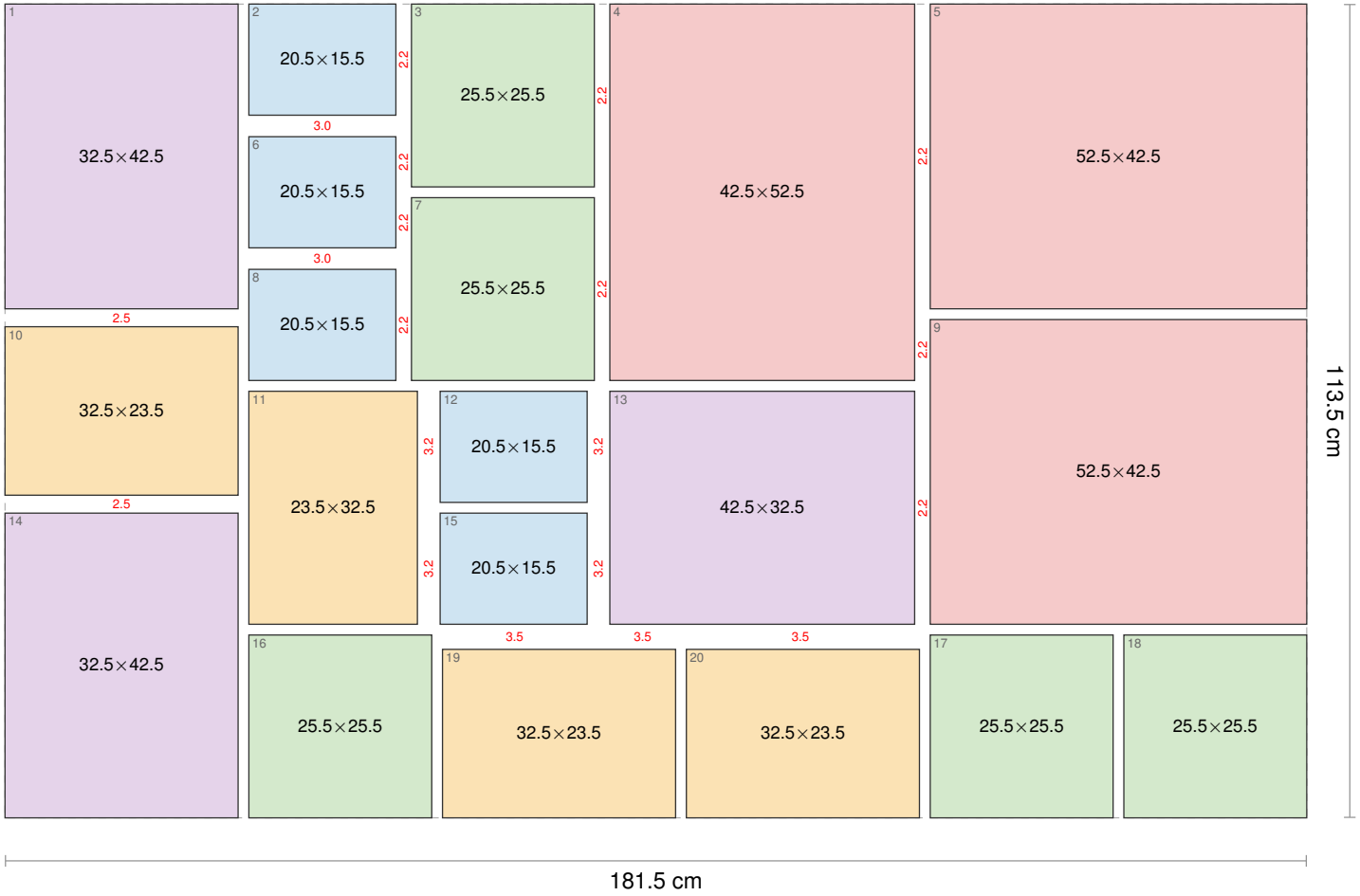
Outer size	181.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.57 cm
Border notches	none
Corners filled	4 / 4
Frames used	20 of 28
Large frames	3, touching 1× (52 cm), min. dist. 44 cm
Entropy	0.72
mix / variety	0.86 / 0.77
stagger / lines	0.50 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	5	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	27.0	0.0
3	32.5 × 23.5	61.0	0.0
4	32.5 × 23.5	95.0	0.0
5	25.5 × 25.5	129.0	0.0
6	25.5 × 25.5	156.0	0.0
7	20.5 × 15.5	0.0	27.0
8	42.5 × 32.5	22.7	27.0
9	32.5 × 42.5	67.3	27.0
10	52.5 × 42.5	102.0	27.0
11	25.5 × 25.5	156.0	27.0
12	20.5 × 15.5	0.0	44.0
13	23.5 × 32.5	158.0	54.0
14	42.5 × 52.5	0.0	61.0
15	20.5 × 15.5	44.7	61.0
16	32.5 × 42.5	67.3	71.0
17	52.5 × 42.5	102.0	71.0
18	20.5 × 15.5	44.7	79.5
19	25.5 × 25.5	156.0	88.0
20	20.5 × 15.5	44.7	98.0

Layout 7 181.5 × 113.5 cm gap 1.5 cm 20 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

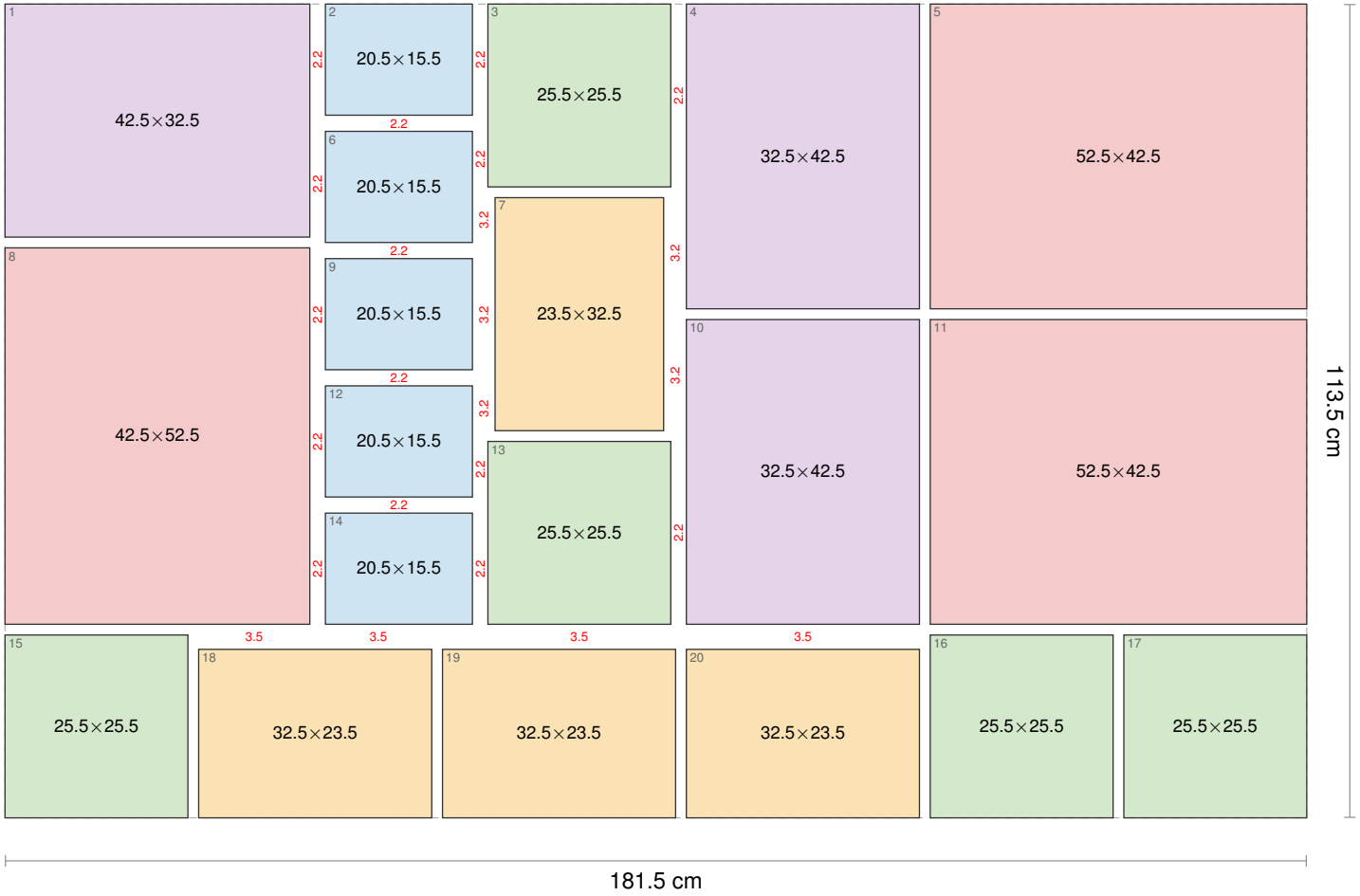
Outer size	181.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.59 cm
Border notches	none
Corners filled	4 / 4
Frames used	20 of 28
Large frames	3, touching 3× (104 cm), min. dist. 44 cm
Entropy	0.73
mix / variety	0.86 / 0.84
stagger / lines	0.49 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	5	9
15.5×20.5	P	0	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	3	5
23.5×32.5	P	1	1
25.5×25.5		5	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	20.5 × 15.5	34.0	0.0
3	25.5 × 25.5	56.7	0.0
4	42.5 × 52.5	84.3	0.0
5	52.5 × 42.5	129.0	0.0
6	20.5 × 15.5	34.0	18.5
7	25.5 × 25.5	56.7	27.0
8	20.5 × 15.5	34.0	37.0
9	52.5 × 42.5	129.0	44.0
10	32.5 × 23.5	0.0	45.0
11	23.5 × 32.5	34.0	54.0
12	20.5 × 15.5	60.7	54.0
13	42.5 × 32.5	84.3	54.0
14	32.5 × 42.5	0.0	71.0
15	20.5 × 15.5	60.7	71.0
16	25.5 × 25.5	34.0	88.0
17	25.5 × 25.5	129.0	88.0
18	25.5 × 25.5	156.0	88.0
19	32.5 × 23.5	61.0	90.0
20	32.5 × 23.5	95.0	90.0

Layout 8 181.5 × 113.5 cm gap 1.5 cm 20 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

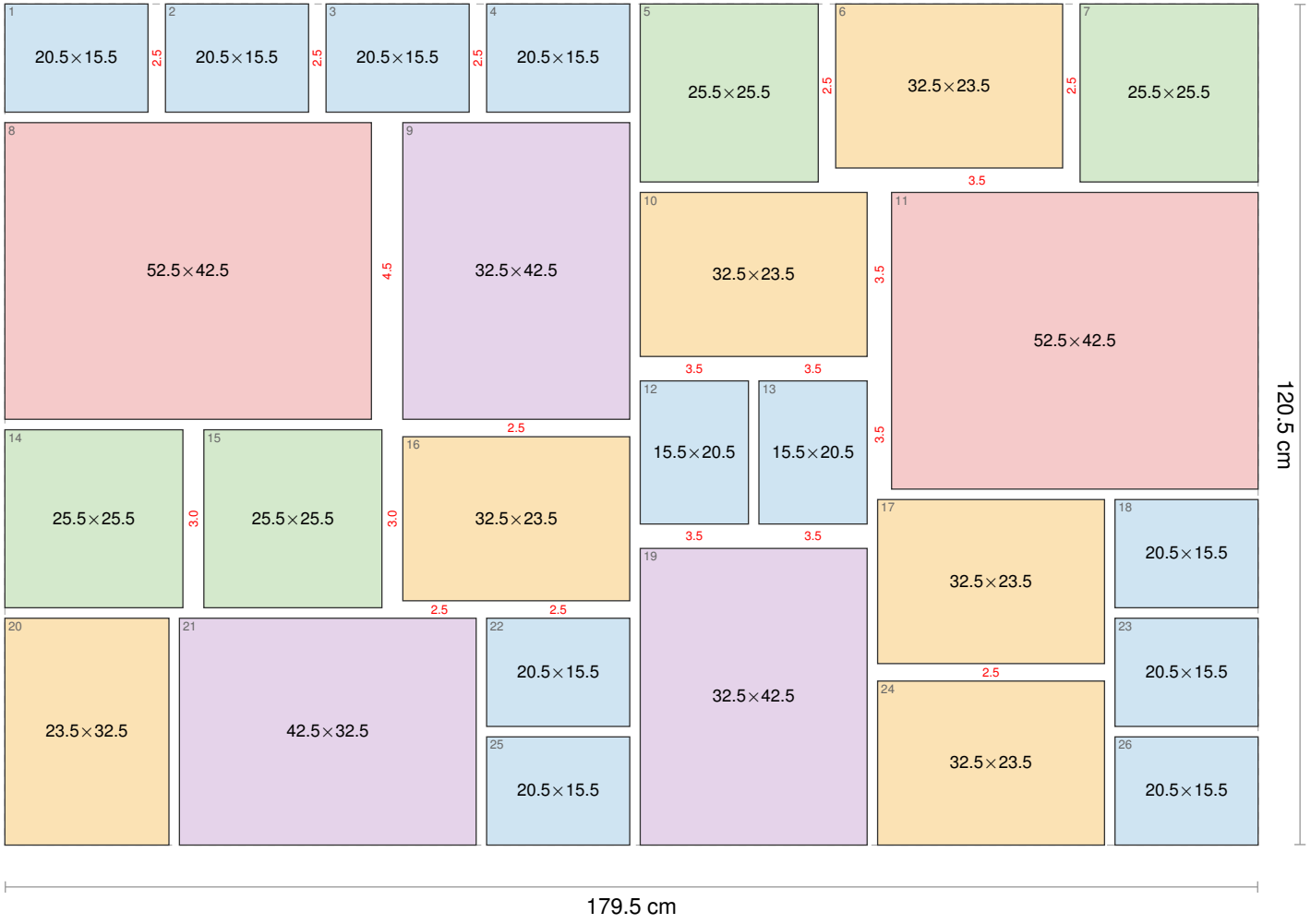
Outer size	181.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.68 cm
Border notches	none
Corners filled	4 / 4
Frames used	20 of 28
Large frames	3, touching 1× (52 cm), min. dist. 44 cm
Entropy	0.64
mix / variety	0.86 / 0.70
stagger / lines	0.50 / 0.50 (3 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	5	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	42.5 × 32.5	0.0	0.0
2	20.5 × 15.5	44.7	0.0
3	25.5 × 25.5	67.3	0.0
4	32.5 × 42.5	95.0	0.0
5	52.5 × 42.5	129.0	0.0
6	20.5 × 15.5	44.7	17.8
7	23.5 × 32.5	68.3	27.0
8	42.5 × 52.5	0.0	34.0
9	20.5 × 15.5	44.7	35.5
10	32.5 × 42.5	95.0	44.0
11	52.5 × 42.5	129.0	44.0
12	20.5 × 15.5	44.7	53.2
13	25.5 × 25.5	67.3	61.0
14	20.5 × 15.5	44.7	71.0
15	25.5 × 25.5	0.0	88.0
16	25.5 × 25.5	129.0	88.0
17	25.5 × 25.5	156.0	88.0
18	32.5 × 23.5	27.0	90.0
19	32.5 × 23.5	61.0	90.0
20	32.5 × 23.5	95.0	90.0

Layout 9 179.5 × 120.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

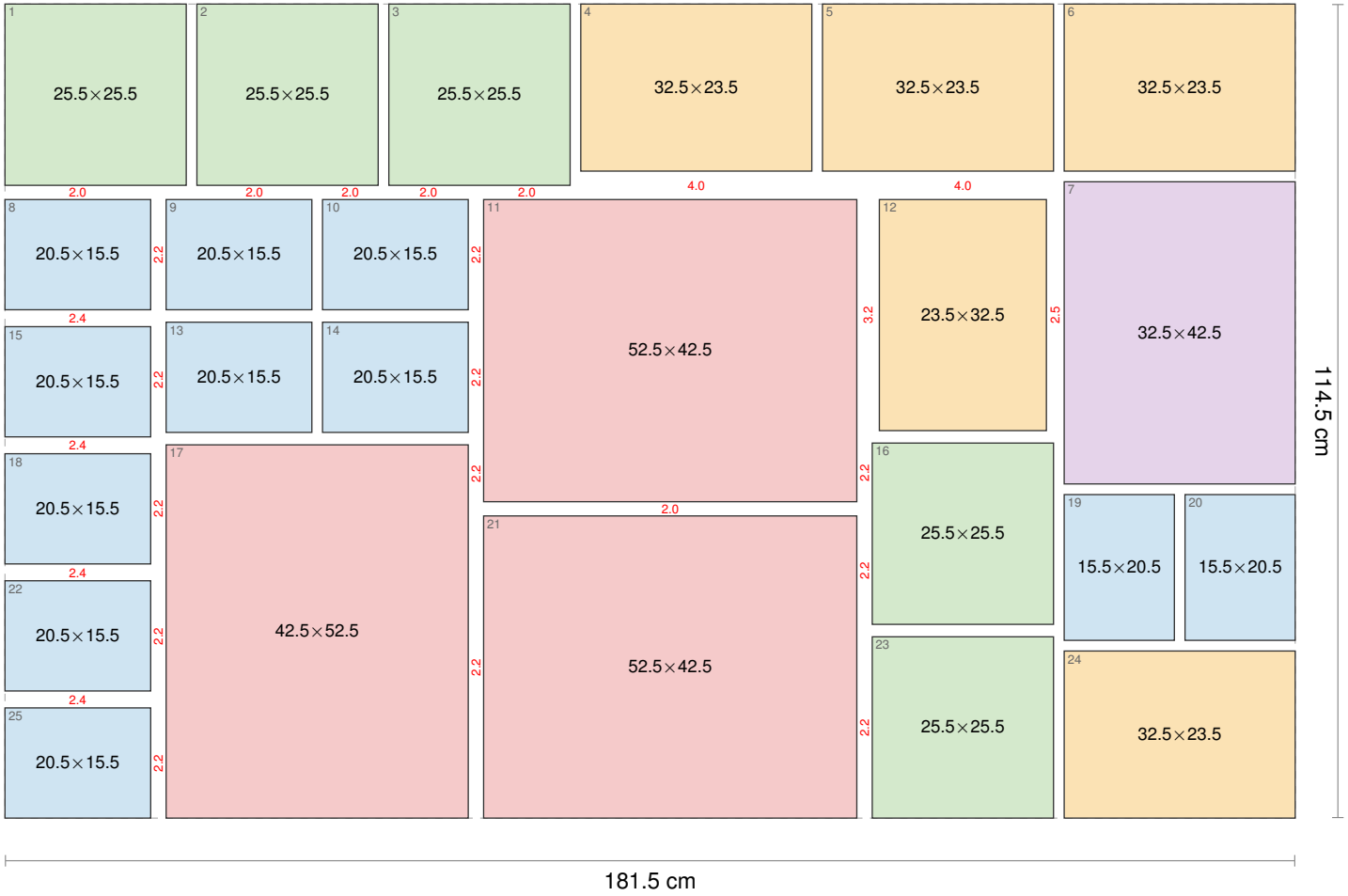
Outer size	179.5 × 120.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.5 cm
Max. deviation	3.0 cm
Mean deviation	0.55 cm
Border notches	none
Corners filled	4 / 4
Frames used	26 of 28
Large frames	2, not touching, min. dist. 127 cm
Entropy	0.78
mix / variety	0.83 / 0.81
stagger / lines	0.50 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	23.0	0.0
3	20.5 × 15.5	46.0	0.0
4	20.5 × 15.5	69.0	0.0
5	25.5 × 25.5	91.0	0.0
6	32.5 × 23.5	119.0	0.0
7	25.5 × 25.5	154.0	0.0
8	52.5 × 42.5	0.0	17.0
9	32.5 × 42.5	57.0	17.0
10	32.5 × 23.5	91.0	27.0
11	52.5 × 42.5	127.0	27.0
12	15.5 × 20.5	91.0	54.0
13	15.5 × 20.5	108.0	54.0
14	25.5 × 25.5	0.0	61.0
15	25.5 × 25.5	28.5	61.0
16	32.5 × 23.5	57.0	62.0
17	32.5 × 23.5	125.0	71.0
18	20.5 × 15.5	159.0	71.0
19	32.5 × 42.5	91.0	78.0
20	23.5 × 32.5	0.0	88.0
21	42.5 × 32.5	25.0	88.0
22	20.5 × 15.5	69.0	88.0
23	20.5 × 15.5	159.0	88.0
24	32.5 × 23.5	125.0	97.0
25	20.5 × 15.5	69.0	105.0
26	20.5 × 15.5	159.0	105.0

Layout 10 181.5 × 114.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

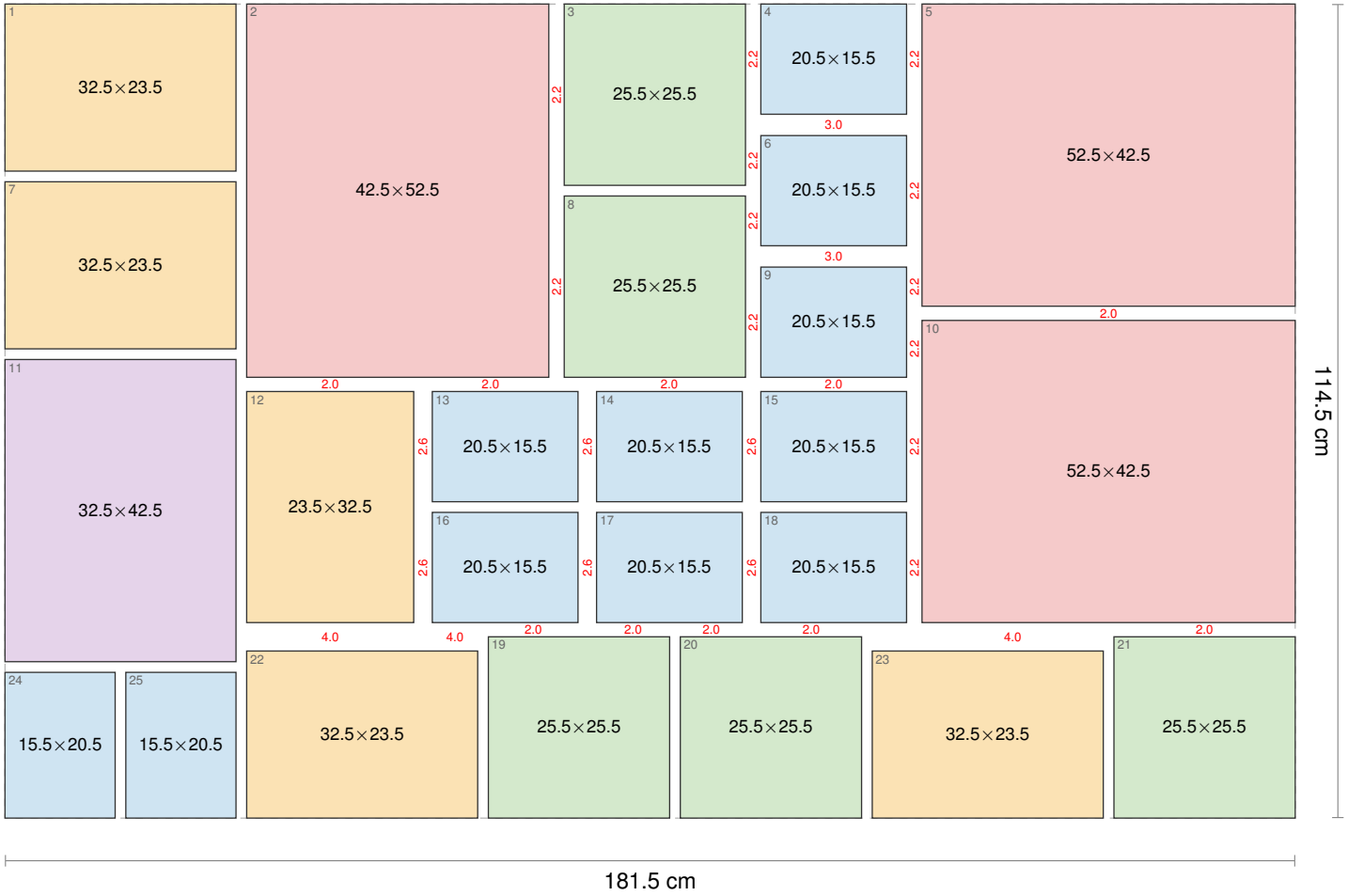
Outer size	181.5 × 114.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.49 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, touching 3× (103 cm), min. dist. 44 cm
Entropy	0.73
mix / variety	0.81 / 0.62
stagger / lines	0.51 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	1	2
	42.5×32.5	L	0	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	27.0	0.0
3	25.5 × 25.5	54.0	0.0
4	32.5 × 23.5	81.0	0.0
5	32.5 × 23.5	115.0	0.0
6	32.5 × 23.5	149.0	0.0
7	32.5 × 42.5	149.0	25.0
8	20.5 × 15.5	0.0	27.5
9	20.5 × 15.5	22.7	27.5
10	20.5 × 15.5	44.7	27.5
11	52.5 × 42.5	67.3	27.5
12	23.5 × 32.5	123.0	27.5
13	20.5 × 15.5	22.7	44.8
14	20.5 × 15.5	44.7	44.8
15	20.5 × 15.5	0.0	45.4
16	25.5 × 25.5	122.0	61.8
17	42.5 × 52.5	22.7	62.0
18	20.5 × 15.5	0.0	63.2
19	15.5 × 20.5	149.0	69.0
20	15.5 × 20.5	166.0	69.0
21	52.5 × 42.5	67.3	72.0
22	20.5 × 15.5	0.0	81.1
23	25.5 × 25.5	122.0	89.0
24	32.5 × 23.5	149.0	91.0
25	20.5 × 15.5	0.0	99.0

Layout 11 181.5 × 114.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

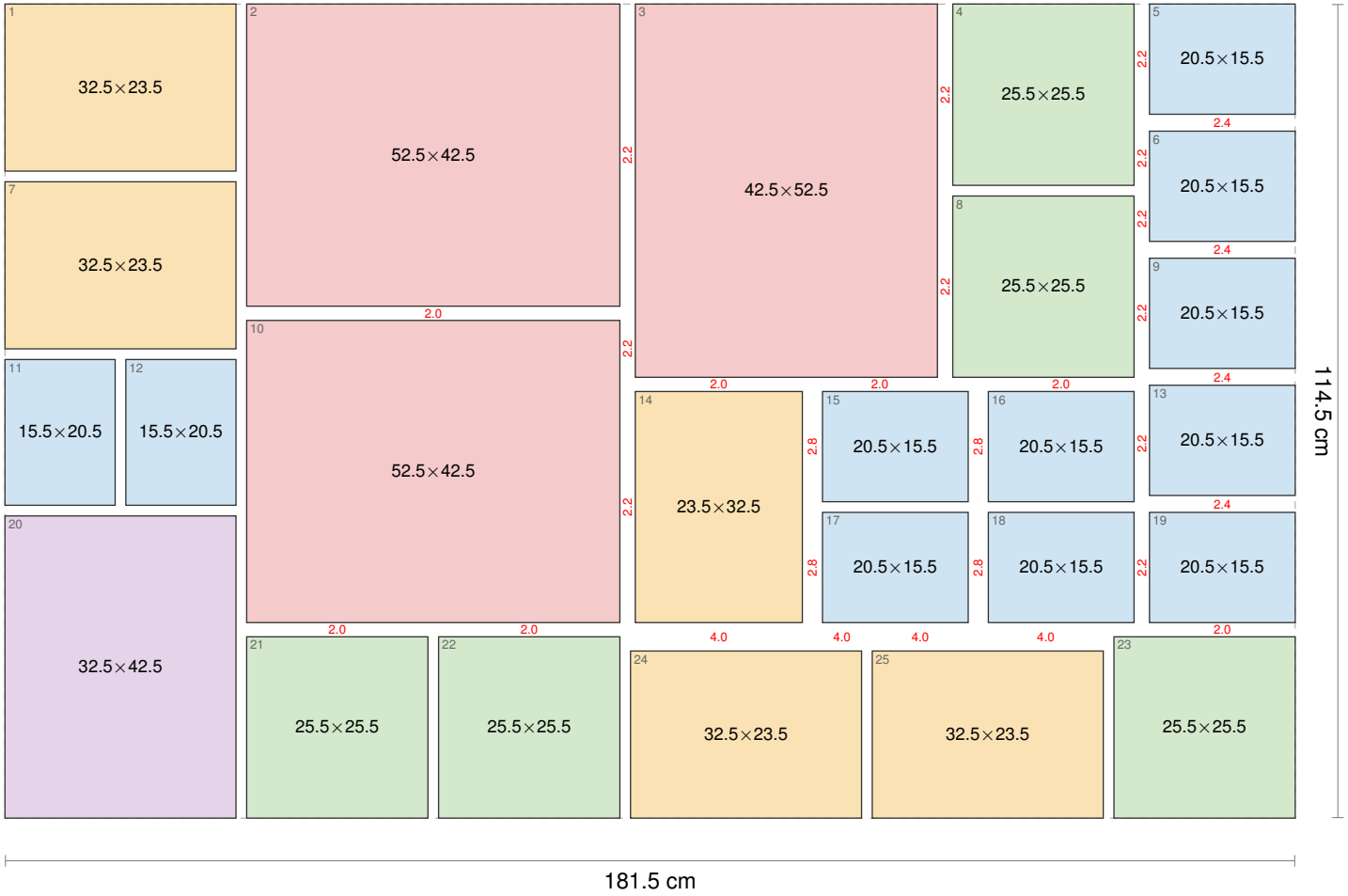
Outer size	181.5 × 114.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.60 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, touching 1× (52 cm), min. dist. 44 cm
Entropy	0.76
mix / variety	0.81 / 0.77
stagger / lines	0.48 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	1	2
	42.5×32.5	L	0	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	42.5 × 52.5	34.0	0.0
3	25.5 × 25.5	78.7	0.0
4	20.5 × 15.5	106.3	0.0
5	52.5 × 42.5	129.0	0.0
6	20.5 × 15.5	106.3	18.5
7	32.5 × 23.5	0.0	25.0
8	25.5 × 25.5	78.7	27.0
9	20.5 × 15.5	106.3	37.0
10	52.5 × 42.5	129.0	44.5
11	32.5 × 42.5	0.0	50.0
12	23.5 × 32.5	34.0	54.5
13	20.5 × 15.5	60.1	54.5
14	20.5 × 15.5	83.2	54.5
15	20.5 × 15.5	106.3	54.5
16	20.5 × 15.5	60.1	71.5
17	20.5 × 15.5	83.2	71.5
18	20.5 × 15.5	106.3	71.5
19	25.5 × 25.5	68.0	89.0
20	25.5 × 25.5	95.0	89.0
21	25.5 × 25.5	156.0	89.0
22	32.5 × 23.5	34.0	91.0
23	32.5 × 23.5	122.0	91.0
24	15.5 × 20.5	0.0	94.0
25	15.5 × 20.5	17.0	94.0

Layout 12 181.5 × 114.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

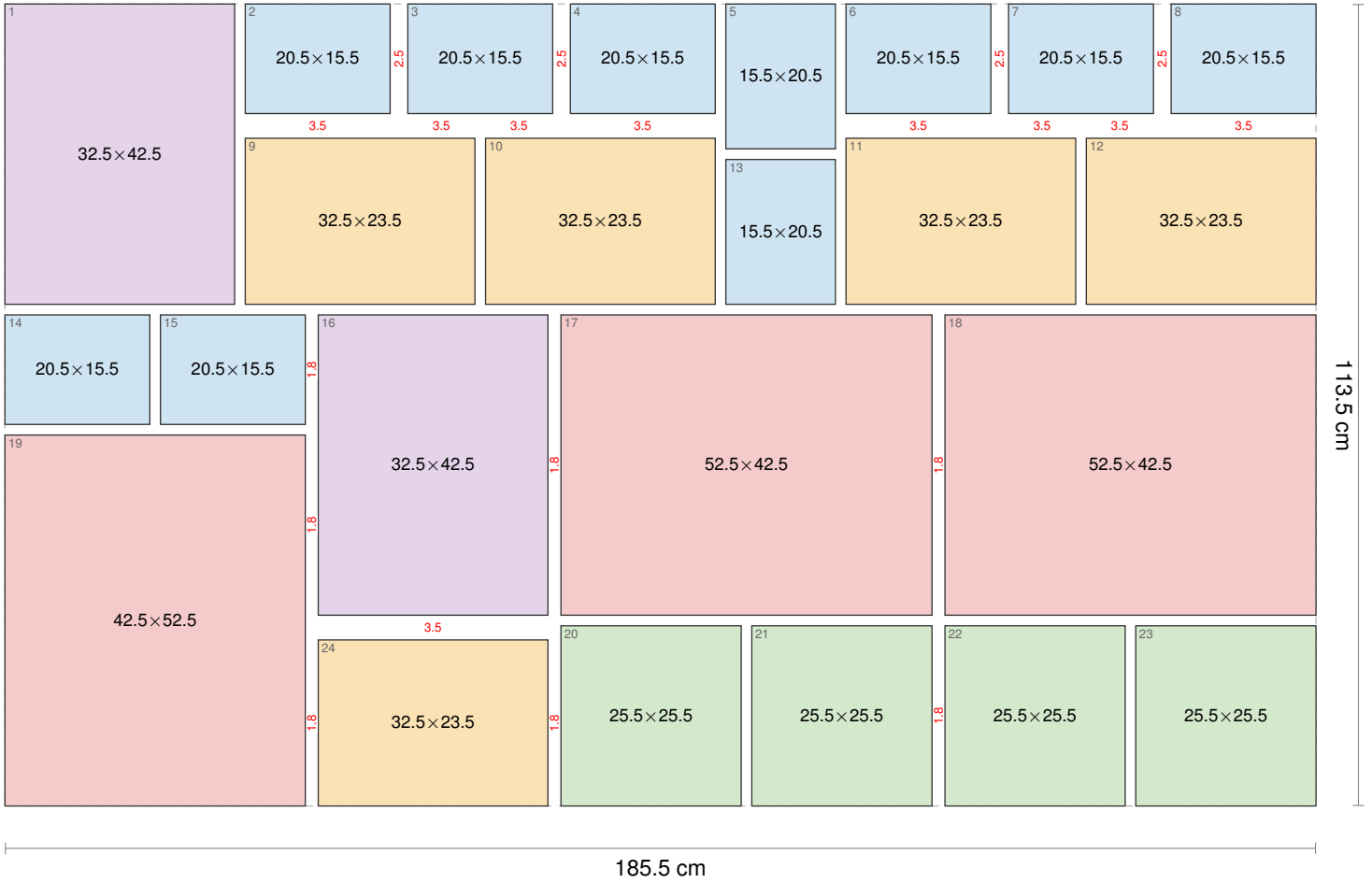
Outer size	181.5 × 114.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.62 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, touching 3× (103 cm), min. dist. 44 cm
Entropy	0.67
mix / variety	0.81 / 0.65
stagger / lines	0.49 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	1	2
	42.5×32.5	L	0	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	52.5 × 42.5	34.0	0.0
3	42.5 × 52.5	88.7	0.0
4	25.5 × 25.5	133.3	0.0
5	20.5 × 15.5	161.0	0.0
6	20.5 × 15.5	161.0	17.9
7	32.5 × 23.5	0.0	25.0
8	25.5 × 25.5	133.3	27.0
9	20.5 × 15.5	161.0	35.8
10	52.5 × 42.5	34.0	44.5
11	15.5 × 20.5	0.0	50.0
12	15.5 × 20.5	17.0	50.0
13	20.5 × 15.5	161.0	53.6
14	23.5 × 32.5	88.7	54.5
15	20.5 × 15.5	115.0	54.5
16	20.5 × 15.5	138.3	54.5
17	20.5 × 15.5	115.0	71.5
18	20.5 × 15.5	138.3	71.5
19	20.5 × 15.5	161.0	71.5
20	32.5 × 42.5	0.0	72.0
21	25.5 × 25.5	34.0	89.0
22	25.5 × 25.5	61.0	89.0
23	25.5 × 25.5	156.0	89.0
24	32.5 × 23.5	88.0	91.0
25	32.5 × 23.5	122.0	91.0

Layout 13 185.5 × 113.5 cm gap 1.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

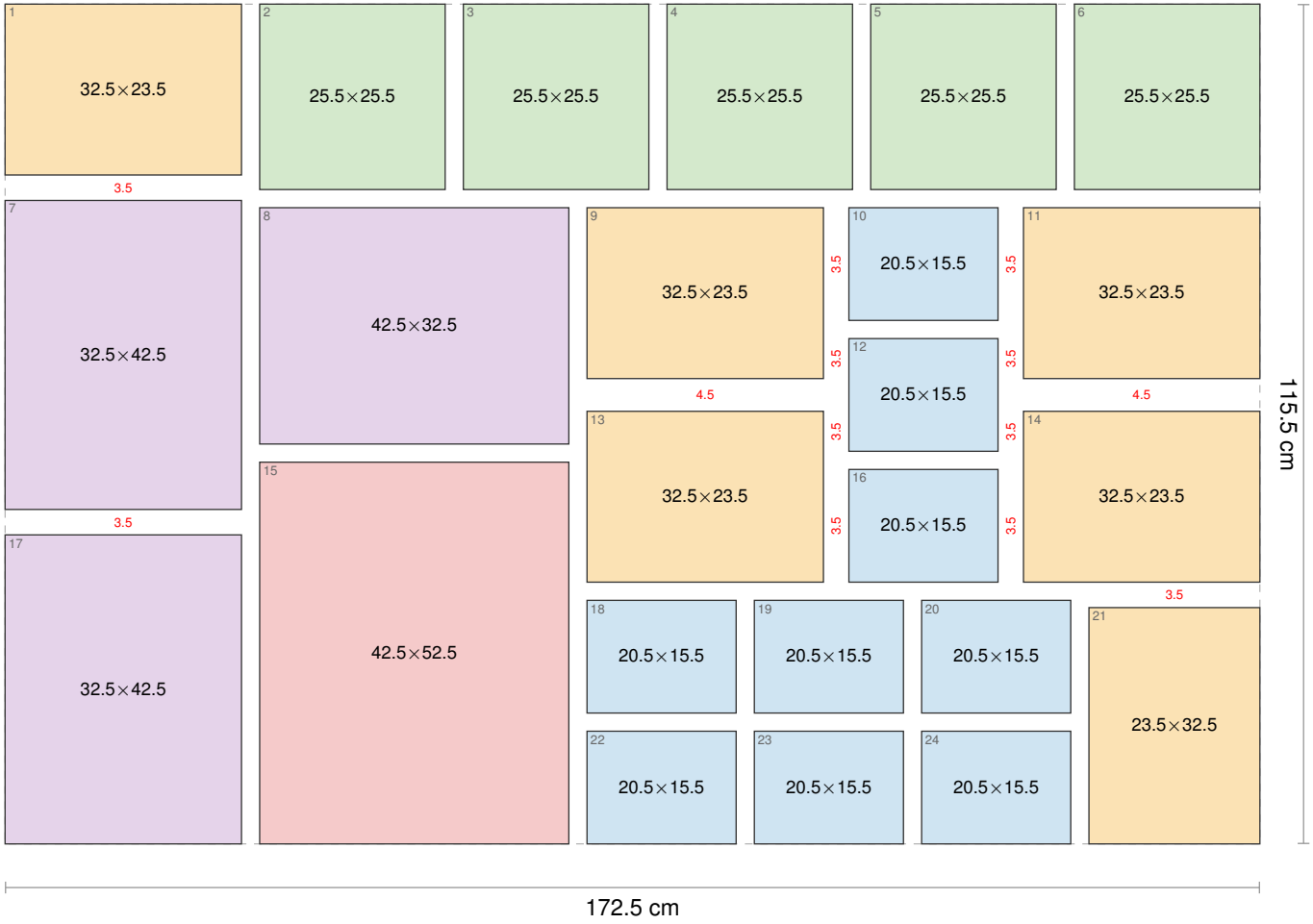
Outer size	185.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.51 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	3, touching 1× (42 cm), min. dist. 54 cm
Entropy	0.74
mix / variety	0.79 / 0.69
stagger / lines	0.47 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	8	9
15.5×20.5	P	2	2
32.5×42.5	P	2	2
42.5×32.5	L	0	1
32.5×23.5	L	5	5
23.5×32.5	P	0	1
25.5×25.5		4	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	20.5 × 15.5	34.0	0.0
3	20.5 × 15.5	57.0	0.0
4	20.5 × 15.5	80.0	0.0
5	15.5 × 20.5	102.0	0.0
6	20.5 × 15.5	119.0	0.0
7	20.5 × 15.5	142.0	0.0
8	20.5 × 15.5	165.0	0.0
9	32.5 × 23.5	34.0	19.0
10	32.5 × 23.5	68.0	19.0
11	32.5 × 23.5	119.0	19.0
12	32.5 × 23.5	153.0	19.0
13	15.5 × 20.5	102.0	22.0
14	20.5 × 15.5	0.0	44.0
15	20.5 × 15.5	22.0	44.0
16	32.5 × 42.5	44.3	44.0
17	52.5 × 42.5	78.7	44.0
18	52.5 × 42.5	133.0	44.0
19	42.5 × 52.5	0.0	61.0
20	25.5 × 25.5	78.7	88.0
21	25.5 × 25.5	105.7	88.0
22	25.5 × 25.5	133.0	88.0
23	25.5 × 25.5	160.0	88.0
24	32.5 × 23.5	44.3	90.0

Layout 14 172.5 × 115.5 cm gap 2.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

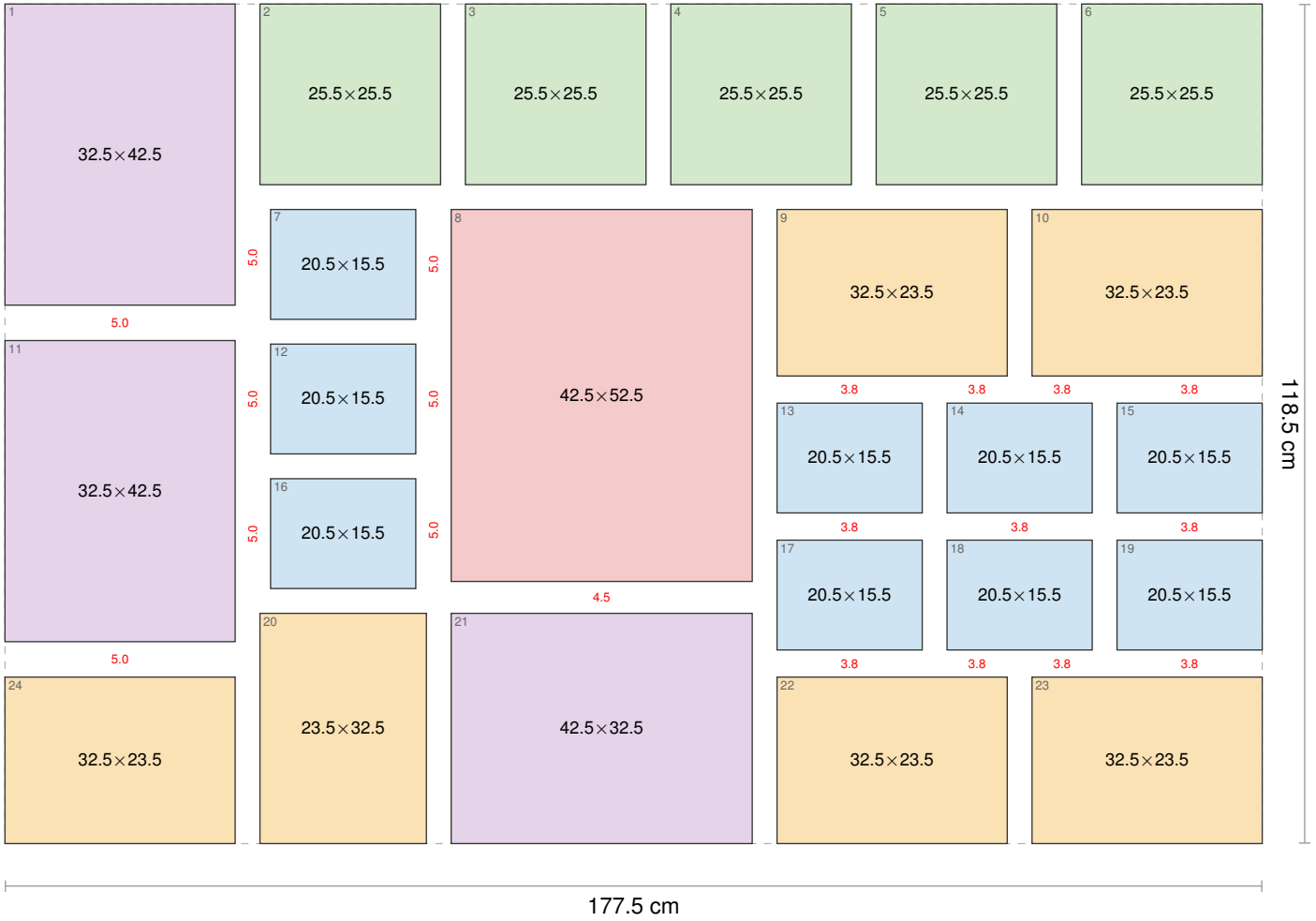
Outer size	172.5 × 115.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 4.5 cm
Max. deviation	2.0 cm
Mean deviation	0.31 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	1
Entropy	0.65
mix / variety	0.74 / 0.65
stagger / lines	0.47 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	25.5 × 25.5	35.0	0.0
3	25.5 × 25.5	63.0	0.0
4	25.5 × 25.5	91.0	0.0
5	25.5 × 25.5	119.0	0.0
6	25.5 × 25.5	147.0	0.0
7	32.5 × 42.5	0.0	27.0
8	42.5 × 32.5	35.0	28.0
9	32.5 × 23.5	80.0	28.0
10	20.5 × 15.5	116.0	28.0
11	32.5 × 23.5	140.0	28.0
12	20.5 × 15.5	116.0	46.0
13	32.5 × 23.5	80.0	56.0
14	32.5 × 23.5	140.0	56.0
15	42.5 × 52.5	35.0	63.0
16	20.5 × 15.5	116.0	64.0
17	32.5 × 42.5	0.0	73.0
18	20.5 × 15.5	80.0	82.0
19	20.5 × 15.5	103.0	82.0
20	20.5 × 15.5	126.0	82.0
21	23.5 × 32.5	149.0	83.0
22	20.5 × 15.5	80.0	100.0
23	20.5 × 15.5	103.0	100.0
24	20.5 × 15.5	126.0	100.0

Layout 15 177.5 × 118.5 cm gap 3.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

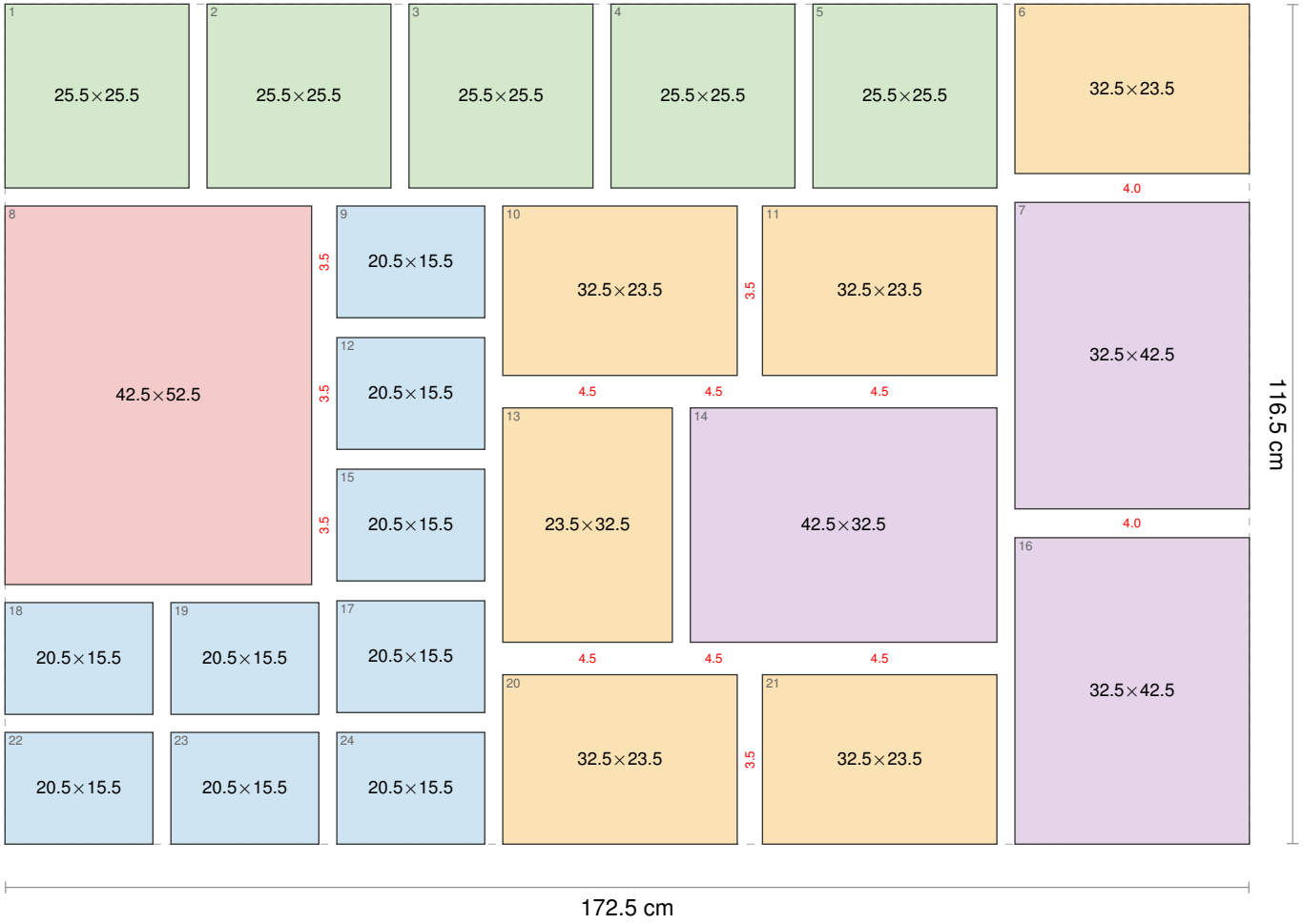
Outer size	177.5 × 118.5 cm
Nominal gap	3.5 cm
Actual gaps	3.5 – 5.0 cm
Max. deviation	1.5 cm
Mean deviation	0.35 cm
Border notches	left 10 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	1
Entropy	0.64
mix / variety	0.74 / 0.62
stagger / lines	0.46 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	25.5 × 25.5	36.0	0.0
3	25.5 × 25.5	65.0	0.0
4	25.5 × 25.5	94.0	0.0
5	25.5 × 25.5	123.0	0.0
6	25.5 × 25.5	152.0	0.0
7	20.5 × 15.5	37.5	29.0
8	42.5 × 52.5	63.0	29.0
9	32.5 × 23.5	109.0	29.0
10	32.5 × 23.5	145.0	29.0
11	32.5 × 42.5	0.0	47.5
12	20.5 × 15.5	37.5	48.0
13	20.5 × 15.5	109.0	56.3
14	20.5 × 15.5	133.0	56.3
15	20.5 × 15.5	157.0	56.3
16	20.5 × 15.5	37.5	67.0
17	20.5 × 15.5	109.0	75.7
18	20.5 × 15.5	133.0	75.7
19	20.5 × 15.5	157.0	75.7
20	23.5 × 32.5	36.0	86.0
21	42.5 × 32.5	63.0	86.0
22	32.5 × 23.5	109.0	95.0
23	32.5 × 23.5	145.0	95.0
24	32.5 × 23.5	0.0	95.0

Layout 16 172.5 × 116.5 cm gap 2.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	172.5 × 116.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 4.5 cm
Max. deviation	2.0 cm
Mean deviation	0.43 cm
Border notches	right 8 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	1
Entropy	0.64
mix / variety	0.74 / 0.61
stagger / lines	0.48 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	28.0	0.0
3	25.5 × 25.5	56.0	0.0
4	25.5 × 25.5	84.0	0.0
5	25.5 × 25.5	112.0	0.0
6	32.5 × 23.5	140.0	0.0
7	32.5 × 42.5	140.0	27.5
8	42.5 × 52.5	0.0	28.0
9	20.5 × 15.5	46.0	28.0
10	32.5 × 23.5	69.0	28.0
11	32.5 × 23.5	105.0	28.0
12	20.5 × 15.5	46.0	46.2
13	23.5 × 32.5	69.0	56.0
14	42.5 × 32.5	95.0	56.0
15	20.5 × 15.5	46.0	64.5
16	32.5 × 42.5	140.0	74.0
17	20.5 × 15.5	46.0	82.8
18	20.5 × 15.5	0.0	83.0
19	20.5 × 15.5	23.0	83.0
20	32.5 × 23.5	69.0	93.0
21	32.5 × 23.5	105.0	93.0
22	20.5 × 15.5	0.0	101.0
23	20.5 × 15.5	23.0	101.0
24	20.5 × 15.5	46.0	101.0

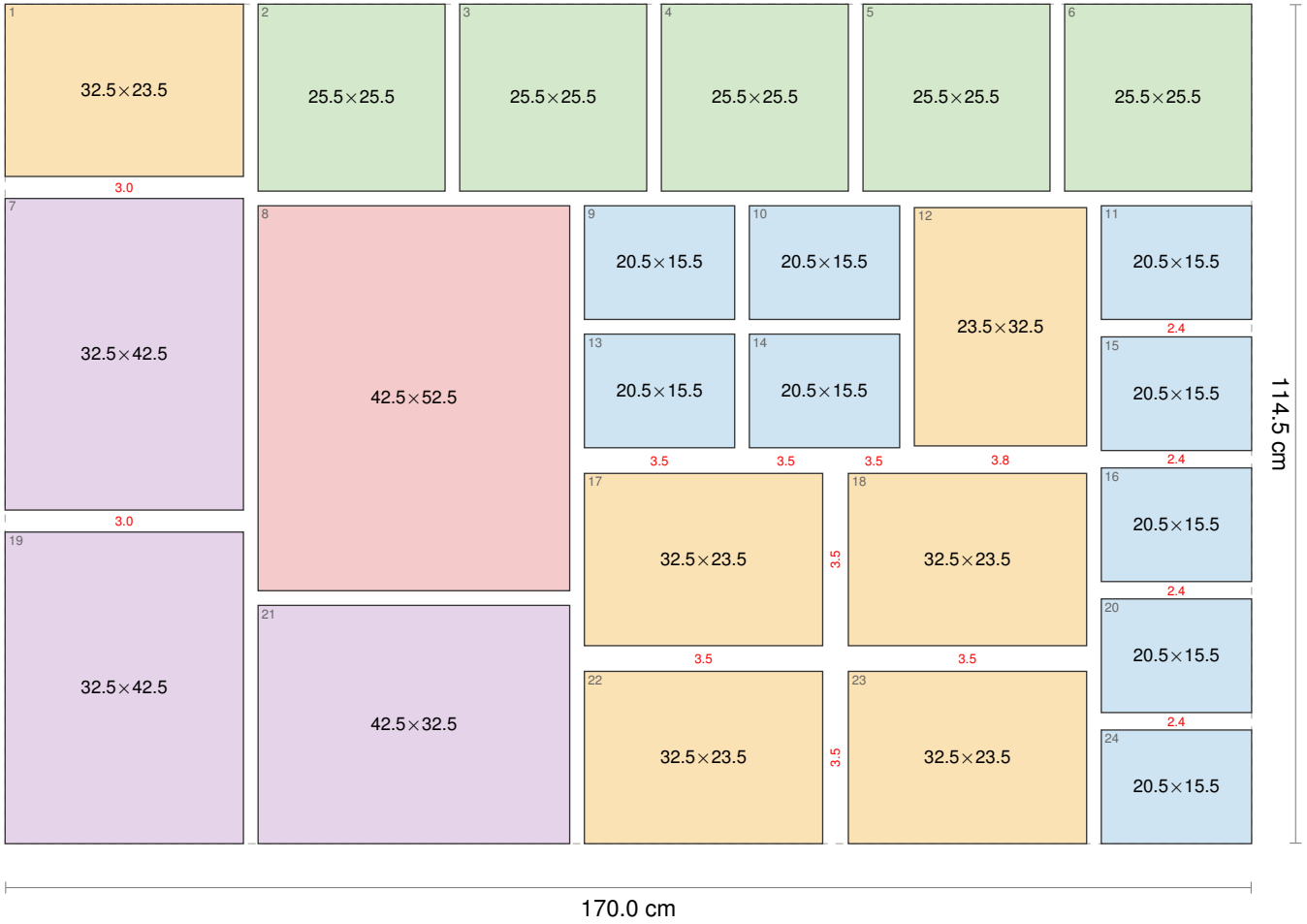
9 free-gap-s1

Identical to the baseline but with another random seed. Comparing it with the baseline shows how reproducible the results of a 20 minute search are.

Parameters that differ from the baseline: seed = 1.
256 runs, 20 min wall-clock budget, 545 candidate states, the 16 best distinct layouts are shown.

#	size (cm)	gap	frames	max dev	mean dev	notches	corners	entropy
1	170.0 × 114.5	2	24	1.8	0.33	4 cm	4/4	0.66
2	170.0 × 114.5	2	24	1.8	0.35	4 cm	4/4	0.71
3	184.0 × 115.0	2	23	2.0	0.57	none	4/4	0.74
4	183.5 × 115.0	2	23	2.2	0.48	none	4/4	0.76
5	190.5 × 115.5	2.5	22	2.0	0.41	none	4/4	0.68
6	171.5 × 116.5	2.5	22	2.0	0.61	none	4/4	0.74
7	170.0 × 115.0	2	19	2.0	0.52	none	4/4	0.71
8	181.5 × 114.5	1.5	25	2.3	0.56	none	4/4	0.68
9	172.0 × 113.5	2	25	2.5	0.41	none	4/4	0.69
10	180.0 × 128.0	3	25	3.0	0.77	none	4/4	0.77
11	180.5 × 127.5	3.5	25	3.1	0.73	none	4/4	0.59
12	172.5 × 115.5	2.5	24	2.0	0.28	4 cm	4/4	0.64
13	170.0 × 115.0	2	24	2.0	0.38	6 cm	4/4	0.70
14	170.5 × 117.5	2.5	24	3.0	0.58	none	4/4	0.77
15	189.5 × 110.5	3.5	24	3.0	0.79	none	4/4	0.76
16	186.5 × 116.5	2.5	23	2.5	0.56	none	4/4	0.58

Layout 1 170.0 × 114.5 cm gap 2 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

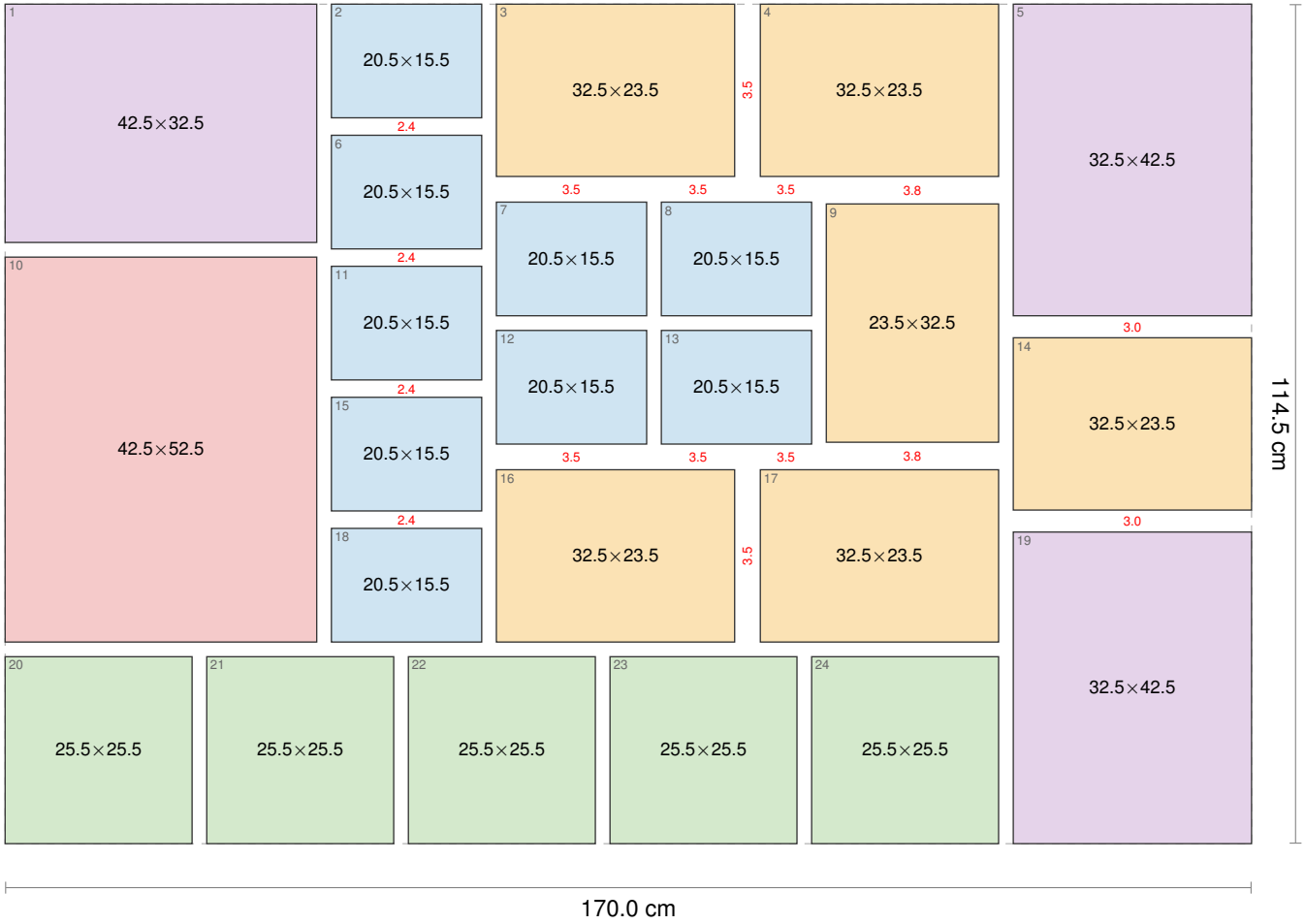
Outer size	170.0 × 114.5 cm
Nominal gap	2 cm
Actual gaps	2.0 – 3.8 cm
Max. deviation	1.8 cm
Mean deviation	0.33 cm
Border notches	bottom 4 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	1
Entropy	0.66
mix / variety	0.74 / 0.68
stagger / lines	0.48 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	25.5 × 25.5	34.5	0.0
3	25.5 × 25.5	62.0	0.0
4	25.5 × 25.5	89.5	0.0
5	25.5 × 25.5	117.0	0.0
6	25.5 × 25.5	144.5	0.0
7	32.5 × 42.5	0.0	26.5
8	42.5 × 52.5	34.5	27.5
9	20.5 × 15.5	79.0	27.5
10	20.5 × 15.5	101.5	27.5
11	20.5 × 15.5	149.5	27.5
12	23.5 × 32.5	124.0	27.8
13	20.5 × 15.5	79.0	45.0
14	20.5 × 15.5	101.5	45.0
15	20.5 × 15.5	149.5	45.4
16	20.5 × 15.5	149.5	63.2
17	32.5 × 23.5	79.0	64.0
18	32.5 × 23.5	115.0	64.0
19	32.5 × 42.5	0.0	72.0
20	20.5 × 15.5	149.5	81.1
21	42.5 × 32.5	34.5	82.0
22	32.5 × 23.5	79.0	91.0
23	32.5 × 23.5	115.0	91.0
24	20.5 × 15.5	149.5	99.0

Layout 2 170.0 × 114.5 cm gap 2 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

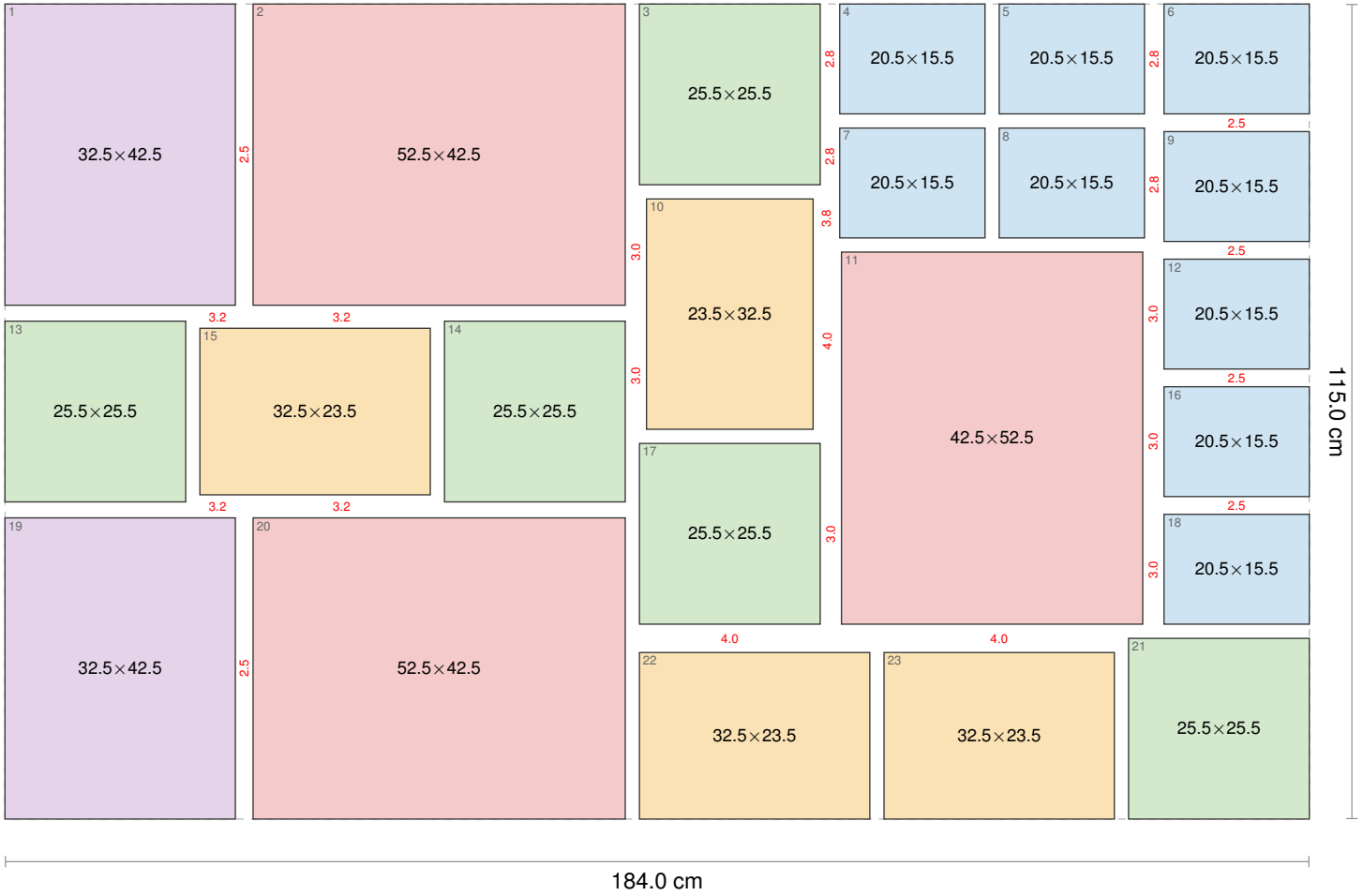
Outer size	170.0 × 114.5 cm
Nominal gap	2 cm
Actual gaps	2.0 – 3.8 cm
Max. deviation	1.8 cm
Mean deviation	0.35 cm
Border notches	top 4 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	1
Entropy	0.71
mix / variety	0.74 / 0.61
stagger / lines	0.48 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	42.5 × 32.5	0.0	0.0
2	20.5 × 15.5	44.5	0.0
3	32.5 × 23.5	67.0	0.0
4	32.5 × 23.5	103.0	0.0
5	32.5 × 42.5	137.5	0.0
6	20.5 × 15.5	44.5	17.9
7	20.5 × 15.5	67.0	27.0
8	20.5 × 15.5	89.5	27.0
9	23.5 × 32.5	112.0	27.2
10	42.5 × 52.5	0.0	34.5
11	20.5 × 15.5	44.5	35.8
12	20.5 × 15.5	67.0	44.5
13	20.5 × 15.5	89.5	44.5
14	32.5 × 23.5	137.5	45.5
15	20.5 × 15.5	44.5	53.6
16	32.5 × 23.5	67.0	63.5
17	32.5 × 23.5	103.0	63.5
18	20.5 × 15.5	44.5	71.5
19	32.5 × 42.5	137.5	72.0
20	25.5 × 25.5	0.0	89.0
21	25.5 × 25.5	27.5	89.0
22	25.5 × 25.5	55.0	89.0
23	25.5 × 25.5	82.5	89.0
24	25.5 × 25.5	110.0	89.0

Layout 3 184.0 × 115.0 cm gap 2 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

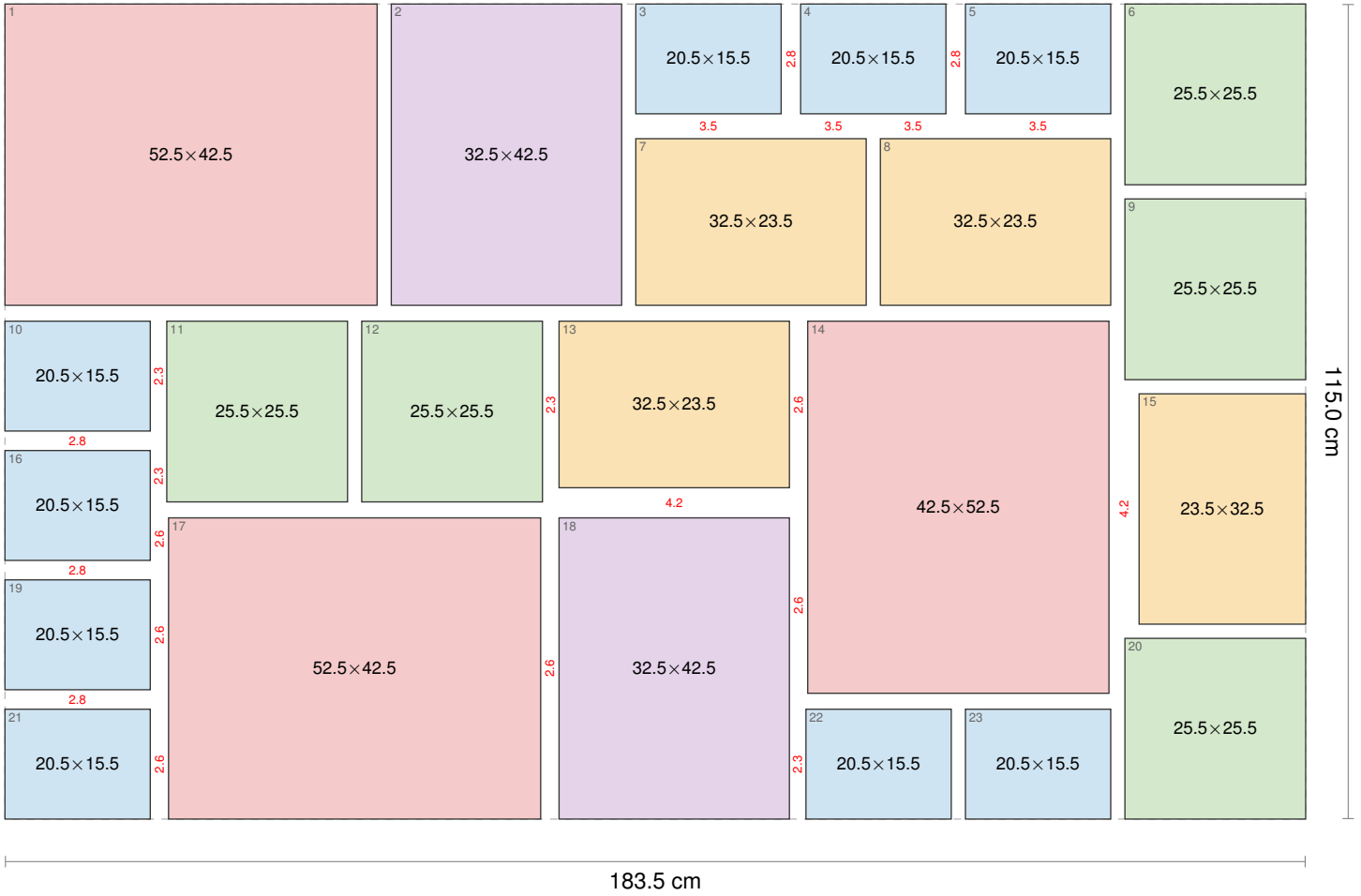
Outer size	184.0 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.0 cm
Max. deviation	2.0 cm
Mean deviation	0.57 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, not touching, min. dist. 72 cm
Entropy	0.74
mix / variety	0.76 / 0.68
stagger / lines	0.54 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	35.0	0.0
3	25.5 × 25.5	89.5	0.0
4	20.5 × 15.5	117.8	0.0
5	20.5 × 15.5	140.2	0.0
6	20.5 × 15.5	163.5	0.0
7	20.5 × 15.5	117.8	17.5
8	20.5 × 15.5	140.2	17.5
9	20.5 × 15.5	163.5	18.0
10	23.5 × 32.5	90.5	27.5
11	42.5 × 52.5	118.0	35.0
12	20.5 × 15.5	163.5	36.0
13	25.5 × 25.5	0.0	44.8
14	25.5 × 25.5	62.0	44.8
15	32.5 × 23.5	27.5	45.8
16	20.5 × 15.5	163.5	54.0
17	25.5 × 25.5	89.5	62.0
18	20.5 × 15.5	163.5	72.0
19	32.5 × 42.5	0.0	72.5
20	52.5 × 42.5	35.0	72.5
21	25.5 × 25.5	158.5	89.5
22	32.5 × 23.5	89.5	91.5
23	32.5 × 23.5	124.0	91.5

Layout 4 183.5 × 115.0 cm gap 2 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

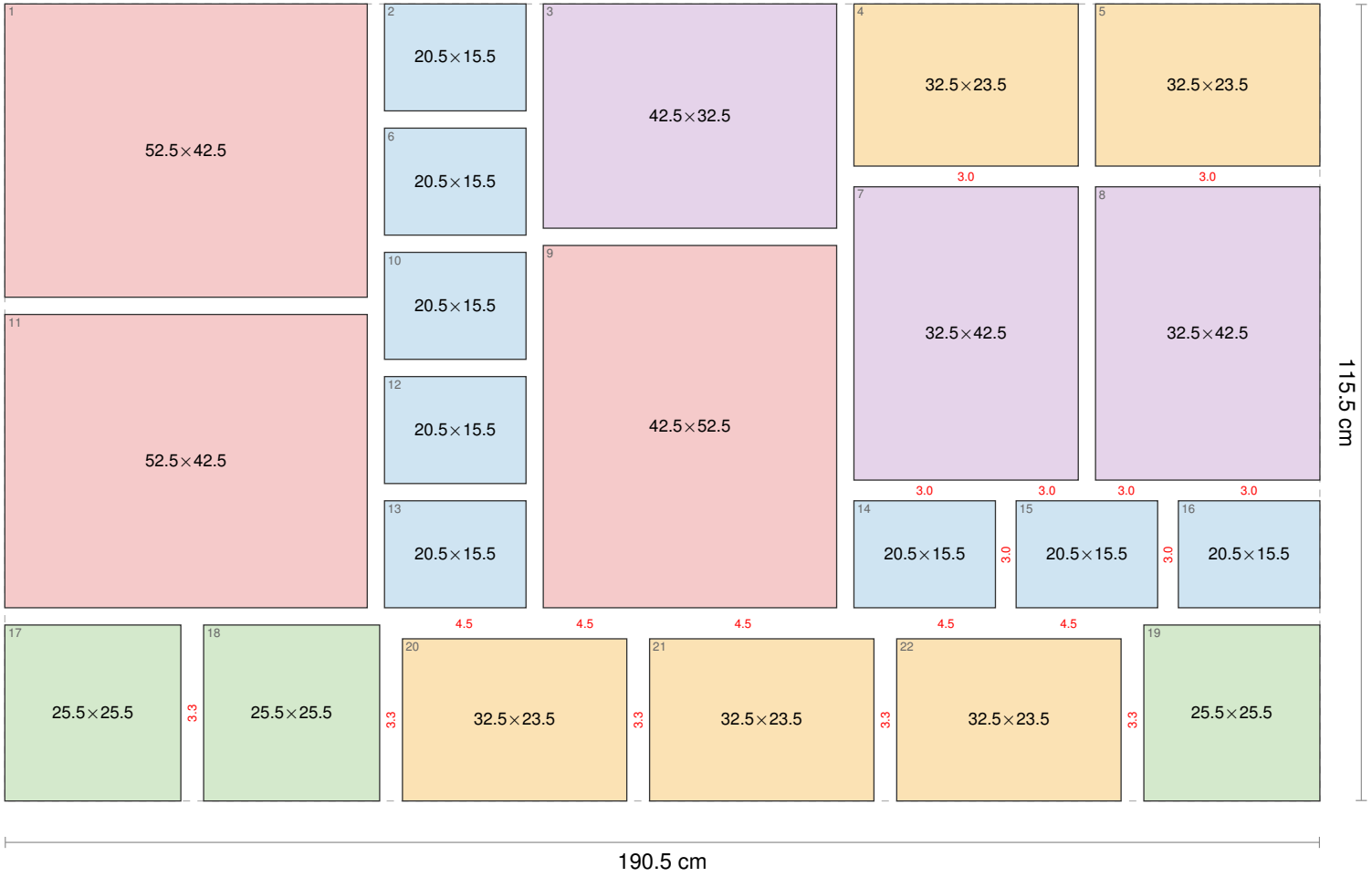
Outer size	183.5 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.2 cm
Max. deviation	2.2 cm
Mean deviation	0.48 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, not touching, min. dist. 76 cm
Entropy	0.76
mix / variety	0.76 / 0.76
stagger / lines	0.54 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 42.5	54.5	0.0
3	20.5 × 15.5	89.0	0.0
4	20.5 × 15.5	112.2	0.0
5	20.5 × 15.5	135.5	0.0
6	25.5 × 25.5	158.0	0.0
7	32.5 × 23.5	89.0	19.0
8	32.5 × 23.5	123.5	19.0
9	25.5 × 25.5	158.0	27.5
10	20.5 × 15.5	0.0	44.8
11	25.5 × 25.5	22.8	44.8
12	25.5 × 25.5	50.3	44.8
13	32.5 × 23.5	78.2	44.8
14	42.5 × 52.5	113.2	44.8
15	23.5 × 32.5	160.0	55.0
16	20.5 × 15.5	0.0	63.0
17	52.5 × 42.5	23.1	72.5
18	32.5 × 42.5	78.2	72.5
19	20.5 × 15.5	0.0	81.2
20	25.5 × 25.5	158.0	89.5
21	20.5 × 15.5	0.0	99.5
22	20.5 × 15.5	113.0	99.5
23	20.5 × 15.5	135.5	99.5

Layout 5 190.5 × 115.5 cm gap 2.5 cm 22 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

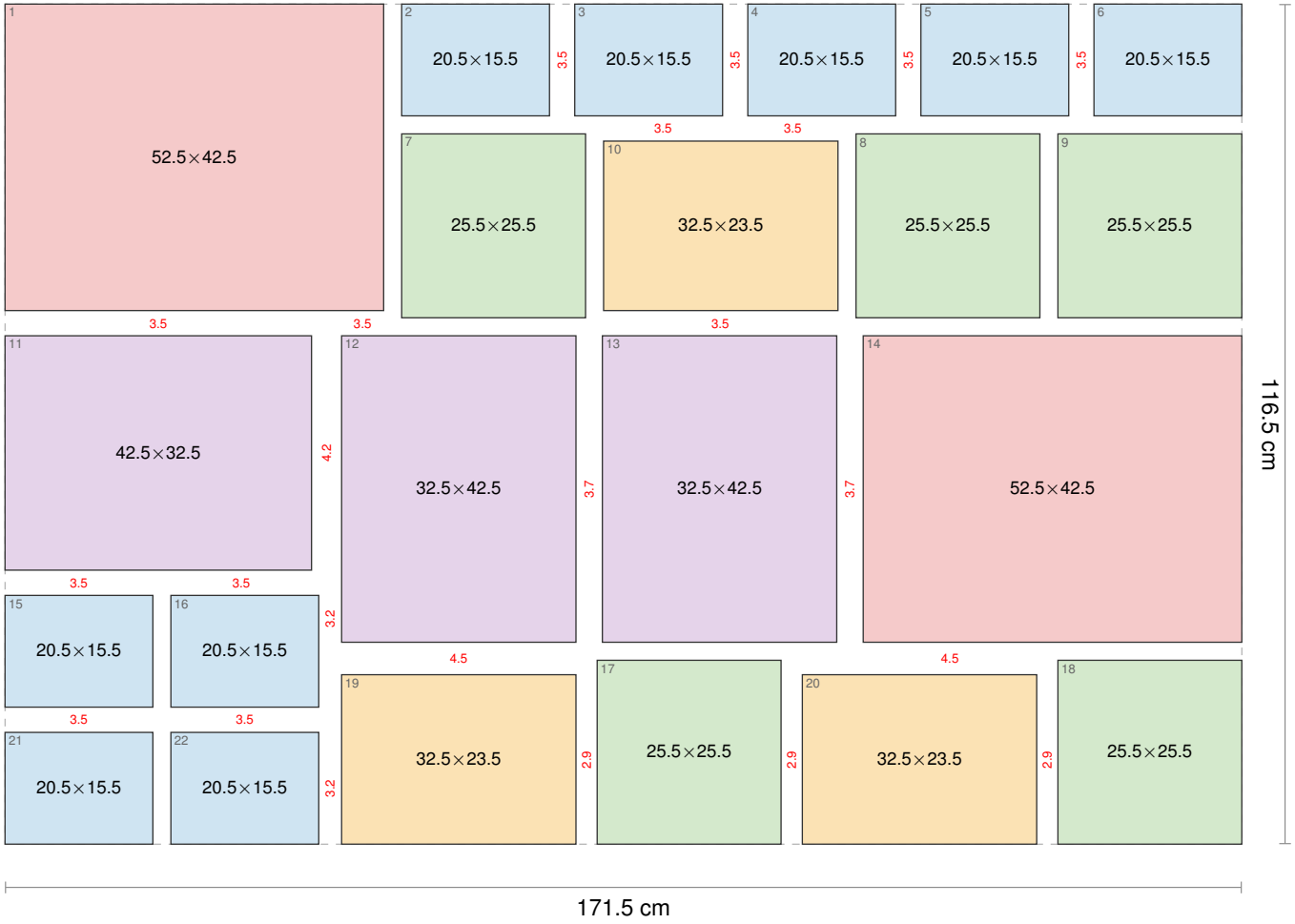
Outer size	190.5 × 115.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 4.5 cm
Max. deviation	2.0 cm
Mean deviation	0.41 cm
Border notches	none
Corners filled	4 / 4
Frames used	22 of 28
Large frames	3, touching 1× (52 cm), min. dist. 45 cm
Entropy	0.68
mix / variety	0.77 / 0.69
stagger / lines	0.50 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	0	1
	25.5×25.5		3	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	20.5 × 15.5	55.0	0.0
3	42.5 × 32.5	78.0	0.0
4	32.5 × 23.5	123.0	0.0
5	32.5 × 23.5	158.0	0.0
6	20.5 × 15.5	55.0	18.0
7	32.5 × 42.5	123.0	26.5
8	32.5 × 42.5	158.0	26.5
9	42.5 × 52.5	78.0	35.0
10	20.5 × 15.5	55.0	36.0
11	52.5 × 42.5	0.0	45.0
12	20.5 × 15.5	55.0	54.0
13	20.5 × 15.5	55.0	72.0
14	20.5 × 15.5	123.0	72.0
15	20.5 × 15.5	146.5	72.0
16	20.5 × 15.5	170.0	72.0
17	25.5 × 25.5	0.0	90.0
18	25.5 × 25.5	28.8	90.0
19	25.5 × 25.5	165.0	90.0
20	32.5 × 23.5	57.6	92.0
21	32.5 × 23.5	93.4	92.0
22	32.5 × 23.5	129.2	92.0

Layout 6 171.5 × 116.5 cm gap 2.5 cm 22 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

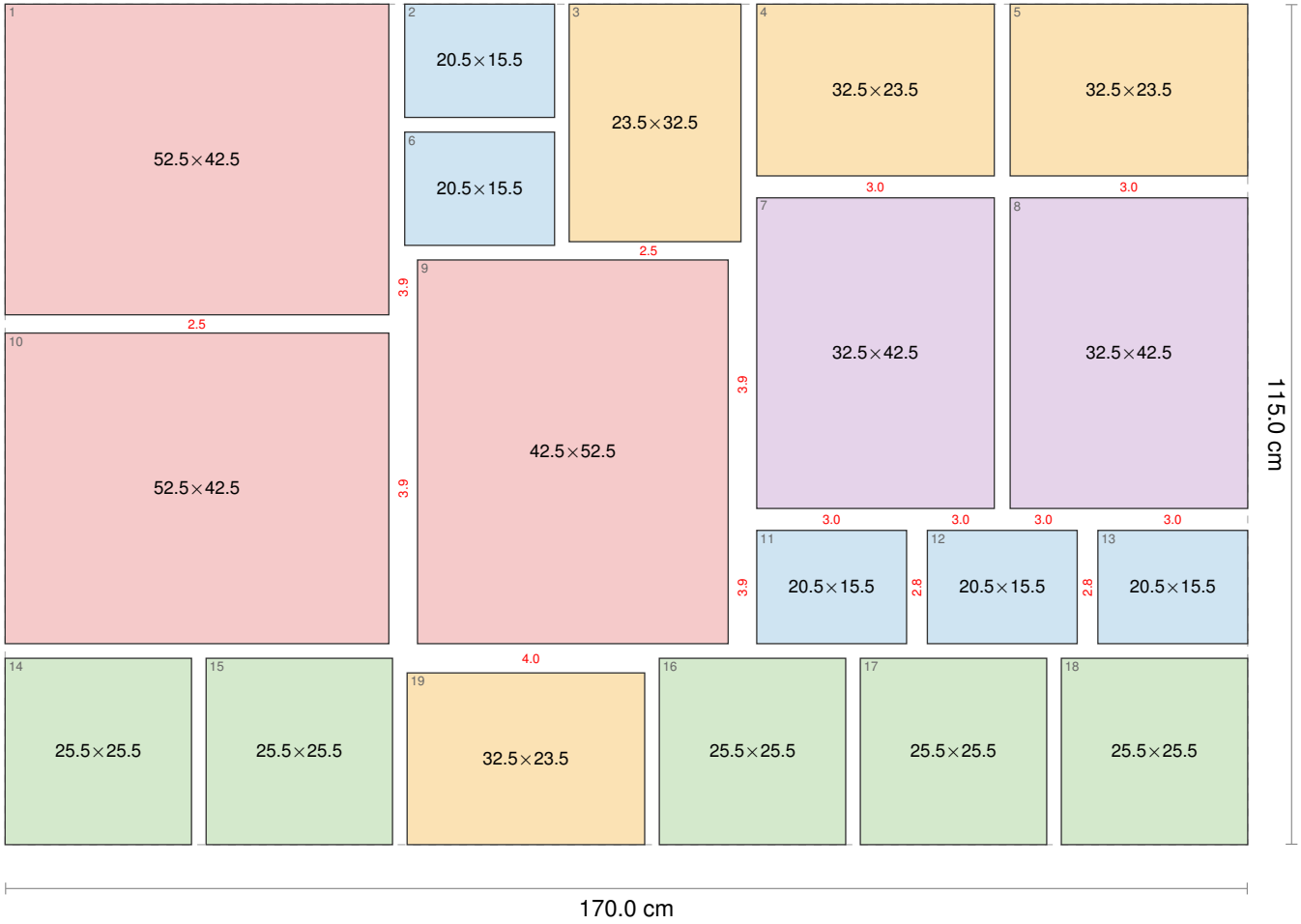
Outer size	171.5 × 116.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 4.5 cm
Max. deviation	2.0 cm
Mean deviation	0.61 cm
Border notches	none
Corners filled	4 / 4
Frames used	22 of 28
Large frames	2, not touching, min. dist. 128 cm
Entropy	0.74
mix / variety	0.71 / 0.74
stagger / lines	0.52 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	3	5
	23.5×32.5	P	0	1
	25.5×25.5		5	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	20.5 × 15.5	55.0	0.0
3	20.5 × 15.5	79.0	0.0
4	20.5 × 15.5	103.0	0.0
5	20.5 × 15.5	127.0	0.0
6	20.5 × 15.5	151.0	0.0
7	25.5 × 25.5	55.0	18.0
8	25.5 × 25.5	118.0	18.0
9	25.5 × 25.5	146.0	18.0
10	32.5 × 23.5	83.0	19.0
11	42.5 × 32.5	0.0	46.0
12	32.5 × 42.5	46.7	46.0
13	32.5 × 42.5	82.8	46.0
14	52.5 × 42.5	119.0	46.0
15	20.5 × 15.5	0.0	82.0
16	20.5 × 15.5	23.0	82.0
17	25.5 × 25.5	82.1	91.0
18	25.5 × 25.5	146.0	91.0
19	32.5 × 23.5	46.7	93.0
20	32.5 × 23.5	110.6	93.0
21	20.5 × 15.5	0.0	101.0
22	20.5 × 15.5	23.0	101.0

Layout 7 170.0 × 115.0 cm gap 2 cm 19 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

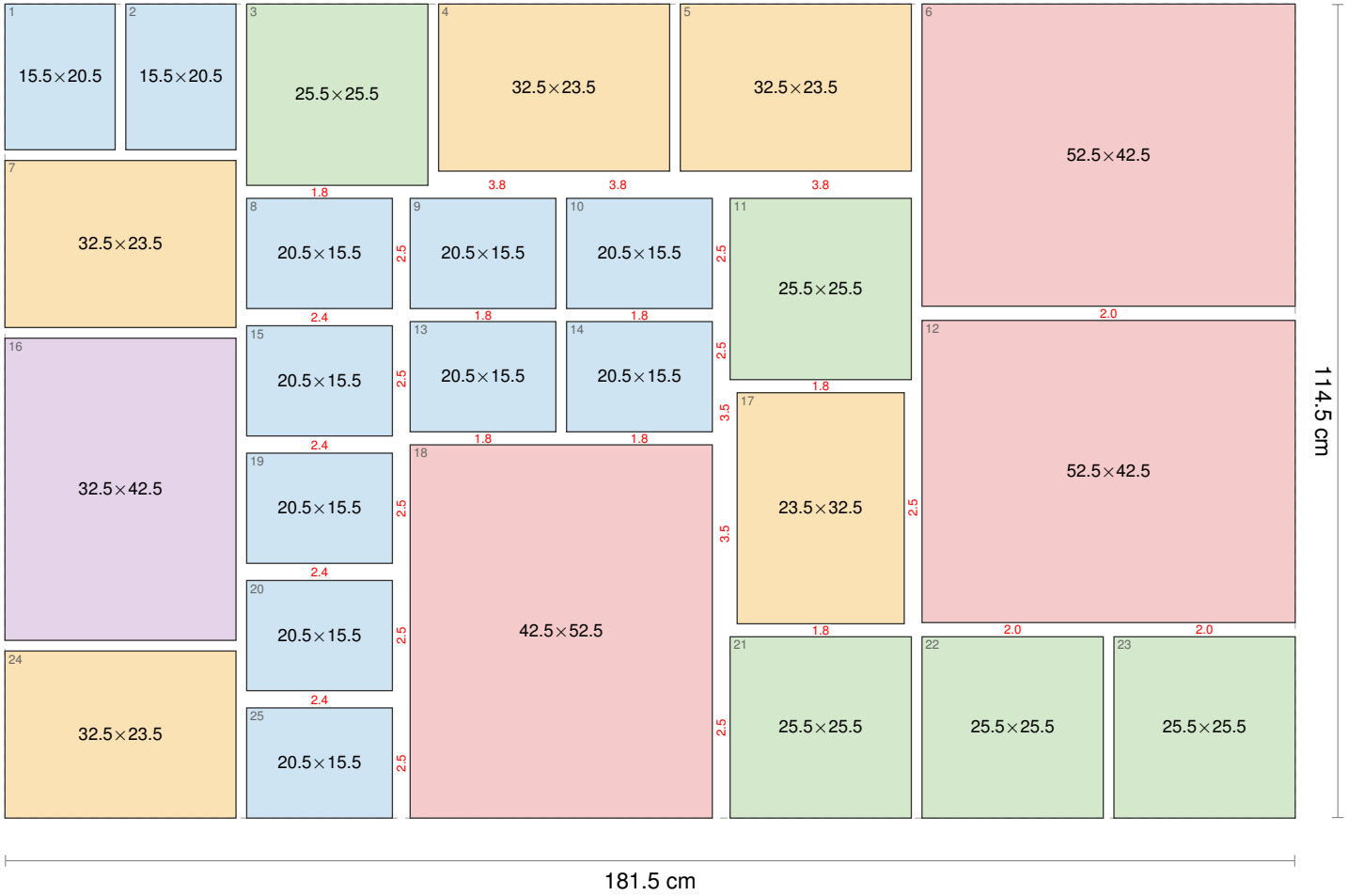
Outer size	170.0 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.0 cm
Max. deviation	2.0 cm
Mean deviation	0.52 cm
Border notches	none
Corners filled	4 / 4
Frames used	19 of 28
Large frames	3, touching 1× (52 cm), min. dist. 45 cm
Entropy	0.71
mix / variety	0.81 / 0.72
stagger / lines	0.55 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	5	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	20.5 × 15.5	54.7	0.0
3	23.5 × 32.5	77.2	0.0
4	32.5 × 23.5	102.8	0.0
5	32.5 × 23.5	137.5	0.0
6	20.5 × 15.5	54.7	17.5
7	32.5 × 42.5	102.8	26.5
8	32.5 × 42.5	137.5	26.5
9	42.5 × 52.5	56.4	35.0
10	52.5 × 42.5	0.0	45.0
11	20.5 × 15.5	102.8	72.0
12	20.5 × 15.5	126.2	72.0
13	20.5 × 15.5	149.5	72.0
14	25.5 × 25.5	0.0	89.5
15	25.5 × 25.5	27.5	89.5
16	25.5 × 25.5	89.5	89.5
17	25.5 × 25.5	117.0	89.5
18	25.5 × 25.5	144.5	89.5
19	32.5 × 23.5	55.0	91.5

Layout 8 181.5 × 114.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

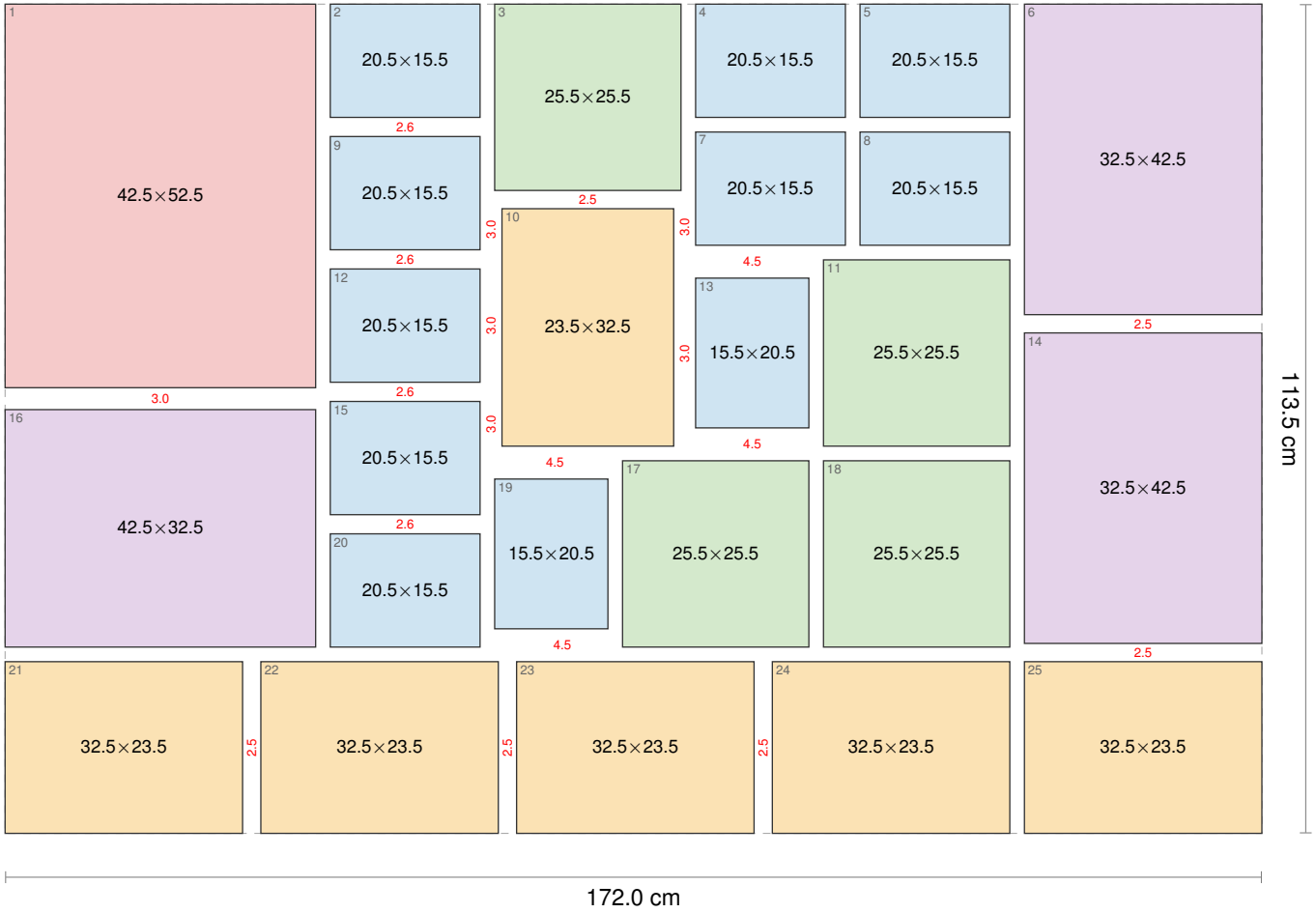
Outer size	181.5 × 114.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.8 cm
Max. deviation	2.3 cm
Mean deviation	0.56 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, touching 1× (52 cm), min. dist. 44 cm
Entropy	0.68
mix / variety	0.81 / 0.66
stagger / lines	0.51 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	1	2
	42.5×32.5	L	0	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	15.5 × 20.5	0.0	0.0
2	15.5 × 20.5	17.0	0.0
3	25.5 × 25.5	34.0	0.0
4	32.5 × 23.5	61.0	0.0
5	32.5 × 23.5	95.0	0.0
6	52.5 × 42.5	129.0	0.0
7	32.5 × 23.5	0.0	22.0
8	20.5 × 15.5	34.0	27.3
9	20.5 × 15.5	57.0	27.3
10	20.5 × 15.5	79.0	27.3
11	25.5 × 25.5	102.0	27.3
12	52.5 × 42.5	129.0	44.5
13	20.5 × 15.5	57.0	44.7
14	20.5 × 15.5	79.0	44.7
15	20.5 × 15.5	34.0	45.2
16	32.5 × 42.5	0.0	47.0
17	23.5 × 32.5	103.0	54.7
18	42.5 × 52.5	57.0	62.0
19	20.5 × 15.5	34.0	63.2
20	20.5 × 15.5	34.0	81.1
21	25.5 × 25.5	102.0	89.0
22	25.5 × 25.5	129.0	89.0
23	25.5 × 25.5	156.0	89.0
24	32.5 × 23.5	0.0	91.0
25	20.5 × 15.5	34.0	99.0

Layout 9 172.0 × 113.5 cm gap 2 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	172.0 × 113.5 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.5 cm
Max. deviation	2.5 cm
Mean deviation	0.41 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	1
Entropy	0.69
mix / variety	0.81 / 0.69
stagger / lines	0.52 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	2	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	1	1
52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	42.5 × 52.5	0.0	0.0
2	20.5 × 15.5	44.5	0.0
3	25.5 × 25.5	67.0	0.0
4	20.5 × 15.5	94.5	0.0
5	20.5 × 15.5	117.0	0.0
6	32.5 × 42.5	139.5	0.0
7	20.5 × 15.5	94.5	17.5
8	20.5 × 15.5	117.0	17.5
9	20.5 × 15.5	44.5	18.1
10	23.5 × 32.5	68.0	28.0
11	25.5 × 25.5	112.0	35.0
12	20.5 × 15.5	44.5	36.2
13	15.5 × 20.5	94.5	37.5
14	32.5 × 42.5	139.5	45.0
15	20.5 × 15.5	44.5	54.4
16	42.5 × 32.5	0.0	55.5
17	25.5 × 25.5	84.5	62.5
18	25.5 × 25.5	112.0	62.5
19	15.5 × 20.5	67.0	65.0
20	20.5 × 15.5	44.5	72.5
21	32.5 × 23.5	0.0	90.0
22	32.5 × 23.5	35.0	90.0
23	32.5 × 23.5	70.0	90.0
24	32.5 × 23.5	105.0	90.0
25	32.5 × 23.5	139.5	90.0

Layout 10 180.0 × 128.0 cm gap 3 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

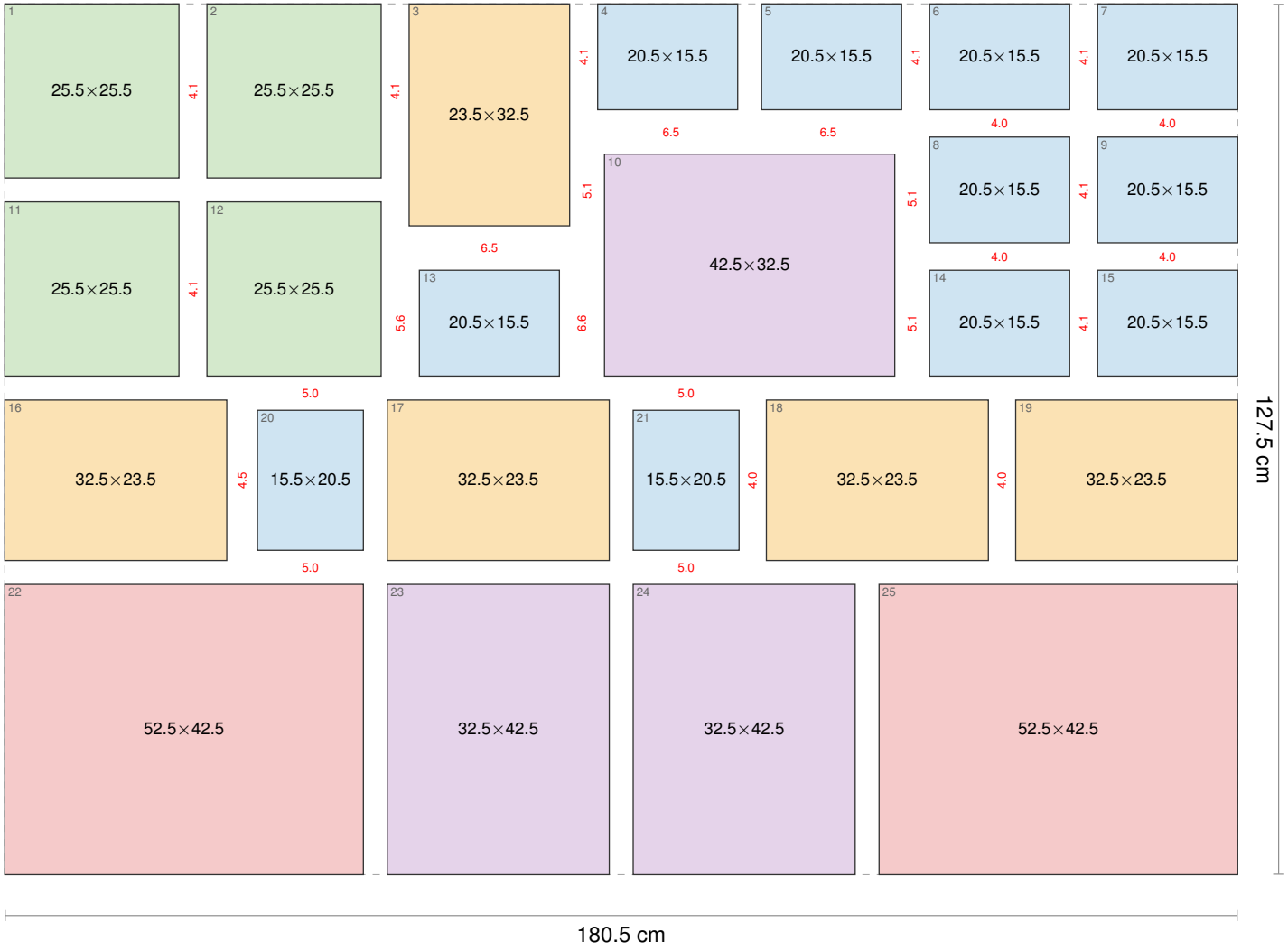
Outer size	180.0 × 128.0 cm
Nominal gap	3 cm
Actual gaps	3.0 – 6.0 cm
Max. deviation	3.0 cm
Mean deviation	0.77 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 92 cm
Entropy	0.77
mix / variety	0.84 / 0.74
stagger / lines	0.51 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 42.5	55.8	0.0
3	52.5 × 42.5	91.7	0.0
4	32.5 × 42.5	147.5	0.0
5	20.5 × 15.5	0.0	45.5
6	20.5 × 15.5	50.0	45.5
7	32.5 × 23.5	119.0	45.5
8	25.5 × 25.5	154.5	45.5
9	23.5 × 32.5	23.5	46.2
10	42.5 × 32.5	73.5	46.2
11	20.5 × 15.5	0.0	64.0
12	20.5 × 15.5	50.0	64.0
13	25.5 × 25.5	154.5	74.0
14	32.5 × 23.5	119.0	75.0
15	15.5 × 20.5	0.0	83.2
16	20.5 × 15.5	18.8	83.2
17	20.5 × 15.5	42.8	83.2
18	20.5 × 15.5	66.7	83.2
19	20.5 × 15.5	93.0	83.2
20	25.5 × 25.5	90.5	102.5
21	25.5 × 25.5	154.5	102.5
22	32.5 × 23.5	18.8	104.5
23	32.5 × 23.5	54.7	104.5
24	32.5 × 23.5	119.0	104.5
25	15.5 × 20.5	0.0	107.5

Layout 11 180.5 × 127.5 cm gap 3.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

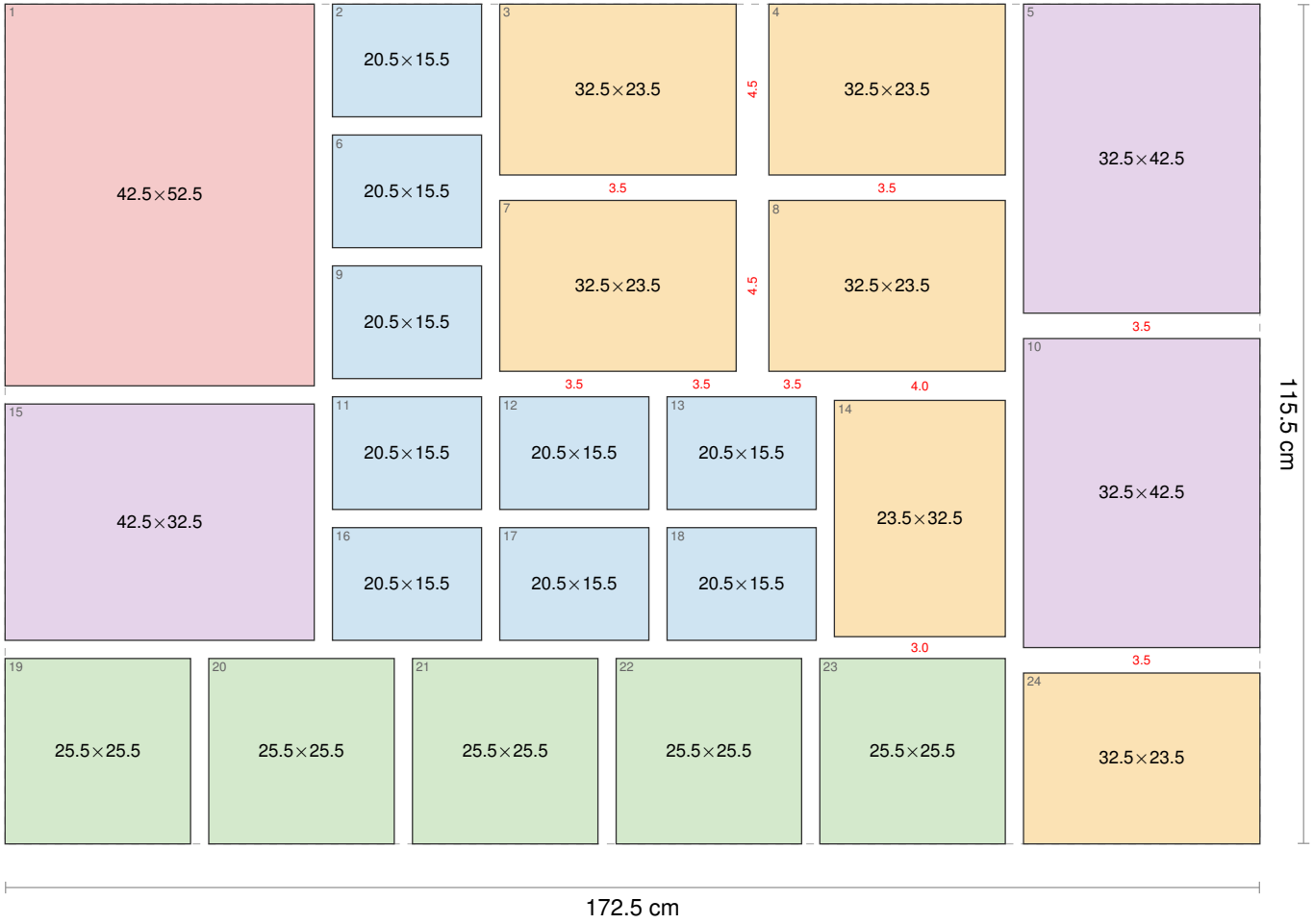
Outer size	180.5 × 127.5 cm
Nominal gap	3.5 cm
Actual gaps	3.5 – 6.6 cm
Max. deviation	3.1 cm
Mean deviation	0.73 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 128 cm
Entropy	0.59
mix / variety	0.83 / 0.56
stagger / lines	0.46 / 0.50 (3 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	29.6	0.0
3	23.5 × 32.5	59.2	0.0
4	20.5 × 15.5	86.8	0.0
5	20.5 × 15.5	110.8	0.0
6	20.5 × 15.5	135.4	0.0
7	20.5 × 15.5	160.0	0.0
8	20.5 × 15.5	135.4	19.5
9	20.5 × 15.5	160.0	19.5
10	42.5 × 32.5	87.8	22.0
11	25.5 × 25.5	0.0	29.0
12	25.5 × 25.5	29.6	29.0
13	20.5 × 15.5	60.7	39.0
14	20.5 × 15.5	135.4	39.0
15	20.5 × 15.5	160.0	39.0
16	32.5 × 23.5	0.0	58.0
17	32.5 × 23.5	56.0	58.0
18	32.5 × 23.5	111.5	58.0
19	32.5 × 23.5	148.0	58.0
20	15.5 × 20.5	37.0	59.5
21	15.5 × 20.5	92.0	59.5
22	52.5 × 42.5	0.0	85.0
23	32.5 × 42.5	56.0	85.0
24	32.5 × 42.5	92.0	85.0
25	52.5 × 42.5	128.0	85.0

Layout 12 172.5 × 115.5 cm gap 2.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

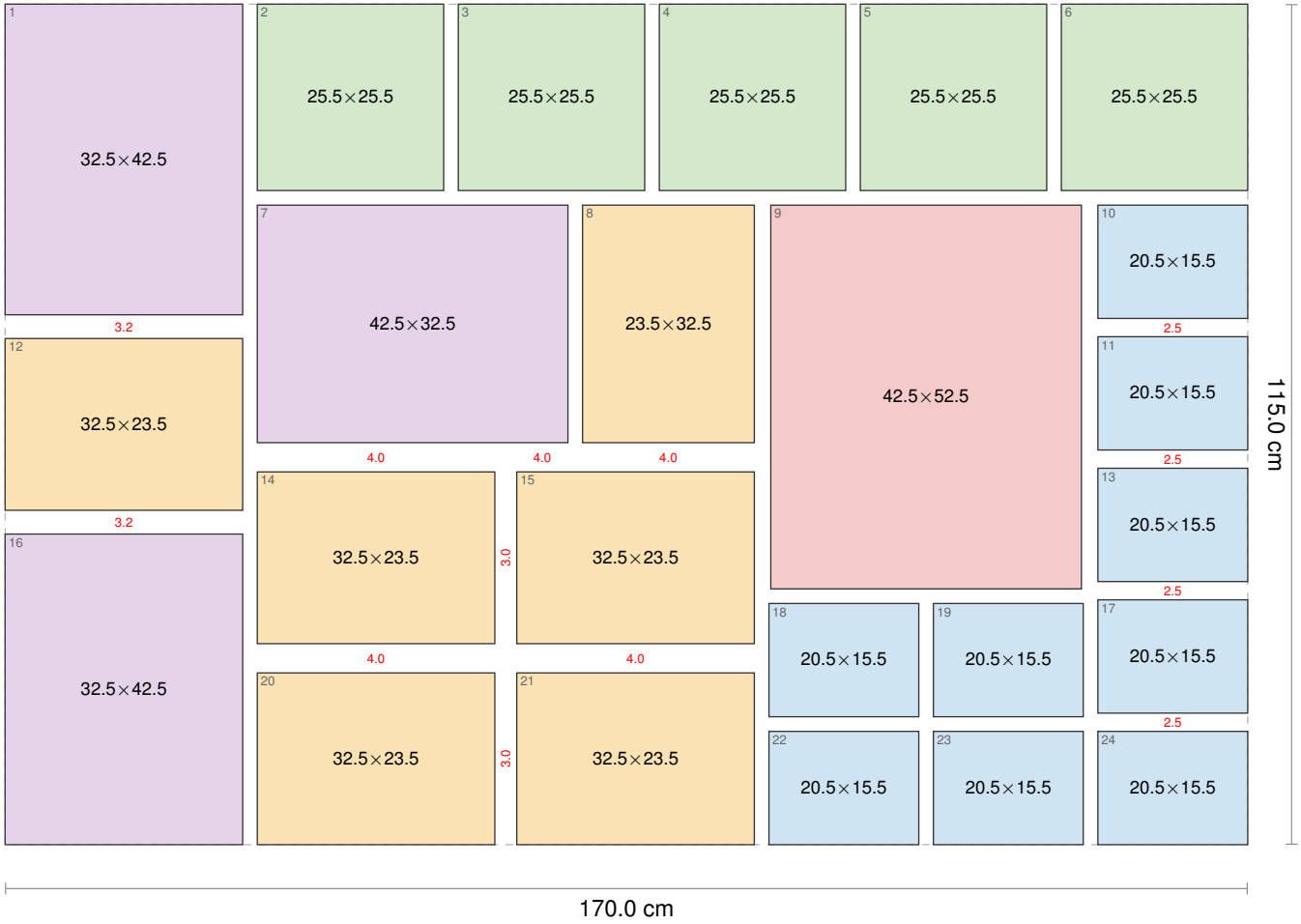
Outer size	172.5 × 115.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 4.5 cm
Max. deviation	2.0 cm
Mean deviation	0.28 cm
Border notches	top 4 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	1
Entropy	0.64
mix / variety	0.74 / 0.61
stagger / lines	0.47 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	42.5 × 52.5	0.0	0.0
2	20.5 × 15.5	45.0	0.0
3	32.5 × 23.5	68.0	0.0
4	32.5 × 23.5	105.0	0.0
5	32.5 × 42.5	140.0	0.0
6	20.5 × 15.5	45.0	18.0
7	32.5 × 23.5	68.0	27.0
8	32.5 × 23.5	105.0	27.0
9	20.5 × 15.5	45.0	36.0
10	32.5 × 42.5	140.0	46.0
11	20.5 × 15.5	45.0	54.0
12	20.5 × 15.5	68.0	54.0
13	20.5 × 15.5	91.0	54.0
14	23.5 × 32.5	114.0	54.5
15	42.5 × 32.5	0.0	55.0
16	20.5 × 15.5	45.0	72.0
17	20.5 × 15.5	68.0	72.0
18	20.5 × 15.5	91.0	72.0
19	25.5 × 25.5	0.0	90.0
20	25.5 × 25.5	28.0	90.0
21	25.5 × 25.5	56.0	90.0
22	25.5 × 25.5	84.0	90.0
23	25.5 × 25.5	112.0	90.0
24	32.5 × 23.5	140.0	92.0

Layout 13 170.0 × 115.0 cm gap 2 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	170.0 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.0 cm
Max. deviation	2.0 cm
Mean deviation	0.38 cm
Border notches	left 6 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	1
Entropy	0.70
mix / variety	0.74 / 0.57
stagger / lines	0.49 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	0	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		5	5
42.5×52.5	P	1	1
52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	25.5 × 25.5	34.5	0.0
3	25.5 × 25.5	62.0	0.0
4	25.5 × 25.5	89.5	0.0
5	25.5 × 25.5	117.0	0.0
6	25.5 × 25.5	144.5	0.0
7	42.5 × 32.5	34.5	27.5
8	23.5 × 32.5	79.0	27.5
9	42.5 × 52.5	104.8	27.5
10	20.5 × 15.5	149.5	27.5
11	20.5 × 15.5	149.5	45.5
12	32.5 × 23.5	0.0	45.8
13	20.5 × 15.5	149.5	63.5
14	32.5 × 23.5	34.5	64.0
15	32.5 × 23.5	70.0	64.0
16	32.5 × 42.5	0.0	72.5
17	20.5 × 15.5	149.5	81.5
18	20.5 × 15.5	104.5	82.0
19	20.5 × 15.5	127.0	82.0
20	32.5 × 23.5	34.5	91.5
21	32.5 × 23.5	70.0	91.5
22	20.5 × 15.5	104.5	99.5
23	20.5 × 15.5	127.0	99.5
24	20.5 × 15.5	149.5	99.5

Layout 14 170.5 × 117.5 cm gap 2.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

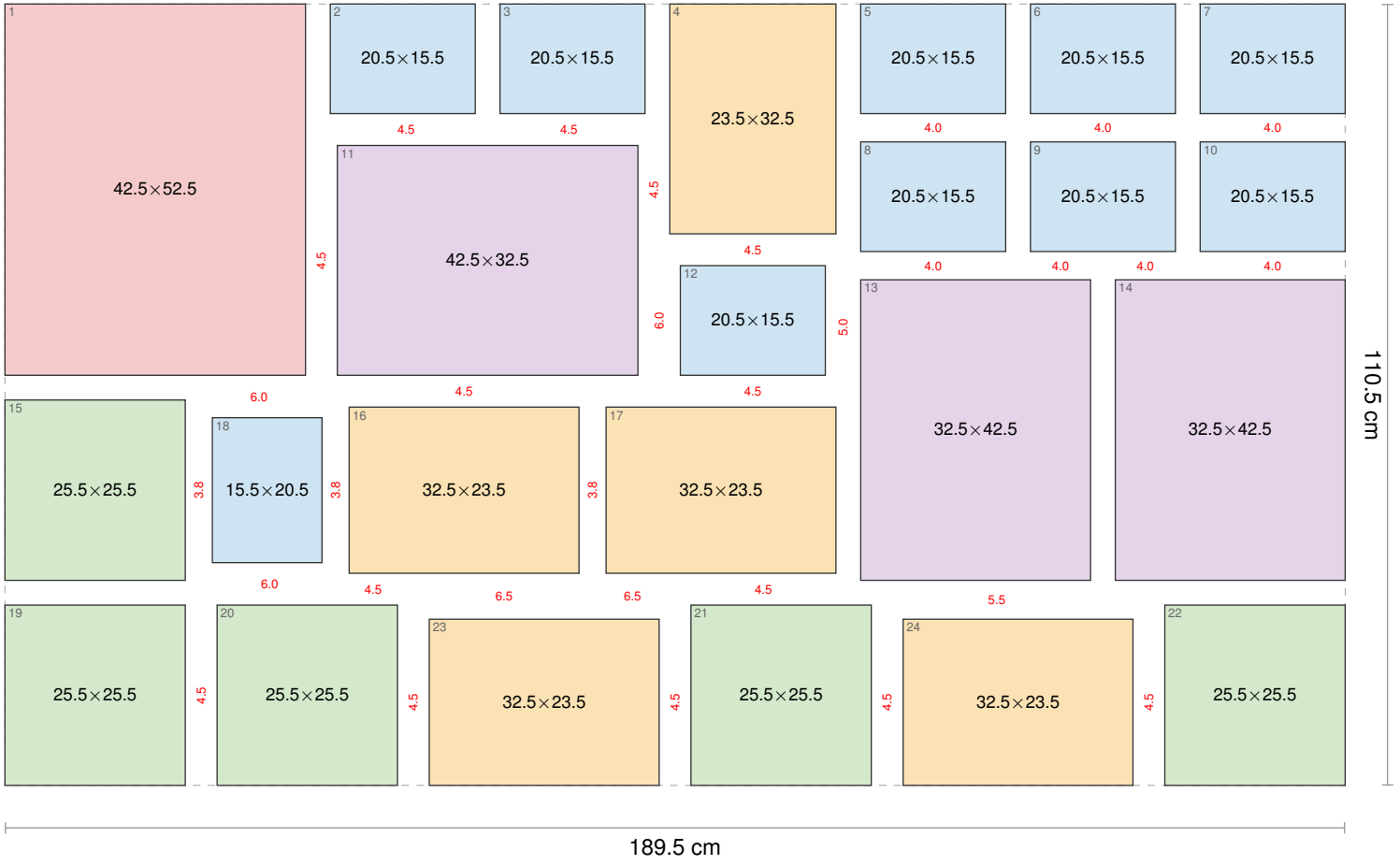
Outer size	170.5 × 117.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.5 cm
Max. deviation	3.0 cm
Mean deviation	0.58 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 90 cm
Entropy	0.77
mix / variety	0.78 / 0.78
stagger / lines	0.51 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	25.5 × 25.5	23.0	0.0
3	32.5 × 42.5	51.0	0.0
4	20.5 × 15.5	86.0	0.0
5	15.5 × 20.5	109.0	0.0
6	15.5 × 20.5	127.0	0.0
7	25.5 × 25.5	145.0	0.0
8	20.5 × 15.5	0.0	19.0
9	20.5 × 15.5	86.0	19.0
10	32.5 × 23.5	109.5	23.0
11	20.5 × 15.5	25.5	28.0
12	23.5 × 32.5	147.0	28.3
13	20.5 × 15.5	0.0	38.0
14	20.5 × 15.5	86.0	38.0
15	25.5 × 25.5	23.0	47.0
16	32.5 × 23.5	51.0	48.0
17	32.5 × 23.5	109.5	49.0
18	20.5 × 15.5	0.0	57.0
19	20.5 × 15.5	86.0	57.0
20	25.5 × 25.5	145.0	63.7
21	52.5 × 42.5	0.0	75.0
22	32.5 × 42.5	55.0	75.0
23	52.5 × 42.5	90.0	75.0
24	25.5 × 25.5	145.0	92.0

Layout 15 189.5 × 110.5 cm gap 3.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	189.5 × 110.5 cm
Nominal gap	3.5 cm
Actual gaps	3.5 – 6.5 cm
Max. deviation	3.0 cm
Mean deviation	0.79 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	1
Entropy	0.76
mix / variety	0.79 / 0.68
stagger / lines	0.55 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	42.5 × 52.5	0.0	0.0
2	20.5 × 15.5	46.0	0.0
3	20.5 × 15.5	70.0	0.0
4	23.5 × 32.5	94.0	0.0
5	20.5 × 15.5	121.0	0.0
6	20.5 × 15.5	145.0	0.0
7	20.5 × 15.5	169.0	0.0
8	20.5 × 15.5	121.0	19.5
9	20.5 × 15.5	145.0	19.5
10	20.5 × 15.5	169.0	19.5
11	42.5 × 32.5	47.0	20.0
12	20.5 × 15.5	95.5	37.0
13	32.5 × 42.5	121.0	39.0
14	32.5 × 42.5	157.0	39.0
15	25.5 × 25.5	0.0	56.0
16	32.5 × 23.5	48.7	57.0
17	32.5 × 23.5	85.0	57.0
18	15.5 × 20.5	29.3	58.5
19	25.5 × 25.5	0.0	85.0
20	25.5 × 25.5	30.0	85.0
21	25.5 × 25.5	97.0	85.0
22	25.5 × 25.5	164.0	85.0
23	32.5 × 23.5	60.0	87.0
24	32.5 × 23.5	127.0	87.0

Layout 16 186.5 × 116.5 cm gap 2.5 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	186.5 × 116.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.0 cm
Max. deviation	2.5 cm
Mean deviation	0.56 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 2× (59 cm), min. dist. 51 cm
Entropy	0.58
mix / variety	0.76 / 0.57
stagger / lines	0.50 / 0.50 (3 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	52.5 × 42.5	28.0	0.0
3	32.5 × 42.5	84.0	0.0
4	20.5 × 15.5	119.0	0.0
5	20.5 × 15.5	142.5	0.0
6	20.5 × 15.5	166.0	0.0
7	20.5 × 15.5	119.0	18.7
8	20.5 × 15.5	142.0	18.7
9	20.5 × 15.5	166.0	18.7
10	25.5 × 25.5	0.0	28.0
11	20.5 × 15.5	166.0	37.2
12	42.5 × 52.5	119.5	37.3
13	32.5 × 42.5	28.0	45.5
14	52.5 × 42.5	64.0	45.5
15	20.5 × 15.5	166.0	55.8
16	23.5 × 32.5	0.0	56.0
17	20.5 × 15.5	166.0	74.3
18	25.5 × 25.5	0.0	91.0
19	25.5 × 25.5	28.0	91.0
20	25.5 × 25.5	56.0	91.0
21	32.5 × 23.5	84.0	93.0
22	32.5 × 23.5	119.0	93.0
23	32.5 × 23.5	154.0	93.0

10 large-apart

Sub-analysis. Like the baseline, but the layout must contain 2 to 3 large frames (outer area of at least 2000 cm^2, i.e. the 52.5 × 42.5 and 42.5 × 52.5 frames) and they are kept apart: two large frames that are direct neighbours are penalised in proportion to the edge they share (barely touching costs little), and the closest pair of large frames is pulled towards a centre distance of at least 70 cm.

Parameters that differ from the baseline: large_area = 2000, large_count = 2-3, large_apart = True, large_min_dist = 70, entropy = 0.
256 runs, 20 min wall-clock budget, 535 candidate states, the 16 best distinct layouts are shown.

#	size (cm)	gap	frames	max dev	mean dev	notches	corners	entropy
1	182.0 × 129.0	4	25	3.0	0.60	none	4/4	0.65
2	191.5 × 113.5	1.5	24	2.5	0.73	4 cm	4/4	0.66
3	184.0 × 115.0	2	23	2.2	0.55	4 cm	4/4	0.68
4	177.5 × 113.5	1.5	21	3.0	0.52	none	4/4	0.66
5	191.5 × 121.5	1.5	26	3.5	0.51	none	4/4	0.69
6	182.5 × 130.5	1.5	26	3.5	0.57	none	4/4	0.81
7	179.5 × 120.5	1.5	26	4.0	0.54	none	4/4	0.69
8	176.5 × 125.5	1.5	25	3.0	1.04	4 cm	4/4	0.81
9	174.5 × 123.5	2.5	24	3.3	0.56	none	4/4	0.74
10	173.5 × 124.5	2.5	24	3.0	0.71	4 cm	4/4	0.81
11	169.0 × 116.5	2	24	2.5	0.69	8 cm	4/4	0.76
12	178.5 × 111.5	1.5	24	3.8	0.55	none	4/4	0.76
13	170.0 × 123.0	2	24	2.8	0.67	8 cm	4/4	0.82
14	174.5 × 120.5	1.5	24	3.0	0.44	7 cm	4/4	0.80
15	189.0 × 123.0	2	24	4.2	0.50	none	4/4	0.62
16	170.5 × 126.5	2.5	23	3.5	0.77	none	4/4	0.72

Layout 1 182.0 × 129.0 cm gap 4 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

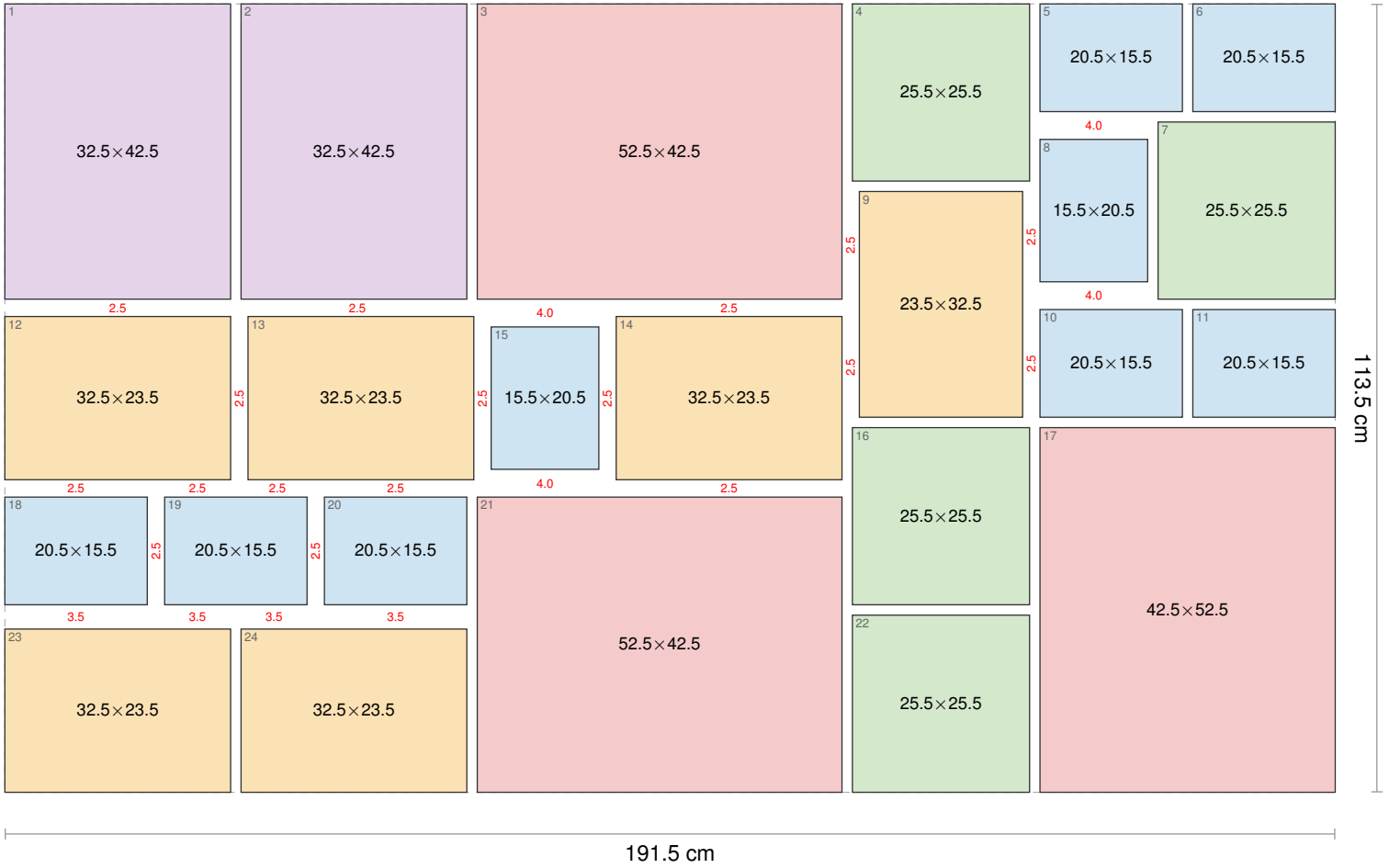
Outer size	182.0 × 129.0 cm
Nominal gap	4 cm
Actual gaps	4.0 – 7.0 cm
Max. deviation	3.0 cm
Mean deviation	0.60 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 132 cm
Entropy	0.65
mix / variety	0.83 / 0.54
stagger / lines	0.48 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 23.5	56.5	0.0
3	32.5 × 23.5	93.0	0.0
4	15.5 × 20.5	129.5	0.0
5	32.5 × 23.5	149.5	0.0
6	32.5 × 42.5	56.5	27.5
7	32.5 × 42.5	93.0	27.5
8	52.5 × 42.5	129.5	27.5
9	32.5 × 23.5	0.0	46.5
10	15.5 × 20.5	37.0	48.0
11	25.5 × 25.5	0.0	74.0
12	25.5 × 25.5	30.3	74.0
13	20.5 × 15.5	136.2	74.0
14	20.5 × 15.5	161.5	74.0
15	23.5 × 32.5	60.6	75.2
16	42.5 × 32.5	88.9	75.2
17	20.5 × 15.5	136.2	93.5
18	20.5 × 15.5	161.5	93.5
19	25.5 × 25.5	0.0	103.5
20	25.5 × 25.5	30.3	103.5
21	20.5 × 15.5	60.6	113.5
22	20.5 × 15.5	85.8	113.5
23	20.5 × 15.5	111.0	113.5
24	20.5 × 15.5	136.3	113.5
25	20.5 × 15.5	161.5	113.5

Layout 2 191.5 × 113.5 cm gap 1.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

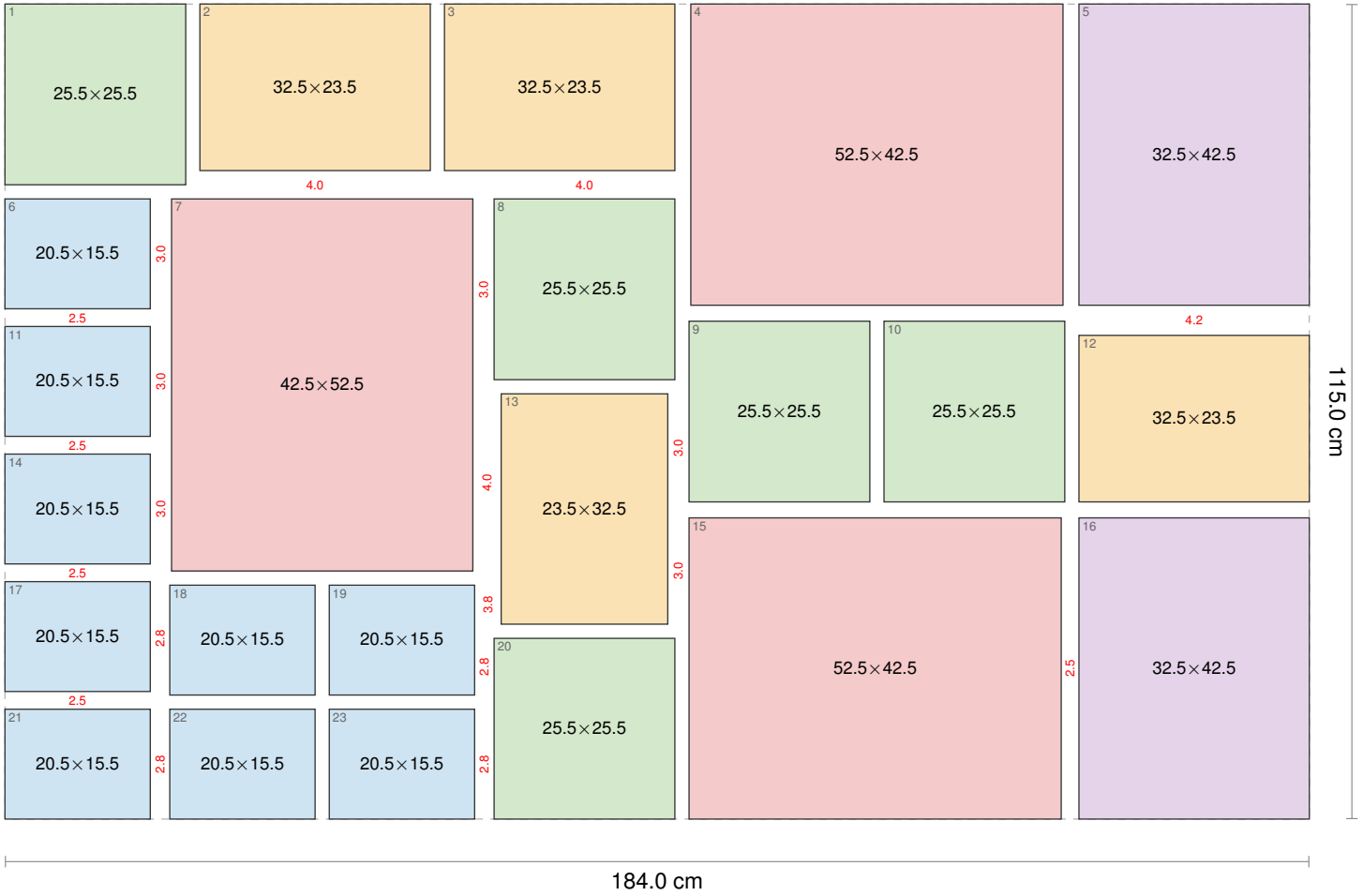
Outer size	191.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.73 cm
Border notches	left 4 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	3, not touching, min. dist. 71 cm
Entropy	0.66
mix / variety	0.85 / 0.80
stagger / lines	0.47 / 0.50 (3 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	32.5 × 42.5	34.0	0.0
3	52.5 × 42.5	68.0	0.0
4	25.5 × 25.5	122.0	0.0
5	20.5 × 15.5	149.0	0.0
6	20.5 × 15.5	171.0	0.0
7	25.5 × 25.5	166.0	17.0
8	15.5 × 20.5	149.0	19.5
9	23.5 × 32.5	123.0	27.0
10	20.5 × 15.5	149.0	44.0
11	20.5 × 15.5	171.0	44.0
12	32.5 × 23.5	0.0	45.0
13	32.5 × 23.5	35.0	45.0
14	32.5 × 23.5	88.0	45.0
15	15.5 × 20.5	70.0	46.5
16	25.5 × 25.5	122.0	61.0
17	42.5 × 52.5	149.0	61.0
18	20.5 × 15.5	0.0	71.0
19	20.5 × 15.5	23.0	71.0
20	20.5 × 15.5	46.0	71.0
21	52.5 × 42.5	68.0	71.0
22	25.5 × 25.5	122.0	88.0
23	32.5 × 23.5	0.0	90.0
24	32.5 × 23.5	34.0	90.0

Layout 3 184.0 × 115.0 cm gap 2 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

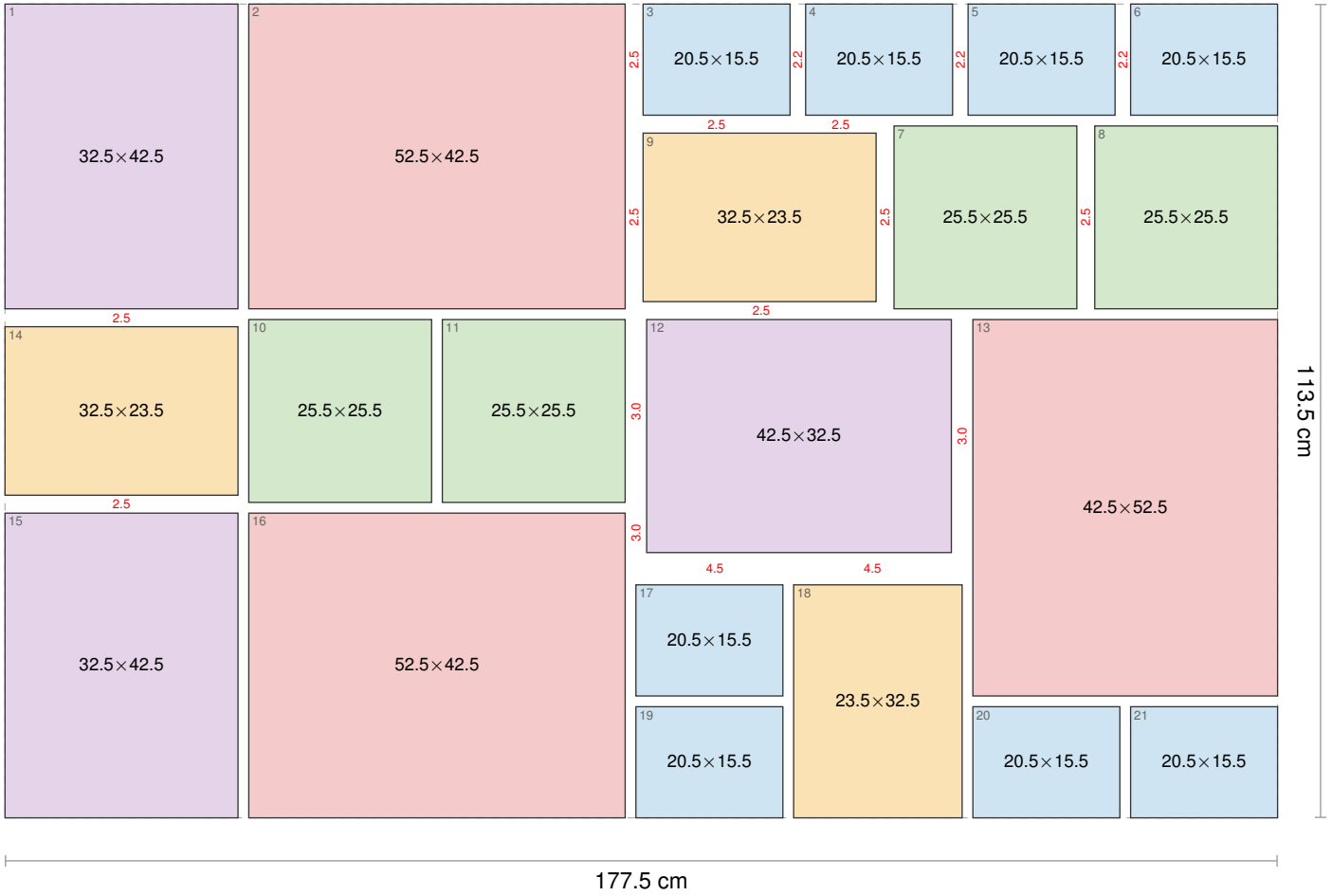
Outer size	184.0 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.2 cm
Max. deviation	2.2 cm
Mean deviation	0.55 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, not touching, min. dist. 73 cm
Entropy	0.68
mix / variety	0.76 / 0.66
stagger / lines	0.54 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	27.5	0.0
3	32.5 × 23.5	62.0	0.0
4	52.5 × 42.5	96.8	0.0
5	32.5 × 42.5	151.5	0.0
6	20.5 × 15.5	0.0	27.5
7	42.5 × 52.5	23.5	27.5
8	25.5 × 25.5	69.0	27.5
9	25.5 × 25.5	96.5	44.8
10	25.5 × 25.5	124.0	44.8
11	20.5 × 15.5	0.0	45.5
12	32.5 × 23.5	151.5	46.8
13	23.5 × 32.5	70.0	55.0
14	20.5 × 15.5	0.0	63.5
15	52.5 × 42.5	96.5	72.5
16	32.5 × 42.5	151.5	72.5
17	20.5 × 15.5	0.0	81.5
18	20.5 × 15.5	23.2	82.0
19	20.5 × 15.5	45.8	82.0
20	25.5 × 25.5	69.0	89.5
21	20.5 × 15.5	0.0	99.5
22	20.5 × 15.5	23.2	99.5
23	20.5 × 15.5	45.8	99.5

Layout 4 177.5 × 113.5 cm gap 1.5 cm 21 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

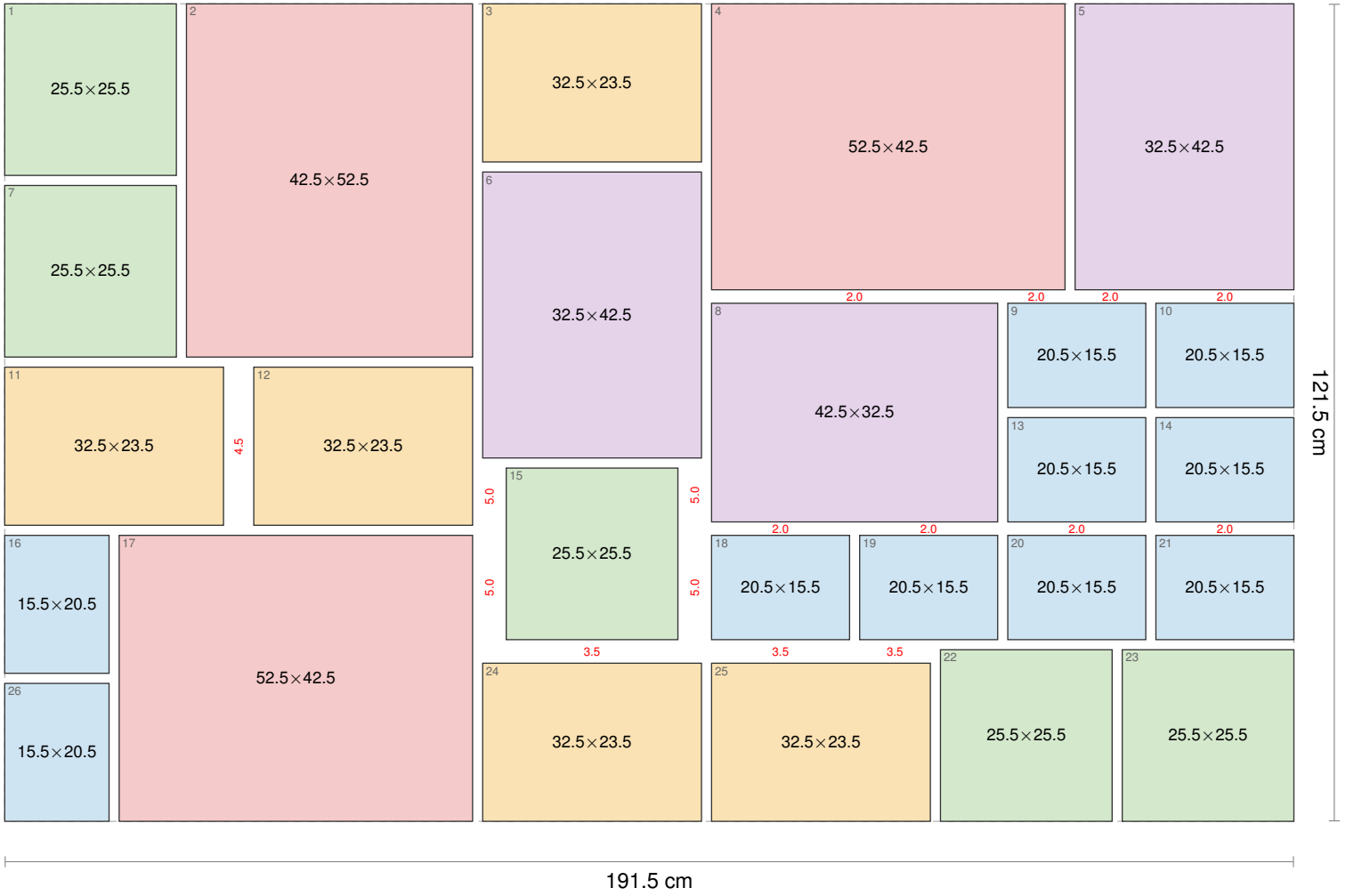
Outer size	177.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.5 cm
Max. deviation	3.0 cm
Mean deviation	0.52 cm
Border notches	none
Corners filled	4 / 4
Frames used	21 of 28
Large frames	3, not touching, min. dist. 71 cm
Entropy	0.66
mix / variety	0.81 / 0.81
stagger / lines	0.50 / 0.50 (3 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	2	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	34.0	0.0
3	20.5 × 15.5	89.0	0.0
4	20.5 × 15.5	111.7	0.0
5	20.5 × 15.5	134.3	0.0
6	20.5 × 15.5	157.0	0.0
7	25.5 × 25.5	124.0	17.0
8	25.5 × 25.5	152.0	17.0
9	32.5 × 23.5	89.0	18.0
10	25.5 × 25.5	34.0	44.0
11	25.5 × 25.5	61.0	44.0
12	42.5 × 32.5	89.5	44.0
13	42.5 × 52.5	135.0	44.0
14	32.5 × 23.5	0.0	45.0
15	32.5 × 42.5	0.0	71.0
16	52.5 × 42.5	34.0	71.0
17	20.5 × 15.5	88.0	81.0
18	23.5 × 32.5	110.0	81.0
19	20.5 × 15.5	88.0	98.0
20	20.5 × 15.5	135.0	98.0
21	20.5 × 15.5	157.0	98.0

Layout 5 191.5 × 121.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

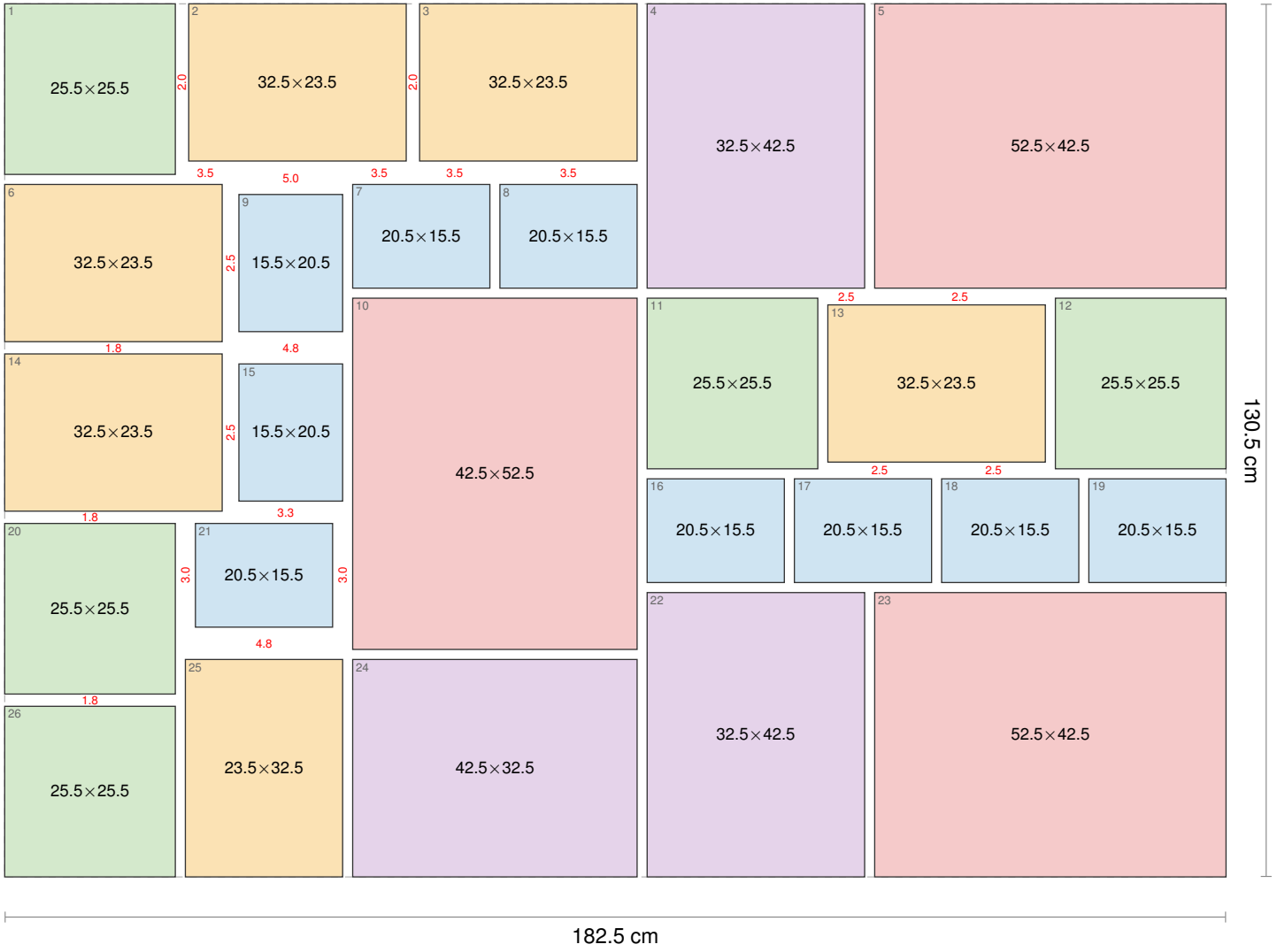
Outer size	191.5 × 121.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 5.0 cm
Max. deviation	3.5 cm
Mean deviation	0.51 cm
Border notches	none
Corners filled	4 / 4
Frames used	26 of 28
Large frames	3, not touching, min. dist. 74 cm
Entropy	0.69
mix / variety	0.84 / 0.71
stagger / lines	0.46 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	8	9
15.5×20.5	P	2	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	0	1
25.5×25.5		5	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	42.5 × 52.5	27.0	0.0
3	32.5 × 23.5	71.0	0.0
4	52.5 × 42.5	105.0	0.0
5	32.5 × 42.5	159.0	0.0
6	32.5 × 42.5	71.0	25.0
7	25.5 × 25.5	0.0	27.0
8	42.5 × 32.5	105.0	44.5
9	20.5 × 15.5	149.0	44.5
10	20.5 × 15.5	171.0	44.5
11	32.5 × 23.5	0.0	54.0
12	32.5 × 23.5	37.0	54.0
13	20.5 × 15.5	149.0	61.5
14	20.5 × 15.5	171.0	61.5
15	25.5 × 25.5	74.5	69.0
16	15.5 × 20.5	0.0	79.0
17	52.5 × 42.5	17.0	79.0
18	20.5 × 15.5	105.0	79.0
19	20.5 × 15.5	127.0	79.0
20	20.5 × 15.5	149.0	79.0
21	20.5 × 15.5	171.0	79.0
22	25.5 × 25.5	139.0	96.0
23	25.5 × 25.5	166.0	96.0
24	32.5 × 23.5	71.0	98.0
25	32.5 × 23.5	105.0	98.0
26	15.5 × 20.5	0.0	101.0

Layout 6 182.5 × 130.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

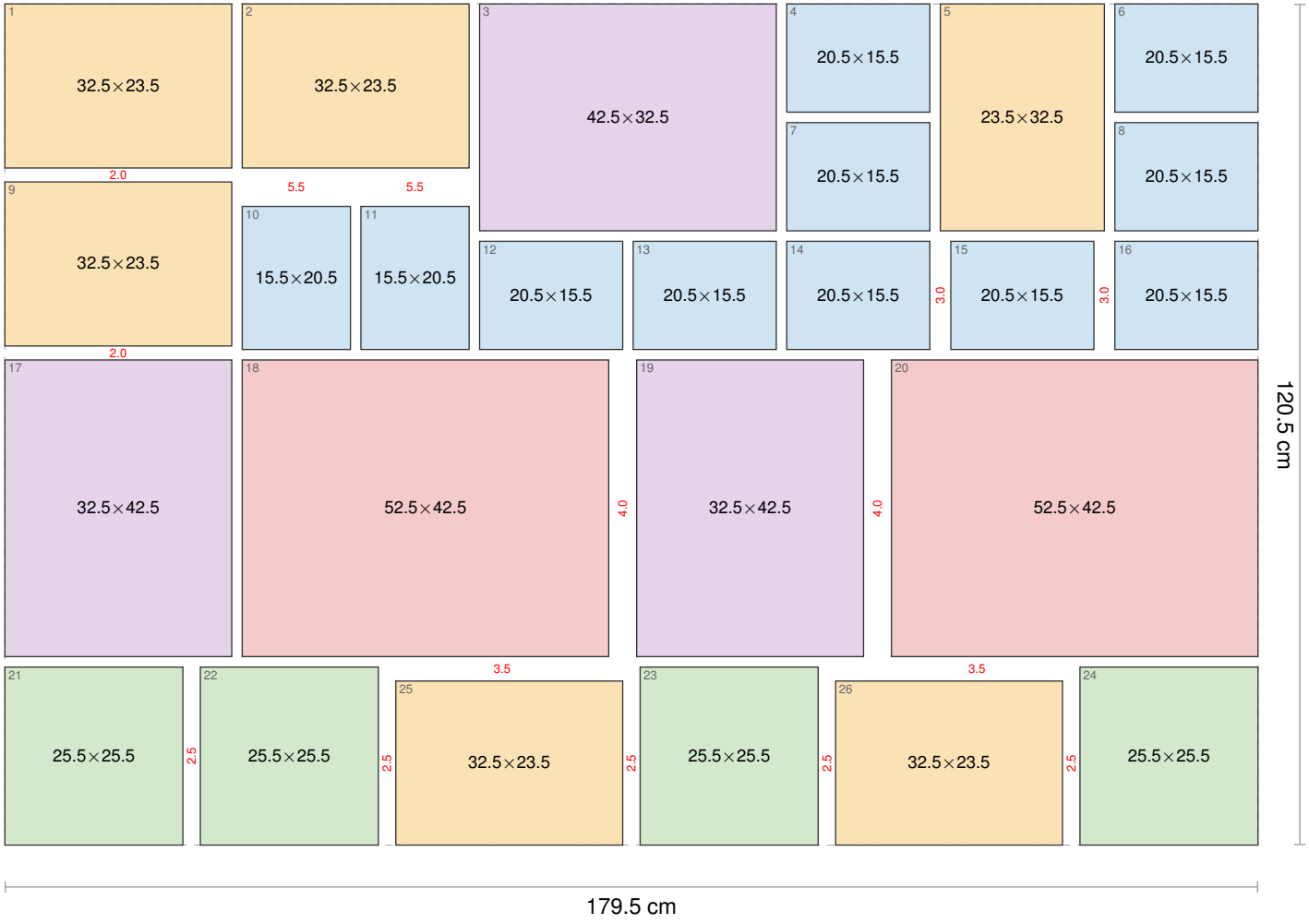
Outer size	182.5 × 130.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 5.0 cm
Max. deviation	3.5 cm
Mean deviation	0.57 cm
Border notches	none
Corners filled	4 / 4
Frames used	26 of 28
Large frames	3, not touching, min. dist. 88 cm
Entropy	0.81
mix / variety	0.89 / 0.84
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	27.5	0.0
3	32.5 × 23.5	62.0	0.0
4	32.5 × 42.5	96.0	0.0
5	52.5 × 42.5	130.0	0.0
6	32.5 × 23.5	0.0	27.0
7	20.5 × 15.5	52.0	27.0
8	20.5 × 15.5	74.0	27.0
9	15.5 × 20.5	35.0	28.5
10	42.5 × 52.5	52.0	44.0
11	25.5 × 25.5	96.0	44.0
12	25.5 × 25.5	157.0	44.0
13	32.5 × 23.5	123.0	45.0
14	32.5 × 23.5	0.0	52.3
15	15.5 × 20.5	35.0	53.8
16	20.5 × 15.5	96.0	71.0
17	20.5 × 15.5	118.0	71.0
18	20.5 × 15.5	140.0	71.0
19	20.5 × 15.5	162.0	71.0
20	25.5 × 25.5	0.0	77.7
21	20.5 × 15.5	28.5	77.7
22	32.5 × 42.5	96.0	88.0
23	52.5 × 42.5	130.0	88.0
24	42.5 × 32.5	52.0	98.0
25	23.5 × 32.5	27.0	98.0
26	25.5 × 25.5	0.0	105.0

Layout 7 179.5 × 120.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

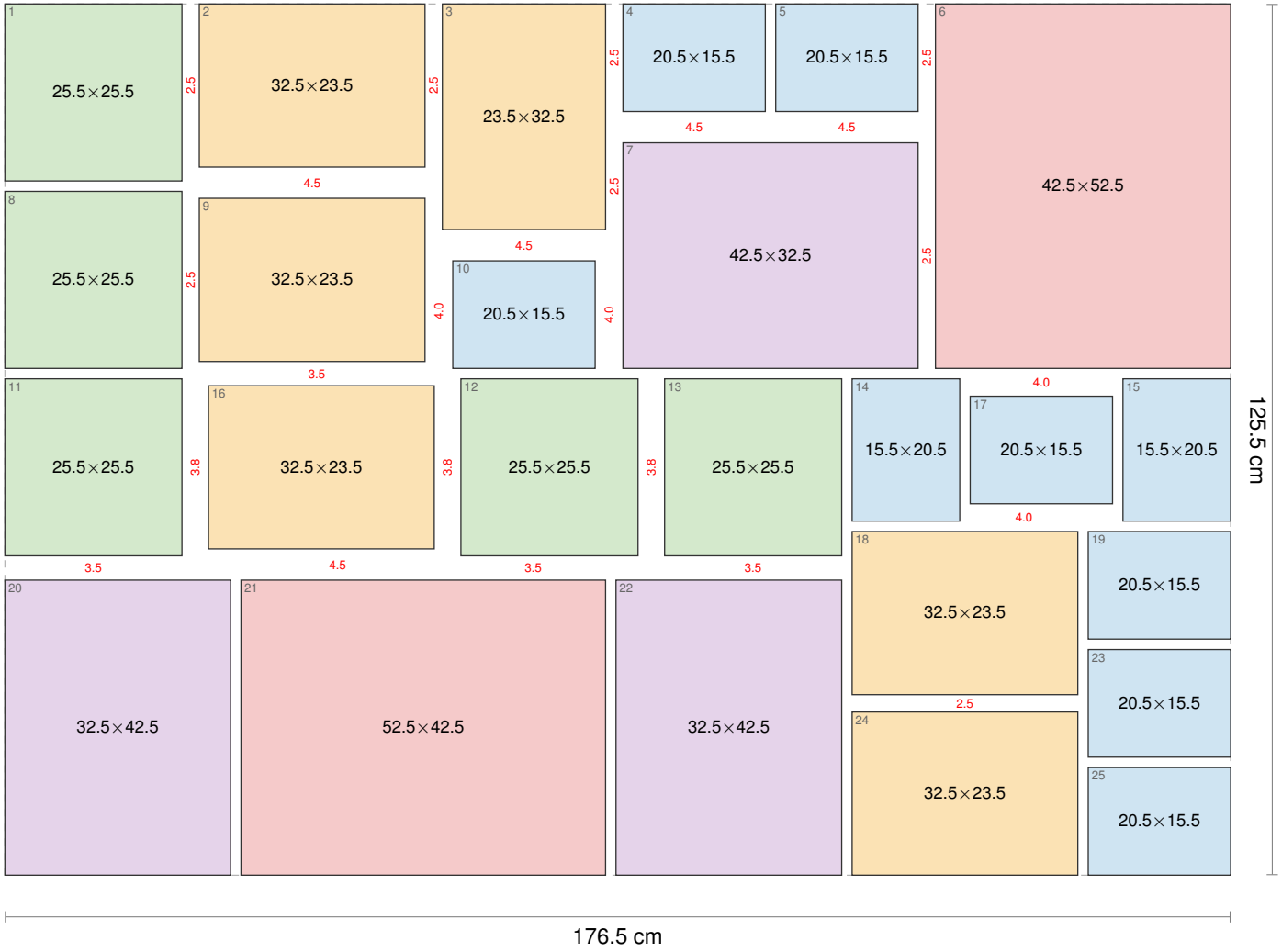
Outer size	179.5 × 120.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 5.5 cm
Max. deviation	4.0 cm
Mean deviation	0.54 cm
Border notches	none
Corners filled	4 / 4
Frames used	26 of 28
Large frames	2, not touching, min. dist. 93 cm
Entropy	0.69
mix / variety	0.83 / 0.75
stagger / lines	0.44 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	32.5 × 23.5	34.0	0.0
3	42.5 × 32.5	68.0	0.0
4	20.5 × 15.5	112.0	0.0
5	23.5 × 32.5	134.0	0.0
6	20.5 × 15.5	159.0	0.0
7	20.5 × 15.5	112.0	17.0
8	20.5 × 15.5	159.0	17.0
9	32.5 × 23.5	0.0	25.5
10	15.5 × 20.5	34.0	29.0
11	15.5 × 20.5	51.0	29.0
12	20.5 × 15.5	68.0	34.0
13	20.5 × 15.5	90.0	34.0
14	20.5 × 15.5	112.0	34.0
15	20.5 × 15.5	135.5	34.0
16	20.5 × 15.5	159.0	34.0
17	32.5 × 42.5	0.0	51.0
18	52.5 × 42.5	34.0	51.0
19	32.5 × 42.5	90.5	51.0
20	52.5 × 42.5	127.0	51.0
21	25.5 × 25.5	0.0	95.0
22	25.5 × 25.5	28.0	95.0
23	25.5 × 25.5	91.0	95.0
24	25.5 × 25.5	154.0	95.0
25	32.5 × 23.5	56.0	97.0
26	32.5 × 23.5	119.0	97.0

Layout 8 176.5 × 125.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

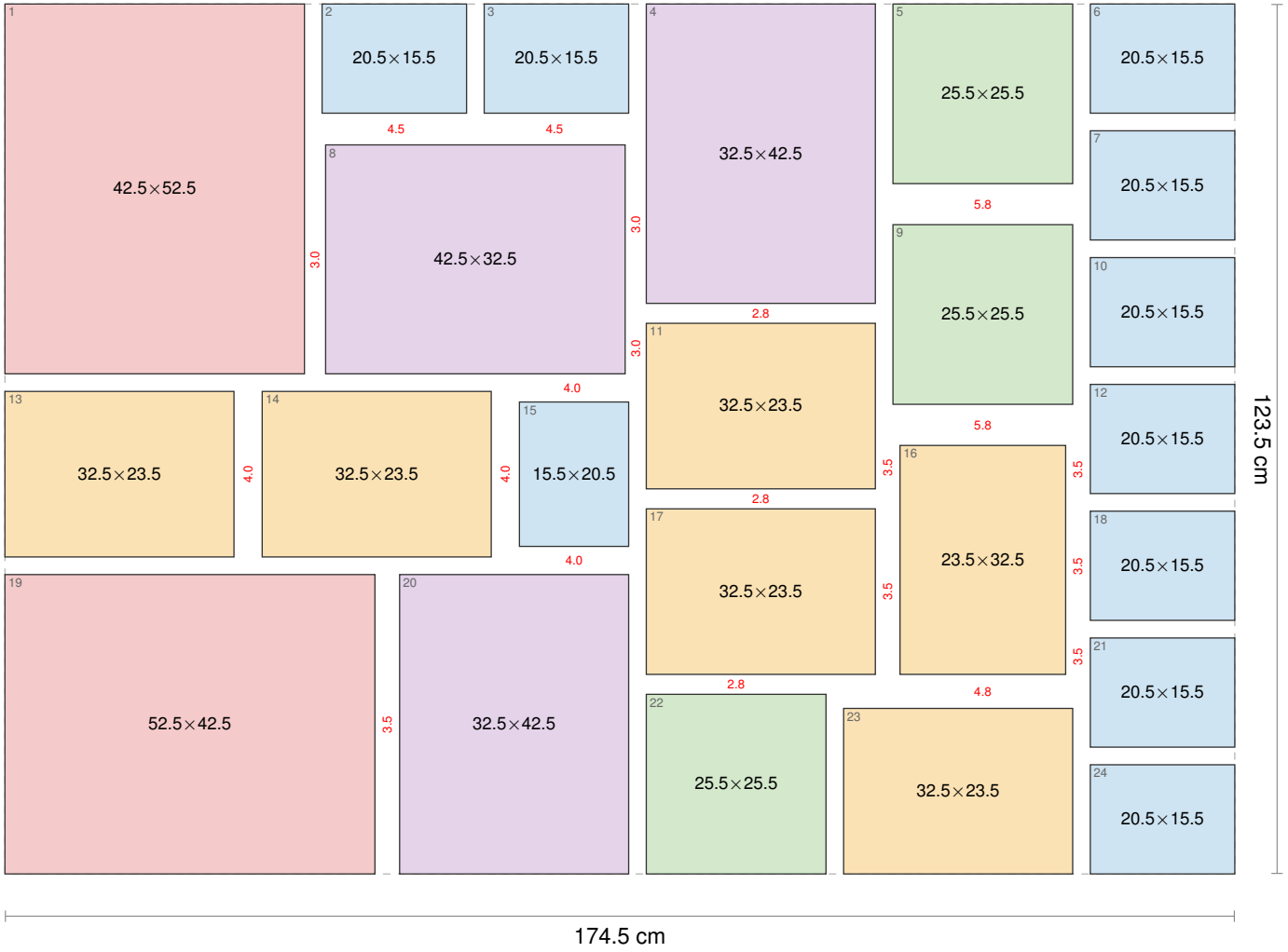
Outer size	176.5 × 125.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.5 cm
Max. deviation	3.0 cm
Mean deviation	1.04 cm
Border notches	left 4 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 123 cm
Entropy	0.81
mix / variety	0.87 / 0.80
stagger / lines	0.57 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	28.0	0.0
3	23.5 × 32.5	63.0	0.0
4	20.5 × 15.5	89.0	0.0
5	20.5 × 15.5	111.0	0.0
6	42.5 × 52.5	134.0	0.0
7	42.5 × 32.5	89.0	20.0
8	25.5 × 25.5	0.0	27.0
9	32.5 × 23.5	28.0	28.0
10	20.5 × 15.5	64.5	37.0
11	25.5 × 25.5	0.0	54.0
12	25.5 × 25.5	65.7	54.0
13	25.5 × 25.5	95.0	54.0
14	15.5 × 20.5	122.0	54.0
15	15.5 × 20.5	161.0	54.0
16	32.5 × 23.5	29.3	55.0
17	20.5 × 15.5	139.0	56.5
18	32.5 × 23.5	122.0	76.0
19	20.5 × 15.5	156.0	76.0
20	32.5 × 42.5	0.0	83.0
21	52.5 × 42.5	34.0	83.0
22	32.5 × 42.5	88.0	83.0
23	20.5 × 15.5	156.0	93.0
24	32.5 × 23.5	122.0	102.0
25	20.5 × 15.5	156.0	110.0

Layout 9 174.5 × 123.5 cm gap 2.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	174.5 × 123.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.8 cm
Max. deviation	3.3 cm
Mean deviation	0.56 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 76 cm
Entropy	0.74
mix / variety	0.83 / 0.80
stagger / lines	0.58 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		3	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	42.5 × 52.5	0.0	0.0
2	20.5 × 15.5	45.0	0.0
3	20.5 × 15.5	68.0	0.0
4	32.5 × 42.5	91.0	0.0
5	25.5 × 25.5	126.0	0.0
6	20.5 × 15.5	154.0	0.0
7	20.5 × 15.5	154.0	18.0
8	42.5 × 32.5	45.5	20.0
9	25.5 × 25.5	126.0	31.3
10	20.5 × 15.5	154.0	36.0
11	32.5 × 23.5	91.0	45.3
12	20.5 × 15.5	154.0	54.0
13	32.5 × 23.5	0.0	55.0
14	32.5 × 23.5	36.5	55.0
15	15.5 × 20.5	73.0	56.5
16	23.5 × 32.5	127.0	62.7
17	32.5 × 23.5	91.0	71.7
18	20.5 × 15.5	154.0	72.0
19	52.5 × 42.5	0.0	81.0
20	32.5 × 42.5	56.0	81.0
21	20.5 × 15.5	154.0	90.0
22	25.5 × 25.5	91.0	98.0
23	32.5 × 23.5	119.0	100.0
24	20.5 × 15.5	154.0	108.0

Layout 10 173.5 × 124.5 cm gap 2.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

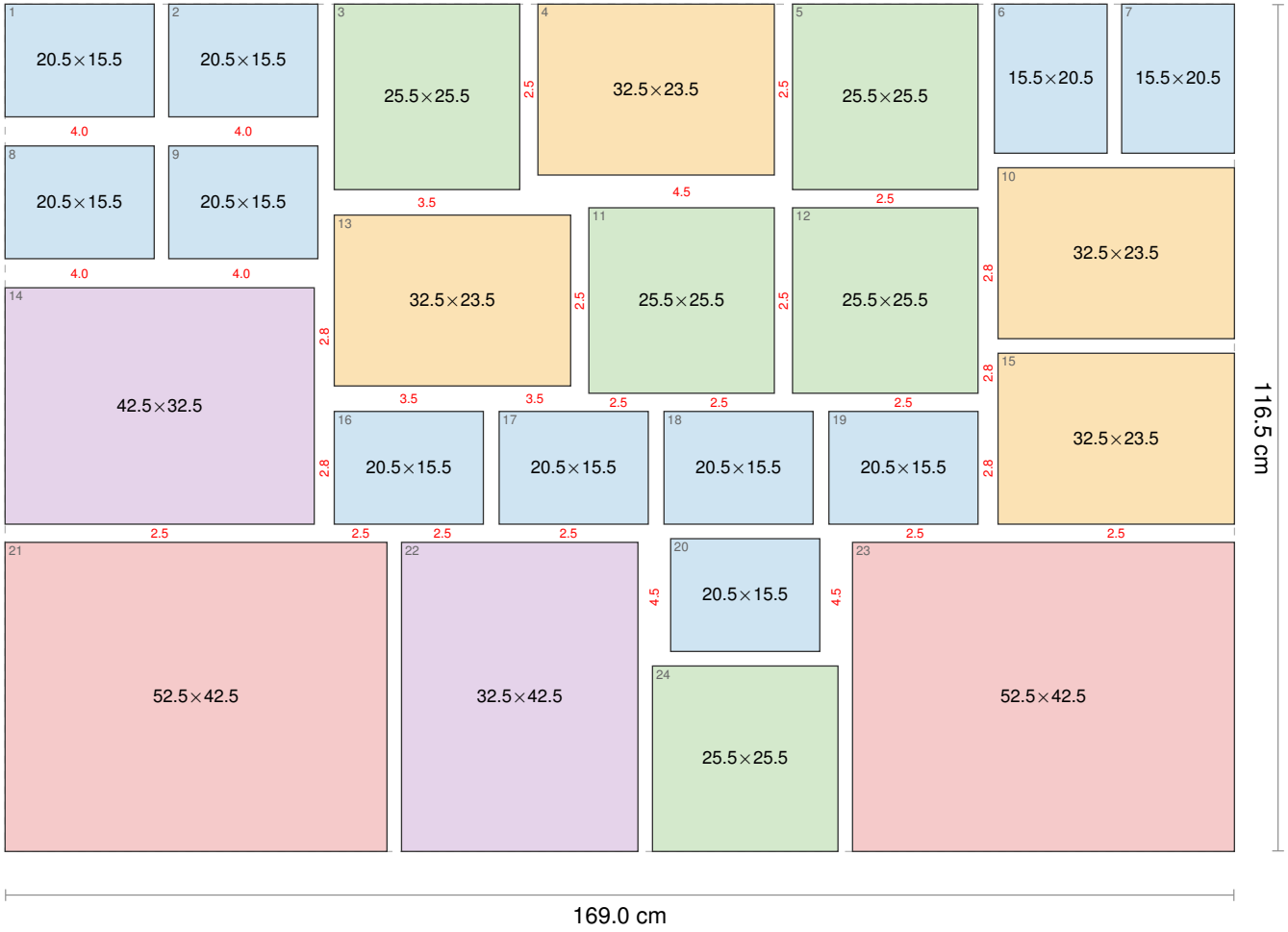
Outer size	173.5 × 124.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.5 cm
Max. deviation	3.0 cm
Mean deviation	0.71 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 85 cm
Entropy	0.81
mix / variety	0.86 / 0.79
stagger / lines	0.57 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		3	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	15.5 × 20.5	35.0	0.0
3	15.5 × 20.5	53.0	0.0
4	32.5 × 23.5	71.0	0.0
5	32.5 × 23.5	106.0	0.0
6	32.5 × 23.5	141.0	0.0
7	25.5 × 25.5	0.0	26.0
8	25.5 × 25.5	30.0	26.0
9	42.5 × 52.5	58.0	26.0
10	23.5 × 32.5	103.5	26.0
11	20.5 × 15.5	130.0	26.0
12	20.5 × 15.5	153.0	26.0
13	42.5 × 32.5	131.0	46.0
14	20.5 × 15.5	0.0	54.5
15	32.5 × 23.5	23.0	54.5
16	20.5 × 15.5	105.0	63.0
17	20.5 × 15.5	0.0	72.7
18	20.5 × 15.5	60.5	81.0
19	32.5 × 42.5	23.0	82.0
20	32.5 × 42.5	86.0	82.0
21	52.5 × 42.5	121.0	82.0
22	20.5 × 15.5	0.0	90.8
23	25.5 × 25.5	58.0	99.0
24	20.5 × 15.5	0.0	109.0

Layout 11 169.0 × 116.5 cm gap 2 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

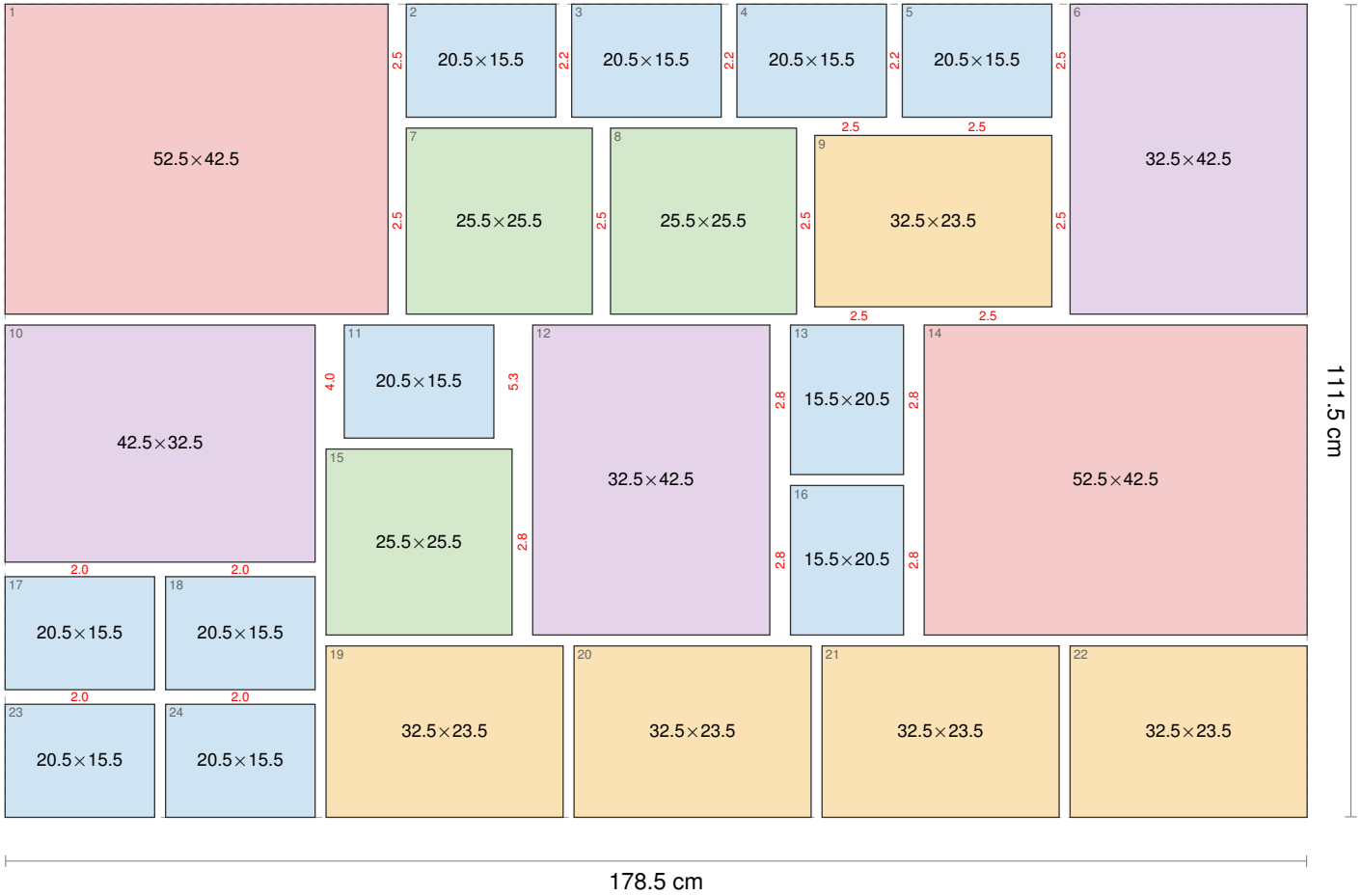
Outer size	169.0 × 116.5 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.5 cm
Max. deviation	2.5 cm
Mean deviation	0.69 cm
Border notches	left 8 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 116 cm
Entropy	0.76
mix / variety	0.76 / 0.73
stagger / lines	0.56 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	1	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	0	1
	25.5×25.5		5	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	22.5	0.0
3	25.5 × 25.5	45.2	0.0
4	32.5 × 23.5	73.2	0.0
5	25.5 × 25.5	108.2	0.0
6	15.5 × 20.5	136.0	0.0
7	15.5 × 20.5	153.5	0.0
8	20.5 × 15.5	0.0	19.5
9	20.5 × 15.5	22.5	19.5
10	32.5 × 23.5	136.5	22.5
11	25.5 × 25.5	80.2	28.0
12	25.5 × 25.5	108.2	28.0
13	32.5 × 23.5	45.2	29.0
14	42.5 × 32.5	0.0	39.0
15	32.5 × 23.5	136.5	48.0
16	20.5 × 15.5	45.2	56.0
17	20.5 × 15.5	67.9	56.0
18	20.5 × 15.5	90.6	56.0
19	20.5 × 15.5	113.2	56.0
20	20.5 × 15.5	91.5	73.5
21	52.5 × 42.5	0.0	74.0
22	32.5 × 42.5	54.5	74.0
23	52.5 × 42.5	116.5	74.0
24	25.5 × 25.5	89.0	91.0

Layout 12 178.5 × 111.5 cm gap 1.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

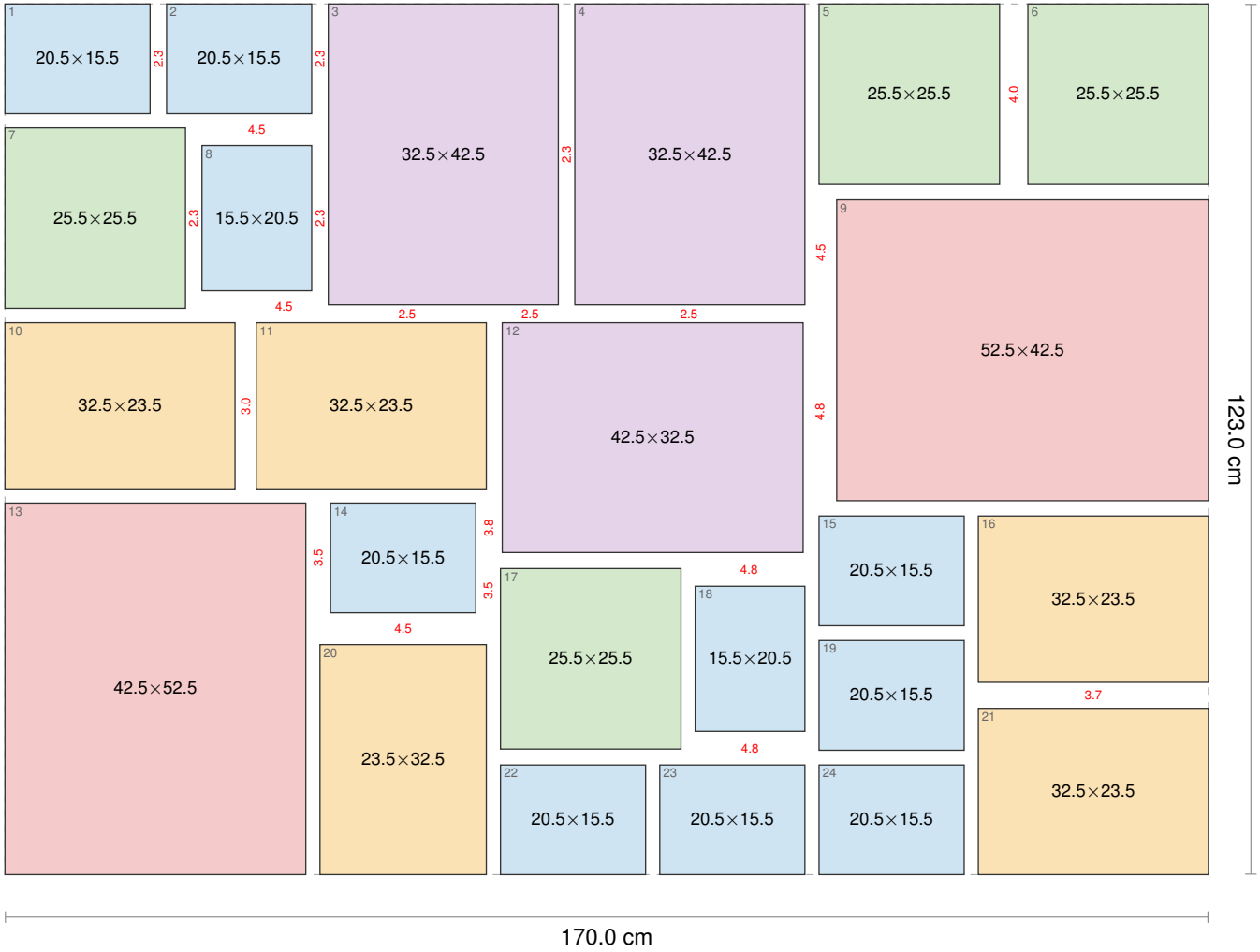
Outer size	178.5 × 111.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 5.3 cm
Max. deviation	3.8 cm
Mean deviation	0.55 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 133 cm
Entropy	0.76
mix / variety	0.78 / 0.71
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	0	1
	25.5×25.5		3	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	20.5 × 15.5	55.0	0.0
3	20.5 × 15.5	77.7	0.0
4	20.5 × 15.5	100.3	0.0
5	20.5 × 15.5	123.0	0.0
6	32.5 × 42.5	146.0	0.0
7	25.5 × 25.5	55.0	17.0
8	25.5 × 25.5	83.0	17.0
9	32.5 × 23.5	111.0	18.0
10	42.5 × 32.5	0.0	44.0
11	20.5 × 15.5	46.5	44.0
12	32.5 × 42.5	72.3	44.0
13	15.5 × 20.5	107.7	44.0
14	52.5 × 42.5	126.0	44.0
15	25.5 × 25.5	44.0	61.0
16	15.5 × 20.5	107.7	66.0
17	20.5 × 15.5	0.0	78.5
18	20.5 × 15.5	22.0	78.5
19	32.5 × 23.5	44.0	88.0
20	32.5 × 23.5	78.0	88.0
21	32.5 × 23.5	112.0	88.0
22	32.5 × 23.5	146.0	88.0
23	20.5 × 15.5	0.0	96.0
24	20.5 × 15.5	22.0	96.0

Layout 13 170.0 × 123.0 cm gap 2 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

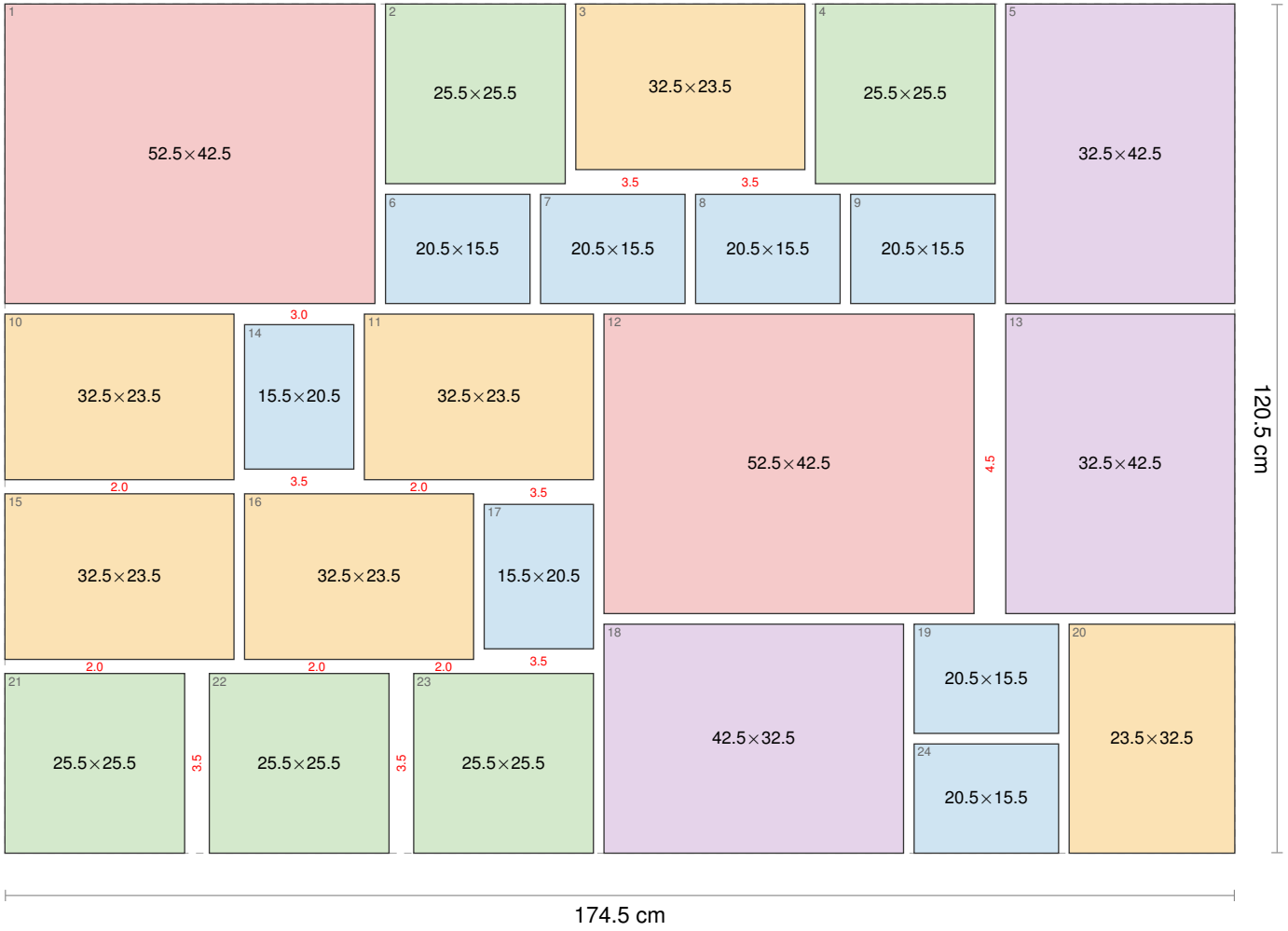
Outer size	170.0 × 123.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.8 cm
Max. deviation	2.8 cm
Mean deviation	0.67 cm
Border notches	top 4 cm, right 4 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 132 cm
Entropy	0.82
mix / variety	0.87 / 0.81
stagger / lines	0.61 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	22.8	0.0
3	32.5 × 42.5	45.7	0.0
4	32.5 × 42.5	80.5	0.0
5	25.5 × 25.5	115.0	0.0
6	25.5 × 25.5	144.5	0.0
7	25.5 × 25.5	0.0	17.5
8	15.5 × 20.5	27.8	20.0
9	52.5 × 42.5	117.5	27.7
10	32.5 × 23.5	0.0	45.0
11	32.5 × 23.5	35.5	45.0
12	42.5 × 32.5	70.2	45.0
13	42.5 × 52.5	0.0	70.5
14	20.5 × 15.5	46.0	70.5
15	20.5 × 15.5	115.0	72.3
16	32.5 × 23.5	137.5	72.3
17	25.5 × 25.5	70.0	79.8
18	15.5 × 20.5	97.5	82.2
19	20.5 × 15.5	115.0	89.9
20	23.5 × 32.5	44.5	90.5
21	32.5 × 23.5	137.5	99.5
22	20.5 × 15.5	70.0	107.5
23	20.5 × 15.5	92.5	107.5
24	20.5 × 15.5	115.0	107.5

Layout 14 174.5 × 120.5 cm gap 1.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

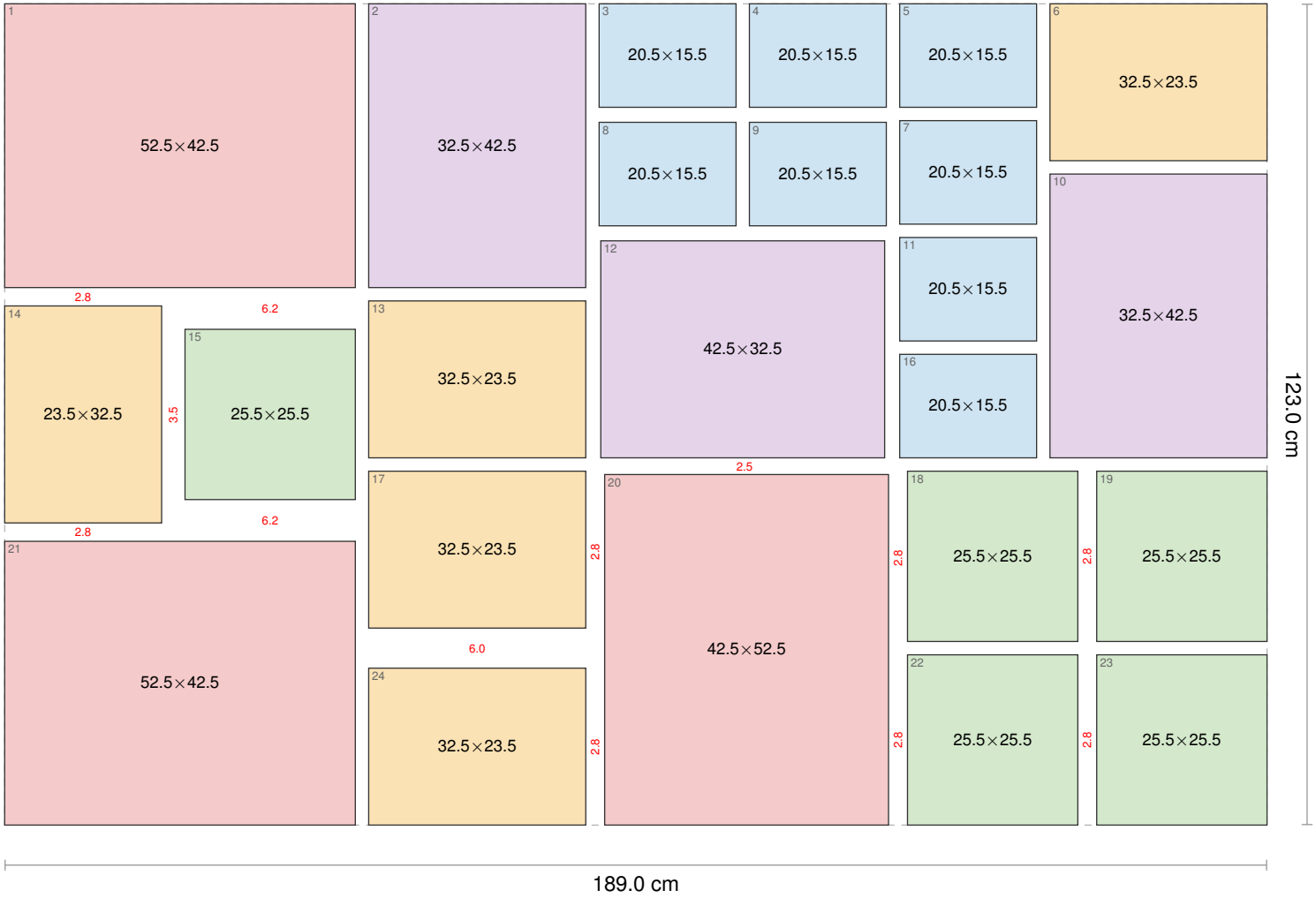
Outer size	174.5 × 120.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.5 cm
Max. deviation	3.0 cm
Mean deviation	0.44 cm
Border notches	bottom 7 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 96 cm
Entropy	0.80
mix / variety	0.86 / 0.79
stagger / lines	0.54 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	6	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	25.5 × 25.5	54.0	0.0
3	32.5 × 23.5	81.0	0.0
4	25.5 × 25.5	115.0	0.0
5	32.5 × 42.5	142.0	0.0
6	20.5 × 15.5	54.0	27.0
7	20.5 × 15.5	76.0	27.0
8	20.5 × 15.5	98.0	27.0
9	20.5 × 15.5	120.0	27.0
10	32.5 × 23.5	0.0	44.0
11	32.5 × 23.5	51.0	44.0
12	52.5 × 42.5	85.0	44.0
13	32.5 × 42.5	142.0	44.0
14	15.5 × 20.5	34.0	45.5
15	32.5 × 23.5	0.0	69.5
16	32.5 × 23.5	34.0	69.5
17	15.5 × 20.5	68.0	71.0
18	42.5 × 32.5	85.0	88.0
19	20.5 × 15.5	129.0	88.0
20	23.5 × 32.5	151.0	88.0
21	25.5 × 25.5	0.0	95.0
22	25.5 × 25.5	29.0	95.0
23	25.5 × 25.5	58.0	95.0
24	20.5 × 15.5	129.0	105.0

Layout 15 189.0 × 123.0 cm gap 2 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	189.0 × 123.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 6.2 cm
Max. deviation	4.2 cm
Mean deviation	0.50 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	3, not touching, min. dist. 80 cm
Entropy	0.62
mix / variety	0.82 / 0.67
stagger / lines	0.50 / 0.50 (3 through)

outer w × h (cm)		used	have
20.5×15.5	L	8	9
15.5×20.5	P	0	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	4	5
23.5×32.5	P	1	1
25.5×25.5		5	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 42.5	54.5	0.0
3	20.5 × 15.5	89.0	0.0
4	20.5 × 15.5	111.5	0.0
5	20.5 × 15.5	134.0	0.0
6	32.5 × 23.5	156.5	0.0
7	20.5 × 15.5	134.0	17.5
8	20.5 × 15.5	89.0	17.8
9	20.5 × 15.5	111.5	17.8
10	32.5 × 42.5	156.5	25.5
11	20.5 × 15.5	134.0	35.0
12	42.5 × 32.5	89.2	35.5
13	32.5 × 23.5	54.5	44.5
14	23.5 × 32.5	0.0	45.2
15	25.5 × 25.5	27.0	48.8
16	20.5 × 15.5	134.0	52.5
17	32.5 × 23.5	54.5	70.0
18	25.5 × 25.5	135.2	70.0
19	25.5 × 25.5	163.5	70.0
20	42.5 × 52.5	89.8	70.5
21	52.5 × 42.5	0.0	80.5
22	25.5 × 25.5	135.2	97.5
23	25.5 × 25.5	163.5	97.5
24	32.5 × 23.5	54.5	99.5

Layout 16 170.5 × 126.5 cm gap 2.5 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	170.5 × 126.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 6.0 cm
Max. deviation	3.5 cm
Mean deviation	0.77 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, not touching, min. dist. 79 cm
Entropy	0.72
mix / variety	0.79 / 0.80
stagger / lines	0.55 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	1	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	0	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	25.5 × 25.5	55.0	0.0
3	32.5 × 42.5	83.0	0.0
4	52.5 × 42.5	118.0	0.0
5	20.5 × 15.5	57.5	28.0
6	25.5 × 25.5	0.0	46.0
7	25.5 × 25.5	64.0	46.0
8	32.5 × 23.5	92.0	46.0
9	25.5 × 25.5	145.0	46.0
10	32.5 × 23.5	28.5	47.0
11	15.5 × 20.5	127.0	48.5
12	42.5 × 52.5	128.0	74.0
13	20.5 × 15.5	0.0	74.5
14	42.5 × 32.5	23.5	74.5
15	20.5 × 15.5	69.0	74.5
16	32.5 × 23.5	92.0	74.5
17	20.5 × 15.5	0.0	92.8
18	20.5 × 15.5	69.0	92.8
19	32.5 × 23.5	92.0	103.0
20	20.5 × 15.5	0.0	111.0
21	20.5 × 15.5	23.0	111.0
22	20.5 × 15.5	46.0	111.0
23	20.5 × 15.5	69.0	111.0

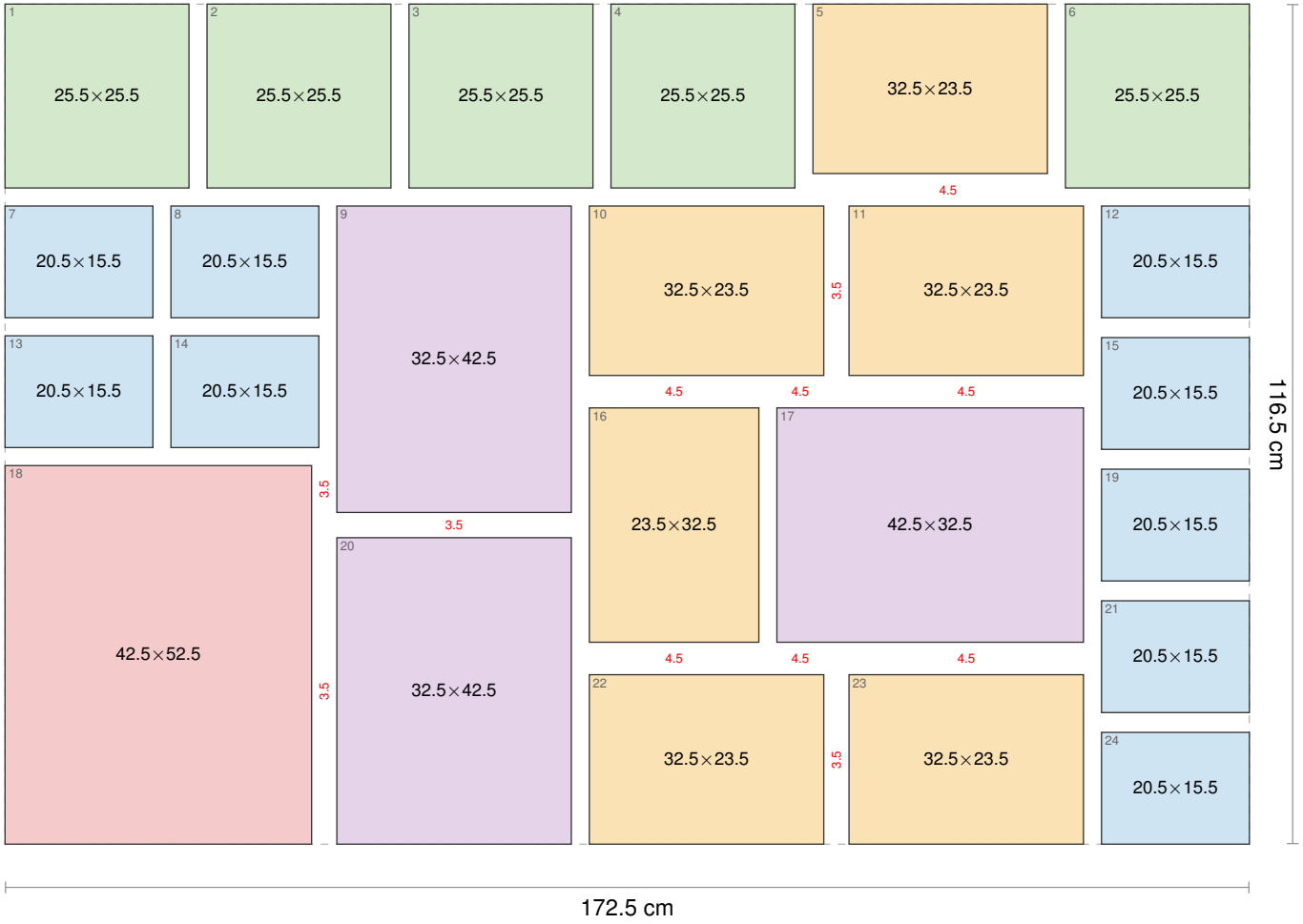
11 entropy

Sub-analysis. Like the baseline, but the cost also rewards the irregularity (“entropy”) of the constellation: a mix of many formats, different formats next to each other, frame edges that do not line up, and few straight gap lines running across the whole collage.

Parameters that differ from the baseline: large_apart = False, large_min_dist = 70, entropy = 40.
256 runs, 20 min wall-clock budget, 530 candidate states, the 16 best distinct layouts are shown.

#	size (cm)	gap	frames	max dev	mean dev	notches	corners	entropy
1	172.5 × 116.5	2.5	24	2.0	0.43	none	4/4	0.72
2	181.5 × 113.5	1.5	20	2.0	0.62	none	4/4	0.78
3	187.5 × 113.5	1.5	25	2.5	0.51	4 cm	4/4	0.80
4	187.5 × 113.5	1.5	25	2.5	0.54	4 cm	4/4	0.79
5	177.5 × 126.5	2.5	25	3.0	0.79	none	4/4	0.79
6	172.5 × 124.5	2.5	23	2.5	0.80	none	4/4	0.75
7	176.0 × 123.0	2	23	3.0	0.45	none	4/4	0.82
8	184.0 × 115.0	2	23	2.0	0.54	6 cm	4/4	0.76
9	177.5 × 113.5	1.5	21	2.0	0.56	6 cm	4/4	0.78
10	170.5 × 115.5	2	21	3.0	0.80	none	4/4	0.79
11	170.5 × 117.5	2.5	19	3.0	0.67	none	4/4	0.84
12	179.5 × 120.5	1.5	26	2.5	0.55	10 cm	4/4	0.80
13	179.5 × 120.5	1.5	26	3.0	0.56	7 cm	4/4	0.79
14	172.5 × 116.5	2.5	25	3.0	0.53	4 cm	4/4	0.78
15	177.0 × 123.0	2	25	2.0	0.64	12 cm	4/4	0.79
16	187.5 × 125.0	2	25	3.8	0.86	none	4/4	0.78

Layout 1 172.5 × 116.5 cm gap 2.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	172.5 × 116.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 4.5 cm
Max. deviation	2.0 cm
Mean deviation	0.43 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	1
Entropy	0.72
mix / variety	0.74 / 0.65
stagger / lines	0.49 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	28.0	0.0
3	25.5 × 25.5	56.0	0.0
4	25.5 × 25.5	84.0	0.0
5	32.5 × 23.5	112.0	0.0
6	25.5 × 25.5	147.0	0.0
7	20.5 × 15.5	0.0	28.0
8	20.5 × 15.5	23.0	28.0
9	32.5 × 42.5	46.0	28.0
10	32.5 × 23.5	81.0	28.0
11	32.5 × 23.5	117.0	28.0
12	20.5 × 15.5	152.0	28.0
13	20.5 × 15.5	0.0	46.0
14	20.5 × 15.5	23.0	46.0
15	20.5 × 15.5	152.0	46.2
16	23.5 × 32.5	81.0	56.0
17	42.5 × 32.5	107.0	56.0
18	42.5 × 52.5	0.0	64.0
19	20.5 × 15.5	152.0	64.5
20	32.5 × 42.5	46.0	74.0
21	20.5 × 15.5	152.0	82.8
22	32.5 × 23.5	81.0	93.0
23	32.5 × 23.5	117.0	93.0
24	20.5 × 15.5	152.0	101.0

Layout 2 181.5 × 113.5 cm gap 1.5 cm 20 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

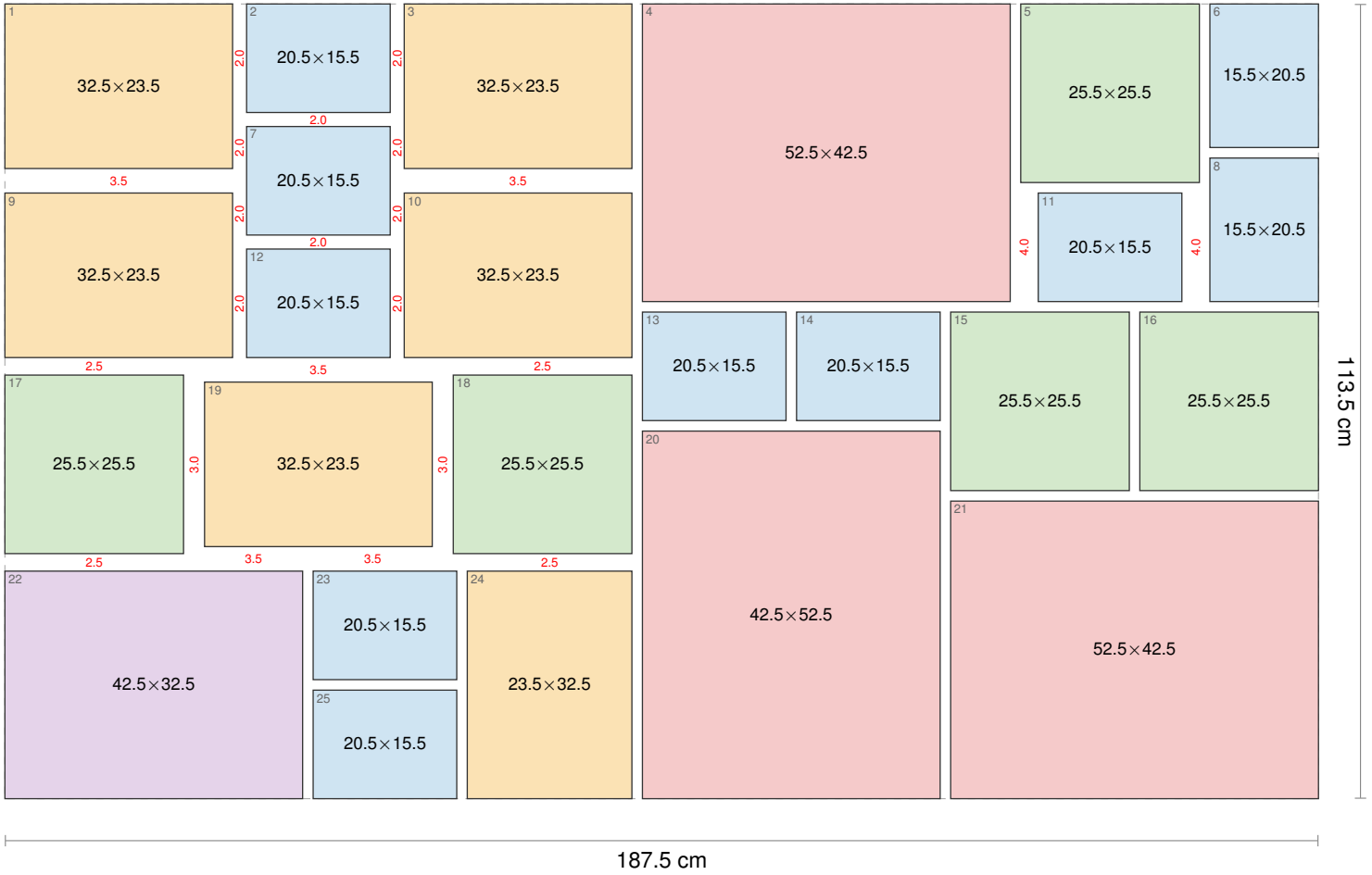
Outer size	181.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.62 cm
Border notches	none
Corners filled	4 / 4
Frames used	20 of 28
Large frames	3, touching 2× (78 cm), min. dist. 44 cm
Entropy	0.78
mix / variety	0.86 / 0.78
stagger / lines	0.49 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	5	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 23.5	54.0	0.0
3	32.5 × 23.5	88.0	0.0
4	32.5 × 23.5	122.0	0.0
5	25.5 × 25.5	156.0	0.0
6	23.5 × 32.5	54.0	27.0
7	42.5 × 32.5	80.7	27.0
8	20.5 × 15.5	126.3	27.0
9	32.5 × 42.5	149.0	27.0
10	52.5 × 42.5	0.0	44.0
11	20.5 × 15.5	126.3	44.0
12	42.5 × 52.5	54.0	61.0
13	20.5 × 15.5	98.7	61.0
14	25.5 × 25.5	121.3	61.0
15	32.5 × 42.5	149.0	71.0
16	20.5 × 15.5	98.7	79.5
17	25.5 × 25.5	0.0	88.0
18	25.5 × 25.5	27.0	88.0
19	25.5 × 25.5	121.3	88.0
20	20.5 × 15.5	98.7	98.0

Layout 3 187.5 × 113.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

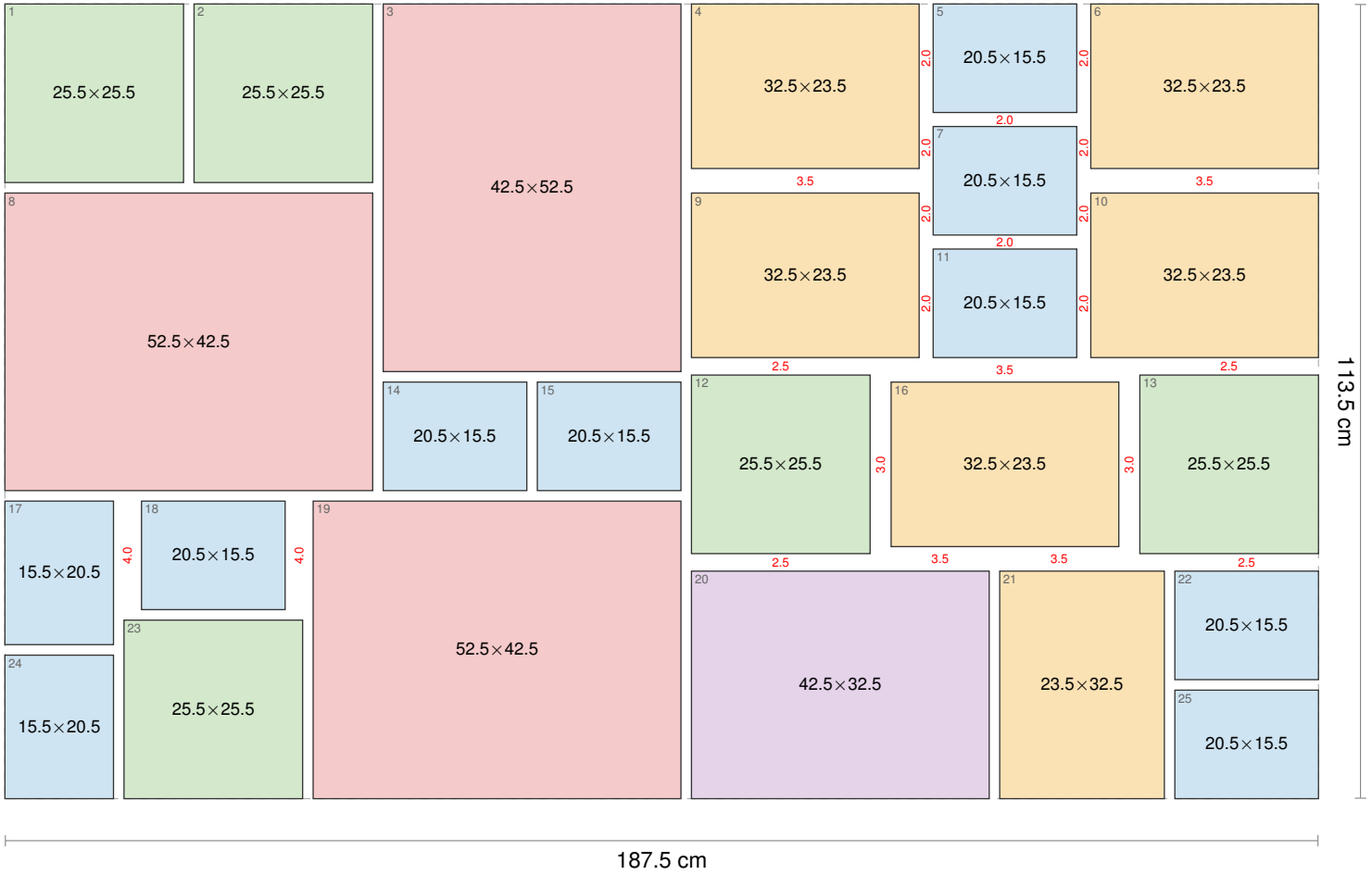
Outer size	187.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.51 cm
Border notches	left 4 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, touching 1× (42 cm), min. dist. 49 cm
Entropy	0.80
mix / variety	0.82 / 0.86
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	0	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	20.5 × 15.5	34.5	0.0
3	32.5 × 23.5	57.0	0.0
4	52.5 × 42.5	91.0	0.0
5	25.5 × 25.5	145.0	0.0
6	15.5 × 20.5	172.0	0.0
7	20.5 × 15.5	34.5	17.5
8	15.5 × 20.5	172.0	22.0
9	32.5 × 23.5	0.0	27.0
10	32.5 × 23.5	57.0	27.0
11	20.5 × 15.5	147.5	27.0
12	20.5 × 15.5	34.5	35.0
13	20.5 × 15.5	91.0	44.0
14	20.5 × 15.5	113.0	44.0
15	25.5 × 25.5	135.0	44.0
16	25.5 × 25.5	162.0	44.0
17	25.5 × 25.5	0.0	53.0
18	25.5 × 25.5	64.0	53.0
19	32.5 × 23.5	28.5	54.0
20	42.5 × 52.5	91.0	61.0
21	52.5 × 42.5	135.0	71.0
22	42.5 × 32.5	0.0	81.0
23	20.5 × 15.5	44.0	81.0
24	23.5 × 32.5	66.0	81.0
25	20.5 × 15.5	44.0	98.0

Layout 4 187.5 × 113.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	187.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.54 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, touching 2× (34 cm), min. dist. 54 cm
Entropy	0.79
mix / variety	0.82 / 0.83
stagger / lines	0.52 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	0	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	27.0	0.0
3	42.5 × 52.5	54.0	0.0
4	32.5 × 23.5	98.0	0.0
5	20.5 × 15.5	132.5	0.0
6	32.5 × 23.5	155.0	0.0
7	20.5 × 15.5	132.5	17.5
8	52.5 × 42.5	0.0	27.0
9	32.5 × 23.5	98.0	27.0
10	32.5 × 23.5	155.0	27.0
11	20.5 × 15.5	132.5	35.0
12	25.5 × 25.5	98.0	53.0
13	25.5 × 25.5	162.0	53.0
14	20.5 × 15.5	54.0	54.0
15	20.5 × 15.5	76.0	54.0
16	32.5 × 23.5	126.5	54.0
17	15.5 × 20.5	0.0	71.0
18	20.5 × 15.5	19.5	71.0
19	52.5 × 42.5	44.0	71.0
20	42.5 × 32.5	98.0	81.0
21	23.5 × 32.5	142.0	81.0
22	20.5 × 15.5	167.0	81.0
23	25.5 × 25.5	17.0	88.0
24	15.5 × 20.5	0.0	93.0
25	20.5 × 15.5	167.0	98.0

Layout 5 177.5 × 126.5 cm gap 2.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

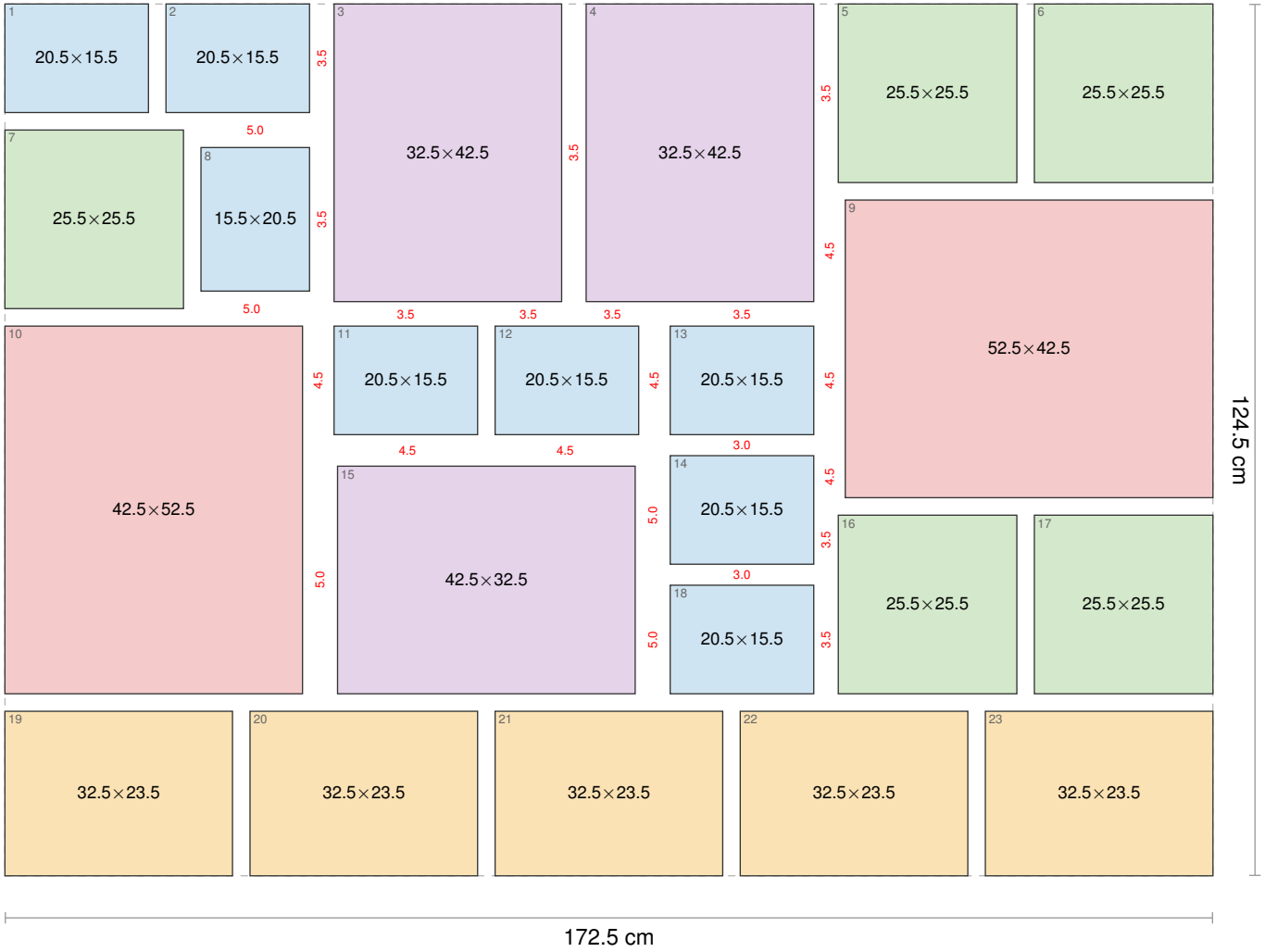
Outer size	177.5 × 126.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.5 cm
Max. deviation	3.0 cm
Mean deviation	0.79 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 90 cm
Entropy	0.79
mix / variety	0.81 / 0.81
stagger / lines	0.56 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	23.7	0.0
3	20.5 × 15.5	47.3	0.0
4	20.5 × 15.5	71.0	0.0
5	32.5 × 23.5	94.0	0.0
6	20.5 × 15.5	129.0	0.0
7	25.5 × 25.5	152.0	0.0
8	25.5 × 25.5	0.0	19.0
9	25.5 × 25.5	66.0	19.0
10	32.5 × 23.5	29.5	20.0
11	20.5 × 15.5	129.0	20.3
12	25.5 × 25.5	152.0	28.0
13	32.5 × 23.5	94.0	29.0
14	20.5 × 15.5	129.0	40.7
15	20.5 × 15.5	0.0	48.0
16	42.5 × 32.5	23.0	48.5
17	23.5 × 32.5	68.0	48.5
18	25.5 × 25.5	152.0	56.0
19	32.5 × 23.5	94.0	58.0
20	15.5 × 20.5	131.5	61.0
21	20.5 × 15.5	0.0	66.0
22	32.5 × 42.5	0.0	84.0
23	52.5 × 42.5	35.0	84.0
24	32.5 × 42.5	90.0	84.0
25	52.5 × 42.5	125.0	84.0

Layout 6 172.5 × 124.5 cm gap 2.5 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

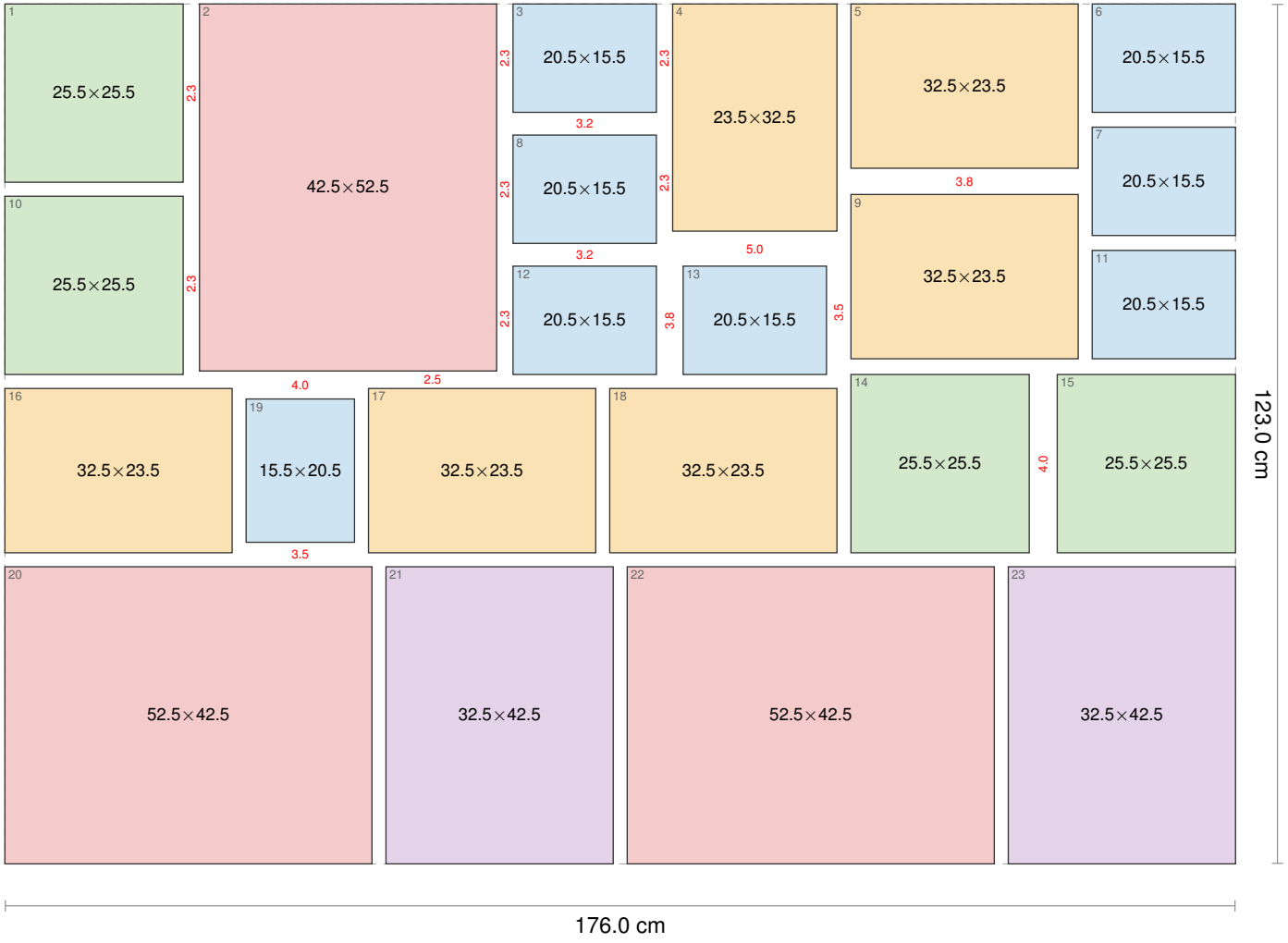
Outer size	172.5 × 124.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.0 cm
Max. deviation	2.5 cm
Mean deviation	0.80 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	2, not touching, min. dist. 127 cm
Entropy	0.75
mix / variety	0.81 / 0.69
stagger / lines	0.52 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	0	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	23.0	0.0
3	32.5 × 42.5	47.0	0.0
4	32.5 × 42.5	83.0	0.0
5	25.5 × 25.5	119.0	0.0
6	25.5 × 25.5	147.0	0.0
7	25.5 × 25.5	0.0	18.0
8	15.5 × 20.5	28.0	20.5
9	52.5 × 42.5	120.0	28.0
10	42.5 × 52.5	0.0	46.0
11	20.5 × 15.5	47.0	46.0
12	20.5 × 15.5	70.0	46.0
13	20.5 × 15.5	95.0	46.0
14	20.5 × 15.5	95.0	64.5
15	42.5 × 32.5	47.5	66.0
16	25.5 × 25.5	119.0	73.0
17	25.5 × 25.5	147.0	73.0
18	20.5 × 15.5	95.0	83.0
19	32.5 × 23.5	0.0	101.0
20	32.5 × 23.5	35.0	101.0
21	32.5 × 23.5	70.0	101.0
22	32.5 × 23.5	105.0	101.0
23	32.5 × 23.5	140.0	101.0

Layout 7 176.0 × 123.0 cm gap 2 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	176.0 × 123.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 5.0 cm
Max. deviation	3.0 cm
Mean deviation	0.45 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, not touching, min. dist. 79 cm
Entropy	0.82
mix / variety	0.83 / 0.89
stagger / lines	0.54 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	42.5 × 52.5	27.8	0.0
3	20.5 × 15.5	72.7	0.0
4	23.5 × 32.5	95.5	0.0
5	32.5 × 23.5	121.0	0.0
6	20.5 × 15.5	155.5	0.0
7	20.5 × 15.5	155.5	17.6
8	20.5 × 15.5	72.7	18.8
9	32.5 × 23.5	121.0	27.2
10	25.5 × 25.5	0.0	27.5
11	20.5 × 15.5	155.5	35.2
12	20.5 × 15.5	72.7	37.5
13	20.5 × 15.5	97.0	37.5
14	25.5 × 25.5	121.0	53.0
15	25.5 × 25.5	150.5	53.0
16	32.5 × 23.5	0.0	55.0
17	32.5 × 23.5	52.0	55.0
18	32.5 × 23.5	86.5	55.0
19	15.5 × 20.5	34.5	56.5
20	52.5 × 42.5	0.0	80.5
21	32.5 × 42.5	54.5	80.5
22	52.5 × 42.5	89.0	80.5
23	32.5 × 42.5	143.5	80.5

Layout 8 184.0 × 115.0 cm gap 2 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

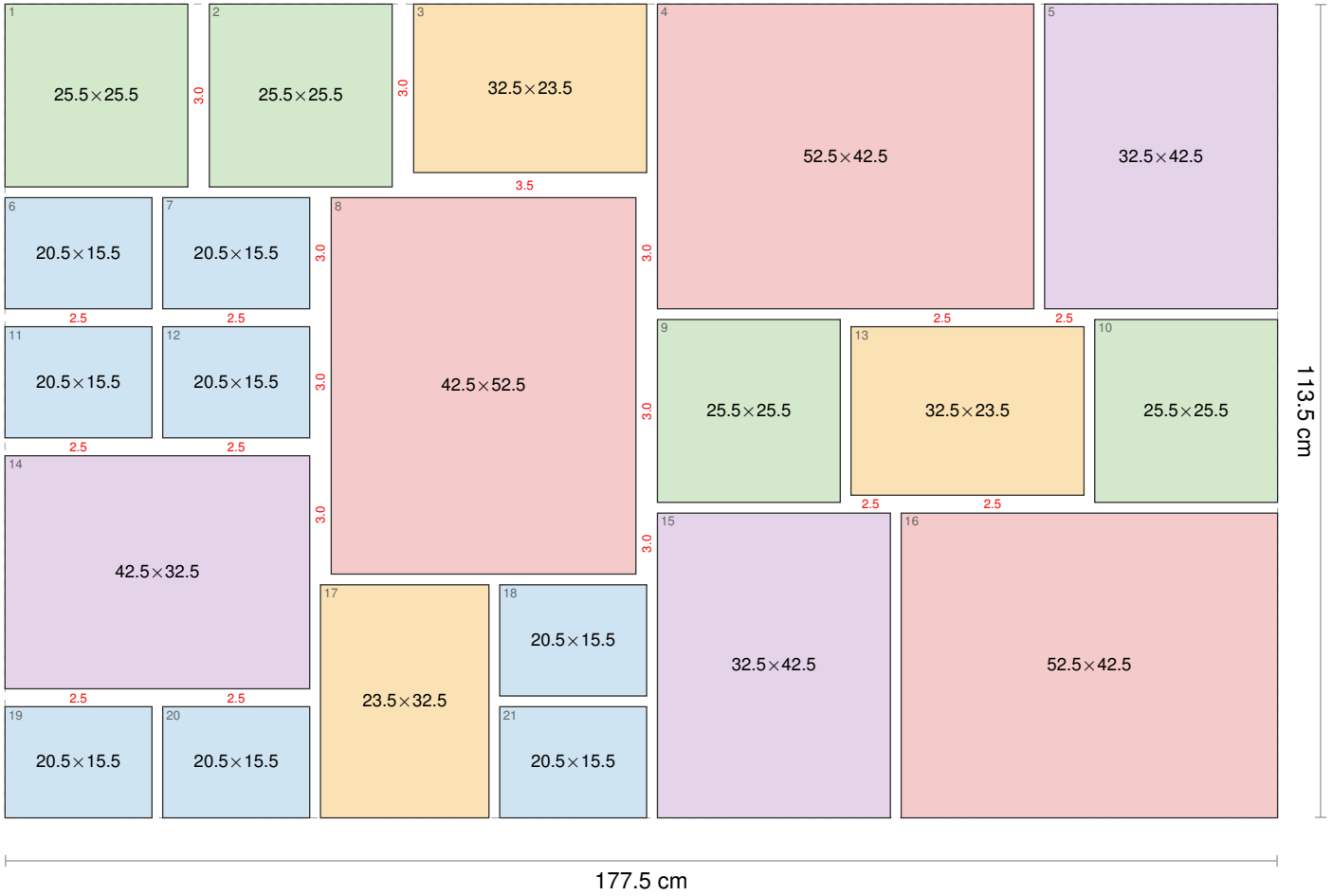
Outer size	184.0 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.0 cm
Max. deviation	2.0 cm
Mean deviation	0.54 cm
Border notches	right 6 cm
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, touching 1× (52 cm), min. dist. 45 cm
Entropy	0.76
mix / variety	0.76 / 0.76
stagger / lines	0.54 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	27.5	0.0
3	25.5 × 25.5	62.0	0.0
4	32.5 × 23.5	89.5	0.0
5	25.5 × 25.5	124.0	0.0
6	32.5 × 42.5	151.5	0.0
7	52.5 × 42.5	0.0	27.5
8	20.5 × 15.5	55.2	27.5
9	23.5 × 32.5	79.3	27.5
10	42.5 × 52.5	106.8	27.5
11	20.5 × 15.5	55.2	45.5
12	32.5 × 42.5	151.5	45.8
13	25.5 × 25.5	78.3	62.0
14	20.5 × 15.5	55.2	63.5
15	52.5 × 42.5	0.0	72.5
16	20.5 × 15.5	55.2	81.5
17	20.5 × 15.5	106.5	82.0
18	20.5 × 15.5	129.0	82.0
19	25.5 × 25.5	78.3	89.5
20	32.5 × 23.5	151.5	91.5
21	20.5 × 15.5	55.2	99.5
22	20.5 × 15.5	106.5	99.5
23	20.5 × 15.5	129.0	99.5

Layout 9 177.5 × 113.5 cm gap 1.5 cm 21 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

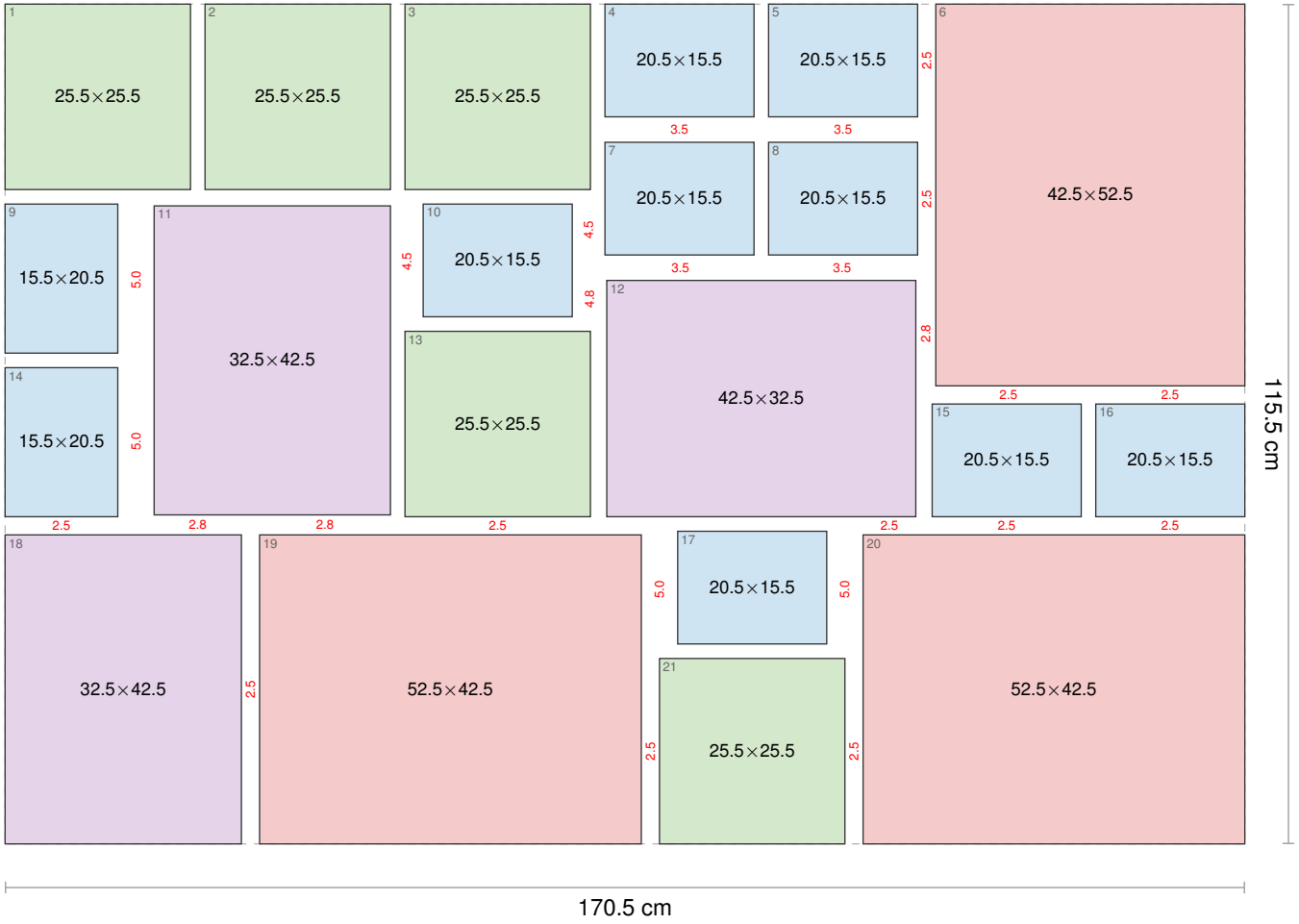
Outer size	177.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 3.5 cm
Max. deviation	2.0 cm
Mean deviation	0.56 cm
Border notches	top 6 cm
Corners filled	4 / 4
Frames used	21 of 28
Large frames	3, not touching, min. dist. 60 cm
Entropy	0.78
mix / variety	0.81 / 0.82
stagger / lines	0.50 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	2	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	28.5	0.0
3	32.5 × 23.5	57.0	0.0
4	52.5 × 42.5	91.0	0.0
5	32.5 × 42.5	145.0	0.0
6	20.5 × 15.5	0.0	27.0
7	20.5 × 15.5	22.0	27.0
8	42.5 × 52.5	45.5	27.0
9	25.5 × 25.5	91.0	44.0
10	25.5 × 25.5	152.0	44.0
11	20.5 × 15.5	0.0	45.0
12	20.5 × 15.5	22.0	45.0
13	32.5 × 23.5	118.0	45.0
14	42.5 × 32.5	0.0	63.0
15	32.5 × 42.5	91.0	71.0
16	52.5 × 42.5	125.0	71.0
17	23.5 × 32.5	44.0	81.0
18	20.5 × 15.5	69.0	81.0
19	20.5 × 15.5	0.0	98.0
20	20.5 × 15.5	22.0	98.0
21	20.5 × 15.5	69.0	98.0

Layout 10 170.5 × 115.5 cm gap 2 cm 21 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	170.5 × 115.5 cm
Nominal gap	2 cm
Actual gaps	2.0 – 5.0 cm
Max. deviation	3.0 cm
Mean deviation	0.80 cm
Border notches	none
Corners filled	4 / 4
Frames used	21 of 28
Large frames	3, not touching, min. dist. 68 cm
Entropy	0.79
mix / variety	0.76 / 0.79
stagger / lines	0.60 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	0	5
	23.5×32.5	P	0	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	25.5 × 25.5	27.5	0.0
3	25.5 × 25.5	55.0	0.0
4	20.5 × 15.5	82.5	0.0
5	20.5 × 15.5	105.0	0.0
6	42.5 × 52.5	128.0	0.0
7	20.5 × 15.5	82.5	19.0
8	20.5 × 15.5	105.0	19.0
9	15.5 × 20.5	0.0	27.5
10	20.5 × 15.5	57.5	27.5
11	32.5 × 42.5	20.5	27.8
12	42.5 × 32.5	82.8	38.0
13	25.5 × 25.5	55.0	45.0
14	15.5 × 20.5	0.0	50.0
15	20.5 × 15.5	127.5	55.0
16	20.5 × 15.5	150.0	55.0
17	20.5 × 15.5	92.5	72.5
18	32.5 × 42.5	0.0	73.0
19	52.5 × 42.5	35.0	73.0
20	52.5 × 42.5	118.0	73.0
21	25.5 × 25.5	90.0	90.0

Layout 11 170.5 × 117.5 cm gap 2.5 cm 19 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

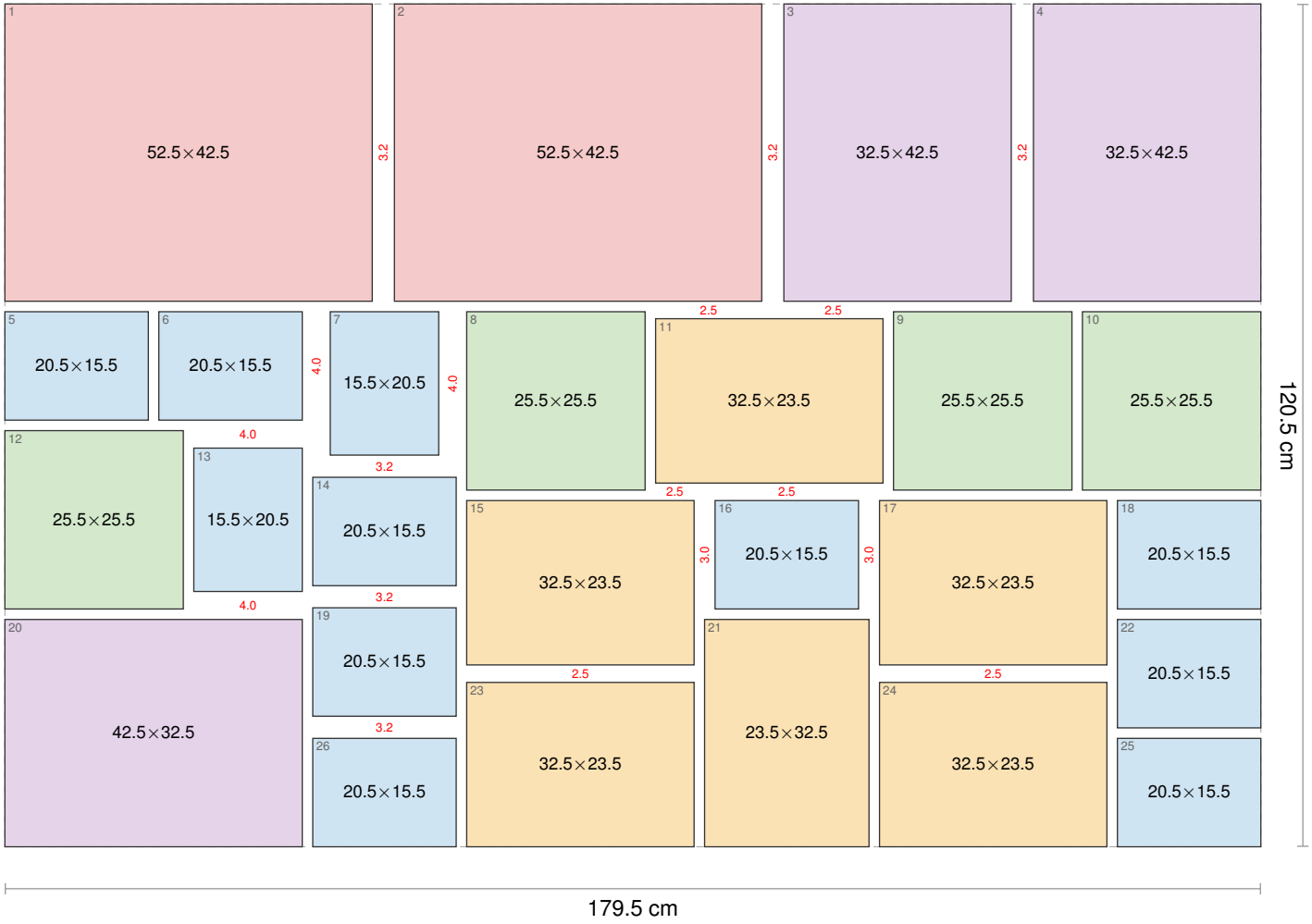
Outer size	170.5 × 117.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.5 cm
Max. deviation	3.0 cm
Mean deviation	0.67 cm
Border notches	none
Corners filled	4 / 4
Frames used	19 of 28
Large frames	3, not touching, min. dist. 70 cm
Entropy	0.84
mix / variety	0.89 / 0.86
stagger / lines	0.63 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	3	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	15.5 × 20.5	0.0	0.0
2	15.5 × 20.5	18.0	0.0
3	32.5 × 42.5	36.0	0.0
4	25.5 × 25.5	71.0	0.0
5	42.5 × 52.5	99.5	0.0
6	25.5 × 25.5	145.0	0.0
7	32.5 × 23.5	0.0	23.0
8	20.5 × 15.5	73.5	28.0
9	25.5 × 25.5	145.0	28.3
10	25.5 × 25.5	36.0	47.0
11	32.5 × 23.5	64.0	48.0
12	32.5 × 23.5	0.0	49.0
13	23.5 × 32.5	147.0	56.7
14	20.5 × 15.5	99.0	57.0
15	20.5 × 15.5	122.0	57.0
16	52.5 × 42.5	0.0	75.0
17	32.5 × 42.5	55.0	75.0
18	52.5 × 42.5	90.0	75.0
19	25.5 × 25.5	145.0	92.0

Layout 12 179.5 × 120.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

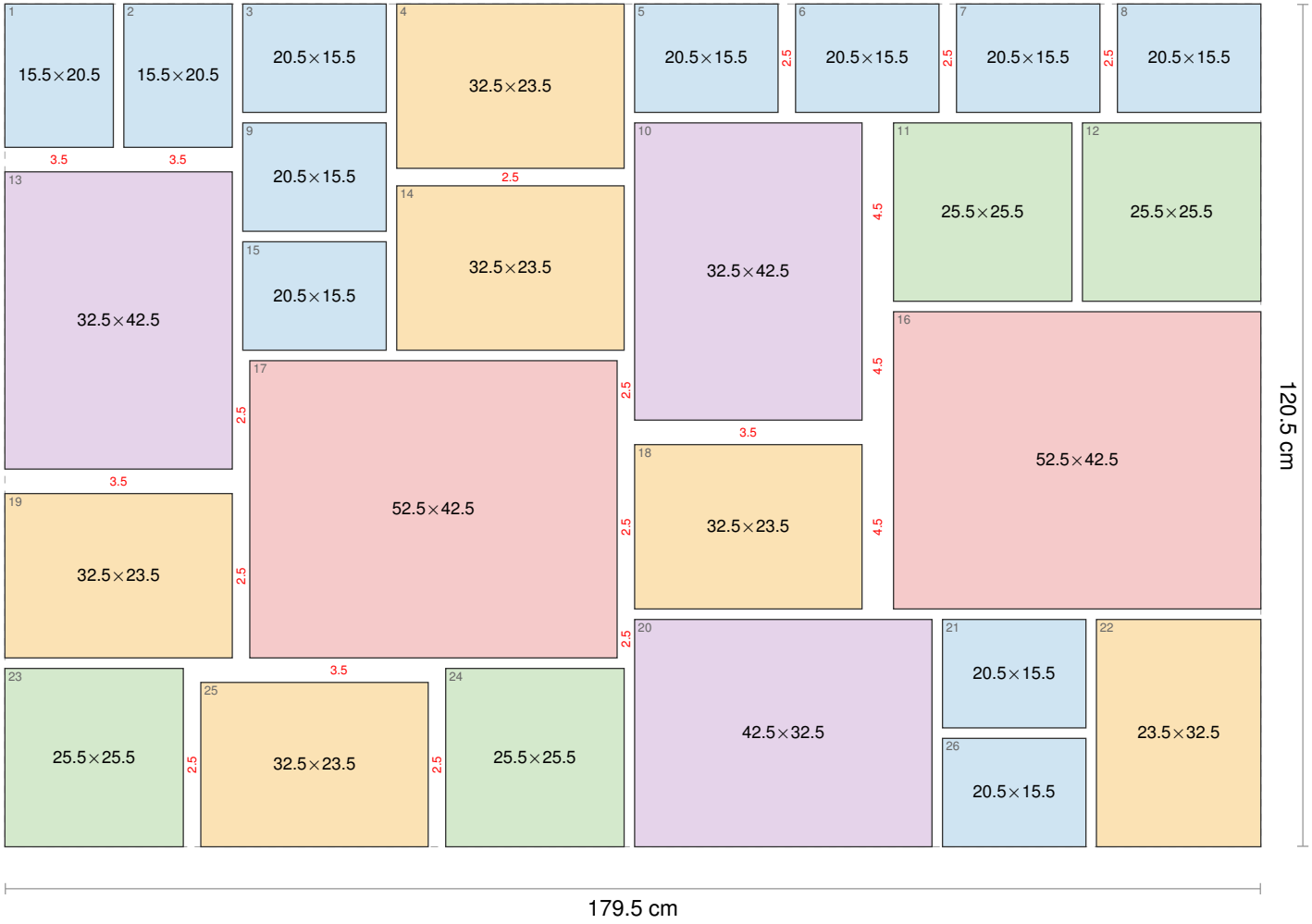
Outer size	179.5 × 120.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.55 cm
Border notches	top 10 cm
Corners filled	4 / 4
Frames used	26 of 28
Large frames	2, not touching, min. dist. 56 cm
Entropy	0.80
mix / variety	0.83 / 0.83
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	52.5 × 42.5	55.7	0.0
3	32.5 × 42.5	111.3	0.0
4	32.5 × 42.5	147.0	0.0
5	20.5 × 15.5	0.0	44.0
6	20.5 × 15.5	22.0	44.0
7	15.5 × 20.5	46.5	44.0
8	25.5 × 25.5	66.0	44.0
9	25.5 × 25.5	127.0	44.0
10	25.5 × 25.5	154.0	44.0
11	32.5 × 23.5	93.0	45.0
12	25.5 × 25.5	0.0	61.0
13	15.5 × 20.5	27.0	63.5
14	20.5 × 15.5	44.0	67.7
15	32.5 × 23.5	66.0	71.0
16	20.5 × 15.5	101.5	71.0
17	32.5 × 23.5	125.0	71.0
18	20.5 × 15.5	159.0	71.0
19	20.5 × 15.5	44.0	86.3
20	42.5 × 32.5	0.0	88.0
21	23.5 × 32.5	100.0	88.0
22	20.5 × 15.5	159.0	88.0
23	32.5 × 23.5	66.0	97.0
24	32.5 × 23.5	125.0	97.0
25	20.5 × 15.5	159.0	105.0
26	20.5 × 15.5	44.0	105.0

Layout 13 179.5 × 120.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

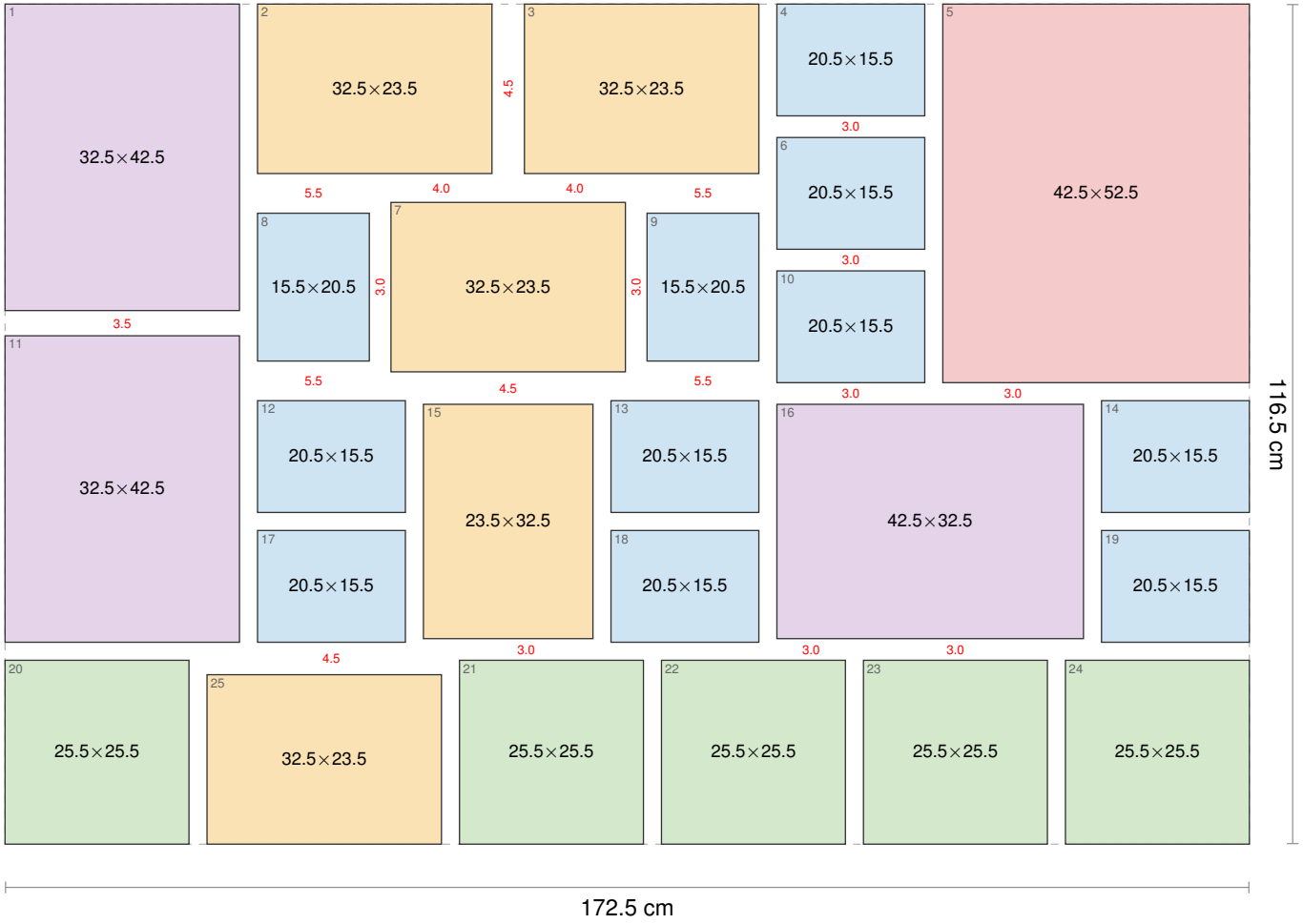
Outer size	179.5 × 120.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.5 cm
Max. deviation	3.0 cm
Mean deviation	0.56 cm
Border notches	left 7 cm
Corners filled	4 / 4
Frames used	26 of 28
Large frames	2, not touching, min. dist. 92 cm
Entropy	0.79
mix / variety	0.83 / 0.80
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	15.5 × 20.5	0.0	0.0
2	15.5 × 20.5	17.0	0.0
3	20.5 × 15.5	34.0	0.0
4	32.5 × 23.5	56.0	0.0
5	20.5 × 15.5	90.0	0.0
6	20.5 × 15.5	113.0	0.0
7	20.5 × 15.5	136.0	0.0
8	20.5 × 15.5	159.0	0.0
9	20.5 × 15.5	34.0	17.0
10	32.5 × 42.5	90.0	17.0
11	25.5 × 25.5	127.0	17.0
12	25.5 × 25.5	154.0	17.0
13	32.5 × 42.5	0.0	24.0
14	32.5 × 23.5	56.0	26.0
15	20.5 × 15.5	34.0	34.0
16	52.5 × 42.5	127.0	44.0
17	52.5 × 42.5	35.0	51.0
18	32.5 × 23.5	90.0	63.0
19	32.5 × 23.5	0.0	70.0
20	42.5 × 32.5	90.0	88.0
21	20.5 × 15.5	134.0	88.0
22	23.5 × 32.5	156.0	88.0
23	25.5 × 25.5	0.0	95.0
24	25.5 × 25.5	63.0	95.0
25	32.5 × 23.5	28.0	97.0
26	20.5 × 15.5	134.0	105.0

Layout 14 172.5 × 116.5 cm gap 2.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

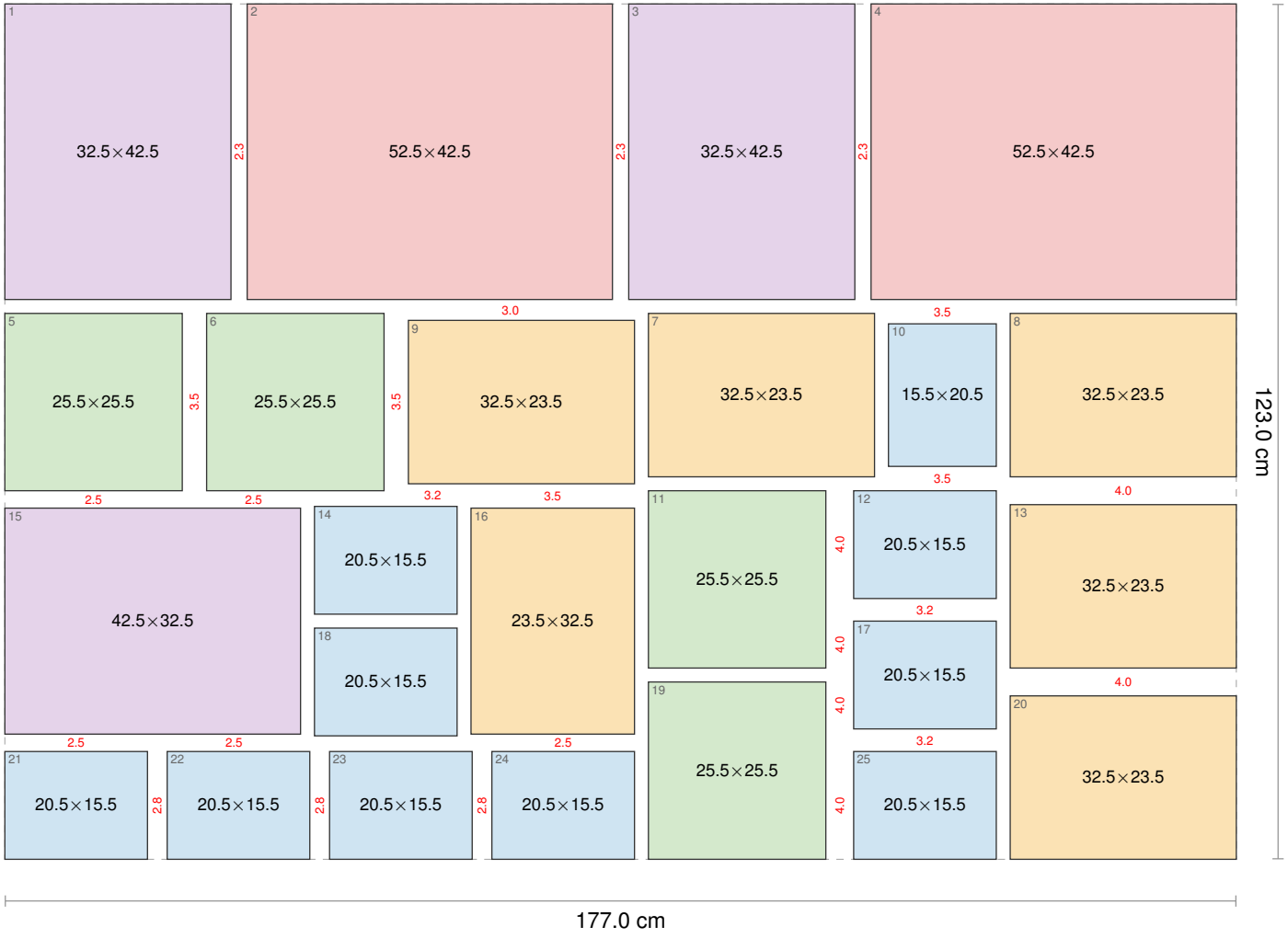
Outer size	172.5 × 116.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.5 cm
Max. deviation	3.0 cm
Mean deviation	0.53 cm
Border notches	top 4 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	1
Entropy	0.78
mix / variety	0.81 / 0.78
stagger / lines	0.52 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	0	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	32.5 × 23.5	35.0	0.0
3	32.5 × 23.5	72.0	0.0
4	20.5 × 15.5	107.0	0.0
5	42.5 × 52.5	130.0	0.0
6	20.5 × 15.5	107.0	18.5
7	32.5 × 23.5	53.5	27.5
8	15.5 × 20.5	35.0	29.0
9	15.5 × 20.5	89.0	29.0
10	20.5 × 15.5	107.0	37.0
11	32.5 × 42.5	0.0	46.0
12	20.5 × 15.5	35.0	55.0
13	20.5 × 15.5	84.0	55.0
14	20.5 × 15.5	152.0	55.0
15	23.5 × 32.5	58.0	55.5
16	42.5 × 32.5	107.0	55.5
17	20.5 × 15.5	35.0	73.0
18	20.5 × 15.5	84.0	73.0
19	20.5 × 15.5	152.0	73.0
20	25.5 × 25.5	0.0	91.0
21	25.5 × 25.5	63.0	91.0
22	25.5 × 25.5	91.0	91.0
23	25.5 × 25.5	119.0	91.0
24	25.5 × 25.5	147.0	91.0
25	32.5 × 23.5	28.0	93.0

Layout 15 177.0 × 123.0 cm gap 2 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

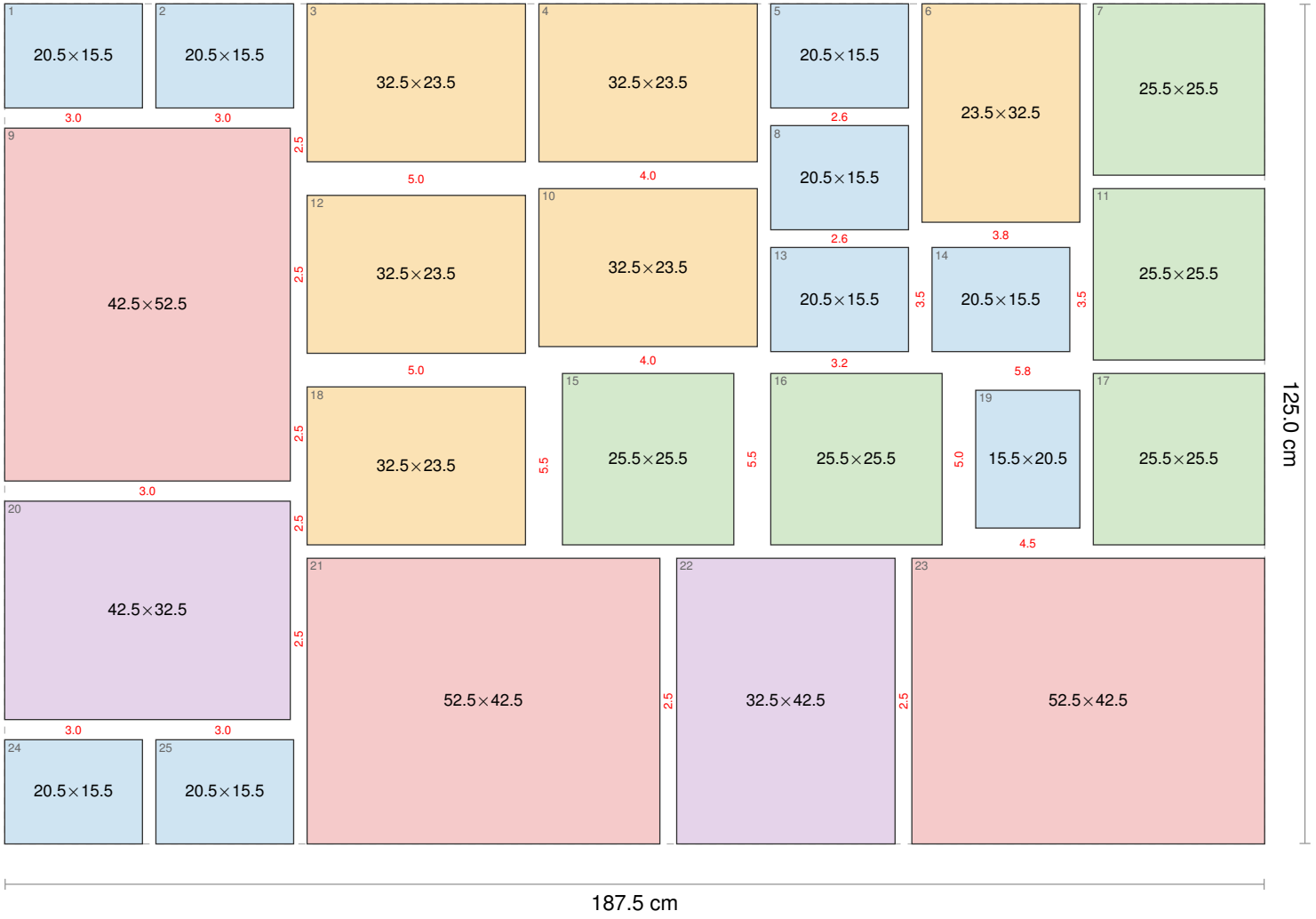
Outer size	177.0 × 123.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.0 cm
Max. deviation	2.0 cm
Mean deviation	0.64 cm
Border notches	bottom 4 cm, right 8 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 90 cm
Entropy	0.79
mix / variety	0.81 / 0.79
stagger / lines	0.54 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	34.8	0.0
3	32.5 × 42.5	89.7	0.0
4	52.5 × 42.5	124.5	0.0
5	25.5 × 25.5	0.0	44.5
6	25.5 × 25.5	29.0	44.5
7	32.5 × 23.5	92.5	44.5
8	32.5 × 23.5	144.5	44.5
9	32.5 × 23.5	58.0	45.5
10	15.5 × 20.5	127.0	46.0
11	25.5 × 25.5	92.5	70.0
12	20.5 × 15.5	122.0	70.0
13	32.5 × 23.5	144.5	72.0
14	20.5 × 15.5	44.5	72.2
15	42.5 × 32.5	0.0	72.5
16	23.5 × 32.5	67.0	72.5
17	20.5 × 15.5	122.0	88.8
18	20.5 × 15.5	44.5	89.8
19	25.5 × 25.5	92.5	97.5
20	32.5 × 23.5	144.5	99.5
21	20.5 × 15.5	0.0	107.5
22	20.5 × 15.5	23.3	107.5
23	20.5 × 15.5	46.7	107.5
24	20.5 × 15.5	70.0	107.5
25	20.5 × 15.5	122.0	107.5

Layout 16 187.5 × 125.0 cm gap 2 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	187.5 × 125.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 5.8 cm
Max. deviation	3.8 cm
Mean deviation	0.86 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, not touching, min. dist. 77 cm
Entropy	0.78
mix / variety	0.84 / 0.78
stagger / lines	0.51 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	1	2
	32.5×42.5	P	1	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	22.5	0.0
3	32.5 × 23.5	45.0	0.0
4	32.5 × 23.5	79.5	0.0
5	20.5 × 15.5	114.0	0.0
6	23.5 × 32.5	136.5	0.0
7	25.5 × 25.5	162.0	0.0
8	20.5 × 15.5	114.0	18.1
9	42.5 × 52.5	0.0	18.5
10	32.5 × 23.5	79.5	27.5
11	25.5 × 25.5	162.0	27.5
12	32.5 × 23.5	45.0	28.5
13	20.5 × 15.5	114.0	36.2
14	20.5 × 15.5	138.0	36.2
15	25.5 × 25.5	83.0	55.0
16	25.5 × 25.5	114.0	55.0
17	25.5 × 25.5	162.0	55.0
18	32.5 × 23.5	45.0	57.0
19	15.5 × 20.5	144.5	57.5
20	42.5 × 32.5	0.0	74.0
21	52.5 × 42.5	45.0	82.5
22	32.5 × 42.5	100.0	82.5
23	52.5 × 42.5	135.0	82.5
24	20.5 × 15.5	0.0	109.5
25	20.5 × 15.5	22.5	109.5

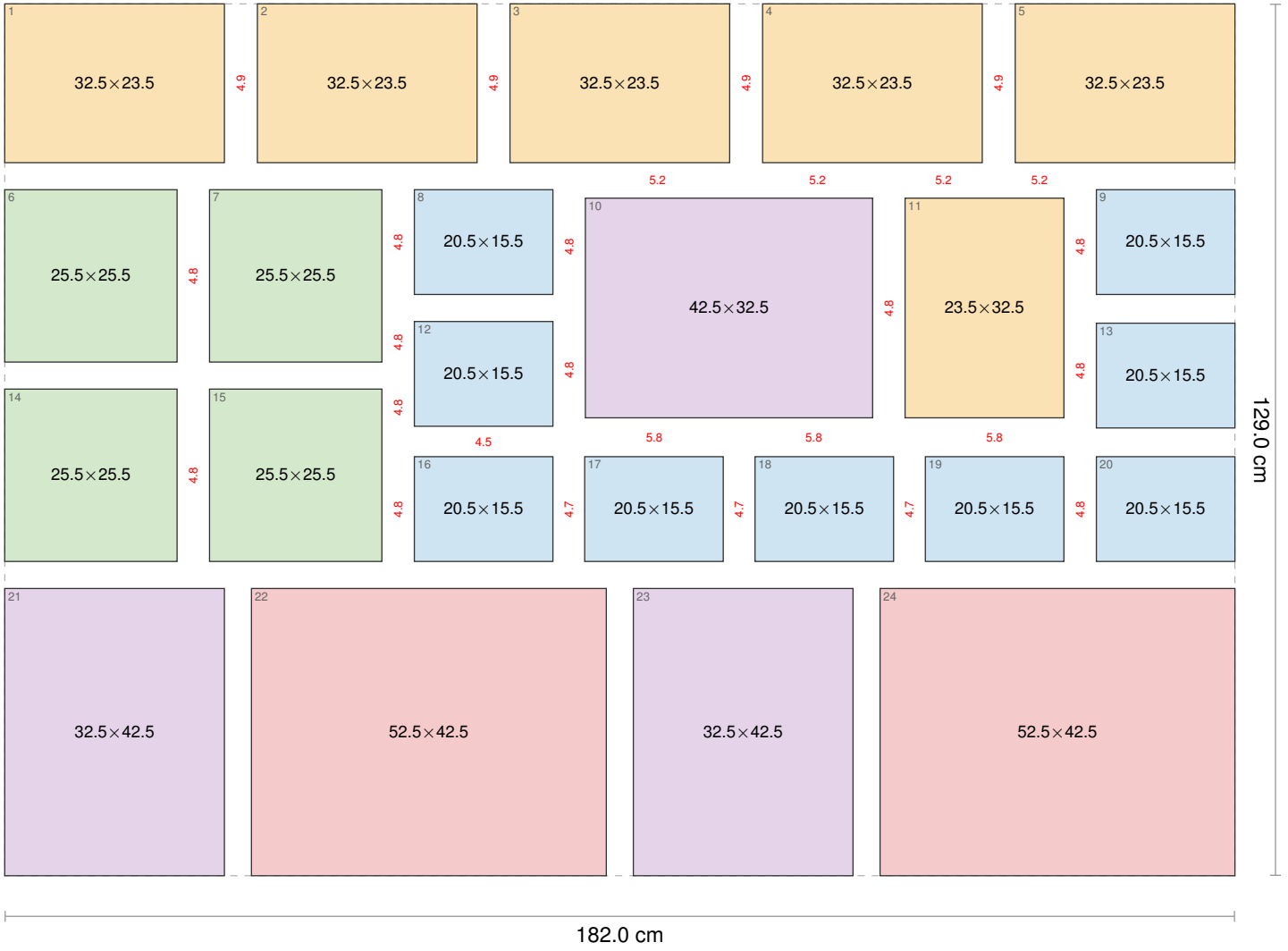
12 large-entropy

Sub-analysis. Both of the above: 2 to 3 large frames kept apart from each other, and the irregularity reward.

Parameters that differ from the baseline: large_area = 2000, large_count = 2-3, large_apart = True, large_min_dist = 70, entropy = 40.
256 runs, 20 min wall-clock budget, 536 candidate states, the 16 best distinct layouts are shown.

#	size (cm)	gap	frames	max dev	mean dev	notches	corners	entropy
1	182.0 × 129.0	4	24	1.8	0.56	none	4/4	0.65
2	184.0 × 115.0	2	20	2.0	0.65	4 cm	4/4	0.75
3	179.5 × 120.5	1.5	26	2.5	0.58	10 cm	4/4	0.67
4	172.5 × 125.5	2.5	25	3.5	0.70	none	4/4	0.75
5	174.5 × 130.5	1.5	25	2.5	0.95	6 cm	4/4	0.78
6	172.5 × 125.5	2.5	25	3.8	0.69	none	4/4	0.78
7	179.5 × 111.5	1.5	25	4.0	0.51	none	4/4	0.79
8	181.5 × 129.5	3.5	24	3.0	0.64	6 cm	4/4	0.77
9	176.5 × 113.5	1.5	23	3.5	0.70	none	4/4	0.79
10	182.5 × 129.5	3.5	23	4.0	0.86	none	4/4	0.82
11	184.5 × 113.5	1.5	23	3.5	0.86	4 cm	4/4	0.78
12	177.5 × 118.5	2.5	22	3.0	0.83	4 cm	4/4	0.72
13	173.5 × 116.5	3	21	3.0	0.63	6 cm	4/4	0.79
14	176.0 × 125.0	2	20	3.0	0.91	4 cm	4/4	0.77
15	179.5 × 120.5	1.5	26	3.0	0.61	8 cm	4/4	0.77
16	179.5 × 120.5	1.5	26	2.5	0.60	13 cm	4/4	0.77

Layout 1 182.0 × 129.0 cm gap 4 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

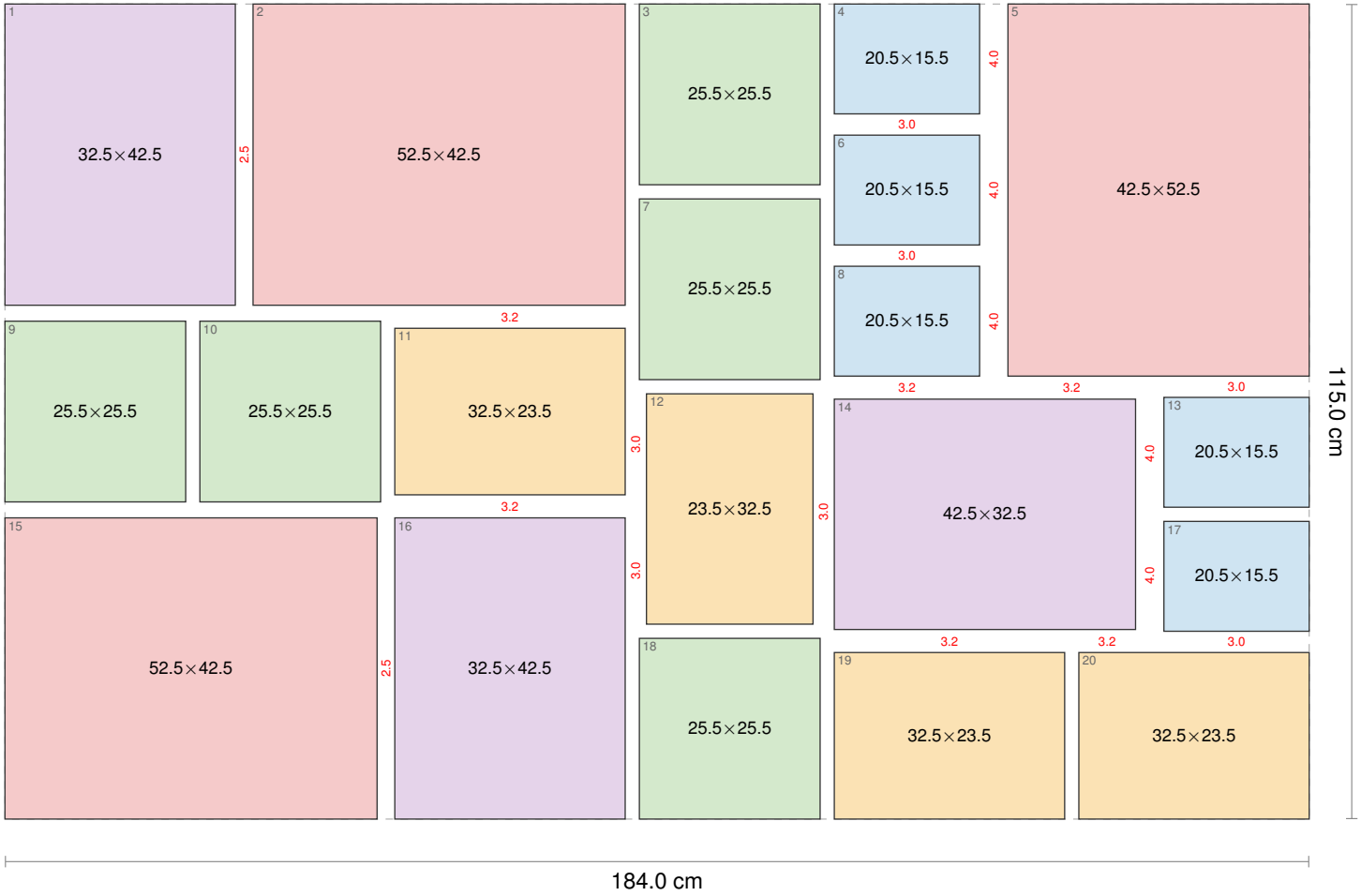
Outer size	182.0 × 129.0 cm
Nominal gap	4 cm
Actual gaps	4.0 – 5.8 cm
Max. deviation	1.8 cm
Mean deviation	0.56 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 93 cm
Entropy	0.65
mix / variety	0.76 / 0.60
stagger / lines	0.48 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	0	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	0	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	32.5 × 23.5	37.4	0.0
3	32.5 × 23.5	74.8	0.0
4	32.5 × 23.5	112.1	0.0
5	32.5 × 23.5	149.5	0.0
6	25.5 × 25.5	0.0	27.5
7	25.5 × 25.5	30.3	27.5
8	20.5 × 15.5	60.6	27.5
9	20.5 × 15.5	161.5	27.5
10	42.5 × 32.5	85.9	28.8
11	23.5 × 32.5	133.2	28.8
12	20.5 × 15.5	60.6	47.0
13	20.5 × 15.5	161.5	47.2
14	25.5 × 25.5	0.0	57.0
15	25.5 × 25.5	30.3	57.0
16	20.5 × 15.5	60.6	67.0
17	20.5 × 15.5	85.8	67.0
18	20.5 × 15.5	111.0	67.0
19	20.5 × 15.5	136.2	67.0
20	20.5 × 15.5	161.5	67.0
21	32.5 × 42.5	0.0	86.5
22	52.5 × 42.5	36.5	86.5
23	32.5 × 42.5	93.0	86.5
24	52.5 × 42.5	129.5	86.5

Layout 2 184.0 × 115.0 cm gap 2 cm 20 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	184.0 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.0 cm
Max. deviation	2.0 cm
Mean deviation	0.65 cm
Border notches	top 4 cm
Corners filled	4 / 4
Frames used	20 of 28
Large frames	3, not touching, min. dist. 81 cm
Entropy	0.75
mix / variety	0.86 / 0.80
stagger / lines	0.59 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	5	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	35.0	0.0
3	25.5 × 25.5	89.5	0.0
4	20.5 × 15.5	117.0	0.0
5	42.5 × 52.5	141.5	0.0
6	20.5 × 15.5	117.0	18.5
7	25.5 × 25.5	89.5	27.5
8	20.5 × 15.5	117.0	37.0
9	25.5 × 25.5	0.0	44.8
10	25.5 × 25.5	27.5	44.8
11	32.5 × 23.5	55.0	45.8
12	23.5 × 32.5	90.5	55.0
13	20.5 × 15.5	163.5	55.5
14	42.5 × 32.5	117.0	55.8
15	52.5 × 42.5	0.0	72.5
16	32.5 × 42.5	55.0	72.5
17	20.5 × 15.5	163.5	73.0
18	25.5 × 25.5	89.5	89.5
19	32.5 × 23.5	117.0	91.5
20	32.5 × 23.5	151.5	91.5

Layout 3 179.5 × 120.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	179.5 × 120.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.58 cm
Border notches	top 10 cm
Corners filled	4 / 4
Frames used	26 of 28
Large frames	2, not touching, min. dist. 91 cm
Entropy	0.67
mix / variety	0.83 / 0.66
stagger / lines	0.45 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 42.5	55.7	0.0
3	52.5 × 42.5	91.3	0.0
4	32.5 × 42.5	147.0	0.0
5	25.5 × 25.5	0.0	44.0
6	25.5 × 25.5	62.7	44.0
7	25.5 × 25.5	126.2	44.0
8	25.5 × 25.5	154.0	44.0
9	32.5 × 23.5	27.8	45.0
10	32.5 × 23.5	91.3	45.0
11	20.5 × 15.5	0.0	71.0
12	20.5 × 15.5	22.0	71.0
13	20.5 × 15.5	44.0	71.0
14	20.5 × 15.5	66.0	71.0
15	20.5 × 15.5	89.5	71.0
16	32.5 × 23.5	147.0	71.0
17	15.5 × 20.5	113.0	72.5
18	15.5 × 20.5	130.0	72.5
19	42.5 × 32.5	0.0	88.0
20	20.5 × 15.5	44.0	88.0
21	20.5 × 15.5	66.0	88.0
22	23.5 × 32.5	88.0	88.0
23	32.5 × 23.5	113.0	97.0
24	32.5 × 23.5	147.0	97.0
25	20.5 × 15.5	44.0	105.0
26	20.5 × 15.5	66.0	105.0

Layout 4 172.5 × 125.5 cm gap 2.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

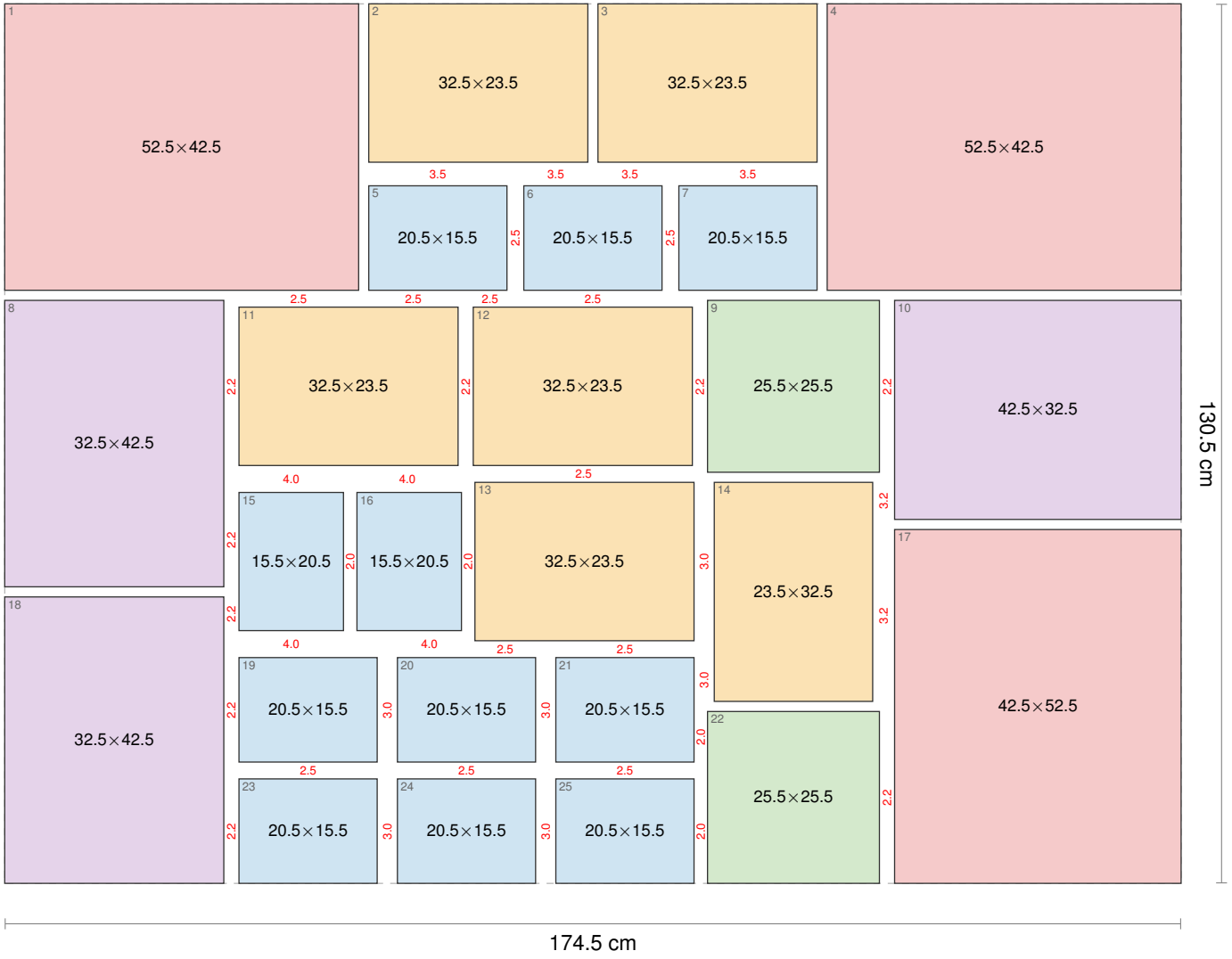
Outer size	172.5 × 125.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 6.0 cm
Max. deviation	3.5 cm
Mean deviation	0.70 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 80 cm
Entropy	0.75
mix / variety	0.79 / 0.73
stagger / lines	0.50 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	32.5 × 23.5	35.0	0.0
3	32.5 × 23.5	70.0	0.0
4	32.5 × 23.5	105.0	0.0
5	32.5 × 42.5	140.0	0.0
6	20.5 × 15.5	35.0	26.0
7	20.5 × 15.5	60.0	26.0
8	52.5 × 42.5	85.0	26.5
9	25.5 × 25.5	55.0	44.0
10	32.5 × 23.5	0.0	46.0
11	32.5 × 23.5	140.0	46.0
12	15.5 × 20.5	35.0	46.5
13	25.5 × 25.5	0.0	72.0
14	20.5 × 15.5	75.3	72.0
15	20.5 × 15.5	100.2	72.0
16	20.5 × 15.5	123.6	72.0
17	25.5 × 25.5	147.0	72.0
18	42.5 × 52.5	28.4	73.0
19	20.5 × 15.5	100.2	91.0
20	20.5 × 15.5	123.6	91.0
21	23.5 × 32.5	73.8	93.0
22	25.5 × 25.5	0.0	100.0
23	25.5 × 25.5	147.0	100.0
24	20.5 × 15.5	100.2	110.0
25	20.5 × 15.5	123.6	110.0

Layout 5 174.5 × 130.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

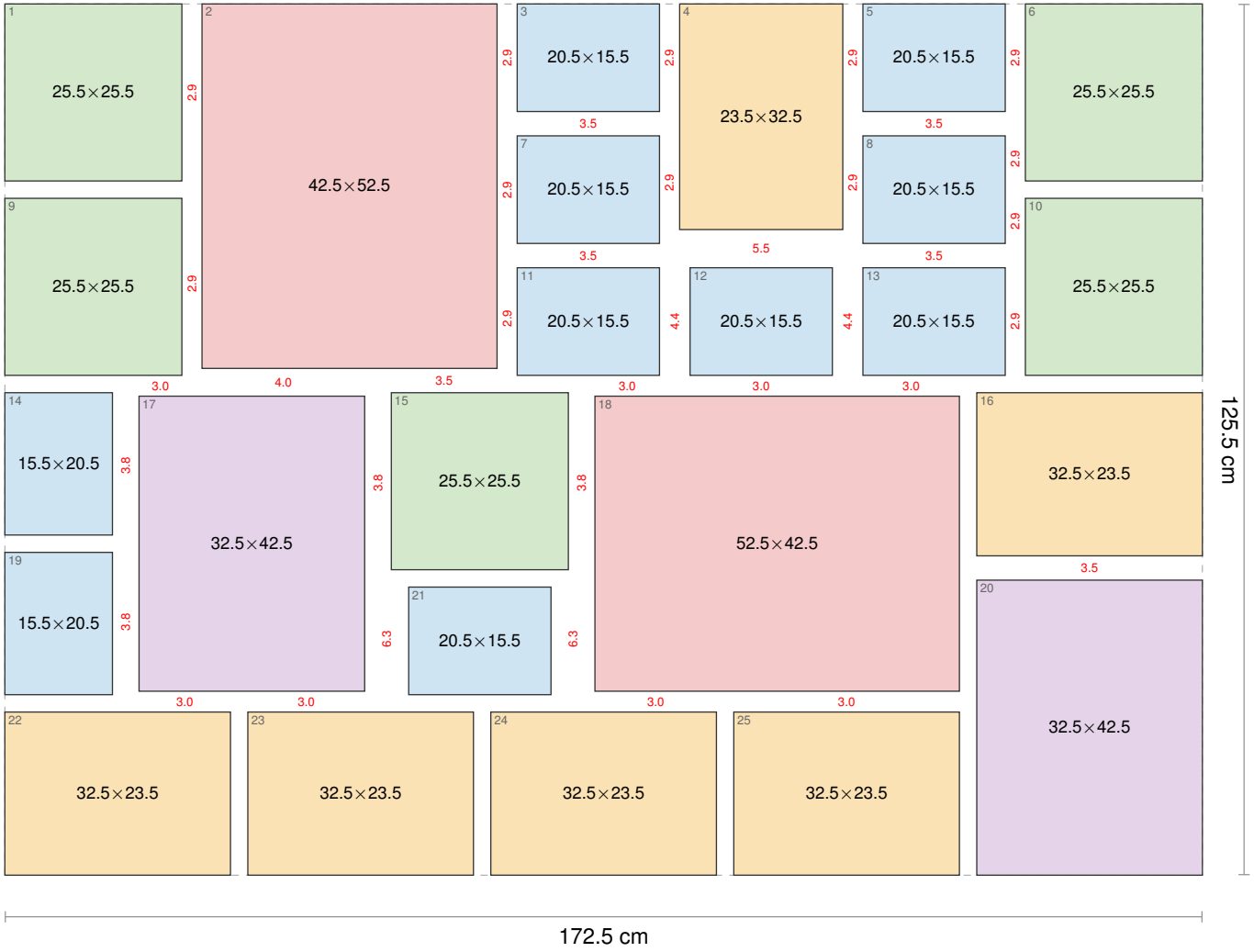
Outer size	174.5 × 130.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.95 cm
Border notches	bottom 6 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, not touching, min. dist. 83 cm
Entropy	0.78
mix / variety	0.86 / 0.74
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		2	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 23.5	54.0	0.0
3	32.5 × 23.5	88.0	0.0
4	52.5 × 42.5	122.0	0.0
5	20.5 × 15.5	54.0	27.0
6	20.5 × 15.5	77.0	27.0
7	20.5 × 15.5	100.0	27.0
8	32.5 × 42.5	0.0	44.0
9	25.5 × 25.5	104.2	44.0
10	42.5 × 32.5	132.0	44.0
11	32.5 × 23.5	34.8	45.0
12	32.5 × 23.5	69.5	45.0
13	32.5 × 23.5	69.8	71.0
14	23.5 × 32.5	105.2	71.0
15	15.5 × 20.5	34.8	72.5
16	15.5 × 20.5	52.2	72.5
17	42.5 × 52.5	132.0	78.0
18	32.5 × 42.5	0.0	88.0
19	20.5 × 15.5	34.8	97.0
20	20.5 × 15.5	58.2	97.0
21	20.5 × 15.5	81.8	97.0
22	25.5 × 25.5	104.2	105.0
23	20.5 × 15.5	34.8	115.0
24	20.5 × 15.5	58.2	115.0
25	20.5 × 15.5	81.8	115.0

Layout 6 172.5 × 125.5 cm gap 2.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

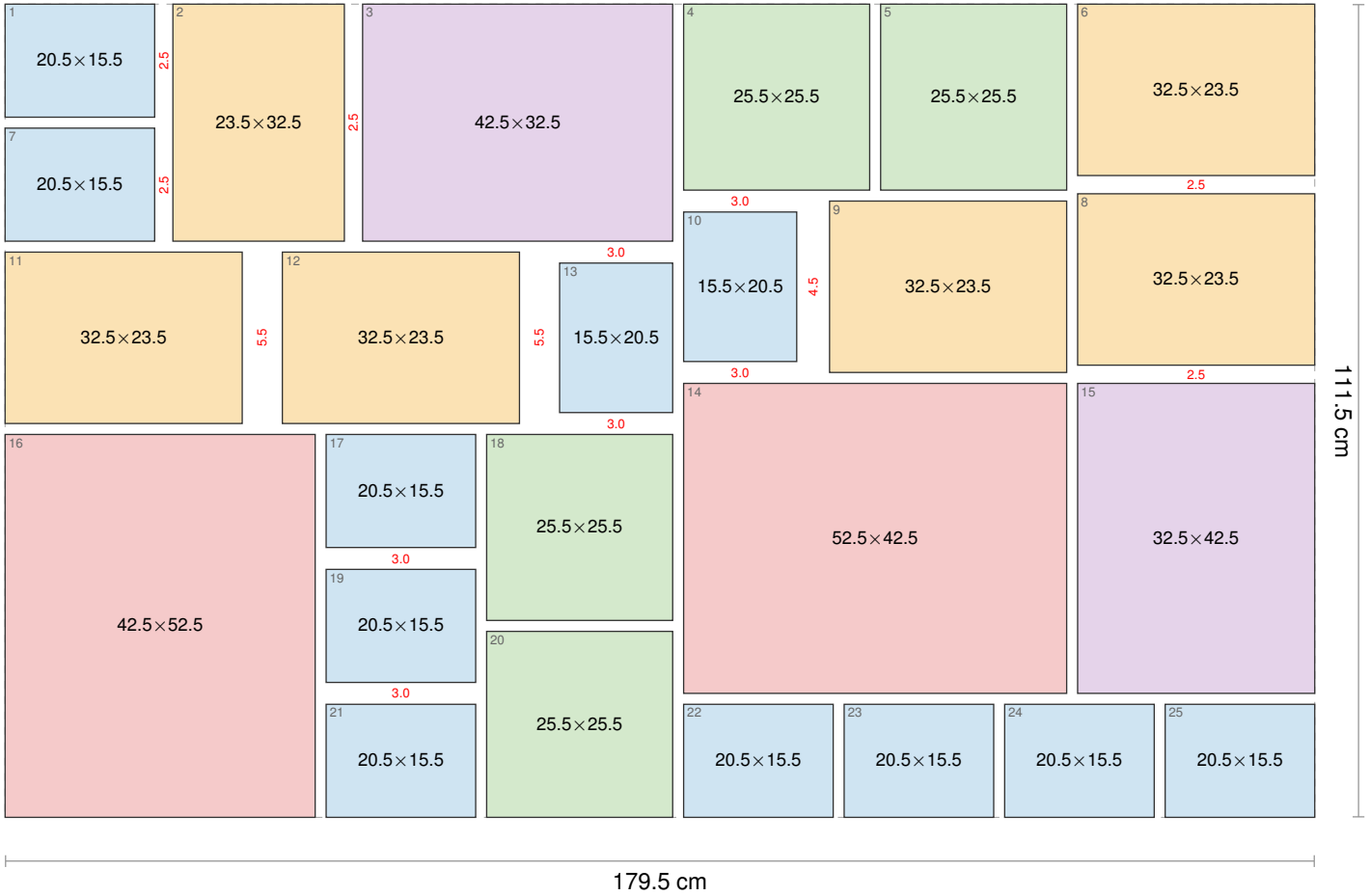
Outer size	172.5 × 125.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 6.3 cm
Max. deviation	3.8 cm
Mean deviation	0.69 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 80 cm
Entropy	0.78
mix / variety	0.82 / 0.77
stagger / lines	0.52 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	42.5 × 52.5	28.4	0.0
3	20.5 × 15.5	73.8	0.0
4	23.5 × 32.5	97.2	0.0
5	20.5 × 15.5	123.6	0.0
6	25.5 × 25.5	147.0	0.0
7	20.5 × 15.5	73.8	19.0
8	20.5 × 15.5	123.6	19.0
9	25.5 × 25.5	0.0	28.0
10	25.5 × 25.5	147.0	28.0
11	20.5 × 15.5	73.8	38.0
12	20.5 × 15.5	98.7	38.0
13	20.5 × 15.5	123.6	38.0
14	15.5 × 20.5	0.0	56.0
15	25.5 × 25.5	55.7	56.0
16	32.5 × 23.5	140.0	56.0
17	32.5 × 42.5	19.3	56.5
18	52.5 × 42.5	85.0	56.5
19	15.5 × 20.5	0.0	79.0
20	32.5 × 42.5	140.0	83.0
21	20.5 × 15.5	58.2	84.0
22	32.5 × 23.5	0.0	102.0
23	32.5 × 23.5	35.0	102.0
24	32.5 × 23.5	70.0	102.0
25	32.5 × 23.5	105.0	102.0

Layout 7 179.5 × 111.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	179.5 × 111.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 5.5 cm
Max. deviation	4.0 cm
Mean deviation	0.51 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 99 cm
Entropy	0.79
mix / variety	0.83 / 0.78
stagger / lines	0.54 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	2	2
32.5×42.5	P	1	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	1	1
52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	23.5 × 32.5	23.0	0.0
3	42.5 × 32.5	49.0	0.0
4	25.5 × 25.5	93.0	0.0
5	25.5 × 25.5	120.0	0.0
6	32.5 × 23.5	147.0	0.0
7	20.5 × 15.5	0.0	17.0
8	32.5 × 23.5	147.0	26.0
9	32.5 × 23.5	113.0	27.0
10	15.5 × 20.5	93.0	28.5
11	32.5 × 23.5	0.0	34.0
12	32.5 × 23.5	38.0	34.0
13	15.5 × 20.5	76.0	35.5
14	52.5 × 42.5	93.0	52.0
15	32.5 × 42.5	147.0	52.0
16	42.5 × 52.5	0.0	59.0
17	20.5 × 15.5	44.0	59.0
18	25.5 × 25.5	66.0	59.0
19	20.5 × 15.5	44.0	77.5
20	25.5 × 25.5	66.0	86.0
21	20.5 × 15.5	44.0	96.0
22	20.5 × 15.5	93.0	96.0
23	20.5 × 15.5	115.0	96.0
24	20.5 × 15.5	137.0	96.0
25	20.5 × 15.5	159.0	96.0

Layout 8 181.5 × 129.5 cm gap 3.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

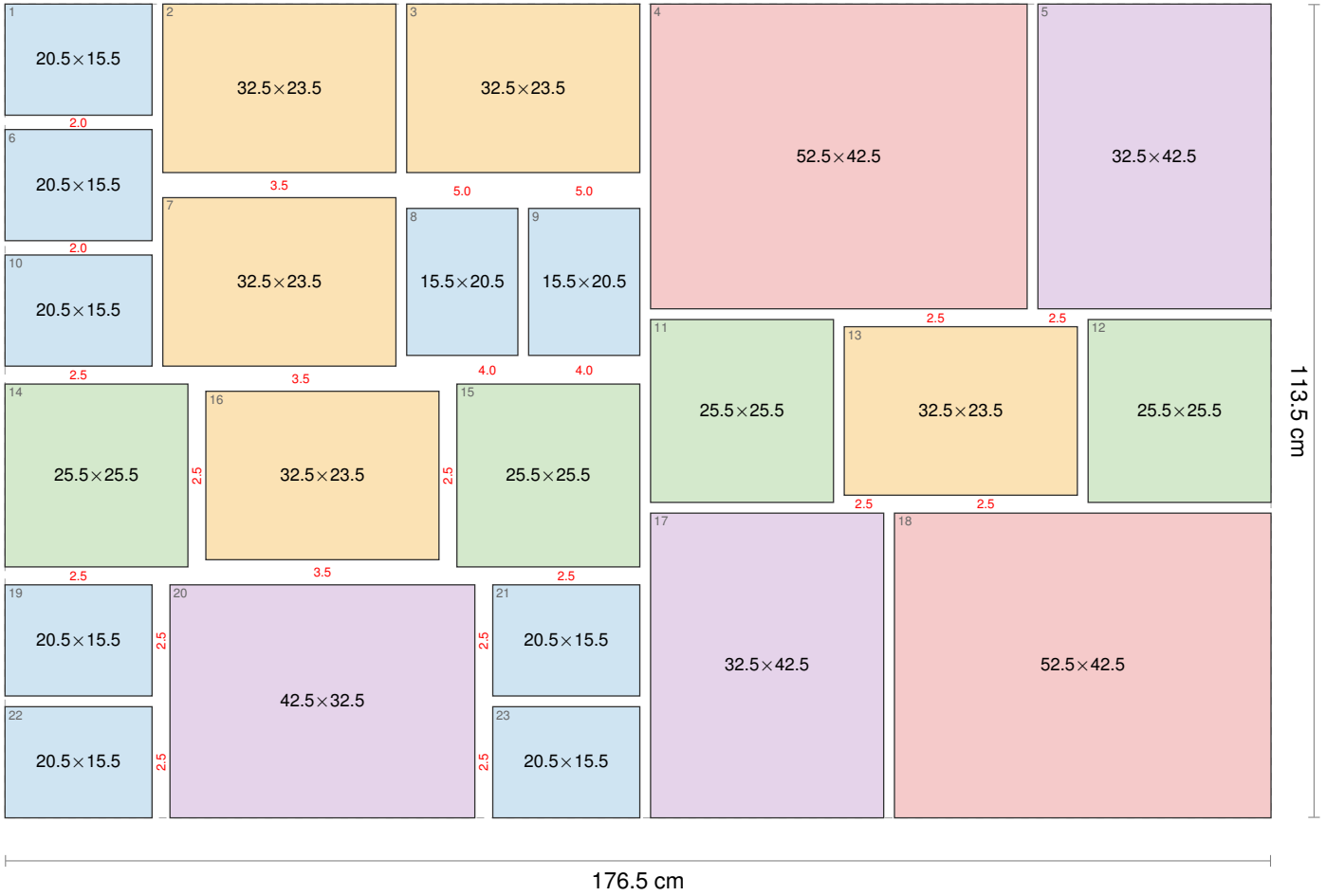
Outer size	181.5 × 129.5 cm
Nominal gap	3.5 cm
Actual gaps	3.5 – 6.5 cm
Max. deviation	3.0 cm
Mean deviation	0.64 cm
Border notches	right 6 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	3, not touching, min. dist. 84 cm
Entropy	0.77
mix / variety	0.76 / 0.79
stagger / lines	0.52 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	1	2
32.5×42.5	P	2	2
42.5×32.5	L	0	1
32.5×23.5	L	5	5
23.5×32.5	P	0	1
25.5×25.5		4	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 42.5	56.5	0.0
3	52.5 × 42.5	93.0	0.0
4	32.5 × 23.5	149.0	0.0
5	32.5 × 42.5	149.0	29.0
6	25.5 × 25.5	0.0	46.0
7	25.5 × 25.5	120.0	46.0
8	32.5 × 23.5	29.0	47.0
9	32.5 × 23.5	84.0	47.0
10	15.5 × 20.5	65.0	48.5
11	20.5 × 15.5	0.0	75.0
12	25.5 × 25.5	70.8	75.0
13	20.5 × 15.5	136.6	75.0
14	20.5 × 15.5	161.0	75.0
15	32.5 × 23.5	100.2	76.0
16	42.5 × 52.5	24.4	77.0
17	20.5 × 15.5	0.0	94.5
18	20.5 × 15.5	136.6	94.5
19	20.5 × 15.5	161.0	94.5
20	25.5 × 25.5	107.2	104.0
21	32.5 × 23.5	70.8	106.0
22	20.5 × 15.5	0.0	114.0
23	20.5 × 15.5	136.6	114.0
24	20.5 × 15.5	161.0	114.0

Layout 9 176.5 × 113.5 cm gap 1.5 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	176.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 5.0 cm
Max. deviation	3.5 cm
Mean deviation	0.70 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	2, not touching, min. dist. 79 cm
Entropy	0.79
mix / variety	0.81 / 0.82
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	0	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	32.5 × 23.5	22.0	0.0
3	32.5 × 23.5	56.0	0.0
4	52.5 × 42.5	90.0	0.0
5	32.5 × 42.5	144.0	0.0
6	20.5 × 15.5	0.0	17.5
7	32.5 × 23.5	22.0	27.0
8	15.5 × 20.5	56.0	28.5
9	15.5 × 20.5	73.0	28.5
10	20.5 × 15.5	0.0	35.0
11	25.5 × 25.5	90.0	44.0
12	25.5 × 25.5	151.0	44.0
13	32.5 × 23.5	117.0	45.0
14	25.5 × 25.5	0.0	53.0
15	25.5 × 25.5	63.0	53.0
16	32.5 × 23.5	28.0	54.0
17	32.5 × 42.5	90.0	71.0
18	52.5 × 42.5	124.0	71.0
19	20.5 × 15.5	0.0	81.0
20	42.5 × 32.5	23.0	81.0
21	20.5 × 15.5	68.0	81.0
22	20.5 × 15.5	0.0	98.0
23	20.5 × 15.5	68.0	98.0

Layout 10 182.5 × 129.5 cm gap 3.5 cm 23 frames



Outer size	182.5 × 129.5 cm
Nominal gap	3.5 cm
Actual gaps	3.5 – 7.5 cm
Max. deviation	4.0 cm
Mean deviation	0.86 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, not touching, min. dist. 84 cm
Entropy	0.82
mix / variety	0.83 / 0.87
stagger / lines	0.57 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	7	9
15.5×20.5	P	1	2
32.5×42.5	P	2	2
42.5×32.5	L	0	1
32.5×23.5	L	4	5
23.5×32.5	P	1	1
25.5×25.5		5	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 42.5	56.7	0.0
3	52.5 × 42.5	93.3	0.0
4	32.5 × 42.5	150.0	0.0
5	25.5 × 25.5	0.0	46.0
6	25.5 × 25.5	67.0	46.0
7	20.5 × 15.5	133.0	46.0
8	25.5 × 25.5	157.0	46.0
9	32.5 × 23.5	30.0	47.0
10	32.5 × 23.5	97.0	47.0
11	15.5 × 20.5	135.5	67.0
12	25.5 × 25.5	157.0	75.0
13	20.5 × 15.5	0.0	76.0
14	32.5 × 23.5	70.0	76.0
15	20.5 × 15.5	107.5	76.0
16	42.5 × 52.5	24.0	77.0
17	20.5 × 15.5	133.0	93.0
18	20.5 × 15.5	0.0	95.0
19	23.5 × 32.5	106.0	97.0
20	25.5 × 25.5	157.0	104.0
21	32.5 × 23.5	70.0	106.0
22	20.5 × 15.5	0.0	114.0
23	20.5 × 15.5	133.0	114.0

Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Layout 11 184.5 × 113.5 cm gap 1.5 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	184.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 5.0 cm
Max. deviation	3.5 cm
Mean deviation	0.86 cm
Border notches	left 4 cm
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, not touching, min. dist. 79 cm
Entropy	0.78
mix / variety	0.77 / 0.82
stagger / lines	0.52 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	8	9
15.5×20.5	P	1	2
32.5×42.5	P	2	2
42.5×32.5	L	0	1
32.5×23.5	L	4	5
23.5×32.5	P	0	1
25.5×25.5		5	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	25.5 × 25.5	54.8	0.0
3	32.5 × 23.5	82.5	0.0
4	32.5 × 23.5	117.2	0.0
5	32.5 × 42.5	152.0	0.0
6	20.5 × 15.5	57.2	27.0
7	20.5 × 15.5	82.5	27.0
8	20.5 × 15.5	105.9	27.0
9	20.5 × 15.5	129.2	27.0
10	32.5 × 23.5	0.0	44.0
11	20.5 × 15.5	36.5	44.0
12	32.5 × 23.5	81.0	44.0
13	20.5 × 15.5	117.5	44.0
14	42.5 × 52.5	142.0	44.0
15	15.5 × 20.5	61.0	45.5
16	25.5 × 25.5	34.0	61.0
17	25.5 × 25.5	115.0	61.0
18	32.5 × 42.5	0.0	71.0
19	52.5 × 42.5	61.0	71.0
20	25.5 × 25.5	34.0	88.0
21	25.5 × 25.5	115.0	88.0
22	20.5 × 15.5	142.0	98.0
23	20.5 × 15.5	164.0	98.0

Layout 12 177.5 × 118.5 cm gap 2.5 cm 22 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

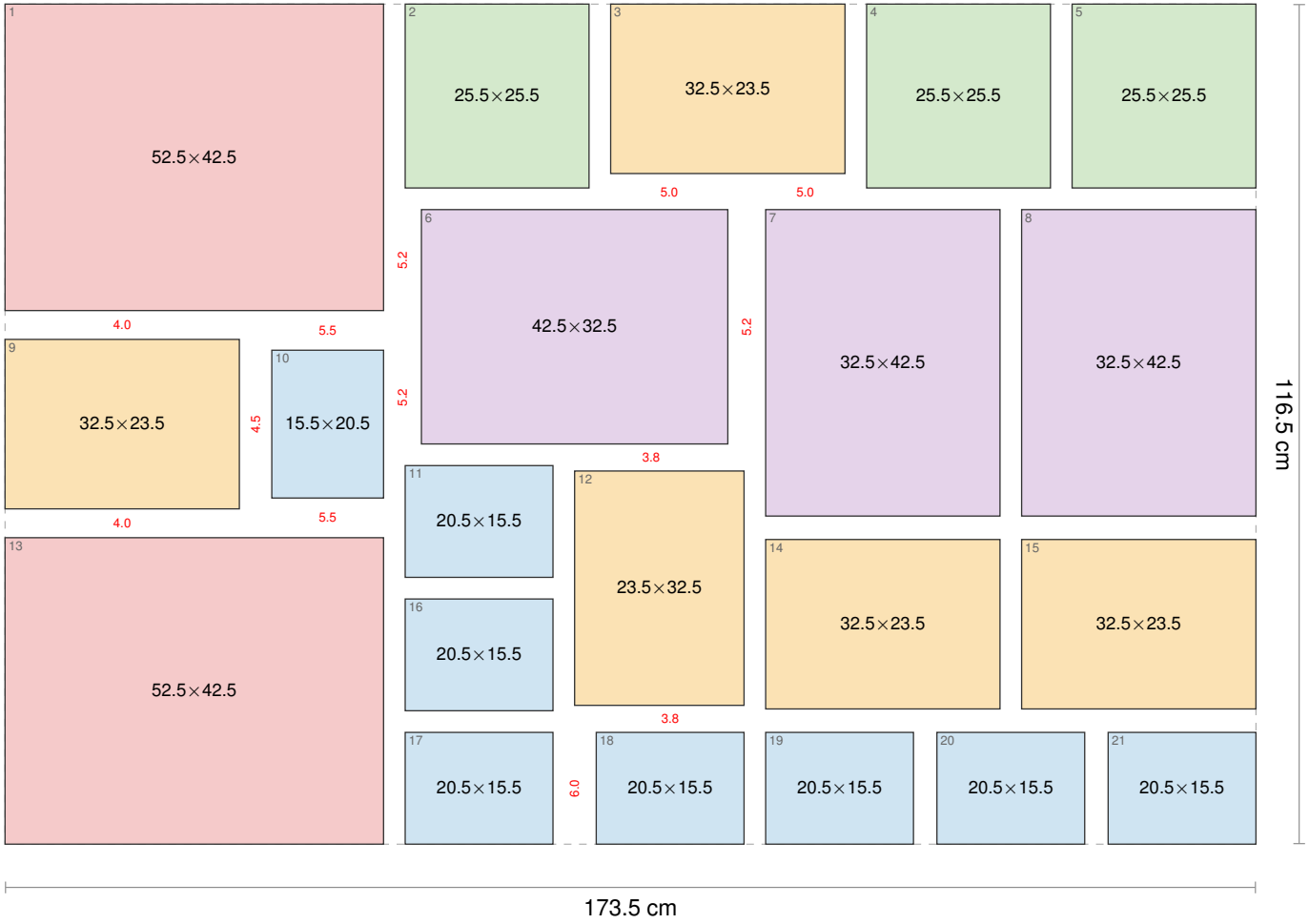
Outer size	177.5 × 118.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.5 cm
Max. deviation	3.0 cm
Mean deviation	0.83 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	22 of 28
Large frames	3, not touching, min. dist. 74 cm
Entropy	0.72
mix / variety	0.85 / 0.70
stagger / lines	0.58 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	2	5
	23.5×32.5	P	1	1
	25.5×25.5		2	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	35.0	0.0
3	32.5 × 42.5	90.0	0.0
4	52.5 × 42.5	125.0	0.0
5	15.5 × 20.5	0.0	45.0
6	15.5 × 20.5	19.0	45.0
7	20.5 × 15.5	61.0	45.0
8	20.5 × 15.5	85.0	45.0
9	20.5 × 15.5	109.0	45.0
10	20.5 × 15.5	133.0	45.0
11	20.5 × 15.5	157.0	45.0
12	20.5 × 15.5	38.0	47.5
13	42.5 × 32.5	106.5	65.0
14	25.5 × 25.5	152.0	65.0
15	42.5 × 52.5	61.0	66.0
16	32.5 × 23.5	0.0	68.0
17	20.5 × 15.5	36.5	68.0
18	23.5 × 32.5	35.0	86.0
19	25.5 × 25.5	152.0	93.0
20	32.5 × 23.5	0.0	95.0
21	20.5 × 15.5	106.0	103.0
22	20.5 × 15.5	129.0	103.0

Layout 13 173.5 × 116.5 cm gap 3 cm 21 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

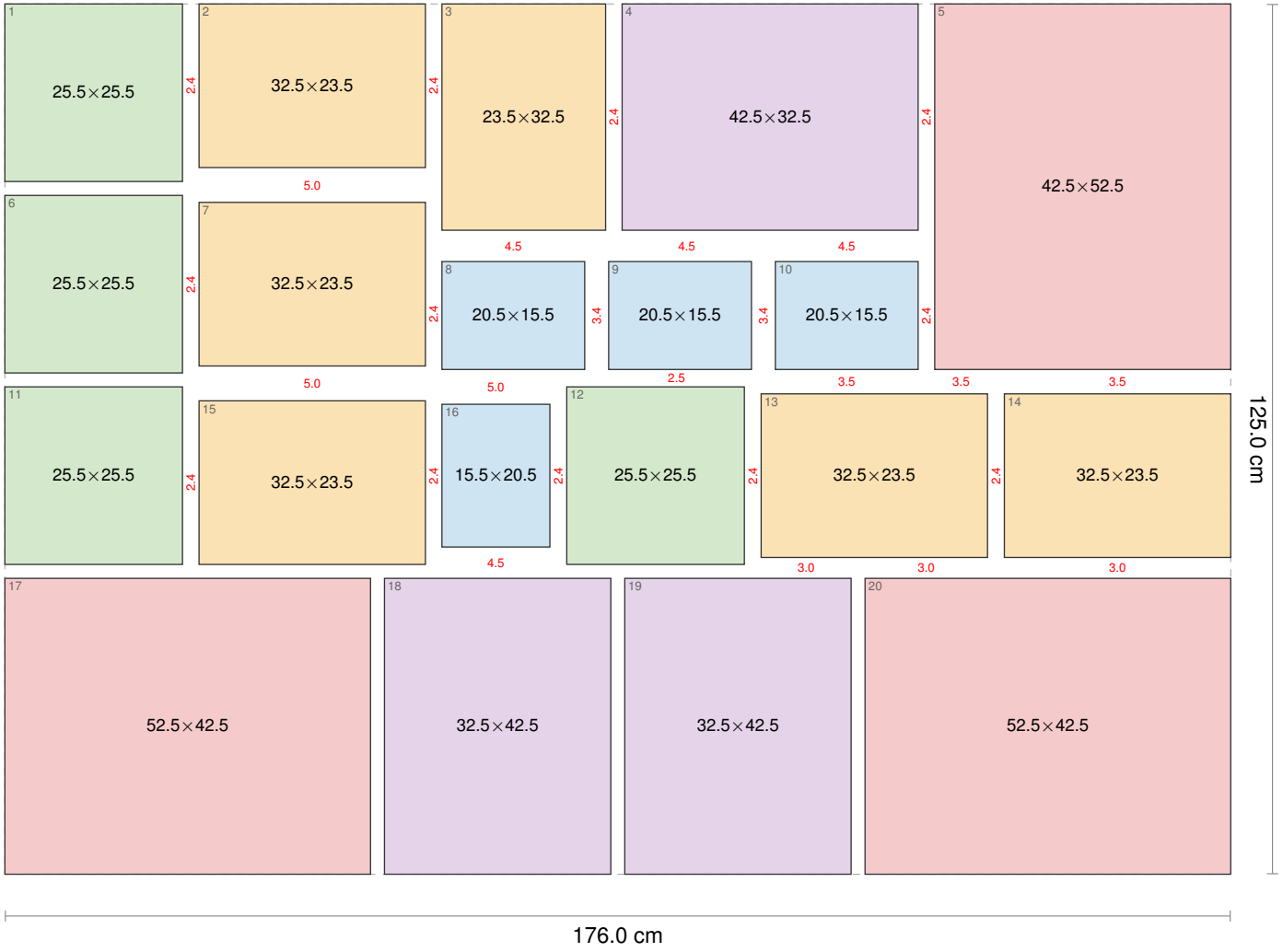
Outer size	173.5 × 116.5 cm
Nominal gap	3 cm
Actual gaps	3.0 – 6.0 cm
Max. deviation	3.0 cm
Mean deviation	0.63 cm
Border notches	bottom 6 cm
Corners filled	4 / 4
Frames used	21 of 28
Large frames	2, not touching, min. dist. 74 cm
Entropy	0.79
mix / variety	0.84 / 0.75
stagger / lines	0.56 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		3	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	25.5 × 25.5	55.5	0.0
3	32.5 × 23.5	84.0	0.0
4	25.5 × 25.5	119.5	0.0
5	25.5 × 25.5	148.0	0.0
6	42.5 × 32.5	57.8	28.5
7	32.5 × 42.5	105.5	28.5
8	32.5 × 42.5	141.0	28.5
9	32.5 × 23.5	0.0	46.5
10	15.5 × 20.5	37.0	48.0
11	20.5 × 15.5	55.5	64.0
12	23.5 × 32.5	79.0	64.8
13	52.5 × 42.5	0.0	74.0
14	32.5 × 23.5	105.5	74.2
15	32.5 × 23.5	141.0	74.2
16	20.5 × 15.5	55.5	82.5
17	20.5 × 15.5	55.5	101.0
18	20.5 × 15.5	82.0	101.0
19	20.5 × 15.5	105.5	101.0
20	20.5 × 15.5	129.2	101.0
21	20.5 × 15.5	153.0	101.0

Layout 14 176.0 × 125.0 cm gap 2 cm 20 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

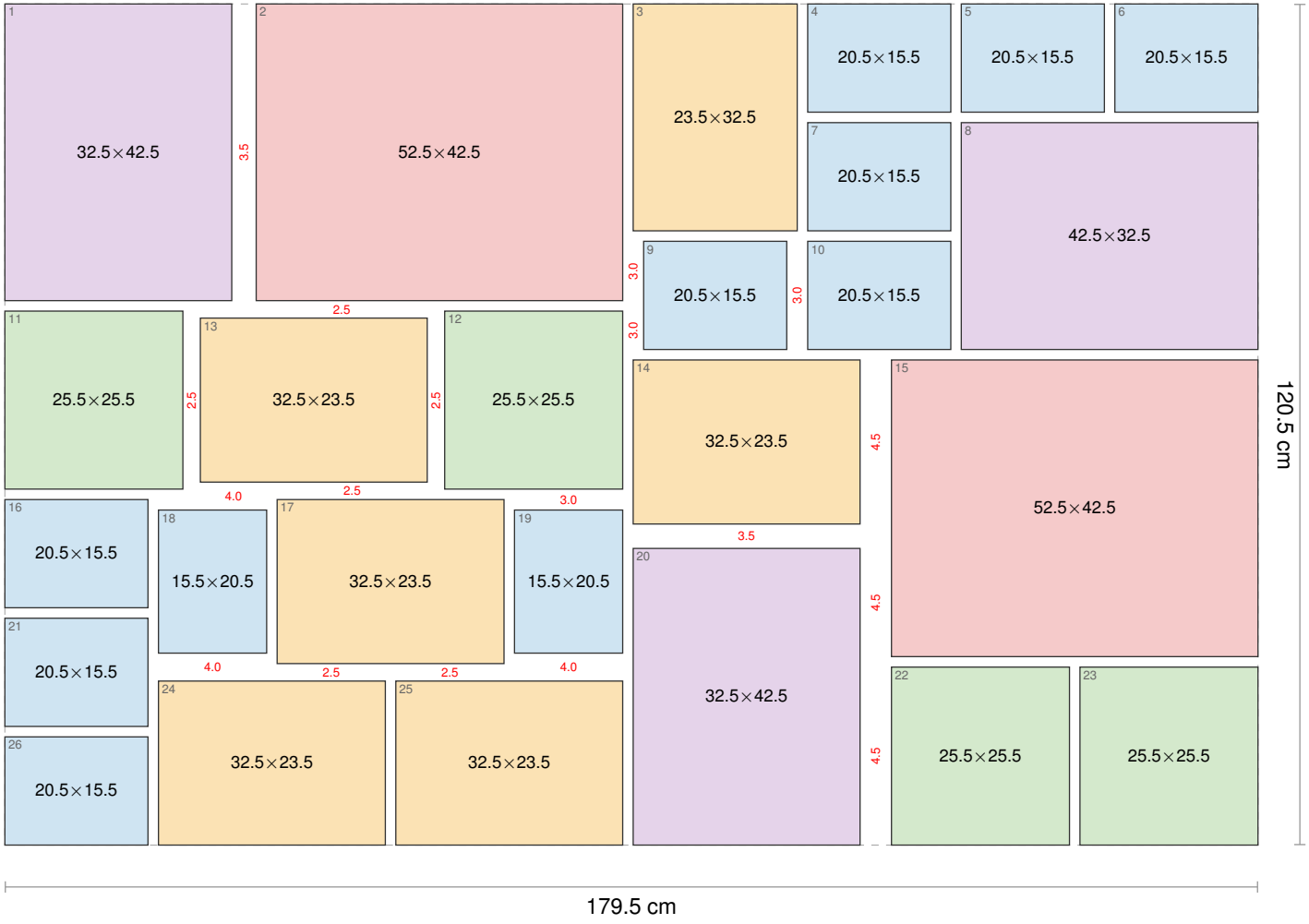
Outer size	176.0 × 125.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 5.0 cm
Max. deviation	3.0 cm
Mean deviation	0.91 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	20 of 28
Large frames	3, not touching, min. dist. 78 cm
Entropy	0.77
mix / variety	0.92 / 0.85
stagger / lines	0.56 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	3	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	27.9	0.0
3	23.5 × 32.5	62.8	0.0
4	42.5 × 32.5	88.6	0.0
5	42.5 × 52.5	133.5	0.0
6	25.5 × 25.5	0.0	27.5
7	32.5 × 23.5	27.9	28.5
8	20.5 × 15.5	62.8	37.0
9	20.5 × 15.5	86.7	37.0
10	20.5 × 15.5	110.6	37.0
11	25.5 × 25.5	0.0	55.0
12	25.5 × 25.5	80.7	55.0
13	32.5 × 23.5	108.6	56.0
14	32.5 × 23.5	143.5	56.0
15	32.5 × 23.5	27.9	57.0
16	15.5 × 20.5	62.8	57.5
17	52.5 × 42.5	0.0	82.5
18	32.5 × 42.5	54.5	82.5
19	32.5 × 42.5	89.0	82.5
20	52.5 × 42.5	123.5	82.5

Layout 15 179.5 × 120.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

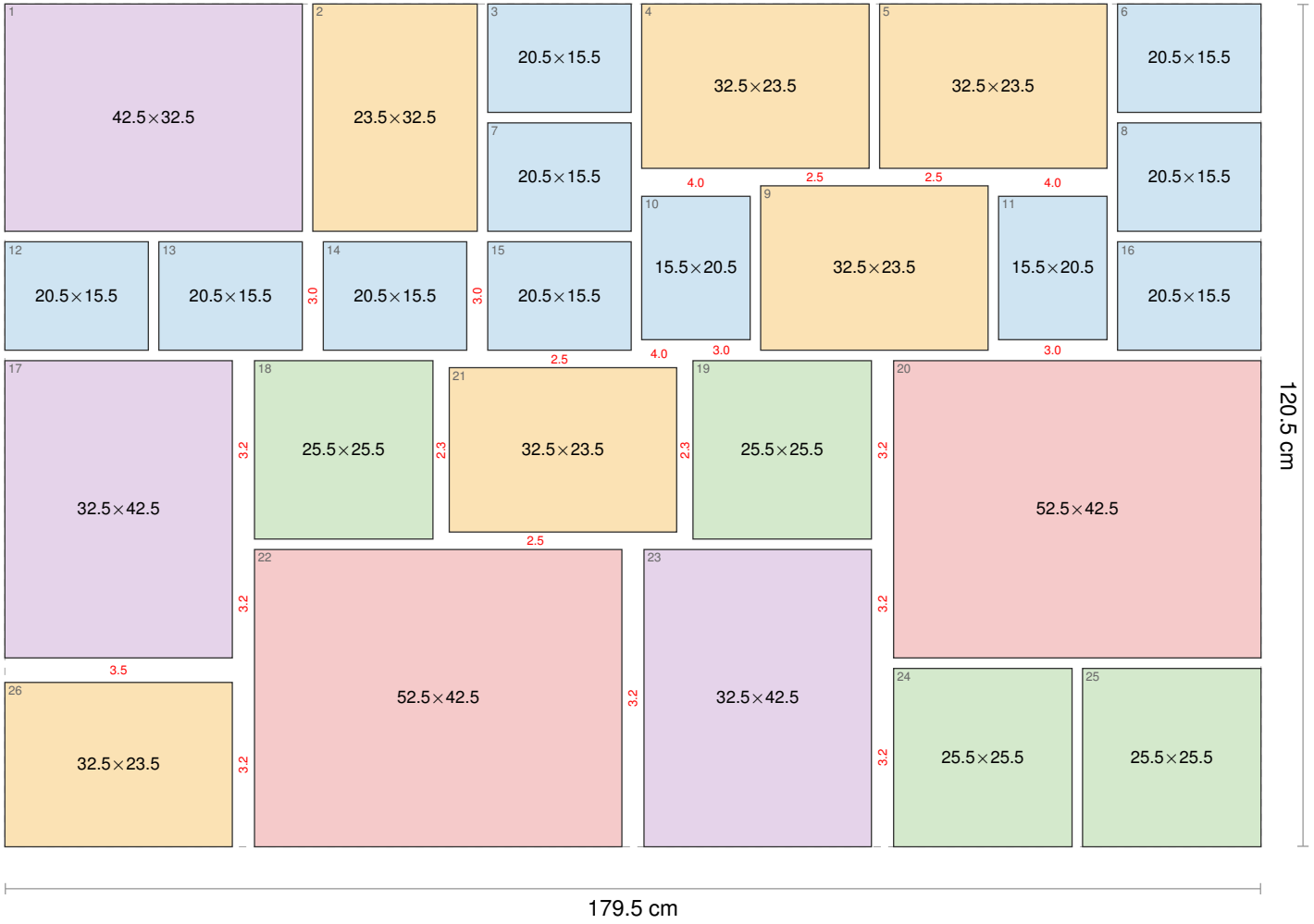
Outer size	179.5 × 120.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.5 cm
Max. deviation	3.0 cm
Mean deviation	0.61 cm
Border notches	top 4 cm, bottom 4 cm
Corners filled	4 / 4
Frames used	26 of 28
Large frames	2, not touching, min. dist. 104 cm
Entropy	0.77
mix / variety	0.83 / 0.73
stagger / lines	0.54 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	36.0	0.0
3	23.5 × 32.5	90.0	0.0
4	20.5 × 15.5	115.0	0.0
5	20.5 × 15.5	137.0	0.0
6	20.5 × 15.5	159.0	0.0
7	20.5 × 15.5	115.0	17.0
8	42.5 × 32.5	137.0	17.0
9	20.5 × 15.5	91.5	34.0
10	20.5 × 15.5	115.0	34.0
11	25.5 × 25.5	0.0	44.0
12	25.5 × 25.5	63.0	44.0
13	32.5 × 23.5	28.0	45.0
14	32.5 × 23.5	90.0	51.0
15	52.5 × 42.5	127.0	51.0
16	20.5 × 15.5	0.0	71.0
17	32.5 × 23.5	39.0	71.0
18	15.5 × 20.5	22.0	72.5
19	15.5 × 20.5	73.0	72.5
20	32.5 × 42.5	90.0	78.0
21	20.5 × 15.5	0.0	88.0
22	25.5 × 25.5	127.0	95.0
23	25.5 × 25.5	154.0	95.0
24	32.5 × 23.5	22.0	97.0
25	32.5 × 23.5	56.0	97.0
26	20.5 × 15.5	0.0	105.0

Layout 16 179.5 × 120.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	179.5 × 120.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.60 cm
Border notches	bottom 10 cm, left 4 cm
Corners filled	4 / 4
Frames used	26 of 28
Large frames	2, not touching, min. dist. 95 cm
Entropy	0.77
mix / variety	0.83 / 0.78
stagger / lines	0.49 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	42.5 × 32.5	0.0	0.0
2	23.5 × 32.5	44.0	0.0
3	20.5 × 15.5	69.0	0.0
4	32.5 × 23.5	91.0	0.0
5	32.5 × 23.5	125.0	0.0
6	20.5 × 15.5	159.0	0.0
7	20.5 × 15.5	69.0	17.0
8	20.5 × 15.5	159.0	17.0
9	32.5 × 23.5	108.0	26.0
10	15.5 × 20.5	91.0	27.5
11	15.5 × 20.5	142.0	27.5
12	20.5 × 15.5	0.0	34.0
13	20.5 × 15.5	22.0	34.0
14	20.5 × 15.5	45.5	34.0
15	20.5 × 15.5	69.0	34.0
16	20.5 × 15.5	159.0	34.0
17	32.5 × 42.5	0.0	51.0
18	25.5 × 25.5	35.7	51.0
19	25.5 × 25.5	98.3	51.0
20	52.5 × 42.5	127.0	51.0
21	32.5 × 23.5	63.5	52.0
22	52.5 × 42.5	35.7	78.0
23	32.5 × 42.5	91.3	78.0
24	25.5 × 25.5	127.0	95.0
25	25.5 × 25.5	154.0	95.0
26	32.5 × 23.5	0.0	97.0